

A MERGED RESEARCH REPORT OF THE SURREAL PROJECT

# Surcomplex Dynamics

## Linearization Thresholds, Periods, and Normal Forms

21 September 2026

### Abstract

Seven independently written manuscripts developed local and global dynamics over complex Hahn fields  $K_\Gamma = \mathbb{C}((t^\Gamma))$  realized inside  $\mathbf{No}[i]$ , with  $\Gamma$  a set-sized ordered abelian group of arbitrary rank. This report is their *union*. Every result any one of them reached survives; results all of them reached are printed once; results proved by genuinely different routes keep each route, marked; and every honest limitation is preserved.

Six of the seven are germ and fixed-point dynamics: multipliers, return iterates, periods, Hamiltonians at the origin of a phase halo. The seventh is the *quasi-periodic* half of the same subject — a cohomological equation on a real torus whose frequency vector has surreal components, with divisors indexed by an integer lattice — and it is not a duplicate of any of the six: none of them contains a resonance lattice, a cohomological equation, a Fourier divisor or a conjugacy to a constant vector field.

Five inequivalent conditions travelled under the single word *resonance* across the seven sources, and six inequivalent conditions travelled under the words *small divisor*; worse, two families of sources used the word *radius* with opposite monotonicity. [Section 2](#) therefore opens the report with a notion table that gives each variant its own name and symbol, and every threshold statement below names which variant its hypothesis uses and in which coefficient category it holds. A threshold proved under one notion is never quoted under another.

The mathematical content: a sharp five-row coefficient-category classification of universal linearization for an exact nonresonant diagonal unitary multiplier, keyed to one exponential divisor rate, with necessity witnessed at a single infinitesimal weight; an exact radius-collapse law  $\text{rad}(h_{k,1}) = R e^{-(k+1)\sigma}$  in the *drifted*-multiplier category, which settles the common-domain germ question there and leaves it open for a fixed multiplier; an exact maximal centered valuation ball read off a *finite return iterate*, with its shell cycles, its cancellation phase diagram and its common-domain lifting; an algebraicity dichotomy and an exact coefficientwise monodromy exponent for parabolic Euler maps, whose infinitesimal part no finite branched cover removes; a complete period classification, explicit moduli and beyond-all-orders invisible moduli for global near-identity maps; a gauge-normalized symplectic Hamiltonian normal form, its full Poisson centralizer, and support-certified domains reaching infinite phase magnitudes; and, on the quasi-periodic side, the theorem that a frequency vector with  $d$  surreal components has **at most**  $d$  arithmetically visible divisor valuations, identified by a canonical descending flag of integer resonance lattices, together with the exact arithmetic criterion for universal solvability of the cohomological equation on one fixed analytic strip, an arbitrary-order analytic lifting obstruction, a unique infinitesimal conjugacy to a constant vector field proved by marked support words rather than by sequential valuation convergence, and its canonical invariant density.

**Status.** This is a merged report assembled from seven AI-assisted, unrefereed research drafts. It is not refereed and not proof-assistant verified; no Lean or other formal development corresponds to any theorem below. Priority is not certified for any statement, and no named published conjecture is claimed solved. The seven sources' own verification programs supply finite exact symbolic checks only; [section 23](#) records exactly what each one does and does not establish, and [section 25](#) carries every limitation stated by every source.

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# 1 What this report is, and how to read it

## 1.1 Seven manuscripts, one body of machinery

The seven source manuscripts were all written independently on 21 September 2026 against a pinned repository snapshot: sources 01–06 against commit `39f2be6667ade51bca2b45daa47e289d69c09764` and source 07 against commit `aa846271b4dcae2c055b216126a87210292ec19b`. They share a single body of machinery — the set-sized Hahn workspace inside  $\mathbf{No}[i]$ , strong summability by well-ordered support and local finiteness, the positive-support lemma of Neumann in finite-word form, halo evaluation by coefficientwise double Taylor sums, Taylor substitution and its near-identity inverse — and each of the seven states that machinery from scratch. The positive-support lemma is stated in all seven. Between 45 and 55 per cent of the combined text of sources 01–06 is the same theory, and source 07 restates the same foundation again.

Sources 01–06 are germ and fixed-point dynamics and were merged first; source 07 is the quasi-periodic half of the subject and was integrated afterwards, as [sections 16 to 18](#). It is genuinely additive rather than a seventh treatment of the same material: grepping sources 01–06 for a resonance lattice, a cohomological equation, a Fourier divisor or a mean-zero conjugacy to a constant vector field returns nothing, and conversely source 07 contains no multiplier, no return iterate and no period. What the two halves do share is this machinery, which is therefore stated once for both, and the coefficient-category architecture of [remark 2.2](#), which is stated once and instantiated twice ([remark 17.2](#)).

They also disagree, in the specific and dangerous way that independent runs at one *foundation* disagree: not by contradicting each other’s theorems, but by attaching the same words to different hypotheses. That is what [section 2](#) exists to prevent.

*Convention 1.1* (The merge rule). A merge is a union, not a selection.

- (i) A statement proved by every source is printed once.
- (ii) A statement proved by genuinely different routes keeps every route, explicitly marked, because the routes carry different resources and different by-products.
- (iii) Anything one source reached that the others did not survives in full.
- (iv) Every honest limitation survives. [Section 25](#) is the consolidated record; dropping one of its items would silently upgrade the report, and that is the worst failure available in an assembly of this kind.
- (v) Nothing is strengthened beyond what a source proves. Where a source records a hypothesis as load-bearing, the hypothesis travels with the statement everywhere it is quoted.

## 1.2 Discipline inherited from companion reports

Three constructions here are inherited rather than restated, and are named by the title and repository path of the companion report that owns them. [Section 22](#) gives the full inheritance ledger. In brief:

- the no-topological-convergence discipline — every set of surcomplex numbers is closed and discrete in the fine order topology, so  $\sum_n t^n = (1 - t)^{-1}$  is a Hahn identity about a summable family and never a limit of partial sums — is the foundation fixed by *Surcomplex analysis: holomorphic functions on  $K = \mathbf{No}[i]$*  (`docs/surcomplex/analysis/`). All seven sources restate it; we cite it and state it once, in [definition 3.1](#).
- the coefficient-ring taxonomy is the four-ring table of *Infinitesimal Analytic Geometry over the Surcomplex Numbers* (`docs/surcomplex/analytic-geometry/`), whose names we adopt verbatim. We add exactly two rows to it, and they are new *rows*, not new rings: see [section 3.2](#).

- the warning that a Hahn constant for a coordinate derivative need not be a constant for the scalar Berarducci–Mantova derivation is the cross-category warning of *Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers* ([docs/surcomplex/differential-equations/](#)). All seven sources derive it independently — source 07 in the form that its torus derivatives are external derivations annihilating  $K_\Gamma$  — and we adopt that report’s wording once, in [warning 3.2](#).

### 1.3 What is offered as new, and what is not

Not new, and credited as classical throughout: the positive-support lemma (Neumann, via Higman’s word well-quasi-order) [2, 3]; algebraic closure of a maximally complete Hahn field [4]; formal logarithms of tangent-to-identity maps, formal flow embedding and semisimple–unipotent splitting [6, 7]; the exponential correspondence for strongly summable derivations of generalized-series algebras [8]; the ordinary homological equation and the Brjuno–Yoccoz theory of small divisors [11, 13, 14, 12]; rank-one ultrametric linearization and its disk estimates [20, 21]; the classical iterative residue and modified-equation coefficients [22, 10]; and classical formal Birkhoff normalization with its convergence–divergence theory [23, 24].

Offered for scrutiny as candidate contributions, in the exact form proved below and with no priority claim: the coefficient-category classification [theorem 6.1](#) with its first-weight necessity mechanism; the exact drifted radius-collapse law [theorem 7.2](#) and the trichotomy [theorem 7.5](#) it yields; the exact finite-return threshold [theorem 8.5](#) with [theorems 8.7, 9.1 and 12.1](#); the algebraicity dichotomy [theorem 13.15](#) with the monodromy formula [theorem 13.11](#) and its torsion obstruction; the period classification [theorem 14.13](#) with [theorems 14.16, 14.18 and 14.26](#); the Hamiltonian package [theorems 15.7, 15.12, 15.16 and 15.21](#); and, from source 07, the finite resonance flag with its single common reciprocal support [theorem 16.5](#), the exact fixed-strip solvability criterion [theorem 17.3](#), the arbitrary-order analytic lifting obstruction [theorem 17.7](#), and the torus normal-form package [theorems 18.1, 18.4 and 18.8](#) and [proposition 18.6](#). Each source’s own bounded novelty assessment is reproduced in [section 25](#). Source 07 adds to the classical list the ordinary Fourier division mechanism with its subexponential criterion, the Liouville construction method, and positive-support Hahn inversion [17, 18, 2, 4], none of which it claims.

## 2 The notion table: five resonances, six small divisors, two radii

This section is the load-bearing apparatus of the merge. Read it before anything else. Every threshold statement in this report carries tags from the tables below, and [section 2.5](#) lists, theorem by theorem, which tags it actually uses.

### 2.1 Why unifying the notation without this table would invert results

The seven sources use the word *resonance* for five inequivalent conditions and the words *small divisor* for six, and two groups of sources use the word *radius* with opposite monotonicity. A merge that harmonized the symbols without separating the notions would produce statements that read correctly and are false: it would apply an exact maximal-ball formula to a multiplier its source explicitly excludes, attach an arithmetic threshold to a theorem proved with no arithmetic hypothesis at all, and reverse the direction of every domain inequality in half the report.

## 2.2 Four notions of resonance

| Tag                                                             | Condition                                                                                                                                                                                                                                                                                   | Where it is the hypothesis, and what it is not                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RES <sub>q</sub><br><i>residue-<br/>resonant of<br/>order q</i> | $\lambda \in K_{\Gamma}^{\times}$ with $v(\lambda) = 0$ , $\lambda$ is <b>never</b> a root of unity, and its residue $\alpha = \bar{\lambda} \in \mathbb{C}^{\times}$ is a root of unity of exact order $q \geq 1$ . Then $c = \lambda^q - 1 \neq 0$ and $\delta = v(c) > 0$ .              | This is the hypothesis of the exact finite-return threshold <a href="#">theorem 8.5</a> , the shell theorem <a href="#">theorem 8.7</a> , the phase diagram <a href="#">theorem 12.1</a> and the common-domain lifting <a href="#">theorem 9.1</a> . It is <b>not</b> $\lambda$ itself being a root of unity — that case is excluded — and it is <b>not</b> a divisor size condition of any kind.                                                                    |
| MR / NR<br><i>multiplier-<br/>resonant</i>                      | For $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ with $ \lambda_j  = 1$ : $\lambda^\alpha = \lambda_j$ for some $ \alpha  \geq 2$ and some $j$ . NR is its negation, nonresonance. In one variable NR reads $\lambda^n \neq \lambda$ for $n \geq 2$ .                               | This is the hypothesis of <a href="#">theorems 6.1, 6.3 and 7.5</a> and of everything built on the homological operator $\mathcal{L}$ of <a href="#">section 5</a> . The multiplier is <i>exactly</i> $\Lambda$ in <a href="#">theorem 6.1</a> and <i>drifted</i> in <a href="#">theorem 7.2</a> ; see <a href="#">warning 2.6</a> .                                                                                                                                 |
| FR<br><i>frequency-<br/>resonant</i>                            | For a frequency vector $\omega \in \mathbb{C}^d$ of ordinary complex constants: $\omega \cdot k = 0$ for some $0 \neq k \in \mathbb{Z}^d$ ; equivalently $\omega \cdot (\alpha - \beta) = 0$ for a monomial $q^\alpha p^\beta$ . Its negation is the hypothesis of the Hamiltonian package. | FR is the exact Hamiltonian analogue of MR, and its negation is what makes the kernel of $L = \{\cdot, H_0\}$ exactly the diagonal monomials $\alpha = \beta$ . It is a condition on <i>ordinary complex</i> numbers, never on Hahn or surreal data.                                                                                                                                                                                                                 |
| PAR<br><i>parabolic</i>                                         | The leading multiplier is exactly 1, perturbed infinitesimally: $F = z + h$ with $v(h) > 0$ , or $G_\varepsilon = w + \varepsilon V(w)$ with $\varepsilon = t^\delta$ , $\delta > 0$ .                                                                                                      | This is the hypothesis of the whole of <a href="#">sections 13 and 14</a> . <b>No divisor hypothesis of any kind is imposed anywhere under PAR.</b> Source 04 says plainly that the word in its own title is not a small-divisor condition, and source 05 says it has no resonance and no small-divisor condition at all because its maps are unipotent. A reader who reads PAR as a small-divisor hypothesis will misattribute every theorem in those two sections. |

| Tag                                                              | Condition                                                                                                                                                                                                                                                                                                  | Where it is the hypothesis, and what it is not                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QR<br><i>lattice-<br/>resonant, for<br/>a Hahn<br/>frequency</i> | For a frequency vector $\Upsilon \in F_{\Gamma}^d = \mathbb{R}((t^{\Gamma}))^d$ whose components are <b>Hahn (hence surreal) scalars</b> : the exact resonance lattice $N_{\Upsilon} = \{k \in \mathbb{Z}^d : k \cdot \Upsilon = 0\}$ is nonzero. Its negation $\neg\text{QR}$ is $N_{\Upsilon} = \{0\}$ . | The flag <a href="#">theorem 16.5</a> and the linear theorem <a href="#">theorem 17.3</a> <i>permit</i> QR; the torus normal form <a href="#">theorem 18.1</a> and all of <a href="#">section 18</a> require $\neg\text{QR}$ . This is <b>exact vanishing of a Hahn scalar</b> , never smallness of anything. It is <b>not</b> FR: that one is a condition on ordinary complex $\omega \in \mathbb{C}^d$ with divisors graded by Taylor degree, whereas QR concerns surreal components with divisors $k \cdot \Upsilon$ graded by the Fourier index $ k $ and of <i>nonzero</i> valuation ( <a href="#">warning 2.9</a> ). |

*Warning 2.1* (The four germ notions do not intersect in the way the word suggests).  $\text{RES}_q$  requires  $\lambda$  *not* to be a root of unity while its residue is one; MR is about exact equality of ordinary unit multipliers; FR is a linear condition on an integer vector; PAR is about the multiplier being 1 up to an infinitesimal. In particular:

- the exact maximal-ball formula of [theorem 8.5](#) needs  $\text{RES}_q$  and must never be applied under PAR or under MR;
- the resonant projection and the retained resonant Hamiltonian algebra of [section 15](#) need FR;
- the parabolic results of [sections 13](#) and [14](#) need none of the three others, and acquire no arithmetic hypothesis by being printed next to them;
- the finite-order splitting of [theorem 19.1](#) needs  $\Lambda^q = I$  — the multiplier itself of finite order — which is *neither*  $\text{RES}_q$  (where  $\lambda$  is not a root of unity) nor mere MR (where resonances exist but need not have finite order).

The fifth notion QR intersects none of these four: it is a condition on an integer lattice attached to a vector of *Hahn* scalars, it has no multiplier and no fixed point, and no theorem stated under  $\text{RES}_q$ , MR, FR or PAR is quoted under it or vice versa.

## 2.3 Six notions of small divisor

The first five concern *ordinary complex* divisors. Every nonzero ordinary complex number, however tiny, has Hahn valuation zero ([remark 2.3](#)), so none of the five is visible to the Hahn valuation, and a proof that keeps only the valuation cannot distinguish them. The sixth,  $\text{SD}_{\text{ss}}$ , is different in one respect and identical in another: its divisors  $k \cdot \Upsilon$  are Hahn scalars of *nonzero* valuation, but the arithmetic condition it imposes is again a condition on ordinary real numbers only — the leading real forms — so that here too the valuation cannot see the arithmetic ([warning 2.11](#)).

| Tag                                                          | Condition                                                                                                                                                                                                                                                                       | Scope                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SD <sub>0</sub><br><i>support only</i>                       | No arithmetic condition whatsoever.<br>Control is by well-ordered positive support and local finiteness alone.                                                                                                                                                                  | Sources 01, and the polynomial and formal rows of <a href="#">theorem 6.1</a> , and the polynomial-layer case (a) of <a href="#">theorem 15.7</a> . Source 01 insists on the point: the Schröder denominators $\nu - \nu^n$ have nonzero ordinary constant terms $\alpha - \alpha^n$ , which may be arbitrarily small in the ordinary modulus and still all have Hahn valuation zero, so no Diophantine, Brjuno or Siegel condition on surreal data appears in <a href="#">section 8</a> at all.                                                                                                                               |
| SD <sub>dr</sub><br><i>drifted exponential rate</i>          | $\sigma(\lambda) = \limsup_{n \rightarrow \infty} \frac{\log^+  \lambda^n - \lambda ^{-1}}{n} \in [0, \infty]$ ,<br><b>together with multiplier drift</b><br>$F'(0) = \lambda + u$ , $v(u) > 0$ .                                                                               | Source 02. The drift is not decoration: it is what creates the higher reciprocal jets $c_{n,k}$ , $k \geq 1$ , whose poles of order $k + 1$ drive the collapse law. <a href="#">Warning 2.6</a> records that removing drift removes the mechanism.                                                                                                                                                                                                                                                                                                                                                                             |
| SD <sub>ex</sub><br><i>exact-multiplier exponential rate</i> | $\tau(\Lambda) = \limsup_{m \rightarrow \infty} \frac{1}{m} \log M_m(\Lambda)$ ,<br>$M_m = \max(1, \max_{ \alpha =m, j}  \lambda^\alpha - \lambda_j ^{-1})$ ,<br>with the multiplier <b>exactly</b> $\Lambda$ : every $f_\gamma$ has $f_\gamma(0) = 0$ and $df_\gamma(0) = 0$ . | Source 03. No infinitesimal linear part is permitted in <a href="#">theorem 6.1</a> ; infinitesimal linear terms <i>are</i> permitted in the identity-reducing flow theorem <a href="#">theorem 4.3</a> .                                                                                                                                                                                                                                                                                                                                                                                                                      |
| SD <sub>fr</sub><br><i>frequency exponential rate</i>        | $\Theta(\omega) = \limsup_{D \rightarrow \infty} M_D(\omega)^{1/D}$ , $M_D = \max(1, \max_{ \alpha + \beta =D, \alpha \neq \beta}  \omega \cdot (\alpha - \beta) ^{-1})$ ,<br>so $\Theta \in [1, \infty]$ .                                                                     | Source 06, where this invariant is called $\Lambda(\omega)$ ; we write $\Theta(\omega)$ to free the symbol $\Lambda$ for the diagonal multiplier matrix of sources 02 and 03. Note the <b>different normalization</b> : $\sigma$ and $\tau$ are logarithmic rates in $[0, \infty]$ while $\Theta$ is their exponential. $\Theta(\omega) < \infty$ is the exact analogue of $\tau < \infty$ ; $\Theta(\omega) = 1$ would be the analogue of $\tau = 0$ and is <b>nowhere required</b> , because source 06's entire coefficients already have the common domain $\mathbb{C}^{2d}$ and so lose nothing to a finite radius factor. |
| SD <sub>∅</sub><br><i>none</i>                               | No divisor condition, and no divisor family to impose one on.                                                                                                                                                                                                                   | Sources 04 and 05. Under PAR the multiplier is $1 + \mu\varepsilon$ or 1, so $\lambda^n - \lambda$ is automatically nonzero for every $n \geq 2$ , formal uniqueness is free, and the entire difficulty is well-ordered support.                                                                                                                                                                                                                                                                                                                                                                                               |

| Tag                                                            | Condition                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Scope                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SD <sub>ss</sub><br><i>stratified</i><br><i>subexponential</i> | For each flag stratum $\Sigma_j$ of <a href="#">theorem 16.5</a> and every real $\varepsilon > 0$ there is $C_{j,\varepsilon} > 0$ with $ \mathbf{a}_j(k) ^{-1} \leq C_{j,\varepsilon} e^{\varepsilon k }$ for $k \in \Sigma_j$ , where $\mathbf{a}_j(k) = k \cdot \varpi_{\gamma_j}$ is the leading <i>real</i> form of the divisor $k \cdot \Upsilon$ on that stratum; equivalently $\limsup_{k \in \Sigma_j} \log^+(1/ \mathbf{a}_j(k) )/ k  = 0$ for every $j$ . | Source 07, where it is called (SS). It is the vanishing-rate condition of <a href="#">remark 2.2</a> on a <i>fourth</i> divisor family ( <a href="#">remark 17.2</a> ), imposed stratum by stratum on the finitely many leading real forms. It therefore corresponds to $\tau = 0$ and $\sigma = 0$ , <b>not</b> to $\tau < \infty$ ; and it says nothing whatever about the surreal size $v(k \cdot \Upsilon) = \gamma_j$ of the divisor itself ( <a href="#">warning 2.11</a> ). |

*Remark 2.2* (One invariant, three names, one category conclusion). Fix a family  $(d_i)_{i \in I}$  of nonzero ordinary complex divisors graded by a degree  $\deg i \in \mathbb{N}$ , and set

$$\kappa = \limsup_{m \rightarrow \infty} \frac{1}{m} \log \max \left( 1, \max_{\deg i = m} |d_i|^{-1} \right).$$

Then  $\sigma$ ,  $\tau$  and  $\log \Theta$  are this one quantity evaluated on three different divisor families:  $\lambda^n - \lambda$  graded by  $n$ ;  $\lambda^\alpha - \lambda_j$  graded by  $|\alpha|$ ; and  $\omega \cdot (\alpha - \beta)$  graded by  $|\alpha| + |\beta|$ . The *category conclusion* is the same in all three instances, and is the single most quotable fact in this report:

| Coefficient category         | Universal criterion                  |
|------------------------------|--------------------------------------|
| polynomial coefficients      | nonresonance only (SD <sub>0</sub> ) |
| formal coefficients          | nonresonance only (SD <sub>0</sub> ) |
| radius-free convergent germs | $\kappa < \infty$                    |
| entire coefficients          | $\kappa < \infty$                    |
| one fixed unchanged polydisk | $\kappa = 0$                         |

The three sources reach this table on three divisor families, in three notations, with three different proofs of necessity. What they do *not* share is the sixth category, common-domain germs; see [section 7](#) and [question 7.10](#).

A *fourth* divisor family,  $k \cdot \Upsilon$  for a Hahn frequency vector graded by the Fourier index  $|k|$ , is added by source 07, and the first row of the table above is instantiated on it as [theorem 17.3](#): “one fixed unchanged polydisk” becomes “one fixed unchanged strip” and “ $\kappa = 0$ ” becomes (SD<sub>ss</sub>), the same vanishing rate imposed on each of the finitely many strata. The table is stated here once and instantiated exactly twice; the two instantiations, and the rows that are **not** instantiated on the quasi-periodic side, are set out in [remark 17.2](#). Reading an uninstantiated row across would assert a theorem no source proves.

*Remark 2.3* (Arithmetic invisible to the Hahn valuation). For every nonzero ordinary divisor,  $v(\lambda^\alpha - \lambda_j) = 0$  and  $v(\omega \cdot (\alpha - \beta)) = 0$ . The distinction among SD<sub>0</sub>, SD<sub>dr</sub>, SD<sub>ex</sub> and SD<sub>fr</sub> therefore comes entirely from the *ordinary complex Taylor growth inside each coefficient function*, not from the valuation. This is exactly why formal and polynomial coefficients always permit normalized linearization under nonresonance while the analytic categories have sharp arithmetic thresholds, and it is why the negative results below never prohibit a conjugacy on an infinitesimal monad: the formal-coefficient conjugacy exists and can be evaluated there ([proposition 6.14](#)).

## 2.4 Two notions of radius, running in opposite directions

| Object                                                                 | Definition                                                                                                                                        | Monotonicity and sources                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| valuation ball<br>$B_r$ , shell $S_r$                                  | $B_r = \{z \in K_\Gamma : v(z) > r\} = t^r \mathfrak{m}$ and<br>$S_r = \{z : v(z) = r\}$ , for $r \in \Gamma$ .                                   | <b>Increasing <math>r</math> makes the ball smaller.</b> Sources 01 ( $B_r$ , $r = \max(\delta - v(b_n))/(n - 1)$ ) and 04 ( $D = \{v(z) > \delta/q\}$ , so larger $\delta/q$ is a smaller disk). $S_r$ is called a bounding shell and is <b>not</b> the topological boundary (valuation balls are clopen) and <b>not</b> a circle for the ordered-surreal modulus. |
| valuation polydisk $D_\rho$                                            | $D_\rho = (t^\rho \mathcal{O}_\Gamma)^{2d} = \{z : v(z_j) \geq \rho \ \forall j\}$ .                                                              | <b>Increasing <math>\rho</math> makes the polydisk smaller;</b> $\rho < 0$ gives domains containing phase points of <i>infinite</i> magnitude. Source 06. Same direction as $B_r$ .                                                                                                                                                                                 |
| ordinary disk /<br>polydisk $\mathbb{D}_R$ ,<br>$\mathbb{D}_R^d$ , $D$ | $\mathbb{D}_R^d = \{z \in \mathbb{C}^d :  z_j  < R\}$ ; $D$ an ordinary connected open subset of $\mathbb{C}$ .                                   | <b>Increasing <math>R</math> makes the domain larger.</b> Sources 02, 03, 05, and the coefficient domains of 04 and 06. This is a domain for the ordinary <i>coefficient functions</i> , never a surcomplex ball.                                                                                                                                                   |
| Taylor radius<br>rad, rad <sub>iso</sub>                               | $\text{rad}(g)^{-1} = \limsup_n  g_n ^{1/n}$ ;<br>$\text{rad}_{\text{iso}}(g)^{-1} =$<br>$\limsup_m (\max_{ \alpha =m,j}  g_{j,\alpha} )^{1/m}$ . | <b>Larger is better.</b> Sources 02 (rad) and 03 (written $\rho$ there; we write rad <sub>iso</sub> , freeing $\rho$ for source 06's rescaling exponent).                                                                                                                                                                                                           |

*Warning 2.4* (Do not identify a valuation ball with an ordinary disk).  $B_r$  and  $\mathbb{D}_R$  are objects of different kinds and their inequalities run in opposite directions. Source 01 states the point sharply: all points of the normalized monad have ordinary residue zero, whereas the ordinary disk  $D$  of [theorem 9.1](#) contains many nonzero ordinary points;  $D$  is a domain of holomorphy for coefficient *functions* and not a claim of a larger centered surcomplex linearization ball. Likewise a Lindahl-type ultrametric disk radius in a fixed rank-one workspace and source 01's maximal centered valuation ball at arbitrary rank are different categories; see [section 22.5](#).

*Warning 2.5* (Support orientation). Throughout this report supports are well ordered in the *increasing* order of  $\Gamma$  and  $v = \min \text{supp}$ , so that  $t^\gamma = \omega^{-\gamma}$  converts increasing Hahn exponents to decreasing surreal monomials. A treatment using the decreasing-support Conway convention directly has every inequality of this report reversed. Silently merging a threshold across the two conventions inverts it.

## 2.5 Tag ledger: which notion each threshold statement uses

| Statement                                               | Resonance        | Divisor         | Coefficient category                  | Domain notion     |
|---------------------------------------------------------|------------------|-----------------|---------------------------------------|-------------------|
| <a href="#">theorem 8.5</a> exact<br>finite-return ball | RES <sub>q</sub> | SD <sub>0</sub> | formal, $\mathbb{C}[[X]]((t^\Gamma))$ | $B_r$ (valuation) |

| Statement                                                    | Resonance           | Divisor                                           | Coefficient category                           | Domain notion                                       |
|--------------------------------------------------------------|---------------------|---------------------------------------------------|------------------------------------------------|-----------------------------------------------------|
| <a href="#">theorem 8.7</a> shell cycles                     | $\text{RES}_q$      | $\text{SD}_0$                                     | ordinary polynomial<br>$P \in \mathbb{C}[X^q]$ | $S_r$ (valuation shell)                             |
| <a href="#">theorem 12.1</a> phase diagram                   | $\text{RES}_q$      | $\text{SD}_0$                                     | formal                                         | $B_r$                                               |
| <a href="#">theorem 9.1</a> common-domain lifting            | $\text{RES}_q$      | $\text{SD}_0$                                     | fixed-polydisk<br>$\mathcal{O}(D)((t^\Gamma))$ | $\mathbb{D}$ (ordinary)                             |
| <a href="#">proposition 10.1</a> no common domain            | NR                  | $\text{SD}_0$ hypothesis, Brjuno used externally  | fails radius-free germ                         | ordinary                                            |
| <a href="#">theorem 8.10</a> return perturbation certificate | $\text{RES}_q$      | $\text{SD}_0$                                     | return coefficients only                       | $B_r$                                               |
| <a href="#">theorem 6.1</a> five-row classification          | NR, exact $\Lambda$ | $\text{SD}_{\text{ex}}$                           | all five rows                                  | $\mathbb{D}_R^d$ , germ, entire, poly, formal       |
| <a href="#">lemma 5.1</a> single-step loss                   | NR                  | $\text{SD}_{\text{ex}}$                           | isotropic Taylor radius                        | $\text{rad}_{\text{iso}}$                           |
| <a href="#">lemma 5.2</a> sharpness                          | NR                  | $\text{SD}_{\text{ex}}$                           | $\mathcal{O}(\mathbb{D}_R^d)$ and entire       | $\text{rad}_{\text{iso}}$                           |
| <a href="#">theorem 6.9</a> word-depth radius bound          | NR                  | $\text{SD}_{\text{ex}}$                           | germ / fixed polydisk                          | $\text{rad}_{\text{iso}}$ , <i>lower bound only</i> |
| <a href="#">theorem 6.3</a> support-controlled lifting       | NR                  | $\text{SD}_{\text{dr}}$ allowed (drift permitted) | four rows                                      | $\mathbb{D}_R$                                      |
| <a href="#">theorem 7.2</a> exact radius collapse            | NR                  | $\text{SD}_{\text{dr}}$ , <b>drift required</b>   | germ yes, common domain no                     | rad                                                 |
| <a href="#">theorem 7.5</a> trichotomy                       | NR                  | $\text{SD}_{\text{dr}}$                           | common domain vs germ                          | $\mathbb{D}_R$                                      |
| <a href="#">theorem 7.6</a> several-variable collapse        | NR                  | $\text{SD}_{\text{dr}}$ , multiplicative detuning | polydisk germ                                  | $\text{rad}_{\text{iso}}$                           |
| <a href="#">theorem 13.15</a> algebraicity dichotomy         | PAR                 | $\text{SD}_\emptyset$                             | $\mathbb{C}(w)((t^\Gamma))$ base               | ordinary universal cover                            |
| <a href="#">theorem 13.11</a> finite-cover obstruction       | PAR                 | $\text{SD}_\emptyset$                             | rational-coefficient                           | ordinary loops                                      |
| <a href="#">theorem 13.22</a> sharp summability disk         | PAR                 | $\text{SD}_\emptyset$                             | formal Taylor family                           | $B_{\delta/q}$ (valuation)                          |
| <a href="#">theorem 14.13</a> period classification          | PAR                 | $\text{SD}_\emptyset$                             | $\mathcal{O}(D)((t^\Gamma))$ , no shrinking    | ordinary $D$                                        |
| <a href="#">theorem 14.16</a> moduli $\mathfrak{m}_K^r$      | PAR                 | $\text{SD}_\emptyset$                             | $\mathcal{O}(D_r)((t^\Gamma))$                 | ordinary punctured plane                            |
| <a href="#">theorem 14.18</a> invisible moduli               | PAR                 | $\text{SD}_\emptyset$                             | $\mathcal{O}(D_r)((t^\Gamma))$                 | ordinary; needs $J_\delta \neq 0$                   |
| <a href="#">theorem 15.5</a> sharp homological criterion     | $\neg \text{FR}$    | $\text{SD}_{\text{fr}}$                           | entire / germ                                  | $\mathbb{C}^{2d}$ / germ radius                     |

| Statement                                                           | Resonance                     | Divisor                                             | Coefficient category                                                               | Domain notion                         |
|---------------------------------------------------------------------|-------------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------|
| <a href="#">theorem 15.7(a)</a><br>polynomial layers                | $\neg$ FR                     | $SD_0$                                              | $\mathcal{P}_{2d}$ polynomial                                                      | finite halo $\mathcal{O}_\Gamma^{2d}$ |
| <a href="#">theorem 15.7(b)</a> entire layers                       | $\neg$ FR                     | $SD_{fr}$                                           | $\mathcal{E}_{2d} = \mathcal{O}(\mathbb{C}^{2d})$                                  | finite halo                           |
| <a href="#">theorem 15.16</a> universal criterion                   | $\neg$ FR                     | $SD_{fr}$                                           | entire; germ at first order                                                        | finite halo                           |
| <a href="#">theorem 15.12</a> Poisson centralizer                   | $\neg$ FR                     | $SD_{fr}$<br>(through 06(b))                        | entire-coefficient                                                                 | finite halo                           |
| <a href="#">theorem 15.21</a> certified domains                     | $\neg$ FR                     | $SD_0$ (poly-nomial input)                          | polynomial                                                                         | $D_\rho$ (valuation)                  |
| <a href="#">theorem 19.1</a><br>finite-order splitting              | $\Lambda^q = I$               | $SD_0$                                              | all five rows                                                                      | $\mathbb{D}_R^d$                      |
| <a href="#">theorem 4.3</a> exp/log correspondence                  | PAR                           | $SD_\emptyset$                                      | $\mathcal{O}(U)$ , germ, entire, poly, formal                                      | arbitrary open $U$                    |
| <a href="#">theorem 4.12</a> centralizer                            | PAR                           | $SD_\emptyset$                                      | $\mathcal{O}(U)$ , $U \subset \mathbb{C}$ connected                                | ordinary $U$                          |
| <a href="#">theorem 4.10</a><br>fixed-point ideals                  | PAR                           | $SD_\emptyset$                                      | any of the five                                                                    | finite halo                           |
| <a href="#">theorem 16.5</a> finite resonance flag                  | QR<br>allowed                 | none used                                           | <i>no coefficient functions:</i> a statement about $F_\Gamma^d$ and $\mathbb{Z}^d$ | none                                  |
| <a href="#">proposition 16.7</a> inverse coefficients               | QR<br>allowed                 | none used                                           | scalar, in $K_\Gamma$                                                              | none                                  |
| <a href="#">proposition 16.9</a><br>arithmetic-free bound           | QR<br>allowed                 | <b>none:</b> no divisor estimate at all             | fixed strip<br>$\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$                | $\mathbb{T}_\rho^d$ (ordinary)        |
| <a href="#">theorem 17.3</a> fixed-strip solvability                | QR<br>allowed                 | $SD_{ss}$ , <b>iff</b>                              | fixed strip                                                                        | $\mathbb{T}_\rho^d$ (ordinary)        |
| <a href="#">corollary 17.4</a> loss $\kappa_\Upsilon$ attained      | QR<br>allowed                 | $SD_{ss}$                                           | fixed strip                                                                        | $\mathbb{T}_\rho^d$                   |
| <a href="#">corollary 17.6</a> higher tails                         | $\neg$ QR for the last clause | $SD_{ss}$ preserved                                 | fixed strip                                                                        | $\mathbb{T}_\rho^d$                   |
| <a href="#">theorem 17.7</a><br>arbitrary-order lifting obstruction | $\neg$ QR                     | $SD_{ss}$ <i>fails</i> ; Liouville                  | entire input, fixed-strip target                                                   | $\mathbb{T}_\rho^2$ , all $\rho$      |
| <a href="#">theorem 18.1</a> torus normal form                      | $\neg$ QR                     | $SD_{ss}$ ; also $v(\mathcal{V}) > \kappa_\Upsilon$ | fixed strip                                                                        | $\mathbb{T}_\rho^d$                   |
| <a href="#">lemma 18.3</a> coordinate inverses                      | —                             | none used                                           | fixed strip                                                                        | $\mathbb{T}_\rho^d$                   |
| <a href="#">theorem 18.4</a> necessity beyond any scale             | $\neg$ QR                     | $SD_{ss}$ <i>fails</i>                              | fixed strip                                                                        | $\mathbb{T}_\rho^d$                   |

| Statement                                                                     | Resonance       | Divisor                                                                     | Coefficient category | Domain notion       |
|-------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------|----------------------|---------------------|
| <a href="#">proposition 18.6</a> failure at $v(\mathcal{V}) = \kappa\Upsilon$ | $\neg\text{QR}$ | $\text{SD}_{\text{ss}}$ holds                                               | fixed strip          | $\mathbb{T}_\rho^d$ |
| <a href="#">theorem 18.8</a> invariant density                                | $\neg\text{QR}$ | $\text{SD}_{\text{ss}}$<br>(through<br><a href="#">theo-<br/>rem 18.1</a> ) | fixed strip          | $\mathbb{T}_\rho^d$ |

Three entries of this ledger deserve to be read twice.

*Warning 2.6* (The drift hypothesis travels with the radius law). [Theorem 7.2](#) and its consequences [corollary 7.4](#) and [theorems 7.5](#) and [7.6](#) require  $\text{SD}_{\text{dr}}$ : the multiplier drifts,  $F'(0) = \lambda + u$  with  $v(u) > 0$ , and two rationally independent positive exponents are used ( $\Gamma_* = \mathbb{Q} + \sqrt{2}\mathbb{Q}$ ,  $u = t$ ,  $v = t^{\sqrt{2}}$ ). Source 02 records that eliminating the drift removes the direct reciprocal-jet mechanism and that it does not settle the fixed-multiplier problem for  $0 < \sigma < \infty$ . The exact law  $\text{rad}(h_{k,1}) = R e^{-(k+1)\sigma}$  must therefore never be quoted without the drift hypothesis, and in particular must not be quoted as an answer in the exact-multiplier category  $\text{SD}_{\text{ex}}$  of [theorem 6.1](#). See [section 7](#) for the resolution of this apparent tension with [question 7.10](#).

*Warning 2.7* ( $\text{RES}_q$  and equal characteristic zero). The divisor cancellation behind [theorem 8.5](#) leaves a factor  $n - 1$  in the denominator ([lemma 8.3](#)). In equal characteristic zero every such nonzero integer is a valuation unit; in mixed characteristic it is not. Source 01 states outright that the exact finite-return formula and the shell-periodicity conclusion must not be transported to  $\mathbb{C}_p$ , and cites  $p$ -adic quadratic-like examples whose linearization bounding sphere carries no periodic point at all [[21](#), Cor. C]. That contrast identifies a hypothesis; it is not a contradiction.

*Warning 2.8* (PAR acquires no arithmetic by proximity). [Sections 13](#) and [14](#) sit between two sections full of divisor thresholds. They have none. Source 04's title word "resonance" means only that the leading multiplier is 1 perturbed by an infinitesimal; source 05 has no eigenvalue ratios at all. Their analogue of a linearization domain is a fixed common ordinary domain with no shrinking whatever, and their analogue of a normal form is a rational time form. A merge of either with a framework carrying a resonance or small-divisor hypothesis would identify two different frameworks.

*Warning 2.9* (Two frequency vectors, two divisor families, two  $\kappa$ 's). This report now contains two frequency vectors and they are objects of different kinds. Conflating them would attach a Hahn-scale statement to an ordinary-complex hypothesis, or the reverse.

- $\omega \in \mathbb{C}^d$  of [section 15](#) is a vector of **ordinary complex constants**. Its divisors are  $\omega \cdot (\alpha - \beta)$  for Taylor multi-indices, graded by the *Taylor degree*  $|\alpha| + |\beta|$ , and they all have Hahn valuation zero ([remark 2.3](#)). Its resonance notion is FR and its divisor notion  $\text{SD}_{\text{fr}}$ .
- $\Upsilon \in F_\Gamma^d = \mathbb{R}((t^\Gamma))^d$  of [sections 16](#) to [18](#) is a vector of **Hahn (hence surreal) scalars**. Its divisors are  $k \cdot \Upsilon$  for  $k \in \mathbb{Z}^d$ , graded by the *Fourier index*  $|k| = |k_1| + \cdots + |k_d|$ , and they have *nonzero* Hahn valuations — that is the entire point of [theorem 16.5](#). Its resonance notion is QR and its divisor notion  $\text{SD}_{\text{ss}}$  ([sections 2.2](#) and [2.3](#)).

Further,  $\kappa$  in [remark 2.2](#) is a generic *logarithmic divisor rate* in  $[0, \infty]$ , whereas  $\kappa\Upsilon$  in [sections 16](#) to [18](#) is an *element of  $\Gamma$* : the valuation of the last visible flag level. They are never compared. The symbol  $\Omega$  continues to denote an infinite element of  $\Gamma$  ([examples 3.7](#) and [12.3](#)) and  $\Omega_F = \text{dz}/W_F$  a time form ([section 14](#)); the frequency vector is  $\Upsilon$  and never  $\Omega$ . The coordinate change  $\Phi_h = \text{id} + h$  of [theorem 18.1](#) is a change of torus coordinates, *not* the flow map  $\Phi_X$  or  $\Phi_c$  of [theorem 4.3](#) and [corollary 4.8](#).

*Warning 2.10* (The visible levels are not a spectrum). The finitely many visible divisor valuations  $\gamma_1 < \dots < \gamma_r$  of [theorem 16.5](#) read like a spectrum of scales and are not one in any sense used in this collection: not an eigenvalue list, not a multiset of purely infinite phases, and not the prime spectrum of a ring. They are values of the valuation  $v$  on a family of Hahn scalars. The word *spectrum* is not used for them anywhere below.

*Warning 2.11* (Arithmetic size and valuation size are incomparable). In [\(16.9\)](#) the quantity  $|\mathfrak{a}_j(k)|$  is an *ordinary real absolute value* while  $v(k \cdot \Upsilon) = \gamma_j$  lies in  $\Gamma$ . Neither bounds the other, and no inequality in one kind of size is used as an estimate for the other anywhere below without the explicit leading-coefficient formula [\(16.9\)](#). A divisor can be infinitesimal (large  $\gamma_j$ ) with completely harmless arithmetic, and a divisor of scale one ( $\gamma_j = 0$ ) can have disastrous arithmetic. This is the torus form of [remark 2.3](#): the arithmetic lives in the ordinary complex coefficient, not in the valuation — with the one difference that here the *divisors themselves* do have nonzero valuations, whereas the divisors of [remark 2.3](#) all have valuation zero.

### 3 Workspaces, supports, coefficient categories, and evaluation

All seven sources set up this material independently. It is stated once here.

#### 3.1 The set-sized workspace inside $\mathbf{No}[i]$

Let  $\Gamma$  be a nonzero ordered abelian group which is a *set*; divisibility is assumed only where it is used and is flagged there. A Hahn series over an ordinary commutative  $\mathbb{C}$ -algebra  $B$  is a formal sum  $A = \sum_{\gamma \in S} t^\gamma A_\gamma$  with  $A_\gamma \in B$  and  $S \subset \Gamma$  well ordered in the increasing order ([warning 2.5](#)), with the standard convolution product. For  $A \neq 0$  put  $v(A) = \min \text{supp } A$  and  $v(0) = +\infty$ ; for a vector or matrix take the minimum over entries. Write

$$\mathcal{H}_\Gamma(B) = B((t^\Gamma)), \quad K_\Gamma = \mathcal{H}_\Gamma(\mathbb{C}) = \mathbb{C}((t^\Gamma)), \quad \mathcal{O}_\Gamma = \{v \geq 0\}, \quad \mathfrak{m}_\Gamma = \{v > 0\},$$

and append a subscript  $+$  or  $> 0$  for strictly positive support. Positive support is a condition on *exponents*, never on the signs of the ordinary complex coefficients. The residue field of  $K_\Gamma$  is canonically  $\mathbb{C}$ , and a bar denotes reduction modulo  $\mathfrak{m}_\Gamma$ ; in particular every nonzero ordinary complex number, including every nonzero integer, has valuation zero. When  $\Gamma$  is divisible,  $K_\Gamma$  is algebraically closed [\[4, Cor. 4\]](#), and algebraic closure is used only in [lemma 8.4](#) and [theorem 11.1](#); the Hamiltonian package of [section 15](#) needs neither closure nor divisibility.

For a set-sized ordered subgroup  $\Gamma \subset \mathbf{No}$  the Conway normal form and the substitution

$$t^\gamma \longmapsto \omega^{-\gamma} \tag{3.1}$$

embed  $K_\Gamma$  as a subfield of  $\mathbf{SC} = \mathbf{No}[i]$ , using Gonshor's normal-form arithmetic [\[5\]](#); the minus sign is what converts increasing Hahn supports to decreasing surreal monomial supports. Every *set* of surcomplex constants lies in one such workspace: collect the exponents of the real and imaginary Conway normal forms of those constants, change sign, and take the additive (if needed  $\mathbb{Q}$ -linear) span. The proper-class field  $\mathbf{No}[i]$  is never formed as a single object; every conclusion about it is read pointwise through a workspace enlarged to contain the exponents of the point and of the input, and all constructions commute with such enlargements ([proposition 4.14](#) and [theorem 11.1](#)). No sum in this report is indexed by a proper class.

**Definition 3.1** (Strong summability). A *set-indexed* family  $(A_i)_{i \in I}$  of Hahn series is *strongly summable* if  $\bigcup_i \text{supp } A_i$  is well ordered and each exponent lies in the support of only finitely many members; its sum is then formed coefficientwise by finite sums. A map is *strongly linear* if it is linear and preserves such families and their sums.

There is no limiting operation in [definition 3.1](#). The identity  $\sum_{j \geq 0} c^j = (1 - c)^{-1}$  for  $c \in \mathfrak{m}_\Gamma$  is a statement about a summable family and multiplication, not about convergence of partial sums in the fine order topology of  $\mathbf{No}$ ; this is the discipline established by the companion report *Surcomplex analysis: holomorphic functions on  $K = \mathbf{No}[i]$*  ([docs/surcomplex/analysis/](#)), which we inherit rather than re-derive. No Banach contraction, Cauchy limit, compactness principle or surcomplex path occurs in any proof below. Ordinary Taylor series of coefficient functions, by contrast, use ordinary complex convergence, and the two summations are never interchanged without a support argument. Cancellation may not rescue an inadmissible family: infinitely many nonzero terms at one exponent already violates [definition 3.1](#) even if some other summation method would assign a value to their complex coefficients.

*Warning 3.2* (The derivative in this report; the cross-category pair). The symbols  $\partial_j$ ,  $\partial_z$ ,  $\partial_w$  differentiate the *ordinary* coordinate and act coefficientwise; every  $t^\gamma$  is a constant for them. They are not a derivation of the surreal coefficient field, and in particular not the Berarducci–Mantova derivation [\[25\]](#). No compatibility with a global composition operator on  $\mathbf{No}$  is asserted. We adopt here, once, the wording of the cross-category warning of the companion report *Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers* ([docs/surcomplex/differential-equations/](#)): a Hahn constant for a coordinate derivative need not be a constant for the scalar Berarducci–Mantova derivation, so  $\varepsilon = \omega^{-1}$  is permitted as a Hahn coefficient of a coordinate problem and prohibited as an infinitesimal phase residue of the scalar derivation. All seven sources derive this distinction independently; it is not restated again below.

### 3.2 The coefficient categories, in the collection’s ring vocabulary

The companion report *Infinitesimal Analytic Geometry over the Surcomplex Numbers* ([docs/surcomplex/analytic-geometry/](#)) fixes a four-ring table and a ring discipline: every numbered statement names the ring over which it is asserted, and no theorem proved of one ring is transferred to another. We adopt its names:

$$\mathcal{O}(D)((t^\Gamma)) \longrightarrow \mathcal{R}_n \subsetneq \mathcal{A}_n \subsetneq \mathcal{F}_n, \quad (3.2)$$

where

- $\mathcal{O}(D)((t^\Gamma))$  is the *fixed-polydisk* ring: one specified, unchanged ordinary domain for every coefficient;
- $\mathcal{R}_n = \varinjlim_{U \ni 0} \mathcal{H}_\Gamma(\mathcal{O}(U))$  is the *common-domain germ* ring: one common domain, then allowed to shrink;
- $\mathcal{A}_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$  is the *radius-free* ring: individually convergent germs, with no common radius required;
- $\mathcal{F}_n = \mathbb{C}[[z_1, \dots, z_n]]((t^\Gamma))$  is the *formal-coefficient* ring.

We add exactly two rows to that table, and they are new *rows*, not new rings: the *entire-coefficient* row  $\mathcal{O}(\mathbb{C}^n)((t^\Gamma))$  (source 06’s  $\mathcal{E}_n$ , also the third row of [theorem 6.1](#)) and the *polynomial-coefficient* row  $\mathbb{C}[z_1, \dots, z_n]((t^\Gamma))$  (source 06’s  $\mathcal{P}_n$ , also the fourth row of [theorem 6.1](#)). Both sit between the fixed-polydisk ring and  $\mathcal{F}_n$  in the obvious way, and neither carries a uniform bound: in the entire

row *no* uniform norm bound is imposed as  $\gamma$  varies, and in the polynomial row *no* uniform degree bound.

We do not reprint the witness that separates the rings; it is the element  $F(z) = \sum_{r \geq 1} t^{rn}/(1 - rz)$  of the analytic-geometry report, whose coefficient at  $r\eta$  has a pole at  $1/r$ . Sources 02, 03 and 04 each re-erect that separation from scratch, with  $\eta = 1$  in source 02 ( $\sum_{m \geq 1} t^m/(1 - mz)$ ) and  $\eta = \delta$  in source 04 ( $\sum_{n \geq 1} t^{n\delta}/(1 - nw)$ ). Those are the same series.

*Convention 3.3* (Ring discipline, inherited). Every numbered statement below names its coefficient category in the vocabulary of (3.2) and the two added rows, and the tag ledger [section 2.5](#) records it. No statement proved of one category is transferred to another. In particular the fixed-polydisk row of [theorem 6.1](#) concerns preservation of a *specified unchanged* polydisk, the radius-free row concerns separate germs, and neither row, nor their combination, classifies  $\mathcal{R}_n$  in the intermediate regime; that is [question 7.10](#).

### 3.3 The support certificate: finite words, not sequential limits

For well ordered  $S \subset \Gamma_{>0}$  let  $S^* = M(S)$  be its additive monoid, including the empty sum 0, and for  $\gamma \in \Gamma$  let

$$\mathcal{W}_S(\gamma) = \{(s_1, \dots, s_k) : k \geq 0, s_i \in S, s_1 + \dots + s_k = \gamma\}$$

be its set of *ordered* words of total weight  $\gamma$ , permutations counted separately.

**Lemma 3.4** (Positive-support lemma of Neumann, in finite-word form). *Let  $S \subset \Gamma_{>0}$  be well ordered. Then  $S^*$  is well ordered, and  $\mathcal{W}_S(\gamma)$  is finite for every  $\gamma \in \Gamma$ ; in particular*

$$\ell_S(\gamma) = \max\{k : (s_1, \dots, s_k) \in \mathcal{W}_S(\gamma)\} \quad (3.3)$$

*is finite for  $\gamma \in S^*$ , with  $\ell_S(0) = 0$ . Moreover, if  $A \subset \Gamma$  is well ordered then  $A + S^*$  is well ordered, and for fixed  $\gamma$  there are only finitely many pairs consisting of  $a \in A$  and a word in  $S$  with  $a + s_1 + \dots + s_k = \gamma$ .*

*Proof.* Order finite words over  $S$  by subsequence embedding with coordinatewise domination. Higman's lemma says this is a well-quasi-order: every infinite sequence of words contains indices  $i < j$  with the earlier word embedded in the later [3]; the special case needed is reproved from scratch in [section A](#). Because all letters are positive, an embedding weakly increases the sum of the letters, and increases it strictly unless the two words are identical: an unused positive letter contributes positively, and a strictly increased matched letter increases the sum. An infinite strictly decreasing sequence in  $S^*$ , after choosing one word per element, would contradict the weak increase; infinitely many distinct words of one fixed weight would contradict the strict increase. This proves the first two assertions.

For the last, assume  $A \neq \emptyset$ , subtract  $a_0 = \min A$ , and apply the first part to the well-ordered positive set  $T = S \cup ((A - a_0) \setminus \{0\})$  at weight  $\gamma - a_0$ : a pair in the assertion is encoded by a word with one distinguished initial letter from  $A - a_0$ , or none if  $a = a_0$ . Well-ordering follows from  $A + S^* \subset a_0 + T^*$ .  $\square$

This is classical Hahn–Neumann mathematics [2, 4] and is not claimed as a contribution. All seven sources state it; sources 02 and 05 each reproduce a minimal-bad-sequence proof of the underlying word order, which is the same argument as [section A](#) and is therefore not printed again. Source 07 uses the *full* finiteness assertion, across all word lengths, together with a *labelled* refinement of it ([section 18.2](#)); both are already contained in [lemma 3.4](#).

For  $\gamma \in S^*$  let  $D_S(\gamma)$  be the finite set of letters occurring in the words of  $\mathcal{W}_S(\gamma)$ .

**Corollary 3.5** (Finite ancestry). *Suppose  $T$  is a  $\mathbb{C}$ -linear operator on coefficient families such that every term of  $T(A)$  adds a nonempty word from  $S$  to an input exponent of  $A$ , with finitely many coefficient operations per word. Then  $\sum_{k \geq 0} c_k T^k(A)$  is strongly summable for any ordinary scalars  $c_k$ , its support lies in  $\text{supp } A + S^*$ , and at a fixed output exponent  $\gamma$  only finitely many input exponents, word lengths and operations contribute — at most  $\ell_S(\gamma)$  applications of  $T$ , and only input data at letters in  $D_S(\gamma)$ . Consequently polynomial identities valid for a nilpotent operator may be used coefficientwise, which is what justifies the geometric, logarithmic and exponential identities below. The same statement holds for operators with Hahn coefficients, for vector and matrix operators, and for a strongly summable input family simultaneously.*

*Proof.* Flatten every intermediate exponent into a word in  $S$  and apply lemma 3.4; lemma 3.4’s last assertion bounds the relevant input weights, and local finiteness of the input family bounds the number of contributing members.  $\square$

**Warning 3.6** (Why a valuation-contraction argument would be incomplete). Writing  $v(T^k A) \geq v(A) + k \min S$  does *not* imply that these lower bounds become cofinal in  $\Gamma$ : for example  $k < \Omega$  for every integer  $k$ . The finite-word certificate of lemma 3.4 and corollary 3.5 establishes summability even when that sequence of bounds never passes a specified higher rank, and every infinite operator series in this report rests on the stronger certificate. Likewise “only finitely many terms below each cutoff” is *not* the criterion: in  $\Gamma = \mathbb{Z}^2$  lexicographic with  $\eta = (0, 1)$  and  $\theta = (1, 0)$  one has  $n\eta < \theta$  for every integer  $n$ , yet  $\sum_{n \geq 0} t^{n\eta}$  is a valid Hahn series with  $(1 - t^\eta) \sum_{n \geq 0} t^{n\eta} = 1$ , while the partial sums do not converge in the valuation topology. Sources 03, 05 and 06 each record this independently.

**Example 3.7** (Infinitely many earlier weights, one relevant word). Let  $\Gamma = \mathbb{Q} + \mathbb{Q}\Omega$  with  $\Omega$  greater than every rational, and  $S = \{1\} \cup \{2 - 1/n : n \geq 2\} \cup \{\Omega\}$ . This  $S$  is well ordered and lies in no finitely generated additive grid, and infinitely many of its members are below  $\Omega$ . Nevertheless  $\mathcal{W}_S(\Omega) = \{(\Omega)\}$ ,  $\ell_S(\Omega) = 1$  and  $\ell_S(\Omega + 2) = 3$ , the words at  $\Omega + 2$  being the three permutations of  $(\Omega, 1, 1)$ : any finite sum of rational letters is rational, and any sum involving two  $\Omega$ ’s exceeds  $\Omega + 2$ . An algorithm computing a coefficient at  $\Omega$  needs the input at  $\Omega$ , not infinitely many rational predecessors. Section 21.5 computes with this  $S$ .

### 3.4 A uniform-monoid support algebra

The finite-return arguments of section 8 need a slightly stronger object than corollary 3.5: one monoid uniform across all Taylor degrees.

**Definition 3.8.** For  $M = \langle S \rangle$  with  $S \subset \Gamma_{>0}$  well ordered, put

$$\mathcal{B}_M = \left\{ \sum_{n \geq 0} h_n X^n \in K_\Gamma[[X]] : \text{supp}(h_n) \subseteq M \text{ for every } n \right\},$$

a subring of  $\mathcal{F}_1 = \mathbb{C}[[X]]((t^\Gamma))$  after regrouping as  $\sum_{\gamma \in M} H_\gamma(X) t^\gamma$  with  $H_\gamma \in \mathbb{C}[[X]]$ . Coefficients need not converge as ordinary complex power series.

The uniformity of the single monoid across all Taylor degrees is load-bearing: an arbitrary member of  $\mathcal{O}_\Gamma[[X]]$  need not lie in any such algebra.

**Lemma 3.9** (Evaluation, inversion, isometry). *Every  $H \in \mathcal{B}_M$  is strongly Hahn-evaluable at every  $u \in \mathfrak{m}_\Gamma$ . If  $H = X + O(X^2)$  then its formal inverse also lies in  $\mathcal{B}_M$ , and the evaluated maps are mutually inverse isometric bijections of  $\mathfrak{m}_\Gamma$ , isometry meaning  $v(H(u) - H(w)) = v(u - w)$ . Formal composition identities between zero-fixing series with integral unit linear coefficients and a common positive-support certificate remain valid after evaluation.*

*Proof.* For  $u \neq 0$  with  $U = \text{supp } u \subset \Gamma_{>0}$ , the supports of  $h_n u^n$  lie in  $\langle S \cup U \rangle$ , and every monomial of  $u^n$  uses at least  $n$  generators from  $U$ , so lemma 3.4 leaves finitely many  $n$  at a fixed exponent: both clauses of definition 3.1 hold. The coefficients of the formal inverse of  $X + \sum_{n \geq 2} h_n X^n$  are integer polynomials in finitely many  $h_n$ , hence supported in  $M$ , and the substitutions in the inverse identities are summable by the same count. For  $u \neq w$  in  $\mathfrak{m}_\Gamma$ ,

$$\frac{H(u) - H(w)}{u - w} = 1 + \sum_{n \geq 2} h_n \sum_{j=0}^{n-1} u^j w^{n-1-j},$$

whose nonconstant summands all have positive valuation and whose total has residue one and valuation zero; this is the isometry. For compositions, expand both levels into products: at a fixed exponent the number of positive generators is finite, and at a fixed formal degree the number of zero-exponent contributions is finite because the substituted series has zero constant term. A unit linear coefficient and its inverse are adjoined by separating their nonzero complex constant terms and expanding the inverse in its positive part.  $\square$

**Example 3.10** (Coefficient-valuation bounds are insufficient at rank  $> 1$ ). Let  $\Gamma = \mathbb{Q} + \mathbb{Q}\varepsilon_0$  with  $0 < \varepsilon_0 < 1/n$  for every positive integer  $n$ . All coefficients of  $A(X) = \sum_{n \geq 2} t^{1/n} X^n$  have positive valuation. Yet at  $u = t^{\varepsilon_0}$  the proposed Taylor sum has exponents  $1/n + n\varepsilon_0$ , which *strictly decrease* as  $n$  grows: it is not a Hahn series. Hence an assertion that “all normalized coefficients are integral” would leave a real gap in an arbitrary-rank linearization proof. The set  $\{1/n : n \geq 2\}$  is not well ordered, so this example does not satisfy the uniform-support hypothesis of lemma 3.9; it is the reason that hypothesis is stated. This is source 01’s observation, and no other source records it.

### 3.5 Halos, monads, and evaluation as a ring homomorphism

For an ordinary open  $U \subset \mathbb{C}^d$  put

$$U_\Gamma^\# = \{a + \xi : a \in U, \xi \in \mathfrak{m}_\Gamma^d\},$$

the *finite halo*; the representation is unique and  $a = \text{st}(a + \xi)$  is the standard part. For  $A = \sum_\gamma t^\gamma A_\gamma$  with all  $A_\gamma$  holomorphic on  $U$ , define

$$A^\#(a + \xi) = \sum_{\gamma, \alpha} t^\gamma \frac{\partial^\alpha A_\gamma(a)}{\alpha!} \xi^\alpha. \quad (3.4)$$

**Proposition 3.11** (Faithful analytic realization). *The family in (3.4) is strongly summable, and (3.4) defines a faithful  $K_\Gamma$ -algebra homomorphism from  $\mathcal{H}_\Gamma(\mathcal{O}(U))$  to functions on  $U_\Gamma^\#$ . It respects coordinate derivatives, substitution by positive near-identity Hahn maps, and infinitesimal Taylor expansions, with the expected coefficients  $(\partial^\alpha A)^\#(z)/\alpha!$ . For germ or formal coefficients use the Taylor coefficients at 0 and restrict the domain to  $\mathfrak{m}_\Gamma^d$ ; polynomial and entire coefficients evaluate on every finite halo, but not automatically at infinite surcomplex points.*

*Proof.* Let  $T = \text{supp } A$  and let  $S$  be the union of the positive supports of the components of  $\xi$ , a finite union of well-ordered positive sets. Every monomial contribution has exponent in  $T + S^*$ , and at a fixed exponent [lemma 3.4](#) permits only finitely many choices of  $\gamma$  and of a word from  $S$ ; the word length bounds  $|\alpha|$  and a finite word has finitely many coordinate assignments. Hence the expanded family is strongly summable. Ordinary Taylor identities for products and derivatives then apply, every regrouping at a Hahn exponent being finite; adding the positive support of a near-identity substitution to  $S$  gives the composition assertion from the ordinary Taylor chain rule. For faithfulness, let  $\gamma_0 = \min \text{supp } A$  for  $A \neq 0$  and pick an ordinary  $a \in U$  with  $A_{\gamma_0}(a) \neq 0$ : the coefficient of  $t^{\gamma_0}$  in  $A^\#(a)$  is then nonzero.  $\square$

At a fixed ordinary Taylor center the numbers  $n!$ , or extremely large inverse small divisors, remain *ordinary complex constants* of valuation zero. Strong summability imposes no real growth condition on them. This single fact explains both the strength of the theorems below and the necessity of stating their domains exactly.

**Lemma 3.12** (Faithfulness of infinitesimal evaluation). *If  $0 \neq A \in \mathcal{F}_1$  and  $\Gamma \neq \{0\}$  is divisible, then  $A(u) \neq 0$  for some  $u \in \mathfrak{m}_\Gamma$ ; evaluation on the whole monad is injective on the formal-coefficient ring.*

*Proof.* Let  $\gamma_0 = \min \text{supp } A$  and let  $n$  be the least degree of the nonzero formal series  $A_{\gamma_0}$ . If another Hahn exponent occurs, let  $\Delta > 0$  be the gap from  $\gamma_0$  to the next one and choose  $s > 0$  with  $ns < \Delta$ ; divisibility supplies such an  $s$  when  $n > 0$ , and any positive  $s$  works when  $n = 0$ . If no other exponent occurs, take any  $s > 0$ . At  $u = t^s$  the exponent  $\gamma_0 + ns$  appears with nonzero coefficient, while higher degrees at  $\gamma_0$  and all other exponents contribute only larger exponents. Evaluation is legitimate by [lemma 3.9](#) after factoring out  $t^{\gamma_0}$ .  $\square$

[Lemma 3.12](#) blocks the evasion of representing a divergent conjugacy by a *different* common-domain family agreeing on all infinitesimals; it is used in [proposition 10.1](#).

## 4 Positive substitution, logarithms, flows, and centralizers

### 4.1 The substitution group

For  $U \subset \mathbb{C}^d$  open set

$$\mathcal{G}_+(U) = \{F(z) = z + u(z) : u \in \mathcal{H}_\Gamma(\mathcal{O}(U))_+^d\}.$$

No condition at  $z = 0$  is imposed, and 0 need not lie in  $U$ . The pullback of  $F = \text{id} + u$  on  $A \in \mathcal{H}_\Gamma(\mathcal{O}(U))$  is

$$C_F A = A \circ F = \sum_{\alpha \in \mathbb{N}^d} \frac{1}{\alpha!} (\partial^\alpha A) u^\alpha, \quad (4.1)$$

every coefficient of which is a finite sum of products and derivatives of functions holomorphic on  $U$ , hence holomorphic on  $U$  regardless of its ordinary norm; the support lies in  $\text{supp } A + S^*$  for  $S$  the union of the supports of the components of  $u$ .

**Proposition 4.1** (Substitution group). *(4.1) defines an associative group law on  $\mathcal{G}_+(U)$ . Every inverse again lies in  $\mathcal{G}_+(U)$ , on the same  $U$ , with support in  $S^* \setminus \{0\}$ . Evaluation gives mutually inverse bijections of  $U_\Gamma^\#$  preserving standard parts and commutes with composition. The same*

statements hold in the common-domain germ, radius-free germ, entire, polynomial and formal categories, with the evaluation domains of [section 3.5](#). The order convention is

$$C_F C_G = C_{G \circ F} : \quad (4.2)$$

pullback reverses composition.

*Proof.* Strong summability is [lemma 3.4](#) and [corollary 3.5](#). At a fixed Hahn exponent only finitely many derivatives and products contribute; introduce a formal parameter for each of the finitely many input support letters involved and discard parameter monomials beyond the required word length. Taylor composition in that finite nilpotent parameter algebra is associative and multiplicative, and each asserted coefficient identity is one of those identities. For the inverse put  $\Delta = C_F - I$ : every term of  $\Delta$  adds a nonempty word from  $S$ , so  $C_F^{-1} = \sum_{k \geq 0} (-\Delta)^k$  is defined coefficientwise and is a two-sided multiplicative inverse. Let  $G = C_F^{-1} z$ , identity plus positive support with support in  $S^*$ . On coordinate polynomials multiplicativity gives  $C_F^{-1} = C_G$ ; at each Hahn weight both operators are finite-order differential operators on an ordinary input, so agreement on polynomial jets extends to all holomorphic inputs, and strong linearity extends it to Hahn inputs. For formal coefficients the finite-jet argument is coefficientwise in  $z$ . Substituting  $z = a + \xi$  and adding  $\text{supp } \xi$  to the control set preserves every identity, and  $\text{st } F(a + \xi) = a$  keeps the evaluated map in  $U_\Gamma^\#$ . Iterates of a single map are unaffected by (4.2), and commuting maps have commuting pullbacks; the convention is retained explicitly to avoid a sign or order ambiguity in logarithms.  $\square$

For maps  $F = \Lambda z + f$  on  $\mathbb{D}_R^d$  with  $\Lambda$  diagonal unitary and  $f$  positive, use instead

$$A \circ F = \sum_{\alpha \in \mathbb{N}^d} \frac{1}{\alpha!} (\partial^\alpha A)(\Lambda z) f^\alpha, \quad (4.3)$$

legitimate because  $\Lambda \mathbb{D}_R^d = \mathbb{D}_R^d$ . For formal coefficients or germs,  $F(0) = 0$  is imposed whenever composition with the nonidentity reduction is used. [Lemma 4.2](#) records the near-identity inverse in the form source 02 uses.

**Lemma 4.2** (Near-identity inverses). *Let  $H = z + U$  be normalized with coefficients in any one of  $\mathcal{O}(\mathbb{D}_R)$ ,  $\mathbb{C}\{z\}$ ,  $\mathbb{C}[[z]]$ ,  $\mathcal{O}(\mathbb{C})$ ,  $\mathbb{C}[z]$ , and positive Hahn support  $B$ . There is a unique normalized inverse  $J = z + V$  with coefficient support in  $B^* \setminus \{0\}$  and coefficients in the same algebra, with  $H \circ J = J \circ H = \text{id}$ .*

*Proof.* In  $H(J) = J + U(J)$  the coefficient  $v_\gamma$  occurs with coefficient one, and every other contribution at exponent  $\gamma$  uses only  $v_\beta$  with  $\beta < \gamma$ ; set  $v_\gamma$  to minus that finite sum and recurse on  $B^*$ . Derivatives, products and finite sums preserve each algebra and the order- $z^2$  normalization. The identical triangular argument gives a left inverse, and associativity, which is coefficientwise a finite Taylor identity, identifies the two.  $\square$

## 4.2 Logarithms are actual coefficientwise vector fields

A positive Hahn vector field on  $U$  is  $X = a_1 \partial_1 + \cdots + a_d \partial_d$  with  $a \in \mathcal{H}_\Gamma(\mathcal{O}(U))_+^d$ , and

$$\exp X = \sum_{k \geq 0} \frac{X^k}{k!}, \quad \log C_F = \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} (C_F - I)^k \quad (4.4)$$

are operators on the Hahn algebra, not numerical logarithms of values of  $F$ .

**Theorem 4.3** (Same-domain exponential–logarithm correspondence). *For every open  $U \subset \mathbb{C}^d$  the assignments  $F \mapsto X_F = \log C_F$  and  $X \mapsto \Phi_X = (\exp X)z$  are inverse bijections between  $\mathcal{G}_+(U)$  and positive Hahn vector fields on  $U$ . All coefficient functions stay holomorphic on the original  $U$ : no shrinking, and no uniform bound on the coefficient family. If  $S$  contains the positive support of the input then the output support lies in  $S^* \setminus \{0\}$ , and each output coefficient has a finite word certificate. For  $F \neq \text{id}$ ,*

$$v(X_F) = v(F - \text{id}), \quad (4.5)$$

*and the leading coefficient of  $X_F z$  equals that of  $F - \text{id}$ . The inverse of  $F$  is  $(\exp(-X_F))z$ . The same assertions hold with common-domain germ, radius-free germ, entire, polynomial or formal coefficients.*

*Proof. Summability.* Put  $\Delta = C_F - I$  and  $S = \text{supp}(F - \text{id})$ . By (4.1),  $\Delta$  adds nonempty words from  $S$ , so both series in (4.4) are strongly summable by corollary 3.5, and every output coefficient is a finite differential expression with coefficients holomorphic on  $U$ .

*The derivation identity.* Introduce an ordinary indeterminate  $s$  and put  $C_F^s = \sum_{k \geq 0} \binom{s}{k} \Delta^k$ . For fixed inputs and a fixed Hahn exponent this is a *polynomial* in  $s$ , not a power series. At every nonnegative integer  $s$  it is the usual integer power of the multiplicative operator  $C_F$ , so the coefficientwise polynomial identity  $C_F^s(AB) = C_F^s(A)C_F^s(B)$  holds identically in  $s$ . Differentiating at  $s = 0$  and using  $\frac{d}{ds} \binom{s}{k} \big|_{s=0} = (-1)^{k+1}/k$  for  $k \geq 1$  gives  $X_F(AB) = X_F(A)B + AX_F(B)$ .

*Why the derivation is a vector field.* At each Hahn weight  $\gamma$  the restriction of  $X_F$  to ordinary coefficient functions is a finite-order differential operator  $D_\gamma$ , and the preceding identity makes it a derivation. Put  $a_{j,\gamma} = D_\gamma z_j$ ; by Leibniz,  $D_\gamma$  agrees with  $\sum_j a_{j,\gamma} \partial_j$  on every coordinate polynomial, and at an ordinary point a finite-order differential operator is determined by a finite Taylor jet, so the two agree on every holomorphic function. For germs or formal series use finite jets at the origin; for polynomials it is immediate. This finite-order step is essential: no classification of arbitrary abstract algebraic derivations of a function ring is invoked.

*Conversely.* If  $X = a \cdot \partial$  is positive, each application adds a positive letter, so  $\exp X$  is strongly summable, and repeated Leibniz  $X^k(AB) = \sum_j \binom{k}{j} X^j(A)X^{k-j}(B)$  makes it multiplicative with inverse  $\exp(-X)$ . Put  $\Phi_X = (\exp X)z = \text{id} + \text{positive}$ . On coordinate polynomials multiplicativity gives  $\exp X = C_{\Phi_X}$ , and the finite-jet argument extends the equality to the whole coefficient algebra and then to Hahn inputs.

*Inverse identities and valuation.*  $\exp(\log(I + T)) = I + T$  and  $\log(\exp T) = T$  hold in a nilpotent operator algebra, and at a specified Hahn coefficient only finitely many words occur, so they hold coefficientwise; this is the bijection. Finally  $X_F z = (F - z) +$  terms with at least two support letters, whose valuation is at least  $2v(F - \text{id})$ , which gives (4.5) and the leading-coefficient assertion. Derivatives and finite products preserve every stated category.  $\square$

*Remark 4.4* (Three sources, one route, three by-products). Sources 03, 04 and 05 each prove theorem 4.3, and all three use the same essential device: the binomial operator family  $\sum_k \binom{s}{k} \Delta^k$ , polynomiality in  $s$  at each fixed Hahn exponent, and a finite-jet identification of the resulting derivation with a coordinate vector field. Being one route, it is printed once, in source 03's several-variable form on an arbitrary open  $U \subset \mathbb{C}^d$ . The routes differ only in by-products, and each by-product survives separately below: source 05 adds that  $v(W_F) = v(h)$  with equal leading coefficient and that inverses are  $\exp(-W_F \partial_z)z$  (both included in theorem 4.3); source 04 adds the exact fixed-point identity lemma 4.5, which is what determines the complete monodromy exponent of section 13; and source 05 adds the Julia equation corollary 4.6 and the intertwining criterion proposition 4.7, which are what make section 14 possible. Source 05 also notes explicitly that the

general algebraic mechanism for exponentiating strongly summable derivations of generalized-series algebras is established work [8] and that the existence of a formal logarithm is not claimed new.

**Lemma 4.5** (Derivative of the logarithmic generator at an ordinary fixed point). *Let  $F = w + \Delta$  with  $\Delta \in \mathcal{H}_1(\mathcal{O}(U))_{>0}$  in one variable,  $X_F = A(w)\partial_w$ . If an ordinary  $a \in U$  satisfies  $F(a) = a$ , then*

$$A(a) = 0, \quad A'(a) = \text{Log } F'(a), \quad (4.6)$$

where  $\text{Log}(1 + u) = \sum_{n \geq 1} (-1)^{n+1} u^n / n$  for  $v(u) > 0$ .

*Proof.* Evaluation at  $a$  annihilates  $N\Phi = (C_F - I)\Phi$ , hence annihilates  $A = Dw$ . With  $\lambda_a = F'(a)$ , the chain rule for Taylor substitution gives  $(N\Phi)'(a) = (\lambda_a - 1)\Phi'(a)$ , so  $(N^r w)'(a) = (\lambda_a - 1)^r$ ; apply the logarithm series, which is a strong sum because  $v(\lambda_a - 1) > 0$  (the identity is immediate if  $\lambda_a = 1$ ).  $\square$

This *exact* identity, and not a truncated modified differential equation, is what fixes the complete monodromy exponent in [section 13](#).

**Corollary 4.6** (Julia equation). *For every  $F \in \mathcal{G}_+(U)$  in one variable, with  $W_F = X_F z$ ,  $W_F(F(z)) = F'(z)W_F(z)$ .*

*Proof.*  $C_F$  commutes with its logarithm; apply  $C_F(W_F \partial_z) = (W_F \partial_z)C_F$  to  $z$ .  $\square$

**Proposition 4.7** (Conjugacy and vector fields). *For  $F_1, F_2, H \in \mathcal{G}_+(U)$  in one variable,  $H \circ F_1 = F_2 \circ H$  if and only if  $W_{F_2}(H) = H'W_{F_1}$ .*

*Proof.* The conjugacy relation is  $C_{F_1}C_H = C_H C_{F_2}$ ; conjugating the summable logarithm gives  $(W_{F_1} \partial_z)C_H = C_H(W_{F_2} \partial_z)$ , and evaluation on  $z$  gives the displayed identity. Conversely that identity and the chain rule give the operator intertwining on the whole algebra, and exponentiation recovers the conjugacy.  $\square$

### 4.3 Times, flows, and unique roots

**Corollary 4.8** (Exact flow law and admissible Hahn times). *Let  $X = a \cdot \partial$  be positive. For every ordinary  $s \in \mathbb{C}$ ,  $\Phi_s = (\exp(sX))z \in \mathcal{G}_+(U)$  and*

$$\Phi_0 = \text{id}, \quad \Phi_{s+r} = \Phi_s \circ \Phi_r, \quad \frac{d}{ds} \Phi_s = a \circ \Phi_s, \quad (4.7)$$

each Hahn coefficient of  $\Phi_s$  being a polynomial in  $s$ . For  $c \in K_\Gamma$  with  $c = 0$  or

$$v(c) + v(a) > 0, \quad (4.8)$$

the map  $\Phi_c = (\exp(cX))z$  is also defined and positive, and the additive flow law holds for such times.

*Proof.* All ordinary scalars have valuation zero, each coefficient involves finitely many powers of  $s$ , and commuting exponentials of multiples of  $X$  satisfy the addition law. Coefficientwise differentiation and commuting  $X$  with its exponential give the last identity. For a Hahn time, (4.8) says exactly that  $cX$  has positive support, since  $c$  is a scalar independent of  $z$ ; apply [theorem 4.3](#) to  $cX$ , and note that the positive supports of two admissible multiples have well-ordered union.  $\square$

Condition (4.8) characterizes when this scalar multiple of a nonzero vector field is positive; it is *not* asserted to be a maximal possible lifetime in a larger class of maps (a constant vector field, for instance, has translations beyond the positive subgroup). Likewise (4.7) is coefficientwise analytic time dependence, not a continuous path in the fine surreal topology.

**Corollary 4.9** (Unique fractional iteration; torsion-freeness). *For every  $F \in \mathcal{G}_+(U)$  and integer  $N \geq 1$  there is a unique  $G \in \mathcal{G}_+(U)$  with  $G^{\circ N} = F$ , namely  $G = (\exp(X_F/N))z$ . The group  $\mathcal{G}_+(U)$  is uniquely divisible and torsion-free.*

*Proof.* Existence is [corollary 4.8](#). If  $G^{\circ N} = F$  then  $C_G^N = C_F$ , and taking logarithms gives  $NX_G = X_F$ , whence uniqueness. If  $F^{\circ N} = \text{id}$  then  $NX_F = 0$ , and characteristic zero gives  $F = \text{id}$ .  $\square$

#### 4.4 Fixed-point ideals, and two routes to iteration invariance

For a coefficient map  $F$  let  $I_{\text{Fix}(F)} = (F_1 - z_1, \dots, F_d - z_d)$  be its fixed-point ideal in the chosen Hahn coefficient algebra. No Noetherianity and no Nullstellensatz is needed to compare such ideals, and none is invoked: ideals are compared through an explicit invertible matrix.

**Theorem 4.10** (Fixed-point ideal of a positive flow). *Let  $X = a \cdot \partial$  be a positive Hahn vector field on  $U$  and let  $c \neq 0$  be an admissible Hahn time. Then*

$$I_{\text{Fix}(\Phi_c)} = (a_1, \dots, a_d) \quad \text{in } \mathcal{H}_\Gamma(\mathcal{O}(U)), \quad (4.9)$$

*and likewise in the other coefficient categories and after localization. In particular all nonzero integer iterates of  $F \in \mathcal{G}_+(U)$  have identical fixed-point ideals; on the evaluated halo there are no nonfixed points of finite period; and whenever a fixed-point quotient or local algebra has finite length, that length is unchanged by nonzero integer iteration.*

*Proof.* Set  $E = cX = b \cdot \partial$  with  $b = ca$  positive, so that  $\Phi_c - z = b + \sum_{k \geq 2} E^k z / k!$ . Since  $E^k z_j = \sum_\ell b_\ell \partial_\ell (E^{k-1} z_j)$ , define

$$B_{j\ell} = \delta_{j\ell} + \sum_{k \geq 2} \frac{\partial_\ell (E^{k-1} z_j)}{k!}, \quad (4.10)$$

strongly summable by [corollary 3.5](#), of the form  $I + \text{positive}$ , hence invertible by its matrix geometric series, and

$$\Phi_c - z = Bb. \quad (4.11)$$

The two coordinate ideals are therefore equal, and since the scalar  $c$  is a unit of  $K_\Gamma$ , the ideal generated by  $b$  is that generated by  $a$ . On evaluation at any halo point  $B$  remains invertible by [proposition 4.1](#), so a point is fixed by  $\Phi_c$  exactly when all components of  $a$  vanish there; apply this with  $c = 1$  and any nonzero integer  $c = n$ . Identical ideals give identical quotient algebras.  $\square$

This theorem concerns maps whose ordinary reduction is the identity. It makes no claim that a general surcomplex map lacks periodic points; in particular rotations of finite order lie outside  $\mathcal{G}_+(U)$  unless they are the identity, and [theorem 8.7](#) exhibits genuine period- $q$  points for a map that is *not* in  $\mathcal{G}_+(U)$ .

*Remark 4.11* (Two routes). Source 03 proves iteration invariance by the explicit matrix (4.10)–(4.11), obtaining ideal equality — scheme structure and finite length — rather than inferring it from equality of zero sets. Source 05 obtains the same invariance as a corollary of unique  $n$ -th roots and the centralizer (corollary 4.9 and theorem 4.12), which gives less (no ideal statement) by a shorter argument but also gives torsion-freeness of the whole group. Both are kept: the ideal statement is not available from the root argument, and the group statement is not available from the matrix.

**Theorem 4.12** (The full positive centralizer in one variable). *Let  $U \subset \mathbb{C}$  be connected, let  $F \in \mathcal{G}_+(U)$  be nonidentity, and write  $X_F = a(z)\partial_z$  with  $\alpha = v(a) > 0$ . Then*

$$\text{Cent}_{\mathcal{G}_+(U)}(F) = \{(\exp(ca\partial_z))z : c \in K_\Gamma, c = 0 \text{ or } v(c) + \alpha > 0\}, \quad (4.12)$$

*and the parametrization is an isomorphism from that additive group of Hahn times onto the centralizer. Zeros of  $a$  need not be removed from  $U$ . No nowhere-zero hypothesis on the leading coefficient of  $a$  is required.*

*Proof.* Let  $G$  be positive with  $X_G = b\partial_z$ . Commuting maps have commuting pullbacks, hence commuting logarithms, all double regroupings being valid over the union of the two positive supports; conversely commuting logarithms have commuting exponentials. Thus  $FG = GF$  iff  $[a\partial_z, b\partial_z] = 0$  iff  $ab' - ba' = 0$ . Embed the coefficient algebra into  $\mathcal{M}(U)((t^\Gamma))$  with  $\mathcal{M}(U)$  the field of ordinary meromorphic functions on the connected  $U$ ; the nonzero series  $a$  is invertible there (factor its leading term and use a positive geometric series), so  $(b/a)' = (ab' - ba')/a^2 = 0$ . Every Hahn coefficient of  $b/a$  is a meromorphic function with zero derivative, hence an ordinary complex constant by connectedness, so  $b/a = c \in K_\Gamma$ . Conversely  $b = ca$  commutes with  $a$  for every scalar  $c$ , and has positive support exactly when  $c = 0$  or  $v(c) + \alpha > 0$ ; theorem 4.3 gives (4.12). Equal maps give  $(c_1 - c_2)a = 0$  and  $a \neq 0$ , so  $c_1 = c_2$ ; corollary 4.8 gives the group assertion. Meromorphic division was used only to identify the ratio: the resulting coefficient  $ca$  is holomorphic on the original  $U$ , including at the zeros of  $a$ .  $\square$

Sources 03 and 05 state this theorem with the same hypotheses, the same conclusion — source 05 writes  $v(c) + \delta > 0$  for  $\delta = v(W_F)$ , which is (4.12) verbatim — the same meromorphic-ratio proof and the same remark that zeros of  $a$  need not be removed. It is therefore printed once, per convention 1.1(i). Two consequences belong to source 05 and survive separately: if the leading coefficient of  $W_F$  is nonzero at  $z_*$ , no nonidentity centralizer element fixes  $z_*$ , so the basepoint normalization in theorem 14.13 removes exactly the centralizer freedom, and the unnormalized conjugacies between two conjugate maps form a torsor under either centralizer. The admissible times  $c$  may have negative valuation provided  $cW_F$  stays positive; the positivity threshold is part of the conclusion, not an unrestricted large-time claim.

*Remark 4.13* (Limits of the centralizer statement). Connectedness is necessary: otherwise one obtains independent scalar series on different ordinary components. In dimension  $d > 1$  the correct general statement is only that commuting positive maps have commuting positive logarithmic vector fields, and those need *not* be scalar multiples of one another; source 03 states that the one-variable meromorphic-ratio argument cannot classify higher-dimensional Lie centralizers and that additional first integrals or commuting geometric structures can occur, and source 05 states that in several variables there is no single reciprocal time one-form encoding a vector field, so the centralizer proof and the finite-period normal form must not be transported. The Hamiltonian centralizer theorem 15.12 is a genuinely different several-variable statement, proved by a different argument in a different algebra.

**Proposition 4.14** (Workspace invariance). *Let  $\Gamma \hookrightarrow \Gamma'$  be an ordered group inclusion. The substitution inverse, the positive logarithm and exponential, and the normalized nonresonant linearizer of [theorem 6.1](#) are unchanged when their inputs are regarded in the larger workspace; their support certificates remain valid; a normalized positive solution constructed in the larger workspace acquires no exponents outside  $\Gamma$ ; and evaluation identities remain valid after adjoining the supports of new halo points.*

*Proof.* The support words use only input exponents and finite addition, and the ordinary coefficient operations are independent of  $\Gamma$ , so the explicit formulas compute the same coefficients in  $\Gamma$  and  $\Gamma'$ , with support already inside  $\Gamma$ . Uniqueness of substitution inverses, of the exponential–logarithm correspondence and of normalized linearization identifies these with the corresponding positive solutions in the larger workspace. Evaluation follows by including the new positive point supports in [lemma 3.4](#).  $\square$

*Warning 4.15* (One recorded exception: the centralizer can grow). [Proposition 4.14](#) does *not* say that the centralizer [\(4.12\)](#) is unchanged by scalar extension: a larger workspace can supply new admissible times  $c$ . Source 03 records this exception explicitly. What is invariant is the particular normalized construction from the original data.

## 5 The homological operator and the radius of one inversion

Fix  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$  with  $|\lambda_j| = 1$  and assume NR:  $\lambda^\alpha \neq \lambda_j$  for  $|\alpha| \geq 2$  and all  $j$ . Let  $B_2^d$  denote coefficient vectors whose Taylor series have no terms of total degree 0 or 1, and put

$$\mathcal{L}h = h(\Lambda z) - \Lambda h(z), \quad (\mathcal{L}h)_{j,\alpha} = (\lambda^\alpha - \lambda_j)h_{j,\alpha}, \quad (5.1)$$

so that under NR the inverse on formal coefficients is

$$(\mathcal{L}^{-1}g)_j(z) = \sum_{|\alpha| \geq 2} \frac{g_{j,\alpha}}{\lambda^\alpha - \lambda_j} z^\alpha. \quad (5.2)$$

It preserves total polynomial degree and acts coefficientwise on Hahn families without changing Hahn exponents. Recall  $\text{rad}_{\text{iso}}(g)^{-1} = \limsup_m (\max_{|\alpha|=m,j} |g_{j,\alpha}|)^{1/m}$ , the radius of absolute convergence on centered equal-radius polydisks (the count  $\binom{m+d-1}{d-1}$  of multi-indices of degree  $m$  is polynomial in  $m$  and does not affect the root test).

**Lemma 5.1** (Single-step analytic loss). *If  $\tau = \tau(\Lambda) < \infty$  then*

$$\text{rad}_{\text{iso}}(\mathcal{L}^{-1}g) \geq e^{-\tau} \text{rad}_{\text{iso}}(g). \quad (5.3)$$

*Hence  $\mathcal{L}^{-1}$  preserves  $\mathcal{O}(\mathbb{D}_R^d)_2^d$  when  $\tau = 0$ , preserves radius-free convergent germs and entire coefficients when  $\tau < \infty$ , and always preserves polynomial and formal coefficients under NR alone.*

*Proof.* By [\(5.2\)](#) the maximal degree- $m$  coefficient of  $\mathcal{L}^{-1}g$  is at most  $M_m$  times that of  $g$ ; take  $m$ -th roots and limsups. For  $\text{rad}_{\text{iso}}(g) = \infty$  the root growth of  $g$  is zero while  $\limsup M_m^{1/m} = e^\tau < \infty$ , so the product again has root growth zero. Polynomial support in the Taylor degree is not changed at all, and for formal coefficients [\(5.2\)](#) divides by nonzero constants.  $\square$

**Lemma 5.2** (Sharpness and analytic failure). *If  $\tau > 0$  and  $0 < R < \infty$ , there is  $\varphi \in \mathcal{O}(\mathbb{D}_R^d)_2^d$  for which  $\mathcal{L}^{-1}\varphi$  is not holomorphic on  $\mathbb{D}_R^d$ ; it can be chosen with Taylor radius exactly  $Re^{-\tau}$ , read as zero when  $\tau = \infty$ . If  $\tau = \infty$  there is even an entire  $\varphi \in \mathcal{O}(\mathbb{C}^d)_2^d$  for which  $\mathcal{L}^{-1}\varphi$  has zero radius.*

*Proof.* Choose increasing degrees  $m_k$  with  $m_k^{-1} \log M_{m_k} \rightarrow \tau$  and for each  $k$  a pair  $(\alpha_k, j_k)$ ,  $|\alpha_k| = m_k$ , realizing the maximal inverse divisor; discard finitely many degrees so that  $M_{m_k} > 1$ . Then

$$\varphi(z) = \sum_{k \geq 1} R^{-m_k} z^{\alpha_k} e_{j_k} \quad (5.4)$$

is holomorphic on  $\mathbb{D}_R^d$ , and its inverse has degree- $m_k$  coefficient of modulus  $R^{-m_k} M_{m_k}$  and no other nonzero coefficient, so its radius is exactly  $Re^{-\tau}$ . For the stronger assertion choose  $m_k$  with  $M_{m_k} \geq \exp(k^2 m_k)$  and put

$$\varphi(z) = \sum_{k \geq 1} \exp(-km_k) z^{\alpha_k} e_{j_k}, \quad (5.5)$$

which converges normally on every closed polydisk once  $R \exp(-k) \rightarrow 0$  and is therefore entire, while its inverse has a degree- $m_k$  coefficient of modulus at least  $\exp((k^2 - k)m_k)$  and Taylor radius zero.  $\square$

These witnesses use only ordinary analytic coefficient functions; no infinitesimal support manipulation is hidden in them. Sources 02, 03 and 06 each construct such a witness; in one variable source 02's forcing is the explicit rational function

$$P_R(z) = \frac{z^2}{1 - z/R} = \sum_{n \geq 2} R^{2-n} z^n, \quad (5.6)$$

for which  $\text{rad}(\mathcal{L}^{-1}P_R) = Re^{-\sigma}$  exactly — all its Taylor coefficients being  $R^{2-n}$ , so no choice of a sparse subsequence is needed — and source 02's entire witness sets  $q_{n_j} = |d_{n_j}|^{1/2}$  along a sequence with  $|d_{n_j}|^{-1/n_j} \rightarrow \infty$ . Source 06's version, [theorem 15.5](#), is on frequency divisors and is stated as a sharp *iff*; source 06's explicit super-Liouville instance is [proposition 15.17](#).

**Proposition 5.3** (A compact-disk estimate, offered only as a starting point). *Suppose  $d = 1$  and  $\sigma = 0$ , and let  $0 < r < s < R$ . Then*

$$\|\mathcal{L}^{-1}q\|_r \leq \|q\|_s \sum_{n \geq 2} \frac{(r/s)^n}{|\lambda^n - \lambda|}, \quad \|q\|_s = \max_{|z| \leq s} |q(z)|, \quad (5.7)$$

*and the numerical series converges.*

*Proof.* Cauchy's estimate gives  $|q_n| \leq \|q\|_s s^{-n}$ ; sum the absolutely convergent Taylor majorant on  $|z| \leq r$ , convergence following from  $\limsup |\lambda^n - \lambda|^{-1/n} = e^\sigma = 1$ .  $\square$

Source 02 offers [\(5.7\)](#) explicitly as a *starting point* for normed questions and not as part of any theorem: it shows the inverse is continuous for the compact-open Fréchet topology, at the cost of a slightly larger compact disk, while the domain of holomorphy remains the full open  $\mathbb{D}_R$ . None of the theorems below imposes or produces such an estimate.

## 6 The coefficient-category classification

### 6.1 The sharp five-row table, for an exact multiplier

Consider

$$F(z) = \Lambda z + f(z), \quad f \in \mathcal{H}_\Gamma(B)_+^d, \quad f_\gamma(0) = 0, \quad df_\gamma(0) = 0, \quad (6.1)$$

the last condition being the exactness of the multiplier: no infinitesimal linear part is permitted at any Hahn exponent (tag  $\text{SD}_{\text{ex}}$ , [section 2.3](#)).

**Theorem 6.1** (Universal coefficient-category classification). *For every nonzero  $\Gamma$  and every  $\Lambda$  satisfying NR, the following table gives necessary and sufficient conditions for every map (6.1) in the indicated category to admit a normalized conjugacy*

$$H(z) = z + h(z), \quad h \in \mathcal{H}_\Gamma(B)_+^d, \quad h_\gamma(0) = 0, \quad dh_\gamma(0) = 0, \quad H \circ F = \Lambda H. \quad (6.2)$$

*Whenever it exists, that conjugacy is unique among normalized positive-support conjugacies and its inverse lies in the same category.*

| Coefficient category $B$                                                               | Exact condition          |
|----------------------------------------------------------------------------------------|--------------------------|
| $\mathcal{O}(\mathbb{D}_R^d)$ , on the <i>same fixed</i> polydisk (fixed-polydisk row) | $\tau(\Lambda) = 0$      |
| $\mathbb{C}\{z_1, \dots, z_d\}$ , no common radius required (radius-free row)          | $\tau(\Lambda) < \infty$ |
| $\mathcal{O}(\mathbb{C}^d)$ , entire coefficients (new row)                            | $\tau(\Lambda) < \infty$ |
| $\mathbb{C}[z_1, \dots, z_d]$ , polynomial coefficients (new row)                      | NR only                  |
| $\mathbb{C}[[z_1, \dots, z_d]]$ , formal coefficients                                  | NR only                  |

*If  $S = \text{supp } f \subset \Gamma_{>0}$  then  $\text{supp } h \subseteq S^* \setminus \{0\}$ . Each  $h_\gamma$  depends on only finitely many coefficient functions of  $f$  — those at letters in  $D_S(\gamma)$  — and on at most  $\ell_S(\gamma)$  ordinary homological inversions. The result is compatible with enlargement of the Hahn workspace (proposition 4.14).*

Necessity is already visible in an input with support  $\{\eta\}$  for any single  $\eta > 0$ . Sufficiency is a genuinely nonlinear arbitrary-support statement: no countability, finite rank, or finitely generated grid assumption is made on  $S$ .

## 6.2 Sufficiency: a support-controlled inverse of the nonlinear operator

Write  $F = \Lambda z + f$ , seek  $H = z + h$ , and set

$$\mathcal{N}_f h = h(\Lambda z + f(z)) - h(\Lambda z) = \sum_{|\beta| \geq 1} \frac{(\partial^\beta h)(\Lambda z)}{\beta!} f^\beta, \quad (6.3)$$

so that Schröder's equation becomes

$$(\mathcal{L} + \mathcal{N}_f)h = -f. \quad (6.4)$$

The map  $F$  is nonlinear, but with  $F$  fixed this equation is *linear in  $h$* , and in the Hahn setting that observation is decisive:  $\mathcal{L}$  does not alter Hahn weights while  $\mathcal{N}_f$  adds positive support words, and both preserve the absence of coordinate terms of degrees zero and one. Whenever  $\mathcal{L}^{-1}$  acts on the category, put  $T = \mathcal{L}^{-1}\mathcal{N}_f$  and

$$h = - \sum_{k \geq 0} (-T)^k \mathcal{L}^{-1} f. \quad (6.5)$$

**Proposition 6.2** (Support-controlled inverse; route A). *Let  $B_2^d$  be one of the five categories of theorem 6.1 and suppose  $\mathcal{L}^{-1}$  preserves it. Then (6.5) defines the unique normalized positive conjugacy; its support lies in  $S^* \setminus \{0\}$ ; at weight  $\gamma$  only terms with*

$$k + 1 \leq \ell_S(\gamma) \quad (6.6)$$

*contribute, and only input functions  $f_\sigma$  with  $\sigma \in D_S(\gamma)$  are involved.*

*Proof.* Each term of  $\mathcal{N}_f$  contains at least one coefficient of  $f$ , hence adds a nonempty word in  $S$ , and the initial input  $\mathcal{L}^{-1}f$  already contains one letter; a term with  $k$  copies of  $T$  therefore contains at least  $k + 1$  letters. [Lemma 3.4](#) and [corollary 3.5](#) give the support inclusion, strong summability, (6.6), and the finiteness of the relevant letters and intermediate sums. Every contributing expression is assembled from finitely many derivatives, products, compositions with the fixed diagonal rotation, and applications of  $\mathcal{L}^{-1}$ , all of which preserve the assumed category, so  $h_\gamma \in B_2^d$ ; the operator geometric identity gives  $(I + T)h = -\mathcal{L}^{-1}f$ , which is (6.4), hence  $H \circ F = \Lambda H$ . For uniqueness let  $h, \tilde{h}$  be normalized positive solutions, not necessarily with the constructed support, and let  $\gamma_0$  be the least weight of  $q = h - \tilde{h} \neq 0$ ; since  $\mathcal{N}_f$  adds positive weights, the coefficient of  $\mathcal{N}_f q$  at  $\gamma_0$  vanishes, so  $\mathcal{L}q_{\gamma_0} = 0$  and NR gives  $q_{\gamma_0} = 0$ , a contradiction. Finally  $H$  has a positive inverse in the same category by [proposition 4.1](#), normalized by differentiating the inverse identity.  $\square$

Applying [lemma 5.1](#) gives sufficiency in all five rows. Note that the polynomial row requires no bound on the inverse divisors at all: a polynomial uses only finitely many of them at any one Hahn coefficient, and the infinitely many nonlinear corrections are separated into different positive Hahn weights.

**Theorem 6.3** (Support-controlled lifting; route B, drift permitted). *Let  $\lambda$  be an irrational unit multiplier ( $d = 1$ ), let  $A \subset \Gamma_{>0}$  be well ordered, and write*

$$F(z) = \lambda z + P(z), \quad P = \sum_{a \in A} t^a f_a, \quad \delta_a = f'_a(0), \quad \delta = \sum_a t^a \delta_a, \quad \Lambda_{\text{dr}} = \lambda + \delta, \quad N = P - \delta z, \quad (6.7)$$

with  $f_a \in z\mathbb{C}[[z]]$ , so that every coefficient of  $N$  vanishes to order at least two and the multiplier is permitted to drift. There is a unique normalized positive-support formal solution  $H = z + U$  of  $H(F(z)) = \Lambda_{\text{dr}} H(z)$ , with

$$\text{supp}(H - z) \subseteq A^* \setminus \{0\}, \quad (6.8)$$

and a normalized compositional inverse with the same bound. Moreover: (i) if  $\sigma = 0$  and all  $f_a$  are holomorphic on  $\mathbb{D}_R$ , all coefficients of  $H$  and  $H^{-1}$  are holomorphic on that same  $\mathbb{D}_R$ ; (ii) if  $\sigma < \infty$  and all  $f_a$  are convergent germs, the coefficients are convergent germs; (iii) if  $\sigma < \infty$  and all  $f_a$  are entire, the coefficients are entire; (iv) if all  $f_a$  are polynomials, all coefficients are polynomials, with no arithmetic assumption beyond NR. No uniform coefficient bound, no lower Archimedean bound on the positive exponents, and no convergence in a complex parameter is assumed.

*Proof.* Substituting  $H = z + U$  and expanding at  $\lambda z$  gives, after cancelling  $\Lambda_{\text{dr}} z$ ,

$$\mathcal{L}U = -N + \delta U - \sum_{\beta > 0} \sum_{k \geq 1} t^\beta \frac{h_\beta^{(k)}(\lambda z)}{k!} P(z)^k, \quad (6.9)$$

whose coefficient at  $\gamma \in A^* \setminus \{0\}$  is

$$\mathcal{L}h_\gamma = -N_\gamma + \sum_{\substack{a \in A, \beta \in A_{>0}^* \\ a + \beta = \gamma}} \delta_a h_\beta - \sum_{\substack{k \geq 1, \beta \in A_{>0}^* \\ a_1, \dots, a_k \in A \\ \beta + a_1 + \dots + a_k = \gamma}} \frac{h_\beta^{(k)}(\lambda z)}{k!} f_{a_1}(z) \cdots f_{a_k}(z). \quad (6.10)$$

Every sum is finite even for a non-Archimedean order: represent  $\beta$  by a word in  $A$  and append the  $a_i$ , then apply [lemma 3.4](#); and every occurring  $\beta$  is strictly less than  $\gamma$ . The right side vanishes to order at least two —  $N_\gamma$  by definition,  $\delta_a h_\beta$  by induction, and in the last line  $h_\beta^{(k)}$  vanishes to order

at least  $\max(2 - k, 0)$  while the product of  $k$  functions  $f_{a_i}$  vanishes to order at least  $k$ . Since  $A^*$  is well ordered and  $\mathcal{L}$  is invertible on  $z^2\mathbb{C}[[z]]$ , (6.10) defines  $h_\gamma$  by well-founded recursion, with all coefficients outside  $A^*$  set to zero. For (i)–(iv), induct on  $\gamma$ : differentiation, rotation, multiplication and finite addition preserve each stated algebra, and lemma 5.1 does the same for the homological inverse — and in case (i) the induction retains *exactly*  $\mathbb{D}_R$  at every exponent rather than replacing it by a sequence of smaller disks. For uniqueness compare two normalized positive-support solutions at the least exponent where they differ, as in proposition 6.2. Lemma 4.2 constructs the inverse in the same algebra, and  $H \circ F \circ H^{-1}(z) = \Lambda_{\text{dr}} z$  is then a Hahn identity.  $\square$

*Remark 6.4* (Two genuinely different routes to a support-controlled linearizer). Proposition 6.2 and theorem 6.3 prove overlapping statements by different means, and both are kept because each carries something the other does not. Route A (source 03) inverts the nonlinear Schröder operator as an *operator geometric series* (6.5), which is what yields the finite word-depth bound (6.6) and hence the radius and degree certificates theorems 6.9 and 6.11; it requires the multiplier to be exactly  $\Lambda$  and works in several variables. Route B (source 02) is an exponent-by-exponent well-founded recursion (6.10), which is what accommodates the *drift* term  $\delta_a h_\beta$ ; source 02 remarks that omitting that term gives the wrong recurrence and misses the obstruction of section 7 entirely. A third route, source 01’s Taylor-degree recurrence with a pre-cancelled infinitesimal divisor, appears as lemmas 8.1 and 8.3 and is what makes the *finite-return* threshold theorem 8.5 exact. Source 02 also records the route-A statement in category-free form: proposition 6.5.

**Proposition 6.5** (Reusable abstract form). *Let  $B$  be any ordinary coefficient algebra closed under differentiation, finite products and rotation by  $\lambda$ , and such that the nonresonant homological inverse preserves  $z^2 B$ . Then every positive Hahn perturbation  $F = \lambda z + \sum_a t^a f_a$  with  $f_a \in zB$  admits a unique normalized linearizer with coefficients in  $B$  and support in the additive monoid generated by the input support, and its inverse has the same properties. No norm on  $B$  is required.*

*Proof.* This is the proof of theorem 6.3 with the coefficient algebra left abstract.  $\square$

The common-disk, germ, entire and polynomial rows are then obtained by checking one closure property. Theorem 7.2 shows that such a closure check is sufficient but that passing to an *inductive limit* of domains cannot simply be interchanged with taking a full Hahn family: for  $0 < \sigma < \infty$  every coefficient belongs to the germ inductive limit while the entire Hahn family does not belong to the common-domain inductive limit. That noncommutation is the algebraic form of the exact radius-collapse phenomenon.

### 6.3 Necessity: the first-weight obstruction

**Proposition 6.6** (The first-weight obstruction). *Fix  $\eta > 0$  and an ordinary coefficient vector  $\varphi$  vanishing to order at least two, and let  $F = \Lambda z + t^\eta \varphi(z)$ . Any normalized positive conjugacy in any of the stated analytic categories has no nonzero coefficients of weight below  $\eta$ , and necessarily*

$$h_\eta = -\mathcal{L}^{-1}\varphi. \quad (6.11)$$

*Later Hahn weights, additional scales, and enlargement of  $\Gamma$  cannot repair this.*

*Proof.* If a coefficient below  $\eta$  were nonzero, take the least nonzero weight of  $h$ : there the right side of (6.4) vanishes and  $\mathcal{N}_f h$  also vanishes because it adds at least  $\eta$ , leaving  $\mathcal{L}h_\gamma = 0$ , contradicting NR. At weight  $\eta$  the same reasoning eliminates  $\mathcal{N}_f h$ , whose input would need weight at most zero, and the equation is exactly (6.11).  $\square$

*Completion of the proof of theorem 6.1.* For the fixed-polydisk row with  $\tau > 0$ , take (5.4) and  $f = t^n \varphi$ : proposition 6.6 forces a coefficient not holomorphic on the specified  $\mathbb{D}_R^d$ , so universal preservation of that polydisk fails. For the radius-free and entire rows with  $\tau = \infty$ , take the entire witness (5.5): the forced  $h_\eta$  has zero complex radius, so it is not even a convergent germ and certainly not entire. Higher Hahn coefficients, extra scales and enlargement of  $\Gamma$  cannot change the equation at this first weight. Polynomial and formal sufficiency was proved without arithmetic restrictions beyond NR, and the support and uniqueness claims are those of proposition 6.2; workspace compatibility is proposition 4.14.  $\square$

*Remark 6.7* (Nonresonance is not cosmetic). If  $\lambda^\alpha = \lambda_j$  for some  $|\alpha| \geq 2$ , take  $\varphi = z^\alpha e_j$ : (6.11) would require  $0 = -1$  in that Taylor coefficient. A universal normalized linearization statement therefore fails under MR even for a polynomial with one infinitesimal weight. What is needed is a different target equation, not division by a zero divisor; theorem 19.1 supplies one for multipliers of finite order, and question 24.3 records that the general infinite-order resonant case is open. The several-variable version of the same remark is source 02's: a first-order forcing  $tz^\alpha e_j$  lies outside the range of  $\mathcal{L}$  at that monomial and cannot be removed by a normalized near-identity change of variable.

**Corollary 6.8** (The entire-coefficient threshold; polynomial universality). *Universal linearization of entire-coefficient perturbations by entire-coefficient normalized coordinate changes holds exactly when the relevant divisor rate is finite. Universal polynomial-coefficient linearization holds for every nonresonant multiplier.*

*Proof.* Positive assertions: theorem 6.1 and theorem 6.3(iii), (iv). For the negative direction with an infinite rate, take the entire forcing  $q$  of lemma 5.2 and  $F = \lambda z + tq(z)$ : the coefficient of  $t$  in the normalized linearizer is  $-\mathcal{L}^{-1}q$ , which is not even a convergent germ.  $\square$

## 6.4 Quantitative certificates at a specified exponent

**Theorem 6.9** (Radius at a specified Hahn exponent). *Suppose  $\tau < \infty$  and  $f = \sum_{\sigma \in S} t^\sigma f_\sigma$  has convergent-germ coefficients. For  $\gamma \in S^* \setminus \{0\}$  choose positive  $r_\sigma \leq \text{rad}_{\text{iso}}(f_\sigma)$  for the finite set  $\sigma \in D_S(\gamma)$  and put  $r_{\text{in}}(\gamma) = \min_{\sigma \in D_S(\gamma)} r_\sigma$ . Then*

$$\text{rad}_{\text{iso}}(h_\gamma) \geq r_{\text{in}}(\gamma) \exp(-\ell_S(\gamma)\tau), \quad (6.12)$$

*which for inputs on one  $\mathbb{D}_R^d$  reads  $\text{rad}_{\text{iso}}(h_\gamma) \geq R \exp(-\ell_S(\gamma)\tau)$ ; entire input gives entire  $h_\gamma$ .*

*Proof.* Consider one term of (6.5) contributing at  $\gamma$ : its input letters lie in  $D_S(\gamma)$ ; derivatives, products and composition with a diagonal unitary map do not reduce the minimum ordinary Taylor radius; each application of  $\mathcal{L}^{-1}$  costs at most  $e^{-\tau}$  by lemma 5.1; and a term with  $k$  copies of  $T$  uses  $k+1$  inversions and has at least  $k+1$  letters, hence at most  $\ell_S(\gamma)$  inversions. Finitely many terms occur at this weight, so the minimum is preserved under their sum.  $\square$

*Warning 6.10* ((6.12) is a lower bound only). The single-inversion factor is sharp by lemma 5.2, but it is *not* asserted that the exponent  $\ell_S(\gamma)\tau$  is ever attained by a nonlinear example; cancellations can improve the radius substantially. In particular (6.12) alone does **not** decide whether one common positive radius survives when  $0 < \tau < \infty$  and word depths are unbounded, and source 03 states explicitly that its coefficientwise lower bound is not a proof that radius collapse occurs. That question is question 7.10; the source that answers it does so in a different category, and section 7 keeps the two apart.

**Theorem 6.11** (Polynomial coefficient degrees). *Suppose every  $f_\sigma$  is polynomial and put  $d_\sigma = \max_j \deg(f_{\sigma,j}) \geq 2$ . Then*

$$\deg h_\gamma \leq 1 + \max_{(\sigma_1, \dots, \sigma_k) \in \mathcal{W}_S(\gamma)} \sum_{i=1}^k (d_{\sigma_i} - 1), \quad (6.13)$$

and if all  $d_\sigma \leq D$  then  $\deg h_\gamma \leq 1 + \ell_S(\gamma)(D - 1)$  uniformly. No small-divisor bound is assumed. For polynomial and formal coefficients the theorem and the linearization construction also hold for any diagonal invertible nonresonant  $\Lambda$ , dropping  $|\lambda_j| = 1$ .

*Proof.*  $\mathcal{L}^{-1}$  preserves total degree; the initial term at one letter  $\sigma$  has degree at most  $d_\sigma = 1 + (d_\sigma - 1)$ . If a term for the word  $w$  has degree at most  $1 + \sum_{\sigma \in w} (d_\sigma - 1)$ , a term of  $\mathcal{N}_f$  differentiates it  $|\beta|$  times and multiplies by  $|\beta|$  further coefficient polynomials, giving degree at most  $1 + \sum_{\sigma \in w} (d_\sigma - 1) + \sum_{i \leq |\beta|} (d_{\sigma_i} - 1)$ ; a vanishing derivative contributes no term. Induct through (6.5) and take the finite maximum at  $\gamma$ . For polynomial and formal coefficients, composition with any invertible diagonal matrix preserves the category and the same coefficient equations apply.  $\square$

**Warning 6.12** (What “finite dependence” does and does not mean). [Proposition 6.2](#) and [theorems 6.9](#) and [6.11](#) bound the number of input *coefficient functions* and analytic operators needed at a specified Hahn weight. This is not an unconditional finite-time algorithm for arbitrary analytic inputs: one application of  $\mathcal{L}^{-1}$  to an arbitrary analytic function can involve infinitely many of its ordinary Taylor coefficients. Only with polynomial input and an effective support certificate does each requested Hahn coefficient become a finite exact algebraic calculation; [section C](#) states exactly what must then be certified. Source 02 adds the practical corollary that a domain certificate cannot be a single global Boolean flag: for the  $u^k v$  coefficient of [theorem 7.2](#) the appropriate certificate is the exact radius  $Re^{-(k+1)\sigma}$ , and reporting merely “each coefficient converges” would erase the decisive failure.

## 6.5 Actual conjugacies on halos and on the monad

**Proposition 6.13** (Actual conjugacy on a finite halo). *Under [theorem 6.1](#) or [theorem 6.3\(i\)](#),  $H^\#$  is a bijection of  $(\mathbb{D}_R^d)_\Gamma^\#$  with inverse  $(H^{-1})^\#$ ; both  $F^\#$  and multiplication by the multiplier are bijections of that halo; and  $H^\#(F^\#(w)) = \Lambda H^\#(w)$  there. With entire or polynomial coefficients the same holds on  $\mathbb{C}^\# = \mathbb{C}^d + \mathfrak{m}_\Gamma^d$ .*

*Proof.* Hahn–Taylor evaluation respects composition whenever the reductions map the base domains into one another, by expanding both sides and regrouping at a fixed exponent ([lemma 3.4](#) reduces this to finite Taylor identities). The reductions of  $H, H^{-1}$  are the identity and preserve  $\mathbb{D}_R^d$ ; the reduction of  $F$  and of the linear normal form is  $z \mapsto \Lambda z$ , which preserves  $\mathbb{D}_R^d$  because  $|\lambda_j| = 1$ . Positive perturbations of the invertible  $\Lambda$  have a Hahn inverse, so the conjugate maps are bijections of the halo.  $\square$

**Proposition 6.14** (The monad needs no arithmetic). *For every irrational unit multiplier, the formal-coefficient solution of [theorem 6.1](#)/[theorem 6.3](#) evaluates on  $\mathfrak{m}_\Gamma^d$  and yields a bijective conjugacy there, with **no** arithmetic hypothesis, even when some ordinary coefficient series have radius zero.*

*Proof.* At  $\xi \in \mathfrak{m}_\Gamma^d$  use the formal Taylor coefficients at 0 in (3.4): a coefficient may be an arbitrary element of  $\mathbb{C}[[z]]$ , since its scalar coefficients all have valuation zero while powers of  $\xi$  carry positive-support factors, and [lemma 3.4](#) makes the family summable. The condition  $F(0) = 0$  keeps  $F(\xi) \in \mathfrak{m}_\Gamma^d$ , and the normalized inverse and conjugacy identities survive evaluation.  $\square$

**Proposition 6.14** is the exact scope limit of every negative result in this report: they concern analyticity of the ordinary coefficient *functions on neighborhoods*, not the absence of every form of surcomplex linearization at infinitesimal arguments. There is no assertion that every algebraic conjugacy of evaluated functions must arise from a coefficient family, and failure in one analytic category prohibits no set-theoretic conjugacy.

## 6.6 Several variables

For  $d \geq 1$ ,  $L = \text{diag}(\lambda_1, \dots, \lambda_d)$  with  $|\lambda_j| = 1$  and NR, put

$$\Delta_n = \min_{|\alpha|=n, 1 \leq j \leq d} |\lambda^\alpha - \lambda_j| > 0, \quad \sigma_d(L) = \limsup_{n \rightarrow \infty} \frac{\log^+ \Delta_n^{-1}}{n}. \quad (6.14)$$

All several-variable assertions below are about the specified *equal-radius* polydisk geometry  $\mathbb{D}_R^d$ ; no claim about an optimal general Reinhardt domain is made or needed.

**Lemma 6.15** (Multivariable homological inverse).  *$\mathcal{L}_L h = h(Lz) - Lh(z)$  is invertible on vector formal series of order at least two, with  $(\mathcal{L}_L^{-1} q)_{\alpha,j} = q_{\alpha,j}/(\lambda^\alpha - \lambda_j)$ . If  $\sigma_d = 0$  it preserves holomorphy on every  $\mathbb{D}_R^d$ ; if  $\sigma_d < \infty$  it preserves radius-free germs and entire functions, a finite nonzero isotropic radius losing at most the factor  $e^{-\sigma_d}$ ; and it preserves polynomial vectors under NR alone.*

*Proof.*  $z^\alpha e_j$  is an eigenvector with eigenvalue  $\lambda^\alpha - \lambda_j$ . Taking maxima at total degree  $n$  gives  $\max_{|\alpha|=n,j} |(\mathcal{L}_L^{-1} q)_{\alpha,j}| \leq \Delta_n^{-1} \max_{|\alpha|=n,j} |q_{\alpha,j}|$ ; apply the root test, the count of degree- $n$  monomials having only polynomial growth.  $\square$

**Theorem 6.16** (Multivariable support-controlled lifting). *Let  $F(z) = Lz + \sum_{a \in A} t^a f_a(z)$  with  $f_a(0) = 0$ ,  $A \subset \Gamma_{>0}$  well ordered, and  $M = DF(0) = L + \sum_a t^a Df_a(0)$ . Under NR there is a unique normalized formal coordinate change  $H = z + U$  with  $DU(0) = 0$  coefficientwise and*

$$H(F(z)) = MH(z), \quad \text{supp}(H - z) \subseteq A^* \setminus \{0\}. \quad (6.15)$$

*If  $\sigma_d = 0$  and all  $f_a$  are holomorphic on  $\mathbb{D}_R^d$ , then  $H$  and its inverse have coefficients holomorphic on that same polydisk; for finite  $\sigma_d$ , radius-free germs and entire coefficients are preserved; polynomial coefficients are preserved for every nonresonant  $L$ . The infinitesimal matrix  $M - L$  need not be diagonal.*

*Proof.* With  $P = F - Lz$ ,  $E = M - L$ ,  $N = P - Ez$ , the vector form of (6.9) is

$$\mathcal{L}_L U = -N + EU - \sum_{\beta > 0} \sum_{k \geq 1} \frac{t^\beta}{k!} D^k h_\beta(Lz)[P(z), \dots, P(z)], \quad (6.16)$$

with  $D^k h$  the symmetric  $k$ -linear derivative. At a fixed Hahn exponent only finitely many terms occur by lemma 3.4, every  $h_\beta$  occurring has strictly smaller exponent, and the right side has total Taylor order at least two by the order count of theorem 6.3. Apply lemma 6.15 by well-founded recursion; the operations preserve the common polydisk because  $L$  does. The least-exponent argument proves uniqueness, and the inverse-coordinate recursion is vectorial with coefficient one on the new unknown, so lemma 4.2 applies with multilinear derivatives.  $\square$

As in proposition 6.13, common-polydisk solutions evaluate to actual conjugacies on  $(\mathbb{D}_R^d)_\Gamma^\#$ .

**Theorem 6.17** (A common coordinate for the centralizer). *Suppose  $d = 1$ ,  $\sigma(\lambda) = 0$ , and  $F$  satisfies [theorem 6.3\(i\)](#) on  $\mathbb{D}_R$  with normalized linearizer  $H$ . Let  $G(z) = \mu z + \sum_{b \in B} t^b g_b(z)$  with  $|\mu| = 1$ ,  $g_b \in z\mathcal{O}(\mathbb{D}_R)$ ,  $B \subset \Gamma_{>0}$  well ordered, and suppose  $G \circ F = F \circ G$ . Then, with  $M = G'(0)$ ,*

$$H \circ G \circ H^{-1}(z) = Mz, \quad (6.17)$$

*with **no** nonresonance or small-divisor assumption on  $\mu$ . Consequently every such  $G$  commuting with  $F$  is linearized by the same  $H$ , any two such maps commute with each other, and  $G'(0)$  determines  $G$  uniquely — in particular  $G'(0) = 1$  forces  $G = \text{id}$ .*

*Proof.* All compositions are legitimate on  $\mathbb{D}_R$  by [lemma 3.4](#) and  $|\lambda| = |\mu| = 1$ . Put  $K = H \circ G \circ H^{-1}$ : it commutes with  $z \mapsto \Lambda_{\text{dr}} z$ , fixes zero and has derivative  $M$  there. Expanding  $K(z) = \sum_{n \geq 1} k_n z^n$  and comparing coefficients in  $K(\Lambda_{\text{dr}} z) = \Lambda_{\text{dr}} K(z)$  gives  $(\Lambda_{\text{dr}}^n - \Lambda_{\text{dr}})k_n = 0$  for  $n \geq 2$ ; the factor is a nonzero Hahn-field element because its exponent-zero coefficient is  $\lambda^n - \lambda \neq 0$ . Hence  $k_n = 0$  for  $n \geq 2$  and  $k_1 = M$ ; at each Hahn exponent the coefficient function has the Taylor series of a linear function, so the identity theorem on  $\mathbb{D}_R$  yields  $K(z) = Mz$  as a common-domain section. Conjugating back and using that two scalar multiplications commute gives the remaining assertions.  $\square$

The same proof works in the radius-free germ category when  $\sigma < \infty$ , and in the polynomial setting whenever the constructed linearizers have polynomial coefficients; in those statements equality is interpreted in the indicated category rather than on a disk that has not been shown to exist. A set-indexed commuting family is treated one member at a time with the same  $H$ ; no unsupported sum over the family is taken.

## 7 The drifted category: exact radius collapse and a trichotomy

This section is the one place in the report where the common-domain germ category is decided, and it is decided *only* in the drifted category  $\text{SD}_{\text{dr}}$ . [Warning 2.6](#) applies to everything in it.

### 7.1 Two independent monomials in one Hahn field

Work in the divisible ordered group

$$\Gamma_* = \mathbb{Q} + \sqrt{2}\mathbb{Q} \subset \mathbb{R}, \quad u = t, \quad v = t^{\sqrt{2}}.$$

The map  $(k, \ell) \mapsto k + \ell\sqrt{2}$  is injective on  $\mathbb{N}^2$  and only finitely many such pairs lie below any fixed real bound, so the image of  $\mathbb{N}^2$  is a well-ordered additive monoid and a double formal series  $\sum_{k, \ell \geq 0} b_{k, \ell} u^k v^\ell$  is an unambiguous Hahn series with that support. This is a use of two independent *exponents*, not of two ordinary complex parameter values. For a fixed finite  $R > 0$  put

$$F_{u, v}(z) = (\lambda + u)z + vP_R(z), \quad P_R \text{ as in (5.6)}, \quad (7.1)$$

so that both perturbation coefficients are holomorphic on  $\mathbb{D}_R$  and the multiplier is  $\Lambda_{\text{dr}} = \lambda + u$ . [Theorem 6.3](#) gives a unique formal linearizer with support in  $\mathbb{N} + \sqrt{2}\mathbb{N}$ . At  $v = 0$  the map is already linear and its normalized linearizer is the identity — either by setting  $v = 0$  in [\(6.10\)](#) or by ordinary formal uniqueness for the multiplier  $\lambda + u$ , every  $(\lambda + u)^n - (\lambda + u)$  having nonzero constant term in  $u$ . Hence its first nontrivial  $v$ -block is

$$H(z) = z + vQ(u, z) + O(v^2), \quad Q(u, z) = \sum_{k \geq 0} u^k h_{k, 1}(z), \quad (7.2)$$

where  $O(v^2)$  means only the blocks of  $v$ -degree at least two and is *not* an asymptotic assertion in an ordinary parameter. Extracting the coefficient of  $v$  in  $H(F) = \Lambda_{\text{dr}} H$  gives  $Q(u, (\lambda + u)z) - (\lambda + u)Q(u, z) = -P_R(z)$  and hence

$$h_{k,1}(z) = - \sum_{n \geq 2} R^{2-n} c_{n,k} z^n, \quad c_{n,k} = [u^k] \frac{1}{(\lambda + u)^n - (\lambda + u)}. \quad (7.3)$$

The radius problem is thereby reduced to an exact estimate of the *reciprocal jets*  $c_{n,k}$  for fixed  $k$  and increasing  $n$ .

## 7.2 Reciprocal jets and dominance of the highest pole

With  $a_{n,1} = n\lambda^{n-1} - 1$  and  $a_{n,j} = \binom{n}{j} \lambda^{n-j}$  for  $j \geq 2$  (and  $\binom{n}{j} = 0$  for  $j > n$ ), one has  $(\lambda + u)^n - (\lambda + u) = d_n + \sum_{j \geq 1} a_{n,j} u^j$  with  $d_n = \lambda^n - \lambda$ , so that for  $k \geq 1$

$$c_{n,k} = \sum_{r=1}^k (-1)^r d_n^{-r-1} \sum_{\substack{j_1 + \dots + j_r = k \\ j_i \geq 1}} a_{n,j_1} \dots a_{n,j_r}, \quad c_{n,0} = d_n^{-1}, \quad (7.4)$$

and the term with  $r = k$  is unique and equals

$$(-1)^k a_{n,1}^k d_n^{-k-1}. \quad (7.5)$$

It is this *highest* pole in the small divisor, and not merely its first power, that drives the successive radius losses. These are exactly the objects source 02's program checks.

**Lemma 7.1** (Uniform reciprocal-jet estimates). *For each fixed  $k \geq 1$  there is  $C_k > 0$  independent of  $n$  with*

$$|c_{n,k}| \leq C_k n^k \max\{1, |d_n|^{-k-1}\}, \quad (7.6)$$

$$|c_{n,k} - (-1)^k a_{n,1}^k d_n^{-k-1}| \leq C_k n^k |d_n|^{-k} \quad \text{when } |d_n| \leq 1. \quad (7.7)$$

Also  $|a_{n,1}| \geq n - 1$ . Consequently, along any sequence with  $n \rightarrow \infty$  and  $d_n \rightarrow 0$ ,

$$c_{n,k} = (-1)^k a_{n,1}^k d_n^{-k-1} (1 + o(1)), \quad (7.8)$$

which in particular excludes cancellation of the highest-pole contribution along the subsequences that control exponential growth.

*Proof.* The reverse triangle inequality gives  $|n\lambda^{n-1} - 1| \geq n - 1$  and  $|a_{n,1}| \leq n + 1$ ; for  $j \geq 2$ ,  $|a_{n,j}| \leq n^j/j!$ . Any product indexed by a composition  $j_1 + \dots + j_r = k$  is therefore bounded by a constant depending only on  $k$  times  $n^k$ , and there are finitely many compositions of  $k$ ; summing (7.4) gives (7.6), since  $|d_n| \leq 2$  and every pole order is at most  $k + 1$ . Removing the  $r = k$  term leaves pole orders at most  $k$ , giving (7.7) for  $|d_n| \leq 1$ . The removed term has modulus at least  $(n - 1)^k |d_n|^{-k-1}$ , and the ratio of the remainder bound to that lower bound is at most  $C_k(n/(n - 1))^k |d_n| \rightarrow 0$  when  $d_n \rightarrow 0$ .  $\square$

### 7.3 The exact collapse law and the trichotomy

**Theorem 7.2** (Exact coefficient-radius collapse; requires drift). *For the family (7.1) with  $\sigma = \sigma(\lambda) > 0$  ( $\sigma = \infty$  allowed), for every  $k \geq 0$ ,*

$$\limsup_{n \rightarrow \infty} |c_{n,k}|^{1/n} = e^{(k+1)\sigma}, \quad (7.9)$$

*and the coefficient of  $u^k v$  in the normalized formal linearizer has exact radius*

$$\boxed{\text{rad}(h_{k,1}) = R e^{-(k+1)\sigma}.} \quad (7.10)$$

*If  $0 < \sigma < \infty$  the full normalized linearizer lies in the radius-free ring  $\mathcal{A}_1$  over  $\Gamma_*$  but **not** in the common-domain germ ring  $\mathcal{R}_1$ ; if  $\sigma = \infty$  it does not even lie in  $\mathcal{A}_1$ .*

*Proof.* For  $k = 0$ , (7.9) is  $\limsup |d_n|^{-1/n} = e^\sigma$ , which follows from the definition of  $\sigma$  because  $|d_n| \leq 2$ , so replacing  $\log |d_n|^{-1}$  by its positive part does not change the limsup after division by  $n$ . For fixed  $k \geq 1$  and finite positive  $\sigma$ , (7.6) gives the upper bound after  $n$ -th roots since  $(C_k n^k)^{1/n} \rightarrow 1$ . Choose a subsequence  $n_j$  with  $n_j^{-1} \log |d_{n_j}|^{-1} \rightarrow \sigma$ ; since  $\sigma > 0$ ,  $d_{n_j} \rightarrow 0$ , so (7.8) applies and

$$|c_{n_j,k}|^{1/n_j} = |a_{n_j,1}|^{k/n_j} |d_{n_j}|^{-(k+1)/n_j} |1 + o(1)|^{1/n_j} \longrightarrow e^{(k+1)\sigma},$$

using  $n_j - 1 \leq |a_{n_j,1}| \leq n_j + 1$ . If  $\sigma = \infty$ , choose  $n_j$  with  $|d_{n_j}|^{-1/n_j} \rightarrow \infty$ : again  $d_{n_j} \rightarrow 0$  and the same asymptotic gives  $|c_{n_j,k}|^{1/n_j} \rightarrow \infty$ . The Cauchy–Hadamard formula in (7.3) now gives (7.10). For  $0 < \sigma < \infty$ , theorem 6.3(ii) says every coefficient of the full nonlinear linearizer is a convergent germ, while the radii (7.10) have infimum zero, so its coefficients cannot all be holomorphic on any  $\mathbb{D}_r$  with  $r > 0$ . For  $\sigma = \infty$  already  $h_{0,1}$  has radius zero. Formal uniqueness in theorem 6.3 shows that no alternative normalized solution, even with additional positive Hahn exponents, can avoid these coefficients.  $\square$

*Remark 7.3* (Why two exponents, and why the first one does not suffice). The first coefficient  $h_{0,1}$  alone loses the radius only by  $e^{-\sigma}$ , a loss compatible with the existence of *some* smaller common disk; it is the higher multiplier jets  $h_{k,1}$  that force losses by every power  $e^{-(k+1)\sigma}$  and thereby exclude such a disk. The argument is therefore strictly stronger than a first-order homological counterexample. The rational independence of 1 and  $\sqrt{2}$  prevents different pairs  $(k, \ell)$  from merging into the same Hahn coefficient and lets the  $v$ -linear block be isolated without contamination by higher  $v$ -powers. Source 02 records explicitly that replacing  $v$  by an integer power of  $u$  would merge blocks and could introduce cancellation, so that the proof would no longer establish the claimed radius for a particular Hahn coefficient, and that it does not silently make that replacement. It also records that the same necessity statement is *not* asserted for a fixed cyclic value group such as  $\mathbb{Z}$ , where different parameter blocks can collide at one Hahn exponent; resolving the effect of those collisions is a separate problem.

**Corollary 7.4** (A finite-support instability). *For every irrational multiplier with  $0 < \sigma < \infty$ , two nonzero positive Hahn coefficients already suffice for a holomorphic perturbation of a rotation to have a radius-free germ linearizer and no common-domain linearizer; the input coefficients can be taken to be rational functions of  $z$ , namely  $z$  at exponent 1 and  $P_R(z)$  at exponent  $\sqrt{2}$ .*

**Theorem 7.5** (The common-domain / radius-free trichotomy). *Let  $\text{CD}(\lambda)$  assert that for every set-sized ordered abelian  $\Gamma$ , every  $R > 0$  and every normalized input (6.7) with coefficients on  $\mathbb{D}_R$*

there is a normalized linearizer in  $\mathcal{R}_1$ ; let  $\text{CG}(\lambda)$  assert that every such input with radius-free germ coefficients has a normalized linearizer in  $\mathcal{A}_1$ . Then for an irrational unit multiplier

$$\text{CD}(\lambda) \iff \sigma(\lambda) = 0, \quad \text{CG}(\lambda) \iff \sigma(\lambda) < \infty, \quad (7.11)$$

and both equivalences survive replacing the universal quantifier over  $\Gamma$  by the single group  $\Gamma_*$ . In the positive common-domain case the linearizer and its inverse preserve the original disk, not merely an unspecified smaller one.

*Proof.* Positive implications: [theorem 6.3\(i\)](#), (ii). For  $0 < \sigma < \infty$ , [theorem 7.2](#) disproves CD within  $\Gamma_*$  while the radius-free implication still holds; for  $\sigma = \infty$  the same example has a nonconvergent coefficient and disproves both.  $\square$

| Arithmetic ( $\text{SD}_{\text{dr}}$ ) | Common ordinary disk ( $\mathcal{R}_1$ )         | Radius-free germ / entire | Polynomial coefficients |
|----------------------------------------|--------------------------------------------------|---------------------------|-------------------------|
| $\sigma = 0$                           | Always, with the same $\mathbb{D}_R$             | Always / always           | Always                  |
| $0 < \sigma < \infty$                  | Not universally; explicit radii collapse to zero | Always / always           | Always                  |
| $\sigma = \infty$                      | Not universally                                  | Neither universally       | Always                  |

Table 5: Universal normalized linearization in the drifted category. Negative common-domain assertions use  $\Gamma_* = \mathbb{Q} + \sqrt{2}\mathbb{Q}$ ; positive assertions hold for every set-sized ordered abelian group. The polynomial column means that each Hahn coefficient is a polynomial, not that the whole coordinate change is a finite polynomial over the Hahn field.

The distinction between the middle two columns of [table 5](#) is not a contradiction: for entire input coefficients a finite exponential loss of Taylor coefficient growth leaves every coefficient entire, whereas for a finite input radius repeated losses can exhaust every positive common radius. The fixed coefficient algebra, and not the mere existence of Hahn sums, determines the answer.

## 7.4 The several-variable collapse

**Theorem 7.6** (Exact multivariable radius collapse). Put  $P_R^{(d)}(z) = \prod_{j=1}^d (1 - z_j/R)^{-1} - 1 - \sum_j z_j/R = \sum_{|\alpha| \geq 2} R^{-|\alpha|} z^\alpha$ , let  $\mathbf{1} = (1, \dots, 1)^T$ , and in the same two-monomial workspace consider

$$F_{u,v}(z) = (1 + u)Lz + vP_R^{(d)}(z)\mathbf{1}, \quad (7.12)$$

the multiplicative detuning  $(1 + u)L$  being what makes the highest reciprocal pole uniform over every component and every multi-index. Assume NR and  $\sigma_d(L) > 0$ . Then for every  $k \geq 0$

$$\text{rad}_{\text{iso}}(h_{k,1}) = R e^{-(k+1)\sigma_d(L)}. \quad (7.13)$$

Consequently the universal common-polydisk-germ property holds exactly when  $\sigma_d = 0$ , and the universal radius-free germ property exactly when  $\sigma_d < \infty$ , when the workspaces include  $\Gamma_*$ .

*Proof.* At  $v = 0$  the map is linear with identity normalized coordinate change, so the coefficient of  $v$  satisfies the homological equation for the diagonal multiplier  $(1 + u)L$ : in component  $j$  at multi-index  $\alpha$  with  $n = |\alpha|$  it is  $-R^{-n}/((1 + u)^n \lambda^\alpha - (1 + u)\lambda_j)$ . Write  $D_{\alpha,j} = \lambda^\alpha - \lambda_j$ ,  $A_{\alpha,j,1} = n\lambda^\alpha - \lambda_j$  and  $A_{\alpha,j,r} = \binom{n}{r} \lambda^\alpha$  for  $r \geq 2$ ; [\(7.4\)](#) remains valid with these replacements,

and since all phases have modulus one,  $|A_{\alpha,j,1}| \geq n-1$  and  $|A_{\alpha,j,r}| \leq n^r/r!$ , so the estimates of [lemma 7.1](#) hold with constants independent not only of  $n$  but also of  $\alpha$  and  $j$ . The upper estimate gives  $\max_{|\alpha|=n,j} |C_{\alpha,j,k}| \leq C_k n^k \max\{1, \Delta_n^{-k-1}\}$ ; for the lower bound choose  $n$  along a subsequence realizing the limsup in (6.14) and at each such degree a pair  $(\alpha, j)$  attaining the finite minimum  $\Delta_n$ , positivity of  $\sigma_d$  ensuring  $\Delta_n \rightarrow 0$ , so that the highest-pole term dominates its remainder uniformly in the chosen multi-indices. Multiplication by  $R^{-n}$  and the definition of  $\text{rad}_{\text{iso}}$  give (7.13). For finite positive  $\sigma_d$ , [theorem 6.16](#) makes every coefficient a convergent germ while the displayed isotropic radii have infimum zero, and every neighborhood of 0 in  $\mathbb{C}^d$  contains an equal-radius polydisk; for infinite  $\sigma_d$  the first  $v$  coefficient already fails to be a germ.  $\square$

## 7.5 Continued fractions: every regime occurs, and none is Brjuno

Let  $\theta$  be irrational with continued-fraction convergents  $p_j/q_j$ , and  $\|x\|_{\mathbb{Z}}$  the distance to the nearest integer. The classical estimates

$$\frac{1}{q_j + q_{j+1}} < \|q_j \theta\|_{\mathbb{Z}} < \frac{1}{q_{j+1}}, \quad \|m\theta\|_{\mathbb{Z}} \geq \|q_j \theta\|_{\mathbb{Z}} \quad (q_j \leq m < q_{j+1}) \quad (7.14)$$

are standard [[12](#), [11](#)].

**Proposition 7.7** (Continued-fraction formula). *For  $\lambda = e^{2\pi i \theta}$ ,*

$$\sigma(\lambda) = \tau(\lambda) = \limsup_{j \rightarrow \infty} \frac{\log q_{j+1}}{q_j}. \quad (7.15)$$

*Proof.* For real  $x$ ,  $4\|x\|_{\mathbb{Z}} \leq |1 - e^{2\pi i x}| \leq 2\pi\|x\|_{\mathbb{Z}}$  — the upper bound from  $|\sin y| \leq |y|$ , the lower from concavity of  $\sin$  on  $[0, \pi/2]$  — and constants do not affect a logarithmic limsup divided by  $m$ . Hence both invariants equal  $\limsup_m m^{-1} \log \|m\theta\|_{\mathbb{Z}}^{-1}$ , with  $m = n-1$  and  $m/(m+1) \rightarrow 1$ . At  $m = q_j$ , (7.14) gives the lower bound  $\log q_{j+1}/q_j$ ; for  $q_j \leq m < q_{j+1}$  the other half gives  $m^{-1} \log \|m\theta\|_{\mathbb{Z}}^{-1} \leq \log(q_j + q_{j+1})/q_j \leq \log q_{j+1}/q_j + (\log 2)/q_j$ , and  $(\log 2)/q_j \rightarrow 0$ .  $\square$

[Equation \(7.15\)](#) is the point at which *one* continued-fraction quantity governs both of the one-variable divisor rates  $\text{SD}_{\text{dr}}$  and  $\text{SD}_{\text{ex}}$ ; sources 02 and 03 prove it independently with the same estimates, so it is printed once. It does *not* identify the two notions: the hypotheses that accompany them, drift versus exactness, remain different ([warning 2.6](#)).

The classical Brjuno condition is

$$B(\theta) = \sum_{j \geq 0} \frac{\log q_{j+1}}{q_j} < \infty, \quad (7.16)$$

strictly stronger than the mere convergence of those summands to zero, which is exactly the vanishing of the rate.

**Theorem 7.8** (Explicit arithmetic separations). *The following continued fractions are explicitly specified, not numerically approximated.*

- (a) Rate zero, not Brjuno (source 03's form). *With  $a_1 = 16$  and  $a_{k+1} = \lceil \exp(q_k/k) \rceil$ ,  $q_{k+1} = a_{k+1}q_k + q_{k-1}$ ,  $q_0 = 1$ ,  $q_1 = 16$ , one has rate 0 and  $B(\theta) = \infty$ .*
- (b) Rate zero, not Brjuno (source 02's form). *With  $a_1 = 2$  and  $a_{j+1} = \lceil \exp(q_j/(j+1)) \rceil$ , the same conclusion holds.*
- (c) Any prescribed finite rate  $s > 0$ .  $a_{j+1} = \lceil e^{sq_j} \rceil$  gives rate exactly  $s$ .

- (d) Infinite rate.  $a_{j+1} = \lceil e^{q_j^2} \rceil$ , or  $a_{k+1} = \lceil \exp(kq_k) \rceil$ , gives rate  $\infty$ .
- (e) Source 01's non-Brjuno witness. With  $q_{-1} = 0$ ,  $q_0 = 1$  and  $a_{n+1} = 2^{q_n^2}$ , one has  $(\log q_{n+1})/q_n \geq q_n \log 2$ , so  $B(\theta) = \infty$  (this one is used in [proposition 10.1](#), where only divergence of the Brjuno sum is needed).

Consequently every row of [theorem 6.1](#) and every regime of [theorem 7.5](#) is a nonempty arithmetic possibility, and the rate-zero rows apply to rotations that are not Brjuno.

*Proof.* In (a), every partial quotient is a finite positive integer, so  $\theta$  is a well-specified irrational and  $q_k \rightarrow \infty$ ; from the recursion  $q_k \exp(q_k/k) \leq q_{k+1} \leq q_k(\exp(q_k/k) + 2)$ , whence

$$\frac{1}{k} + \frac{\log q_k}{q_k} \leq \frac{\log q_{k+1}}{q_k} \leq \frac{1}{k} + \frac{\log q_k}{q_k} + \frac{\log(1 + 2e^{-q_k/k})}{q_k}.$$

Both outer expressions tend to zero, so the rate is zero by [proposition 7.7](#), while the middle expression is at least  $1/k$ , so [\(7.16\)](#) diverges. Case (b) is identical with  $j + 1$  in place of  $k$  and  $\log 3$  in place of  $\log(1 + 2e^{-q_k/k})$ . In (c) and (d) the same estimates give the quotient tending to  $s$ , respectively to  $q_j + o(1)$  or  $k + o(1)$ . In (e) the recursion gives  $q_{n+1} \geq a_{n+1}q_n = 2^{q_n^2}q_n$ , so  $\log q_{n+1} \geq q_n^2 \log 2$ .  $\square$

*Remark 7.9* (No contradiction with Brjuno–Yoccoz). By the Brjuno–Yoccoz theorem the ordinary quadratic  $Q_\lambda(w) = \lambda w + w^2$  is locally analytically linearizable exactly for Brjuno rotations [[13](#), [14](#)]. For the multipliers of [theorem 7.8\(a\)](#), (b) it is therefore *not* analytically linearizable, while every positive Hahn perturbation with coefficients on a fixed disk *is* linearizable on that same disk. This weakens nothing: the ordinary nonlinear part in  $Q_\lambda$  is of size one, whereas the nonlinear part in [theorem 6.1](#) has strictly positive Hahn support and zero ordinary reduction, and the resulting conjugacy is a Hahn family, not a convergent ordinary family under a numerical specialization of  $t$ . Sources 01, 02 and 03 all invoke Brjuno–Yoccoz for exactly this separation, as an external classical theorem which none of them reproves.

## 7.6 Resolution: which category answers which question

**Question 7.10** (Common-domain germs for a finite positive rate; *open* in the exact-multiplier category). Assume  $0 < \tau(\Lambda) < \infty$  and an *exact* multiplier ([\(6.1\)](#), tag  $\text{SD}_{\text{ex}}$ ). Characterize the positive perturbations whose coefficients are holomorphic on one common ordinary polydisk and whose unique radius-free linearizer has coefficients holomorphic on *some* common, possibly smaller, polydisk.

*Warning 7.11* (The two sources answer different questions; do not average them). Source 03 poses [question 7.10](#) as open and proves only the word-depth lower bound [\(6.12\)](#), stating explicitly that this does not prove radius collapse ([warning 6.10](#)). Source 02 proves outright *failure*:  $\text{CD}(\lambda)$  holds iff  $\sigma(\lambda) = 0$ , with the exact law [\(7.10\)](#), so for  $0 < \sigma < \infty$  the linearizer is a radius-free coefficient germ with no common domain whatsoever.

This is not a contradiction, and each source says so. Source 02's negative direction **requires** multiplier drift  $F'(0) = \lambda + u$  and two rationally independent positive exponents, both of which source 03 excludes: source 03's multiplier is *exactly*  $\Lambda$ , with no infinitesimal linear part at any Hahn exponent. The resolution adopted here, and the only one consistent with both proofs:

- [Theorems 7.2](#) and [7.5](#) are the answer **in the drifted category**  $\text{SD}_{\text{dr}}$ . There [question 7.10](#) is settled, negatively, and the law [\(7.10\)](#) is exact.

- [Question 7.10](#) remains **open** in the fixed-multiplier category  $\text{SD}_{\text{ex}}$ . Source 02 itself records that eliminating drift removes the direct reciprocal-jet mechanism and that a different counterexample, or a positive cancellation theorem, would be needed.
- The two must not be averaged into a single “intermediate case” claim, and (7.10) must never be quoted without its drift hypothesis.

For the weaker requirement of retaining the *original* disk, one positive parameter is already enough: [lemma 5.2](#) and the first-order equation give failure whenever the rate is positive. The unresolved issue concerns the existence of *some* smaller common neighborhood for every input.

## 8 The exact finite-return threshold and its shell cycles

Everything in this section is under  $\text{RES}_q$  and  $\text{SD}_0$ :  $\lambda$  is *never* a root of unity, its residue  $\alpha = \bar{\lambda}$  has exact finite order  $q$ , and **no** arithmetic condition of any kind is imposed. [Warnings 2.1](#), [2.4](#) and [2.7](#) apply throughout. The ambient  $\Gamma$  is divisible here, so that  $K_\Gamma$  is algebraically closed and rational division of exponents is available.

### 8.1 Two support-controlled linearization lemmas; route C

For  $T(X) = \nu X + \sum_{j=2}^D d_j X^j$  and  $H(X) = X + \sum_{n \geq 2} h_n X^n$  with  $\nu \neq 0$  not a root of unity, comparison of degree  $n$  in  $H(T(X)) = \nu H(X)$  gives the triangular recurrence

$$(\nu - \nu^n)h_n = \sum_{j=1}^{n-1} h_j [X^n]T(X)^j, \quad h_1 = 1, \quad (8.1)$$

whose denominators are nonzero, so (8.1) defines a unique normalized formal conjugacy over any field of characteristic zero. What remains is to control all of its coefficients *simultaneously*. This is route C of [remark 6.4](#).

**Lemma 8.1** (Nonresonant residue). *Suppose  $T \in \mathcal{O}_\Gamma[X]$ ,  $v(\nu) = 0$ , and  $\alpha = \bar{\nu}$  is not a root of unity in  $\mathbb{C}$ . Then the normalized formal conjugacy and its inverse belong to  $\mathcal{B}_\mathbf{M}$  for a positive-support monoid  $\mathbf{M}$  depending only on the finitely many coefficients of  $T$ .*

*Proof.* Write  $\nu = \alpha + e$  with  $e \in \mathfrak{m}_\Gamma$  and let  $S$  be the finite union of the positive supports of  $e$  and of the nonlinear coefficients  $d_j$ . For  $n \geq 2$  the element  $\nu - \nu^n$  has nonzero constant term  $\alpha - \alpha^n$ , so its inverse is a formal power series in  $e$  with ordinary complex coefficients and support in  $\langle \text{supp } e \rangle$ . Then (8.1) proves inductively that every  $h_n$  is integral with support in  $\mathbf{M} = \langle S \rangle$ , and the formal inverse has the same property by [lemma 3.9](#).  $\square$

*Remark 8.2* (No Diophantine bound, and why that is a real statement). There is *no* lower bound on the ordinary complex numbers  $\alpha - \alpha^n$  anywhere in [lemma 8.1](#). Those constants may be arbitrarily small in the ordinary modulus and still all have Hahn valuation zero ([remark 2.3](#)), which is exactly what makes this a  $\text{SD}_0$  statement. It is not the assertion that “all small divisors are units, so all analytic problems disappear”: [proposition 10.1](#) shows what is lost, and [example 3.10](#) shows that coefficient-valuation bounds alone are insufficient at rank  $> 1$ .

**Lemma 8.3** (Divisible displacement: cancelling a common infinitesimal divisor). *Let  $0 \neq c \in \mathfrak{m}_\Gamma$ ,  $A \in X^2 \mathcal{O}_\Gamma[X]$  and*

$$T(X) = (1 + c)X + cA(X) = X + cV(X), \quad V = X + A. \quad (8.2)$$

Then  $1 + c$  is not a root of unity, and the normalized formal conjugacy of  $T$  to  $(1 + c)X$ , together with its inverse, has all coefficients supported in one positive-support monoid generated by  $\text{supp } c$  and the positive parts of the coefficients of  $A$ .

*Proof.* For a positive integer  $n$  the first nonzero term of  $(1 + c)^n - 1$  is  $nc$ , all remaining terms having strictly larger valuation, so the multiplier is not a root of unity. For  $n \geq 2$ ,

$$(1 + c) - (1 + c)^n = -c(n - 1)u_n(c), \quad u_n(0) = 1, \quad u_n \in \mathbb{Q}[c]. \quad (8.3)$$

For  $j < n$  the coefficient  $[X^n]T(X)^j$  contains at least one nonlinear term, hence is divisible by  $c$  as a polynomial in  $c$  and the coefficients of  $A$ . Substituting this and (8.3) into (8.1) cancels one factor  $c$  before any valuation estimate is made. The remaining division is by the nonzero ordinary integer  $n - 1$  — a valuation unit in equal characteristic zero, which is where [warning 2.7](#) enters — and by  $u_n(c)$ , whose inverse is supported in  $\langle \text{supp } c \rangle$ . Induction on  $n$  gives one common monoid, a finite union of well-ordered positive supports; inversion of  $H$  uses only polynomials in its coefficients.  $\square$

This cancellation is the whole reason the exact threshold is read from a *return* map: multiplying all the small divisors in (8.1) together would lose the common factor present in every nonlinear numerator. Source 01 states this explicitly, and no other source performs this cancellation.

**Lemma 8.4** (Roots on the unit shell). *Let  $Q \in \mathcal{O}_\Gamma[X]$  with  $Q(0) = 1$  and suppose  $P = \bar{Q}$  has degree  $M > 0$ . Then  $Q$  has no root in  $\mathfrak{m}_\Gamma$  and exactly  $M$  roots of valuation zero, counted with algebraic multiplicity; for each nonzero root  $a \in \mathbb{C}$  of  $P$ , the roots of  $Q$  with residue  $a$  have total multiplicity equal to the multiplicity of  $a$  in  $P$ .*

*Proof.* A root of positive valuation is impossible because the constant term would strictly dominate. Factor in the algebraically closed  $K_\Gamma$  as  $Q(X) = \prod_j (1 - X/\beta_j)$  with roots repeated by multiplicity; every  $v(\beta_j) \leq 0$  so all factors are integral, a factor with  $v(\beta_j) < 0$  reduces to 1, and one with  $v(\beta_j) = 0$  reduces to  $1 - X/\bar{\beta}_j$ . Reducing this finite factorization proves all assertions; a simple root of  $P$  corresponds to exactly one simple root of  $Q$  in that residue class.  $\square$

## 8.2 The exact maximal ball

Let

$$f(z) = \lambda z + \sum_{n=2}^d a_n z^n \in K_\Gamma[z], \quad d \geq 2, \quad a_d \neq 0, \quad v(\lambda) = 0, \quad (8.4)$$

with  $\lambda$  not a root of unity, and put  $\eta(f) = \max_{a_n \neq 0} (-v(a_n)/(n - 1))$ ,  $\alpha = \bar{\lambda}$ . All maxima are finite maxima over nonzero polynomial coefficients.

**Theorem 8.5** (Exact finite-return linearization). *Define  $r = r(f)$  as follows. If  $\alpha$  is not a root of unity, set  $r = \eta(f)$ . If  $\alpha$  has exact order  $q \geq 1$  (tag  $\text{RES}_q$ ), write*

$$g(z) = f^{\circ q}(z) = \lambda^q z + \sum_{n=2}^{d^q} b_n z^n, \quad c = \lambda^q - 1, \quad \delta = v(c) > 0, \quad (8.5)$$

and set

$$r = \max_{b_n \neq 0} \frac{\delta - v(b_n)}{n - 1} \quad (8.6)$$

— read off the actual coefficients of the finite  $q$ -th return iterate, including all cancellations — in which case  $r > \eta(f)$ .

Let  $H(z) = z + \sum_{n \geq 2} h_n z^n$  be the unique normalized formal solution of  $H \circ f = \lambda H$ . Then  $H$  and its formal inverse are strongly Hahn-evaluable on  $B_r$  and their evaluated maps are mutually inverse isometries of  $B_r$ ; in particular  $f : B_r \rightarrow B_r$  is conjugate to multiplication by  $\lambda$ . More strongly, with  $\rho = t^r$ ,

$$H_r(X) = \rho^{-1} H(\rho X) = X + \sum_{n \geq 2} h_n \rho^{n-1} X^n \quad (8.7)$$

and its inverse have a uniform positive-support certificate, so in particular  $v(h_n) + (n-1)r \geq 0$  for  $n \geq 2$  — though the certificate is strictly stronger than that inequality.

No strictly larger centered valuation ball admits an injective conjugacy of  $f$  to multiplication by  $\lambda$ . In the nonresonant-residue case the obstruction is a nonzero  $p \in S_r$  with  $f(p) = 0$ ; in the  $\text{RES}_q$  case it is a nonzero  $p \in S_r$  of exact period  $q$ .

*Proof. Nonresonant residue.* Take  $\rho = t^\eta$  and  $F(X) = \rho^{-1} f(\rho X)$ ; its nonlinear coefficients have valuations  $v(a_n) + (n-1)\eta \geq 0$  with at least one equal to zero, so  $F \in \mathcal{O}_\Gamma[X]$  with nonlinear reduction. By lemmas 3.9 and 8.1 its normalized conjugacy and inverse are strongly summable mutually inverse isometries of  $\mathfrak{m}_\Gamma$ , and scaling back gives every existence and support assertion. For maximality apply lemma 8.4 to  $Q(X) = F(X)/(\lambda X) = 1 + \sum_{n \geq 2} (a_n \rho^{n-1}/\lambda) X^{n-1}$ , whose reduction is nonconstant:  $Q$  has a unit root  $u$ , and  $p = \rho u \in S_\eta$  satisfies  $f(p) = 0 = f(0)$ . Any centered ball strictly containing  $B_\eta$  contains that whole shell, and an injective conjugacy there would give  $\lambda H(p) = H(f(p)) = H(f(0)) = \lambda H(0)$ .

*The resonant threshold lies inside the injectivity scale.* With  $\alpha$  of order  $q$ , the multiplier  $\lambda^q$  has residue one and is not one because  $\lambda$  is not a root of unity, so  $c \neq 0$  and  $\delta > 0$ . Scaling by  $\sigma_0 = t^\eta$ , the polynomial  $F_\eta(X) = \sigma_0^{-1} f(\sigma_0 X)$  is integral with reduction of degree at least two; reduction commutes with polynomial iteration, so  $\sigma_0^{-1} g(\sigma_0 X) = \bar{F}_\eta^{\circ q}(X)$  also has degree at least two, and some  $n \geq 2$  satisfies  $v(b_n) + (n-1)\eta = 0$ . For that index  $(\delta - v(b_n))/(n-1) = \eta + \delta/(n-1) > \eta$ , so  $r > \eta$ . At the final scale  $\rho = t^r$  the polynomial  $F(X) = \rho^{-1} f(\rho X)$  has reduction exactly  $\alpha X$ : every nonlinear coefficient acquires positive valuation. It maps  $\mathfrak{m}_\Gamma$  to itself and preserves the residue direction of a unit up to multiplication by  $\alpha$ .

*Linearizing the return and recovering  $f$ .* At  $\rho = t^r$  write

$$G(X) = \rho^{-1} g(\rho X) = (1+c)X + cA(X), \quad A(X) = \sum_{n \geq 2} \frac{b_n \rho^{n-1}}{c} X^n \in X^2 \mathcal{O}_\Gamma[X], \quad (8.8)$$

integrality being exactly (8.6). Lemma 8.3 constructs the normalized conjugacy  $J$  of  $G$  to  $(1+c)X$  with a uniform support certificate. It remains to see that  $J$  linearizes  $F$  and not merely its  $q$ -th iterate: in the formal power-series group  $T = J \circ F \circ J^{-1}$  commutes with multiplication by  $\mu = 1+c$  because  $F$  commutes with  $G = F^{\circ q}$ , so writing  $T(X) = \sum_{n \geq 1} \tau_n X^n$  the commutation says  $\tau_n(\mu^n - \mu) = 0$ ; since  $\mu$  is not a root of unity all  $\tau_n$  with  $n \geq 2$  vanish, and the linear coefficient is  $\lambda$ , so  $T(X) = \lambda X$  and  $J$  is the unique normalized conjugacy of  $F$ . Adjoining the positive supports of the finitely many coefficients of  $F$  to the certificate, all compositions are legitimate by lemma 3.9, and scaling by  $\rho$  gives the existence, isometry and coefficient assertions.

*The exact-period obstruction.* Define the integral polynomial

$$Q(X) = \frac{G(X) - X}{cX} = 1 + \sum_{n \geq 2} \frac{b_n \rho^{n-1}}{c} X^{n-1}. \quad (8.9)$$

At least one nonlinear coefficient of  $G$  attains equality in (8.6), so  $P = \bar{Q}$  is nonconstant, and lemma 8.4 gives a unit root  $u$ ; then  $p = \rho u \in S_r$  and  $g(p) = p$ . The least period  $k$  of  $p$  divides  $q$ ; every point of its finite orbit lies on  $S_r$  and reduction of the normalized orbit multiplies by  $\alpha$  at each step, so if  $a = \bar{u} \neq 0$  then  $\alpha^k a = a$  and  $q \mid k$ , whence  $k = q$ . On any larger centered ball an injective conjugacy would give  $H(p) = H(f^{\circ q}(p)) = \lambda^q H(p)$ , and  $\lambda^q \neq 1$  forces  $H(p) = 0 = H(0)$ .  $\square$

*Remark 8.6* (Exactly what is exact). Theorem 8.5 determines the maximal *centered valuation ball* only: not the full pointwise summability locus of the formal series (exceptional evaluations on the bounding shell are allowed), not a maximal non-ball invariant domain, and not every analytic continuation of the conjugacy. The formal-coefficient family lands in  $\mathcal{F}_1$ ; theorem 9.1 is what reaches the fixed-polydisk row, and only under  $\text{RES}_q$ .

### 8.3 The shell polynomial and the closest cycles

Retain  $r, \rho, c, Q$  and put

$$P(X) = \bar{Q}(X), \quad M = \deg P = \max\{n - 1 : v(b_n) + (n - 1)r = \delta\}, \quad (8.10)$$

a finite ordinary complex polynomial with  $P(0) = 1$ : the initial polynomial on the first face of the return equation seen from the origin.

**Theorem 8.7** (Resonant shell cycles). *(i)  $f^{\circ q}(z) = z$  has exactly  $M$  nonzero solutions on  $S_r$ , counted with algebraic multiplicity; every such solution has exact period  $q$ ; and there are no nonzero periodic points in  $B_r$ .  
(ii)  $P(\alpha X) = P(X)$ , i.e.  $P \in \mathbb{C}[X^q]$ ; in particular  $q \mid M$ .  
(iii) If  $P$  is squarefree there is one simple point  $p_a = \rho(a + u_a)$ ,  $u_a \in \mathfrak{m}_\Gamma$ , for every root  $a$  of  $P$ ; these form exactly  $M/q$  distinct cycles, and  $f(p_a) = p_{\alpha a}$ .  
(iv) For each such simple point,*

$$\text{res} \left( \frac{(f^{\circ q})'(p_a) - 1}{c} \right) = aP'(a), \quad v((f^{\circ q})'(p_a) - 1) = \delta. \quad (8.11)$$

*Proof.* Counting and residue classes follow from lemma 8.4 applied to  $Q$ ; exact period  $q$  was proved in theorem 8.5; a nonzero periodic point in  $B_r$  would be sent by the injective conjugacy to a nonzero periodic point of multiplication by  $\lambda$ , which has none. For the symmetry write  $F = \rho^{-1}f(\rho X)$  and  $G = F^{\circ q} = X + cW(X)$  with  $W = XQ$ . The identity  $F \circ G = G \circ F$  gives  $(F(X + cW(X)) - F(X))/c = W(F(X))$ , an identity of integral polynomials whose reduction leaves only the first Taylor term on the left since  $c \in \mathfrak{m}_\Gamma$ ; as  $\bar{F} = \alpha X$  this is  $\alpha \bar{W}(X) = \bar{W}(\alpha X)$ , and substituting  $\bar{W} = XP$  gives  $P(\alpha X) = P(X)$ , so only degrees divisible by  $q$  occur. When  $P$  is squarefree, uniqueness in each residue class gives  $f(p_a) = p_{\alpha a}$ , and the action of  $\alpha$  on  $\mathbb{C}^\times$  has orbits of length  $q$ . Finally  $G'(X) = 1 + cW'(X)$ , and at  $p_a = \rho u$  with  $\bar{u} = a$  the derivative of the unscaled iterate equals  $G'(u)$ , whose reduction is  $\bar{W}'(u) = P(a) + aP'(a) = aP'(a) \neq 0$ .  $\square$

*Remark 8.8* (Multiplicity is not the number of distinct points). If  $P$  has a multiple root, theorem 8.7 gives a multiplicity count only. It does *not* say that several actual periodic points have collided: higher Hahn coefficients can split one repeated initial root into distinct points with the same residue direction, and in that situation  $v((f^{\circ q})'(p) - 1) = \delta$  need not hold, because the displayed residue may vanish. Example 12.2 exhibits the degeneracy concretely.

**Corollary 8.9** (A finite obstruction certificate). *In the  $\text{RES}_q$  case the data  $q, c, r, P$  certify exact maximality without computing a single coefficient of the infinite conjugacy. At most  $d^q - 1$  shell points occur with multiplicity, and their count is a positive multiple of  $q$ .*

*Proof.* The return polynomial has degree  $d^q$  and  $M = \deg P \leq d^q - 1$ ; exactness and positivity are [theorems 8.5](#) and [8.7](#).  $\square$

[Section B](#) states the accompanying four-step finite algebraic procedure and *exactly* what it is relative to.

## 8.4 A finite-return perturbation certificate

**Theorem 8.10** (Stability above the first return face). *Let  $f$  satisfy the  $\text{RES}_q$  hypotheses of [theorem 8.5](#) with data  $q, c, \delta, r, P$ , and let  $\tilde{f}$  be another nonlinear polynomial with the same multiplier  $\lambda$  and return coefficients  $\tilde{b}_n$  (pad either list with zeros). Suppose*

$$v(\tilde{b}_n - b_n) + (n - 1)r > \delta \quad \text{for every nonzero difference.} \quad (8.12)$$

*Then  $r(\tilde{f}) = r(f)$  and the shell polynomial of  $\tilde{f}$  is also  $P$ . If  $P$  is squarefree and at least one coefficient changes, put*

$$\epsilon = \min_{\tilde{b}_n \neq b_n} (v(\tilde{b}_n - b_n) + (n - 1)r - \delta) > 0; \quad (8.13)$$

*the closest cycles have a canonical matching by residue direction, matched points satisfy*

$$v(\tilde{p}_a - p_a) \geq r + \epsilon, \quad (8.14)$$

*the matching intertwines  $f$  and  $\tilde{f}$  on these finite cycle sets, and their leading multiplier ratios in [\(8.11\)](#) agree.*

*Proof.* At  $\rho = t^r$  the two normalized displacement polynomials are  $Q$  as in [\(8.9\)](#) and  $\tilde{Q}$  with  $\tilde{b}_n$  in place of  $b_n$ . The hypothesis says every coefficient of  $\tilde{Q} - Q$  has positive valuation, so  $\tilde{Q}$  is integral with the same nonconstant reduction  $P$ ; the inequalities defining the maximum hold at  $r$  for all  $\tilde{b}_n$  and at least one remains an equality, so  $r(\tilde{f}) = r$ . For a simple root  $a$  of  $P$  let  $u = p_a/\rho$  and  $\tilde{u} = \tilde{p}_a/\rho$ , both of residue  $a$ . When  $u \neq \tilde{u}$  the divided difference  $(Q(u) - Q(\tilde{u}))/(\rho(u - \tilde{u}))$  is integral with nonzero residue  $P'(a)$ ; moreover  $Q(u) = 0$  and  $Q(\tilde{u}) = (\tilde{Q} - Q)(\tilde{u})$ , of valuation at least  $\epsilon$ , whence  $v(u - \tilde{u}) \geq \epsilon$ , which is [\(8.14\)](#) (it is automatic if the two roots coincide). The direction rule  $a \mapsto \alpha a$  and [theorem 8.7](#) give the rest.  $\square$

**Warning 8.11** (The certificate is sufficient, not necessary; and it is about the return). A perturbation may change the first return face and nevertheless preserve its maximal slope, so [\(8.12\)](#) is sufficient only. It is also a condition on the *return* polynomial: it must not be silently replaced by the same inequalities on the original coefficients of  $f$ . Where coefficients are available only to a specified precision, [theorem 8.10](#) says which strict inequalities certify that the computed first face is already correct; at a tie or a possible cancellation more precision may be necessary, and treating a truncated zero as an exact zero would invalidate the maximum.

## 9 Common-domain lifting: one leading coefficient, two instantiations

The support arguments of [section 8](#) establish more than formal existence but less than ordinary analyticity of every coefficient function. Under  $\text{RES}_q$  a further argument supplies exactly that regularity, and it lands in the fixed-polydisk row of [section 3.2](#).

**Theorem 9.1** (Coherent realization of resonant linearization). *Under the  $\text{RES}_q$  hypotheses of [theorem 8.5](#), the scaled conjugacy  $H_r$  of (8.7) has an expansion*

$$H_r(X) = \sum_{\gamma \in \mathbf{M}} H_\gamma(X) t^\gamma \quad \text{in } \mathcal{O}(D)((t^\Gamma)) \quad (9.1)$$

for a single ordinary complex disk  $D$  centered at zero, with  $\mathbf{M}$  a positive-support monoid,  $H_0(0) = 0$ ,  $H'_0(0) = 1$ , and  $H_\gamma \in X^2\mathcal{O}(D)$  for  $\gamma > 0$ . One may take any sufficiently small disk on which the shell polynomial  $P$  has no zero, and the leading coefficient is

$$H_0(X) = X \exp \left( \int_0^X \left( \frac{1}{sP(s)} - \frac{1}{s} \right) ds \right), \quad (9.2)$$

whose integrand is holomorphic at zero. After an ordinary shrinking, the formal inverse of  $H_r$  also has all Hahn coefficient functions holomorphic on one common ordinary neighborhood.

*Proof.* Use  $G = X + cV$  from (8.8), so  $V = X + A = XQ$  and  $V_0 := \bar{V} = XP(X)$ ; all coefficients of the normalized conjugacy lie in a common monoid  $\mathbf{M}$  by [lemma 8.3](#). Taylor expansion of the formal identity  $H_r(X + cV) = (1 + c)H_r(X)$ , followed by cancellation of  $c$ , gives

$$VH'_r + \sum_{k \geq 2} \frac{c^{k-1}}{k!} V^k H_r^{(k)} = H_r. \quad (9.3)$$

This is first read coefficientwise in  $X$ , where only finitely many terms contribute at each degree because  $V$  has order one; it may also be regrouped at a fixed Hahn exponent, because the term of index  $k$  uses at least  $k - 1$  positive generators from  $c$ , so [lemma 3.4](#) makes that regrouping finite. At exponent zero, (9.3) becomes

$$V_0 H'_0 = H_0. \quad (9.4)$$

Choose an ordinary disk  $D$  centered at zero on which  $P$  has no zero; since  $P(0) = 1$  the function  $1/(XP(X)) - 1/X$  has a removable singularity at zero, and (9.2) defines the unique solution of (9.4) with  $H'_0(0) = 1$ , with  $H_0/X$  nonvanishing on  $D$ .

Now let  $\gamma > 0$  in the well-ordered monoid  $\mathbf{M}$  and assume  $H_\beta$  constructed on this same disk for every  $\beta < \gamma$ . Isolating the terms containing  $H_\gamma$  in (9.3) gives

$$LH_\gamma := V_0 H'_\gamma - H_\gamma = R_\gamma, \quad R_\gamma = -[t^\gamma] \left( (V - V_0) H'_r + \sum_{k \geq 2} \frac{c^{k-1}}{k!} V^k H_r^{(k)} \right). \quad (9.5)$$

Every appearance of a coefficient of  $H_r$  on the right is multiplied by a positive exponent, so only  $H_\beta$  with  $\beta < \gamma$  occurs, and [lemma 3.4](#) makes  $R_\gamma$  a finite expression in earlier coefficient functions and their derivatives — hence holomorphic on the same  $D$ , with no new shrinking. Moreover  $V - V_0$  vanishes to order at least two, since the coefficient of  $X$  in  $V$  is exactly one, and the terms with

$k \geq 2$  likewise, so  $R_\gamma \in X^2\mathcal{O}(D)$ . Variation of the constant in (9.4) gives the unique normalized solution

$$H_\gamma(X) = H_0(X) \int_0^X \frac{R_\gamma(s)}{V_0(s)H_0(s)} ds, \quad (9.6)$$

whose apparent singularity at zero is removable (numerator of order at least two, denominator of order exactly two) and whose denominator is nonzero elsewhere on  $D$ ; thus  $H_\gamma \in X^2\mathcal{O}(D)$ . Transfinite induction on  $\mathbf{M}$  constructs all coefficients on one fixed disk, and they have the same formal Taylor expansions as the unique solution of (8.1), so this analytic family is the regrouping of the  $H_r$  already constructed, not a different conjugacy. For the inverse, first shrink to a neighborhood on which  $H_0$  is biholomorphic; the exponent-zero coefficient of the inverse is  $H_0^{-1}$ , and at any positive exponent the coefficient equation for composition is linear in the unknown with coefficient  $H_0'(H_0^{-1}(Y))$ , which never vanishes on a sufficiently small fixed neighborhood, all other terms depending on earlier exponents through finitely many ordinary derivatives and products.  $\square$

*Remark 9.2* (Why the differential recursion is genuinely triangular). It would be *incorrect* to justify (9.5) by saying that  $(k-1)v(c) \rightarrow +\infty$ : that can fail to be cofinal in an arbitrary-rank value group (warning 3.6). The correct statement is that at a *specified* exponent  $\gamma$  infinitely many decompositions into generators from one well-ordered positive support are impossible, which simultaneously controls the Taylor index, the Hahn products and the number of earlier coefficients entering the right side. The construction is transfinite, but each individual coefficient equation has finite dependence.

*Warning 9.3* (No uniform norm bound on the coefficient family). [Theorem 9.1](#) asserts *no* uniform ordinary norm bound on  $(H_\gamma)$ : the coefficient functions may be arbitrarily large with arbitrarily large derivatives. What is uniform is their ordinary domain of holomorphy and the Hahn support certificate. The same disclaimer is made independently by sources 02, 03, 05 and 06 for their own coefficient families.

## 9.1 The same formula, twice, and why both instantiations survive

**Proposition 9.4** (The leading-coefficient exponential integral). *Let  $U$  be a simply connected ordinary domain containing 0, and let  $V_0 \in \mathcal{O}(U)$  satisfy  $V_0(0) = 0$ ,  $V_0'(0) = \mu \neq 0$  and  $V_0(w) \neq 0$  for  $w \in U \setminus \{0\}$ . The unique solution of  $V_0 h_0' = \mu h_0$  on  $U$  with  $h_0(0) = 0$ ,  $h_0'(0) = 1$  is*

$$h_0(w) = w \exp \left( \int_0^w \left( \frac{\mu}{V_0(\zeta)} - \frac{1}{\zeta} \right) d\zeta \right), \quad (9.7)$$

whose integrand is holomorphic at 0 and whose quotient  $h_0/w$  is nonvanishing on  $U$ .

*Proof.*  $\mu/V_0 - 1/w$  has a removable singularity at 0 because  $(V_0/w)(0) = \mu$ ; on a simply connected  $U$  its primitive vanishing at 0 exists and is unique, and exponentiating gives a solution with the stated normalization, unique because the logarithmic derivative determines  $h_0$  up to a constant fixed at 0.  $\square$

Formula (9.7) is *one* formula. It is source 01's (9.2) with  $V_0 = XP(X)$  and  $\mu = 1$ , and it is source 04's leading Koenigs coordinate with  $V_0 = V$  the Euler polynomial and  $\mu = V'(0)$ , and both sources prove it by the same transfinite variation-of-constants recursion — source 01's (9.6), source 04's (13.6). It is therefore printed once, per [convention 1.1\(i\)](#). What does *not* merge is the pair of instantiations, and both survive:

- under  $\text{RES}_q$ ,  $V_0 = XP(X)$  where  $P$  is the *shell polynomial* of (8.10), an ordinary polynomial determined by the first face of the return equation. Its nonzero roots are the closest periodic directions in the valued field, and its *absence* of zeros on a small ordinary disk is what permits (9.4) to be solved there. One finite polynomial thus plays two different roles, and (9.2) is the precise bridge. This identification of  $P$  is unavailable under PAR.
- under PAR,  $V_0 = V$  is the perturbing field itself and  $\mu$  is its derivative at 0, which is what makes the *global* factorization [corollary 13.9](#) and the monodromy exponents [lemma 13.7](#) available. That factorization is unavailable under  $\text{RES}_q$ , where  $V_0$  is not the input.

Concretely, source 01's three regimes give

$$H_0(X) = \begin{cases} X(1 + X^2)^{-1/2}, & \sigma_0 < \delta/2, \\ X(1 - X^2)^{-1/6}(1 + 2X^2)^{-1/3}, & \sigma_0 = \delta/2, \\ X(1 - 2X^4)^{-1/4}, & \sigma_0 > \delta/2, \end{cases} \quad (9.8)$$

with branches equal to one at zero (see [theorem 12.1](#)); differentiating their logarithms verifies  $XP(X)H'_0(X) = H_0(X)$  directly. Source 04's instantiation gives  $h_0 = w/(1 + w)$  for  $V = w + w^2$  and  $h_0 = w(1 + w^q)^{-1/q}$  for  $V = w(1 + w^q)$ ; see [section 13](#).

## 10 Nonresonant Hahn linearization need not be coherent

The extra conclusion of [theorem 9.1](#) cannot be imported into [lemma 8.1](#). The obstruction is ordinary small-divisor divergence in the exponent-zero coefficient, *not* a failure of Hahn summability.

**Proposition 10.1** (A divergent ordinary linearizer survives on the monad). *There is an ordinary complex polynomial  $f(X) = \alpha X + X^2$  with  $\alpha$  not a root of unity whose normalized formal conjugacy is strongly Hahn-evaluable and isometric on  $\mathfrak{m}_\Gamma$  in every nontrivial workspace  $K_\Gamma$ , yet does not lie in the radius-free ring  $\mathcal{A}_1 = \mathbb{C}\{X\}((t^\Gamma))$  and so has no common-domain representative as a normalized formal conjugacy.*

*Proof.* Choose  $\alpha = e^{2\pi i\theta}$  with  $\theta$  irrational and non-Brjuno — for instance [theorem 7.8\(e\)](#). By Yoccoz's quadratic theorem the germ  $\alpha X + X^2$  is analytically linearizable exactly when  $\theta$  is Brjuno [\[13\]](#), so its unique normalized formal linearizer  $H \in \mathbb{C}[[X]]$  has zero ordinary radius of convergence. All its coefficients lie in the constant field, so its Hahn support is  $\{0\}$  and [lemma 3.9](#) makes  $H$  and its formal inverse strongly evaluable isometries of  $\mathfrak{m}_\Gamma$ , the exact maximal ball being  $B_0$  by [theorem 8.5](#). But the coefficient at Hahn exponent zero is the divergent ordinary series  $H$  itself, so the family is not in  $\mathcal{A}_1$ , let alone in the fixed-polydisk or common-domain germ rings.  $\square$

[Lemma 3.12](#) rules out an evasion by change of representation: a different common-domain family cannot represent the divergent conjugacy as the same function on all infinitesimals. Likewise a normalized common-domain conjugacy satisfying the same functional identity would satisfy it formally, and uniqueness in (8.1) would force the same divergent exponent-zero coefficient.

**Warning 10.2** (What [proposition 10.1](#) does and does not say). There is no conflict with Yoccoz's theorem: an ordinary divergent series can be Hahn-summable at an infinitesimal argument because each successive power introduces a new positive exponent. That is a change of analytic category, not a proof of classical analytic linearizability. Moreover the coherent conclusion of [theorem 9.1](#) is **resonant-only**, and [proposition 10.1](#) establishes failure of a *universal* common-domain assertion in the nonresonant case — it is explicitly **not** an if-and-only-if classification, and not every nonresonant case is noncoherent.

## 11 Workspace invariance and the full surcomplex class

**Theorem 11.1** (Workspace invariance for the finite-return package). *Let  $\Gamma \subseteq \Gamma'$  be divisible ordered abelian groups and view  $K_\Gamma \subset K_{\Gamma'}$ . For  $f \in K_\Gamma[z]$  satisfying [theorem 8.5](#), the threshold  $r$ , the normalized formal conjugacy and the shell polynomial are unchanged by the extension. The conjugacy and its inverse strongly evaluate at every point of  $B_r(K_{\Gamma'})$ , including points absent from  $K_\Gamma$ . The finite shell root set, algebraic multiplicities, exact periods and multiplier formulas gain no new points and do not change, and the common-domain coefficient functions of [theorem 9.1](#) are also unchanged.*

*Proof.* The embedding preserves valuation, residue and all finite coefficient arithmetic, so the return polynomial and the finite maximum (8.6) are identical, and (8.1) gives the same formal coefficients in both fields. For a new point  $z \in B_r(K_{\Gamma'})$ , the normalized input  $z/t^r$  has positive well-ordered support in  $\Gamma'$ , whose union with the constructed certificate is still well ordered, so [lemma 3.9](#) applies unchanged. Every relevant root is a root of a polynomial over the algebraically closed  $K_\Gamma$ , which already factors completely into linear factors in  $K_\Gamma[z]$ , and the same finite factorization remains valid in  $K_{\Gamma'}[z]$ : no additional roots, no change of multiplicity. The remaining assertions concern the same formal coefficients, the same ordinary residue polynomial and the same uniquely normalized differential recursion.  $\square$

**Corollary 11.2** (Transfer to the surcomplex numbers). *Let  $f \in \mathbf{SC}[z] = \mathbf{No}[i][z]$  be a nonlinear polynomial fixing zero, with unit multiplier for the natural Hahn valuation and that multiplier not a root of unity. Then the formulas and conclusions of [theorems 8.5](#), [8.7](#), [8.10](#) and [9.1](#) hold on the corresponding class-sized ball in  $\mathbf{SC}$ , interpreted pointwise by set-sized Hahn sums, and the threshold and finite shell cycles are independent of the workspace used to calculate them.*

*Proof.* Place the finitely many coefficients of  $f$  in one divisible workspace and construct its sequence of formal conjugacy coefficients there. For a particular argument, enlarge the workspace to include it and apply [theorem 11.1](#); two choices agree in their common larger workspace and so define one class function pointwise. The finite obstructing root set already lies in the initial algebraically closed workspace. No sum in this construction is indexed by a proper class.  $\square$

[Corollary 11.2](#) may be read in a class theory such as NBG or with definable classes in ZFC; it requires no proper-class polynomial ring as a set and does not replace Hahn summation by fine-topological convergence. The analogous statements for the Hamiltonian package are [corollary 15.9](#), and for the several-variable linearizer [proposition 4.14](#).

## 12 Cancellation: a tuned cubic and a two-parameter phase diagram

The exact threshold (8.6) is read off the *actual* return coefficients  $b_n$ , including the cancellations that occur when  $f^{\circ q}$  is composed. This subsection exhibits a family in which that distinction is strictly consequential: any bound depending only on  $a = \max_{n \geq 2} |a_n|^{1/(n-1)}$ ,  $q$  and  $|1 - \lambda^q|$  — the shape of the general lower radius of the rank-one antecedent [20, Thm. 1.1] — throws the improvement away.

### 12.1 A tangent binomial, for calibration

For  $f(z) = (1+c)z + az^m$  with  $0 \neq c \in \mathfrak{m}_\Gamma$ ,  $a \neq 0$ ,  $m \geq 2$ , one has  $q = 1$  and  $r = (v(c) - v(a))/(m-1)$ . The nonzero fixed points are exactly  $p^{m-1} = -c/a$ , all  $m-1$  of them simple and on  $S_r$ , and at each

the multiplier is known *exactly*, not merely to leading order:

$$f'(p) = 1 + c + map^{m-1} = 1 - (m-1)c. \quad (12.1)$$

With  $A = \text{res}(ap^{m-1}/c)$  the shell polynomial is  $P(X) = 1 + AX^{m-1}$  and  $H_0(X) = X(1 + AX^{m-1})^{-1/(m-1)}$  on the ordinary branch normalized to one at zero. This already shows why a degree-independent small-divisor loss can be wasteful.

## 12.2 A cubic with two suppressed return terms

Let  $\delta > 0$ ,  $e = t^\delta$ , and

$$f_e(z) = (-1 + e)z + z^2 - z^3. \quad (12.2)$$

Its residue multiplier has order  $q = 2$ ,  $\eta(f_e) = 0$ , and  $c = \lambda^2 - 1 = e(e-2)$  has valuation  $\delta$ . The nonlinearity of the second iterate is not guessed from the original quadratic term; its exact coefficients are

| $n$ | $b_n$ in $f_e^{\circ 2}$ | $n$ | $b_n$ in $f_e^{\circ 2}$ |
|-----|--------------------------|-----|--------------------------|
| 2   | $e(e-1)$                 | 6   | $6(e-1)$                 |
| 3   | $-e(e-2)(e-1)$           | 7   | $-3(e-2)$                |
| 4   | $-e(3e-4)$               | 8   | $-3$                     |
| 5   | $3e^2 - 9e + 4$          | 9   | $1$                      |

At  $e = 0$  the return displacement factors as

$$f_0^{\circ 2}(z) - z = z^5(z^2 - 2z + 2)(z^2 - z + 2), \quad (12.3)$$

so that the cubic *and* quartic terms have both disappeared; their actual valuations, not the separate sizes of the terms that cancelled, are what enter (8.6). The degree-five coefficient is a unit and the exact maximum is

$$\boxed{r(f_e) = \delta/4.} \quad (12.4)$$

For  $\rho = t^{\delta/4}$  one gets  $P(X) = 1 - 2X^4$ : four simple closest period-two points in two cycles, with residue directions satisfying  $a^4 = 1/2$ ; since  $aP'(a) = -4$ ,

$$(f_e^{\circ 2})'(p_a) = 1 - 4c + o_v(c) = 1 + 8e + o_v(e), \quad (12.5)$$

where  $R = o_v(c)$  means  $v(R) > v(c)$ , and  $H_0(X) = X(1 - 2X^4)^{-1/4}$ .

For comparison, the untuned quadratic  $(-1 + e)z + z^2$  has threshold  $\delta/2$  and shell polynomial  $1 + X^2$ . The cubic (12.2) therefore admits an *exactly larger* linearization ball with the same multiplier and the same basic coefficient scale  $\eta = 0$ ; the difference is produced entirely by cancellation in a finite return.

## 12.3 The two-parameter transition

The cancellation can be broken on a second valuation scale. Let

$$f_{e,s}(z) = (-1 + e)z + z^2 + (-1 + s)z^3, \quad e = t^\delta, \quad s = t^{\sigma_0}, \quad \delta, \sigma_0 > 0, \quad (12.6)$$

with no Archimedean relationship assumed between the two positive exponents.

**Theorem 12.1** (Cancellation phase diagram). *For (12.6),*

$$r(f_{e,s}) = \max\left\{\frac{\delta - \sigma_0}{2}, \frac{\delta}{4}\right\}, \quad (12.7)$$

and the shell polynomial, cycle count and leading return-multiplier ratios are

| Regime                | $P(X)$                                 | Two-cycles | $aP'(a)$ |
|-----------------------|----------------------------------------|------------|----------|
| $\sigma_0 < \delta/2$ | $1 + X^2$                              | one        | $-2$     |
| $\sigma_0 = \delta/2$ | $1 + X^2 - 2X^4 = (1 - X^2)(1 + 2X^2)$ | two        | $-6, -3$ |
| $\sigma_0 > \delta/2$ | $1 - 2X^4$                             | two        | $-4$     |

Every listed initial root is simple. In the middle row the ratio  $-6$  occurs for  $a^2 = 1$  and  $-3$  for  $a^2 = -1/2$ . The three leading coherent coordinates are (9.8).

*Proof.* For the general cubic  $f(z) = (-1 + e)z + z^2 + bz^3$ , direct composition gives

$$\begin{aligned} b_2 &= e(e - 1), & b_3 &= (e - 1)(be^2 - 2be + 2b + 2), \\ b_4 &= 3be^2 - 4be + b + 1, & b_5 &= b(3be^2 - 6be + 3b + 3e - 1), \\ b_6 &= b(6be - 5b + 1), & b_7 &= 3b^2(be - b + 1), \\ b_8 &= 3b^3, & b_9 &= b^4. \end{aligned} \quad (12.8)$$

Substituting  $b = -1 + s$ , the degree-three coefficient has leading terms  $-2e - 2s$  and the degree-four coefficient has leading terms  $4e + s$ ; because  $e$  and  $s$  have leading coefficient one, no cancellation occurs between those pairs even when  $\delta = \sigma_0$ . Hence  $v(b_2) = \delta$ ,  $v(b_3) = v(b_4) = \min\{\delta, \sigma_0\}$  and  $v(b_n) = 0$  for  $5 \leq n \leq 9$ , so the maximum in (8.6) is the larger of  $(\delta - \min\{\delta, \sigma_0\})/2$  and  $\delta/4$ , which is (12.7). If  $\sigma_0 < \delta/2$ , take  $r = (\delta - \sigma_0)/2$ : only the normalized degree-three return coefficient has nonzero residue in  $Q - 1$ , and that residue is one, so  $P = 1 + X^2$ . If  $\sigma_0 > \delta/2$ , take  $r = \delta/4$ : only degree five survives, with residue  $4/(-2) = -2$ , so  $P = 1 - 2X^4$ . At equality both terms survive and  $P = 1 + X^2 - 2X^4$ . All three are squarefree, so the cycle counts follow from theorem 8.7, and reduction of  $XP'(X)$  at the roots gives  $-2$ ,  $-4$  and, in the middle case,  $2a^2 - 8a^4 = -6$  for  $a^2 = 1$  and  $-3$  for  $a^2 = -1/2$ .  $\square$

**Example 12.2** (A repeated initial root at the transition). Replace  $s$  in the equality regime by  $\kappa t^{\delta/2}$  with  $\kappa \in \mathbb{C}^\times$ . The same calculation gives  $P_\kappa(X) = 1 + \kappa X^2 - 2X^4$ , and as a polynomial in  $Y = X^2$ ,

$$\text{Disc}_Y(1 + \kappa Y - 2Y^2) = \kappa^2 + 8.$$

For  $\kappa^2 + 8 \neq 0$  the four shell points are simple in distinct residue directions. For  $\kappa^2 + 8 = 0$  the initial polynomial has repeated roots — an *initial-direction degeneracy*, and not by itself an assertion of an actual multiple periodic point (remark 8.8); determining the finer splitting there requires higher Hahn terms of the return equation. The exact threshold remains  $\delta/4$ , since the degree-four term of  $P_\kappa$  cannot vanish.

**Example 12.3** (A genuinely higher-rank example). Let  $\Gamma = \mathbb{Q} + \mathbb{Q}\Omega$  with  $\Omega > n$  for every ordinary positive integer  $n$ , and

$$f(z) = (1 + t^\Omega)z + t^3 z^2 + z^4. \quad (12.9)$$

Here  $q = 1$ ,  $\eta = 0$  and  $r = \max\{\Omega - 3, \Omega/3\} = \Omega - 3$ . At  $\rho = t^{\Omega-3}$  the shell polynomial is  $P(X) = 1 + X$ , and there is one closest nonzero fixed point  $p = \rho(-1 + u)$ ,  $u \in \mathfrak{m}_\Gamma$ , with  $f'(p) = 1 - t^\Omega + o_v(t^\Omega)$ .

The other two nonzero fixed points are much farther from zero: all three solve  $z^3 + t^3z + t^\Omega = 0$ , and factoring off  $p$  gives  $z^3 + t^3z + t^\Omega = (z - p)(z^2 + pz + p^2 + t^3)$ ; scaling the quadratic factor by  $z = t^{3/2}Y$  and dividing by  $t^3$  gives reduction  $Y^2 + 1$ , because  $v(p) = \Omega - 3$  dominates every ordinary rational. Those two roots therefore have valuation  $3/2$  and normalized residue directions  $\pm i$ .

This example *cannot* be faithfully described by a real-valued valuation preserving the order of all of  $\Gamma$ : sending  $\Omega$  to a finite real number would fail to preserve  $\Omega > n$  for all  $n$ . The exact calculation uses only ordered-group comparisons and [lemma 3.4](#). Finally, the same threshold persists if  $t^\Omega$  in the linear term is replaced by  $c = t^\Omega(1 + \sum_{j \geq 2} t^{2^{-1/j}})$ , a legitimate Hahn coefficient whose positive support is well ordered although infinitely many exponents lie below 2 before the shift by  $\Omega$ : the leading valuation remains  $\Omega$  and the normalized shell polynomial remains  $1 + X$ . The theorem is therefore *not* restricted to coefficients with discrete or finitely generated exponent support. Both halves of this example are specific to source 01.

## 13 Parabolic dynamics I: algebraicity and hidden monodromy

Everything in this section is under PAR and  $\text{SD}_\emptyset$  ([warning 2.8](#)): the leading multiplier is exactly 1 perturbed by an infinitesimal, and no divisor condition is imposed anywhere. Here  $\Gamma$  is divisible and set-sized — divisibility is load-bearing, because it is what keeps the resonant rescaling  $s = t^{\delta/q}$  inside the chosen workspace — and  $\varepsilon = t^\delta$  with  $\delta > 0$  is a single Hahn monomial, **not** a small nonzero complex number. Ordinary powers of  $\varepsilon$  form the Hahn subring  $\mathbb{C}[[\varepsilon]]$ , and every  $O(\varepsilon^k)$  below means *divisibility* by  $\varepsilon^k$  in a formal series ring, never an analytic estimate uniform in a numerical parameter.

### 13.1 The common-domain coefficient ring and the algebraicity base

Fix an ordinary connected open  $U \subset \mathbb{C}$  and work in  $\mathcal{A}(U) = \mathcal{O}(U)((t^\Gamma))$ , the fixed-polydisk row of [section 3.2](#) in one variable: one well-ordered exponent support, every coefficient holomorphic on that *same*  $U$ , derivatives and primitives coefficientwise with no estimate uniform in the exponent. A series with nowhere-zero leading coefficient on  $U$  is invertible in  $\mathcal{A}(U)$  ([lemma 13.2](#)). Write  $\mathcal{A}^{>0}(U)$  for the positive-valuation part. The algebraicity base is

$$\mathcal{B} = \mathbb{C}(w)((t^\Gamma)), \quad (13.1)$$

Hahn series with *arbitrary rational-function* coefficients.

**Warning 13.1** (The base field is deliberately large).  $\mathcal{B}$  is **not**  $K_\Gamma(w)$ . Its elements may have coefficient functions whose extra poles vary with the Hahn exponent, so there is no common pole-free subdomain for all of  $\mathcal{B}$ , and proving transcendence over  $\mathcal{B}$  is strictly stronger than proving it over  $K_\Gamma(w)$ . A merge must not weaken (13.1) to  $K_\Gamma(w)$ . The mechanism that makes the larger base available is that an ordinary deck transformation fixes *every rational coefficient function separately*, so no degree bound, no finite support and no common bound on pole sets is needed.

The ambient field for algebraicity is

$$\mathcal{L} = \mathcal{M}(\tilde{X})((t^\Gamma)), \quad \mathcal{B} \hookrightarrow \mathcal{L}, \quad (13.2)$$

where  $X = \mathbb{C} \setminus Z(V)$ ,  $p : \tilde{X} \rightarrow X$  is its *ordinary* universal cover and rational functions are pulled back along  $p$ . A deck transformation acts on each coefficient by ordinary analytic continuation and leaves Hahn exponents fixed, hence is a strong field automorphism of  $\mathcal{L}$  fixing  $\mathcal{B}$  pointwise. No covering-space topology on  $\mathbf{No}[i]$  is introduced, and no topological convergence in the surcomplex plane is used.

**Lemma 13.2** (A sufficient unit criterion). *If  $f = t^{\lambda_0}(a + u)$  with  $a \in \mathcal{O}(U)$  nowhere zero and  $u \in \mathcal{A}^{>0}(U)$ , then  $f$  is invertible in  $\mathcal{A}(U)$  with  $f^{-1} = t^{-\lambda_0}a^{-1} \sum_{n \geq 0} (-u/a)^n$ , and every coefficient of the inverse is holomorphic on the original  $U$ .*

*Proof.* The family is summable by lemma 3.4, multiplication by  $1 + u/a$  telescopes coefficientwise to 1, and powers of  $a^{-1}$  are holomorphic on  $U$ , so no new singularity appears.  $\square$

No converse characterization of the units of  $\mathcal{A}(U)$  is claimed; only the stated criterion is used. Every nonzero series is invertible in  $\mathcal{M}(U)((t^\Gamma))$ .

### 13.2 Common-domain Hahn linearization

**Theorem 13.3** (Common-domain Hahn linearization). *Let  $U \subset \mathbb{C}$  be simply connected with  $0 \in U$ , let  $V \in \mathcal{O}(U)$  satisfy  $V(0) = 0$ ,  $V'(0) = \mu \neq 0$  and  $V(w) \neq 0$  on  $U \setminus \{0\}$ , and let  $E \in \mathcal{A}^{>0}(U)$  satisfy  $E(0) = 0$ . Put*

$$F(w) = w + t^\delta(V(w) + E(w)), \quad \lambda = F'(0), \quad \mathbf{M} = \langle \{\delta\} \cup \text{supp } E \rangle. \quad (13.3)$$

*There is a unique  $H \in \mathcal{A}(U)$  with*

$$H \circ F = \lambda H, \quad H(0) = 0, \quad H'(0) = 1, \quad (13.4)$$

*its support is contained in  $\mathbf{M}$ , every coefficient is holomorphic on  $U$ , and its coefficient at exponent zero is  $h_0$  of (9.7). All constructions commute with enlargement of the set-sized value group.*

*Proof.* Theorem 4.3 gives  $\log C_F = A\partial_w$  with  $A \in \mathcal{A}^{>0}(U)$ . Every word in  $\Delta = t^\delta(V + E)$  has an exponent which, after removing one initial  $\delta$ , lies in  $\mathbf{M}$ , so  $A = t^\delta W$  with  $W = V + W_+$ ,  $\text{supp } W \subseteq \mathbf{M}$ ,  $W_+$  of positive valuation and all coefficients holomorphic on  $U$ . Put  $c = \text{Log } \lambda$  and  $\alpha_c = t^{-\delta}c$ ; lemma 4.5 gives  $W(0) = 0$  and  $W'(0) = \alpha_c$ , whose leading coefficient is  $\mu$ . Coefficientwise division by  $w$  is possible since  $W(0) = 0$ , and the leading coefficient of  $W/w$  is  $V/w$ , nowhere zero on  $U$  including at 0, so  $W/w$  is invertible in  $\mathcal{A}(U)$  by lemma 13.2. Hence

$$R(w) := \frac{\alpha_c}{W(w)} - \frac{1}{w} = \frac{\alpha_c/(W(w)/w) - 1}{w} \quad (13.5)$$

belongs to  $\mathcal{A}(U)$ , its numerator vanishing coefficientwise at 0 because  $(W/w)(0) = \alpha_c$ , with support in  $\mathbf{M}$ . Every coefficient of  $R$  has a unique primitive vanishing at 0 since  $U$  is simply connected; write  $B(w) = \int_0^w R$ ,  $B = B_0 + B_+$  with  $B_0$  the coefficient at exponent zero, and set

$$H(w) = we^{B_0(w)} \sum_{n \geq 0} \frac{B_+(w)^n}{n!}, \quad (13.6)$$

whose exponential is certified by lemma 3.4 and preserves the support bound  $\mathbf{M}$ , using the same  $U$  at every stage. Then  $H(0) = 0$ ,  $H'(0) = 1$  and  $H'/H = \alpha_c/W$  away from 0, with the cross-multiplied identity everywhere, so  $DH = cH$ ; as  $D$  fixes Hahn constants, exponentiation gives  $C_F H = \text{Exp}_H(c)H = \lambda H$ , and the exponent-zero part of (13.6) is (9.7). For uniqueness, if  $\tilde{H}$  satisfies (13.4) then  $N^r \tilde{H} = (\lambda - 1)^r \tilde{H}$  so  $D\tilde{H} = c\tilde{H}$ ; the quotient  $\tilde{H}/H = (\tilde{H}/w)/(H/w)$  lies in  $\mathcal{A}(U)$  because the leading coefficient of  $H/w$  is the nowhere-zero  $e^{B_0}$ , the differential equations force its derivative to vanish ( $\mathcal{A}(U)$  being an integral domain for connected  $U$ ), every coefficient is therefore constant, and its value at 0 is 1. Finally all outputs are formed from input supports by finite words, coefficientwise differentiation and uniquely normalized integration, none of which depends on the ambient group beyond containing the indicated supports.  $\square$

*Warning 13.4* (What kind of embedding was proved). [Theorem 13.3](#) constructs a strong logarithm and solves a coefficientwise ordinary differential equation. It does **not** assert convergence of the series at a nonzero ordinary complex value of  $t^\delta$ , nor analytic embedding of every numerical near-identity map in an autonomous flow; those are different questions, and the Gelfreich–Vieiro reference is cited precisely to mark that line [7]. Nor does it assert that the global evaluated map  $H^\#$  is injective on all of  $U_\Gamma^\#$ ; injectivity is proved only on the infinitesimal disk of the resonant model ([theorem 13.22](#)). There is also no ordinary invariant-domain hypothesis:  $C_F$  denotes the Taylor operation (4.1), not a numerical self-map of  $U$ , and at a point  $a + \eta \in U_\Gamma^\#$  the positive displacement preserves the standard part.

**Corollary 13.5** (Universal leading profile at a resonant scale). *Let  $q \geq 1$ ,  $a \in \mathbb{C}^*$ , and let  $\phi(z) = az^{q+1} + \sum_{n>q+1} a_n z^n$  be an ordinary convergent germ. Choose  $c_0 \in \mathbb{C}^*$  with  $ac_0^q = 1$ , set  $s = c_0 t^{\delta/q}$ , and let  $f(z) = (1 + \varepsilon)z + \phi^\#(z)$  at infinitesimal  $z$ . After  $z = sw$ , the normalized Hahn coordinate on any simply connected ordinary  $U$  containing 0 but no root of  $1 + w^q$  has leading coefficient*

$$w(1 + w^q)^{-1/q}, \quad (13.7)$$

with the branch normalized at 0.

*Proof.* Taylor evaluation of the convergent germ at  $sw$  is strongly meaningful for every finite  $w$ , and direct rescaling gives

$$s^{-1}f(sw) = w + \varepsilon \left( w + w^{q+1} + \sum_{n>q+1} a_n c_0^{n-1} t^{\delta(n-q-1)/q} w^n \right),$$

whose final sum has positive Hahn support and polynomial coefficients on such a  $U$ . Apply [theorem 13.3](#) with  $V = w(1 + w^q)$ ; integration of  $1/V - 1/w$  gives  $-q^{-1} \log(1 + w^q)$ .  $\square$

[Corollary 13.5](#) asserts a leading profile and a common-domain construction. It explicitly does **not** extend the exact monodromy formula or the sharp disk theorem below to arbitrary higher-order perturbations.

### 13.3 Polynomial Euler maps and their exact logarithmic residues

Specialize to

$$G_\varepsilon(w) = w + \varepsilon V(w), \quad V \in \mathbb{C}[w], \quad V(0) = 0, \quad V'(0) = \mu \neq 0, \quad (13.8)$$

so [theorem 13.3](#) with  $E = 0$  constructs  $H_\varepsilon \in \mathcal{O}(U)[[\varepsilon]]$ . The polynomial hypothesis supplies structure the general theorem does not have.

**Proposition 13.6** (The generator gains no new pole). *The logarithmic generator of (13.8) has the form*

$$\log C_{G_\varepsilon} = \varepsilon W_\varepsilon(w) \partial_w, \quad W_\varepsilon = V B_\varepsilon, \quad B_\varepsilon \in 1 + \varepsilon \mathbb{C}[w][[\varepsilon]]. \quad (13.9)$$

Setting  $\alpha_\varepsilon = \text{Log}(1 + \mu\varepsilon)/\varepsilon$  and  $\Omega_\varepsilon = \alpha_\varepsilon/W_\varepsilon$ , every coefficient of  $\Omega_\varepsilon$  is a rational function of  $w$  with poles only at zeros of  $V$ , of order at most the multiplicity in  $V$ .

*Proof.* For a polynomial  $p$ ,  $(C_{G_\varepsilon} - I)p = \sum_{k \geq 1} \varepsilon^k V^k p^{(k)}/k!$ , every coefficient of which is polynomial and divisible by  $V$ ; the same holds of each  $N^r w$ , and at a fixed power of  $\varepsilon$  only finitely many  $r$  occur, so the logarithm has polynomial coefficients all divisible by  $V$ , its first term being  $\varepsilon V$ . The inverse of  $B_\varepsilon$  has polynomial coefficients by the formal geometric series, so  $\Omega_\varepsilon$  is a formal series of polynomials divided by the single polynomial  $V$ .  $\square$

The divisor assertion is stronger than rationality of each coefficient: higher terms of the modified vector field cannot introduce *new* spatial pole locations into the logarithmic derivative of the coordinate. Finite expansion of the operator logarithm gives, for the record,

$$W_\varepsilon = V - \frac{\varepsilon}{2}VV' + \varepsilon^2\left(\frac{1}{12}V^2V'' + \frac{1}{3}V(V')^2\right) + O(\varepsilon^3), \quad (13.10)$$

obtained from  $Nw = \varepsilon V$ ,  $N^2w = \varepsilon^2VV' + \frac{\varepsilon^3}{2}V^2V'' + O(\varepsilon^4)$ ,  $N^3w = \varepsilon^3(V(V')^2 + V^2V'') + O(\varepsilon^4)$  substituted in  $N - N^2/2 + N^3/3$ . **No assertion about all coefficients is inferred from this truncation**; the exact identity [lemma 13.7](#) is proved separately.

**Lemma 13.7** (Exact residue at a simple fixed point). *At a simple zero  $a$  of  $V$ , with  $\nu_a = V'(a)$ ,*

$$\text{Res}_{w=a}(\Omega_\varepsilon(w) dw) = \rho_a(\varepsilon) = \frac{\text{Log}(1 + \mu\varepsilon)}{\text{Log}(1 + \nu_a\varepsilon)}, \quad \rho_0 = 1. \quad (13.11)$$

*Proof.* [Lemma 4.5](#) at  $a$  gives  $\varepsilon W'_\varepsilon(a) = \text{Log } G'_\varepsilon(a) = \text{Log}(1 + \nu_a\varepsilon)$ . Since  $a$  is a simple zero of  $V$ , the factorization  $W_\varepsilon = VB_\varepsilon$  makes it a simple zero over  $\mathbb{C}[[\varepsilon]]$  as well, so the residue of  $\alpha_\varepsilon/W_\varepsilon$  is  $\alpha_\varepsilon/W'_\varepsilon(a)$ .  $\square$

These are *coefficientwise ordinary* residues of a rational differential. They are not local residues at displaced surcomplex poles; the fixed points of this polynomial family are exactly the ordinary zeros of  $V$ , so no displacement ambiguity arises.

**Proposition 13.8** (The first correction is an explicit logarithm). *On the common domain  $U$ ,*

$$H_\varepsilon = h_0 \left[ 1 + \frac{\mu\varepsilon}{2} \log\left(\frac{V}{\mu h_0}\right) + O(\varepsilon^2) \right], \quad (13.12)$$

*the quotient  $V/(\mu h_0)$  being holomorphic, nowhere zero and equal to 1 at 0 after removing the common zero, so that its logarithm is uniquely fixed by vanishing at 0.*

*Proof.* Since  $\alpha_\varepsilon = \mu - \mu^2\varepsilon/2 + O(\varepsilon^2)$ , [\(13.10\)](#) gives  $\Omega_\varepsilon = \mu/V + \frac{\mu\varepsilon}{2}(V' - \mu)/V + O(\varepsilon^2)$ , whose coefficient at  $\varepsilon$  is holomorphic at 0 and equals  $\frac{\mu}{2}(\log(V/(\mu h_0)))'$  because  $h'_0/h_0 = \mu/V$ . Integrate with the normalization at 0 and exponentiate.  $\square$

**Corollary 13.9** (Separation of the branching factors). *Suppose  $V$  has degree  $d \geq 2$  with distinct zeros  $0, a_1, \dots, a_{d-1}$ . Partial fractions taken separately at each  $\varepsilon$  coefficient give*

$$\Omega_\varepsilon(w) = Q_\varepsilon(w) + \frac{1}{w} + \sum_{j=1}^{d-1} \frac{\rho_{a_j}(\varepsilon)}{w - a_j}, \quad Q_\varepsilon \in \varepsilon^2\mathbb{C}[w][[\varepsilon]], \quad (13.13)$$

*and with  $P_\varepsilon(w) = \int_0^w Q_\varepsilon$ , on branches normalized near 0,*

$$H_\varepsilon(w) = w \text{Exp}_H(P_\varepsilon(w)) \prod_{j=1}^{d-1} (1 - w/a_j)^{\rho_{a_j}(\varepsilon)}, \quad P_\varepsilon \in \varepsilon^2\mathbb{C}[w][[\varepsilon]], \quad P_\varepsilon(0) = 0. \quad (13.14)$$

*The factor  $\text{Exp}_H(P_\varepsilon)$  and its inverse have polynomial coefficients and belong to  $\mathcal{B}$ ; in particular all nontrivial coefficientwise monodromy is carried by the displayed power factors.*

*Proof.* The coefficients at  $\varepsilon^0$  and  $\varepsilon^1$  have numerator degree smaller than  $d$  by (13.10) and proposition 13.8, so  $Q_\varepsilon \in \varepsilon^2 \mathbb{C}[w][[\varepsilon]]$ . For  $\rho = \rho_0 + \rho_+$  define the power on a chosen ordinary logarithm branch by  $(1 - w/a)^\rho = \exp(\rho_0 \log(1 - w/a)) \text{Exp}_H(\rho_+ \log(1 - w/a))$ . The logarithmic derivative of the right side of (13.14) is (13.13) and its value and derivative at 0 have the required normalization, so uniqueness in theorem 13.3 proves the identity; at a fixed power of  $\varepsilon$  the exponential of  $P_\varepsilon$  uses only finitely many products of polynomials.  $\square$

### 13.4 The torsion test and the finite-cover obstruction

**Lemma 13.10** (An infinitesimal phase has no torsion). *For a finite Hahn constant  $c = c_0 + c_+$  with  $c_0 \in \mathbb{C}$  and  $v(c_+) > 0$ ,*

$$\text{Exp}_H(c) = 1 \iff c_+ = 0 \text{ and } c_0 \in 2\pi i\mathbb{Z}.$$

*Consequently, for  $\rho \in \mathbb{C} + \mathfrak{m}_\Gamma$ , the element  $\text{Exp}_H(2\pi i\rho)$  has finite multiplicative order if and only if  $\rho$  is an ordinary rational number.*

*Proof.* The coefficient at exponent zero in  $\text{Exp}_H(c)$  is  $e^{c_0}$ , so equality to 1 first requires  $c_0 \in 2\pi i\mathbb{Z}$ ; if  $c_+ \neq 0$  then  $\text{Exp}_H(c_+) - 1 = c_+ + c_+^2/2 + \dots$  has the same leading term as  $c_+$ , all higher powers having strictly larger valuation, so it cannot vanish. The last assertion follows from  $\text{Exp}_H(2\pi i\rho)^m = \text{Exp}_H(2\pi im\rho)$ .  $\square$

This single lemma converts “the exponent has a nonzero infinitesimal part” into “no finite branched cover works”.

**Theorem 13.11** (Exact monodromy and obstruction to finite ramification). *At every simple zero  $a$  of  $V$ , continuation of  $H_\varepsilon$  around a positive simple loop about  $a$  is multiplication by*

$$M_a = \text{Exp}_H(2\pi i\rho_a), \quad \rho_a = \frac{\text{Log}(1 + \mu\varepsilon)}{\text{Log}(1 + V'(a)\varepsilon)}. \quad (13.15)$$

*The multiplier has finite order exactly when  $V'(a) = \mu$ , in which case it is 1. When  $V'(a) \neq \mu$ , no finite ordinary branched cover at  $a$  makes the continued coordinate single-valued.*

*Proof.* Near  $a$  the rational-coefficient differential decomposes as  $\Omega_\varepsilon dw = \rho_a dw/(w - a) +$  a holomorphic-coefficient differential; a positive circuit changes a coefficientwise primitive by  $2\pi i\rho_a$ , so exponentiation multiplies  $H_\varepsilon$  by  $M_a$ . Writing  $\nu = V'(a)$ , formal division of the two logarithms gives

$$\rho_a = \frac{\mu}{\nu} + \frac{\mu(\nu - \mu)}{2\nu}\varepsilon + O(\varepsilon^2), \quad (13.16)$$

so if  $\nu \neq \mu$  the coefficient of  $\varepsilon$  is nonzero and lemma 13.10 makes  $M_a$  of infinite order, while if  $\nu = \mu$  the ratio is exactly 1. A finite branched cover has a positive integer ramification index  $m$  at a point above  $a$ , and in suitable local coordinates a circuit upstairs projects to  $m$  circuits downstairs, so its monodromy multiplier is  $M_a^m \neq 1$  in the infinite-order case. This applies in particular to the covers used to make an ordinary algebraic function single-valued.  $\square$

The obstruction is already visible at order  $\varepsilon^1$  whenever  $\nu \neq \mu$ : it is not necessary to inspect infinitely many unknown correction terms to decide whether finite ramification will work.

**Lemma 13.12** (A nonlinear polynomial cannot have one slope at all its zeros). *If a polynomial of degree  $d \geq 2$  has  $d$  distinct zeros then  $\sum_{V(a)=0} 1/V'(a) = 0$ . In particular, if  $V'(0) = \mu \neq 0$ , not all slopes at its zeros can equal  $\mu$ .*

*Proof.* Partial fractions give  $1/V(w) = \sum_a (V'(a)(w-a))^{-1}$ ; the coefficient of  $1/w$  at infinity on the left is zero because  $d \geq 2$ , and on the right it is the displayed sum. If every derivative were  $\mu$  the sum would be  $d/\mu \neq 0$ .  $\square$

### 13.5 The Hahn-algebraicity dichotomy

**Lemma 13.13** (Reduction of an algebraic Hahn element). *If  $Y \in \mathcal{L}$  has valuation zero with leading coefficient  $y_0$  and  $Y$  is algebraic over  $\mathcal{B}$ , then  $y_0$  is algebraic over  $\mathbb{C}(w)$ .*

*Proof.* Take a nonzero relation  $\sum_{j=0}^m b_j Y^j = 0$  with  $b_j \in \mathcal{B}$ , let  $\gamma$  be the least valuation among the nonzero  $b_j$ , and multiply by  $t^{-\gamma}$ : all coefficients then have nonnegative valuation and at least one has nonzero coefficient at zero, and taking the coefficient at zero yields a nonzero polynomial over  $\mathbb{C}(w)$  annihilating  $y_0$  — multiplication reduces correctly because all series in the normalized relation have nonnegative valuation.  $\square$

**Lemma 13.14** (A higher-order logarithmic pole forces transcendence). *Let  $r$  be a rational function with a pole of order at least two at  $a$ . Any nonzero local solution of  $y'/y = r$  is transcendental over  $\mathbb{C}(w)$ .*

*Proof.* On a small punctured sector, integration gives  $y(w) = u(w)(w-a)^\beta \exp(\sum_{j=1}^{k-1} c_j(w-a)^{-j})$  with  $c_{k-1} \neq 0$ ,  $u$  holomorphic and nonzero and  $k \geq 2$  the pole order. Along a suitable ray ending at  $a$  the real part of the leading expression  $c_{k-1}(w-a)^{-(k-1)}$  is positive and dominates the lower terms, so  $|y(w)|$  grows faster than every power of  $|w-a|^{-1}$  along that ray. An algebraic function cannot: after dividing a polynomial relation by its leading coefficient, its coefficients are meromorphic and hence bounded by a fixed power of  $|w-a|^{-1}$  on a small punctured disk, and the elementary bound  $|y| \leq 1 + \max_{j < m} |b_j|$  for a root of  $y^m + \sum_{j < m} b_j y^j$  gives a polynomial growth bound for every root.  $\square$

**Theorem 13.15** (Polynomial Euler maps: the algebraicity dichotomy). *Let  $V \in \mathbb{C}[w]$  with  $V(0) = 0$ ,  $V'(0) = \mu \neq 0$ . On every simply connected ordinary  $U$  containing 0 and no other zero of  $V$ , equation (13.18) below has a unique solution  $H_\varepsilon \in \mathcal{O}(U)[[\varepsilon]]$ , whose coefficients continue to the universal cover of  $X = \mathbb{C} \setminus Z(V)$ , and*

$$H_\varepsilon \text{ is algebraic over } \mathcal{B} = \mathbb{C}(w)((t^\Gamma)) \iff \deg V = 1, \quad (13.17)$$

where

$$H_\varepsilon(w + \varepsilon V(w)) = (1 + \mu\varepsilon)H_\varepsilon(w), \quad H_\varepsilon(0) = 0, \quad H'_\varepsilon(0) = 1, \quad (13.18)$$

composition being Taylor substitution in the positive Hahn displacement  $\varepsilon V(w)$ . At a simple zero  $a$  of  $V$ , positive continuation once around  $a$  multiplies  $H_\varepsilon$  by  $M_a$  of (13.15). If  $V$  is nonlinear and all its zeros are simple, at least one zero has infinite-order monodromy, which no finite branched cover removes.

*Proof.* Existence, uniqueness and the common-domain assertion are theorem 13.3 with  $E = 0$ ; continuation and the monodromy formula are proposition 13.6, lemma 13.7, and theorem 13.11. If  $V$  is linear it is  $\mu w$  and  $H_\varepsilon = w$  solves the equation, an element of  $\mathcal{B}$ . Assume  $\deg V \geq 2$ .

*Route 1: a growth bound (needed for multiple zeros).* If  $V$  has a multiple zero then  $\mu/V$  has a pole of order at least two there, and  $h'_0/h_0 = \mu/V$ , so lemma 13.14 makes  $h_0$  transcendental over  $\mathbb{C}(w)$ ; since  $H_\varepsilon$  has valuation zero in  $\mathcal{L}$ , lemma 13.13 excludes algebraicity over  $\mathcal{B}$ .

*Route 2: an infinite monodromy orbit (needed for simple zeros).* If every zero is simple, [lemma 13.12](#) supplies a zero  $a$  with  $V'(a) \neq \mu$ . Let  $\tau_a$  be the corresponding deck action on  $\mathcal{L}$ ; it fixes  $\mathcal{B}$  and satisfies  $\tau_a^n(H_\varepsilon) = M_a^n H_\varepsilon$  for  $n \geq 0$ . By [theorem 13.11](#) these elements are all distinct,  $H_\varepsilon \neq 0$  and  $M_a$  of infinite order; if  $H_\varepsilon$  were algebraic over  $\mathcal{B}$  they would all be roots of one nonzero polynomial over  $\mathcal{B}$ , impossible in a field.  $\square$

*Remark 13.16* (Two routes, neither substituting for the other). Source 04 states explicitly that its two proof branches — the elementary growth bound for algebraic functions, and the infinite-monodromy-orbit argument — meet in the resonant model but do not logically substitute for one another. Both are therefore kept in [theorem 13.15](#): route 1 is what covers multiple zeros, where there is no simple-pole residue to work with, and route 2 is what covers simple zeros, where  $\mu/V$  has only first-order poles and [lemma 13.14](#) does not apply. A proof discussing only simple-pole residues would miss [example 13.20](#) entirely, even though the dichotomy remains true there.

The argument works over the larger base  $\mathcal{B}$  precisely because ordinary deck transformations fix *every rational coefficient function separately* ([warning 13.1](#)).

**Proposition 13.17** (Classification of algebraic leading coordinates). *For a nonlinear polynomial  $V$  as above,  $h_0$  is algebraic over  $\mathbb{C}(w)$  if and only if all zeros of  $V$  are simple and*

$$\frac{\mu}{V'(a)} \in \mathbb{Q} \quad \text{for every zero } a \text{ of } V, \quad (13.19)$$

in which case

$$h_0(w) = w \prod_{a \neq 0} (1 - w/a)^{\mu/V'(a)}. \quad (13.20)$$

Whenever these conditions hold, the first correction  $h_1$  in  $H_\varepsilon = h_0 + \varepsilon h_1 + \dots$  is already transcendental over  $\mathbb{C}(w)$ .

*Proof.* Multiple zeros are excluded by [lemma 13.14](#). With all zeros simple, the leading part of (13.14) is (13.20); if all exponents are rational, taking a common denominator gives a polynomial relation, so  $h_0$  is algebraic. Conversely continuation of an algebraic function gives a finite orbit, so the local multipliers of (13.20) must have finite order, which for an ordinary complex exponent is rationality. For the last assertion choose  $a$  with  $\nu = V'(a) \neq \mu$  by [lemma 13.12](#), put  $r_0 = \mu/\nu \in \mathbb{Q}$  and a positive integer  $m$  with  $mr_0 \in \mathbb{Z}$ ; after  $m$  circuits about  $a$  the function  $h_0$  returns to itself, while by [proposition 13.8](#) the same continuation changes  $h_1$  by  $2\pi i m \frac{\mu}{2} (1 - r_0) h_0 \neq 0$ . Repeated blocks of  $m$  circuits therefore produce infinitely many distinct branches of  $h_1$ , and an algebraic function has only finitely many.  $\square$

**Example 13.18** (Rational leading coordinate, logarithmic correction). For  $V = w + w^2$ ,  $h_0 = w/(1+w)$  and  $h_1 = \frac{w}{1+w} \log(1+w)$ . At the fixed point  $-1$  the exact exponent is  $\text{Log}(1+\varepsilon)/\text{Log}(1-\varepsilon) = -1 + \varepsilon + O(\varepsilon^2)$ , so the leading multiplier is 1 while the full multiplier begins  $1 + 2\pi i \varepsilon + O(\varepsilon^2)$ : the obstruction is present even without any branching in the leading function. A rational leading coordinate does not even predict the *branching type* of the first correction.

**Example 13.19** (One trivial nonzero-root multiplier). For  $V = w(1-w)(1-2w)$  one has  $\mu = 1$ ,  $V'(1) = 1$ ,  $V'(1/2) = -1/2$ , and

$$h_0 = \frac{w(1-w)}{(1-2w)^2}, \quad h_1 = \frac{3}{2} h_0 \log(1-2w).$$

The exact multiplier at 1 is 1, whereas at  $1/2$  it is  $1 + 3\pi i\varepsilon + O(\varepsilon^2)$  and has infinite order. Thus [theorem 13.11](#) does *not* claim branching at every nonzero fixed point; [lemma 13.12](#) guarantees at least one obstruction, which suffices for transcendence over the whole base.

**Example 13.20** (Why multiple zeros need a separate argument). For  $V = w(1 + w)^2$ ,  $\mu = 1$  and  $h_0(w) = \frac{w}{1+w} \exp(\frac{1}{1+w} - 1)$ . The pole of  $\mu/V$  at  $-1$  has order two, so the leading solution has an essential singularity and is already transcendental — [remark 13.16](#).

### 13.6 The resonant model: exact scale, isometry, continuation

Fix an ordinary integer  $q \geq 1$  and let  $f(z) = (1 + \varepsilon)z + z^{q+1}$  with  $\varepsilon = t^\delta$ ,  $s = t^{\delta/q}$  (divisibility of  $\Gamma$  placing  $s$  in  $K_\Gamma$ ). Under  $z = sw$ ,

$$g(w) = s^{-1}f(sw) = (1 + \varepsilon)w + \varepsilon w^{q+1} = w + \varepsilon V_q(w), \quad V_q(w) = w(1 + w^q). \quad (13.21)$$

Let  $\Psi_\varepsilon$  be its normalized common-domain coordinate;  $U = \{|w| < 1\}$  is one admissible domain, and larger simply connected slit domains avoiding the roots of  $1 + w^q$  are also possible. Since  $g(\zeta w) = \zeta g(w)$  for a  $q$ -th root of unity  $\zeta$ , uniqueness gives  $\Psi_\varepsilon(\zeta w) = \zeta \Psi_\varepsilon(w)$  and hence

$$\Psi_\varepsilon(w) = \sum_{m \geq 0} b_m(\varepsilon) w^{qm+1}, \quad b_0 = 1, \quad b_m \in \mathbb{C}[[\varepsilon]], \quad (13.22)$$

initially as the Taylor series of each ordinary coefficient function; its evaluation as a Hahn family at a specific  $w$  is a separate question. The leading solution is

$$\Psi_0(w) = w(1 + w^q)^{-1/q}, \quad b_m(0) = (-1)^m \frac{(1/q)_m}{m!} \neq 0, \quad (13.23)$$

with  $(x)_m = x(x+1) \cdots (x+m-1)$ .

**Proposition 13.21** (Exact finite recurrence). *With  $\lambda = 1 + \varepsilon$ , for  $m \geq 1$*

$$b_m = - \frac{\sum_{\substack{0 \leq j < m \\ m-j \leq qj+1}} b_j \binom{qj+1}{m-j} \lambda^{qj+1-(m-j)} \varepsilon^{m-j}}{\lambda^{qm+1} - \lambda}, \quad (13.24)$$

*which uniquely constructs  $b_m \in \mathbb{C}[[\varepsilon]]$  and has leading recurrence  $qm b_m(0) = -(q(m-1)+1)b_{m-1}(0)$ .*

*Proof.* In  $g(w)^{qj+1} = w^{qj+1}(\lambda + \varepsilon w^q)^{qj+1}$  the coefficient of  $w^{qm+1}$  is the binomial expression displayed; the term  $j = m$  contributes  $\lambda^{qm+1} b_m$  and the right side of  $\Psi_\varepsilon \circ g = \lambda \Psi_\varepsilon$  contributes  $\lambda b_m$ . The denominator is  $qm\varepsilon + O(\varepsilon^2)$  while every numerator term is divisible by  $\varepsilon$ , so induction gives  $b_m \in \mathbb{C}[[\varepsilon]]$ ; only  $j = m-1$  survives after division by  $\varepsilon$  and reduction modulo  $\varepsilon$ , which also checks (13.23) directly.  $\square$

The normalized formal coordinate of the original map is

$$h(z) = s\Psi_\varepsilon(z/s) = \sum_{m \geq 0} \varepsilon^{-m} b_m(\varepsilon) z^{qm+1}, \quad (13.25)$$

a formal power series in  $z$  over  $K_\Gamma$  — *not* a Hahn family already asserted summable at every  $z$  — whose nonzero coefficients have the exact valuations  $v(\varepsilon^{-m} b_m(\varepsilon)) = -m\delta$ .

**Theorem 13.22** (Sharp domain and valuative isometry). *The family of Taylor terms in (13.25) is strongly summable at  $z \in K_\Gamma$  if and only if  $z = 0$  or*

$$v(z) > \delta/q. \quad (13.26)$$

*On  $D = \{v(z) > \delta/q\}$  the resulting function is a bijection  $h : D \rightarrow D$  satisfying  $h \circ f = \lambda h$  and*

$$v(h(x) - h(y)) = v(x - y) \quad (x, y \in D). \quad (13.27)$$

*No strictly larger centered valuative ball containing  $D$  admits an injective zero-fixing conjugacy to multiplication by  $\lambda$ .*

*Proof.* Let  $z \neq 0$ ,  $\beta = v(z)$ : the  $m$ -th nonzero Taylor term has valuation exactly

$$v(\varepsilon^{-m} b_m z^{qm+1}) = \beta + m(q\beta - \delta). \quad (13.28)$$

**Two distinct failure mechanisms.** If  $q\beta = \delta$ , every term has a nonzero coefficient at the *same* exponent  $\beta$ , so *local finiteness* fails. If  $q\beta < \delta$ , the displayed leading exponents form a strictly descending sequence in the union of supports, so *well-ordering* fails. This proves necessity with no convergence terminology. If  $q\beta > \delta$ , put  $w = z/s$  with  $v(w) > 0$ : all coefficients  $b_m$  have nonnegative  $\varepsilon$ -support, and a term of spatial degree  $qm + 1$  contributes a word of at least that length from  $\text{supp } w$  together with a nonnegative number of  $\delta$ 's, so [lemma 3.4](#) makes the double family strongly summable, and multiplication by  $s$  preserves this.

For invertibility, formal reversion of  $\Psi_\varepsilon = w + O(w^{q+1})$  gives  $\Theta_\varepsilon \in \mathbb{C}[[\varepsilon]][[w]]$  with linear coefficient 1 and no constant term, requiring no division except by a unit; the same positive-word argument evaluates it at every  $w \in \mathfrak{m}_\Gamma$ , and the formal identities become strong identities there, both maps preserving  $\mathfrak{m}_\Gamma$  because their difference from the identity has strictly greater valuation at a nonzero infinitesimal input. For the isometry put  $u = x/s$ ,  $v_1 = y/s$ :

$$\Psi_\varepsilon(u) - \Psi_\varepsilon(v_1) = (u - v_1) \left[ 1 + \sum_{m \geq 1} b_m(\varepsilon) \sum_{j=0}^{qm} u^{qm-j} v_1^j \right],$$

whose bracket is a strong sum equal to 1 plus a positive-valuation element, each nonconstant summand having valuation at least  $q \min\{v(u), v(v_1)\} > 0$  when both inputs are nonzero (a zero input is handled with the vanishing terms omitted). Finally every nonzero solution of  $z^q = -\varepsilon$  is a fixed point of  $f$  of valuation  $\delta/q$ , lying in  $K_\Gamma$  because its constant factor is a complex  $q$ -th root of  $-1$ ; any strictly larger centered ball containing  $D$  contains all elements at valuation  $\delta/q$ , hence such a fixed point in addition to 0, and an injective conjugacy  $k$  would satisfy  $k(z) = \lambda k(z)$  at every fixed point, mapping both to 0 since  $\lambda \neq 1$ .  $\square$

**Warning 13.23** (The maximality statement is about centered balls). [Theorem 13.22](#) concerns centered valuative balls defined by a valuation threshold, under either boundary convention. It does **not** exclude continuation to a non-ball domain avoiding the obstructing fixed points, and does not assert that every coefficientwise continuation is injective. Those distinctions matter at the critical scale, which is exactly where [section 13.7](#) operates.

### 13.7 Continuation at a scale where the Taylor family fails

Let  $w_0 \in U$  be an ordinary nonzero point. At  $z = sw_0$  we have  $v(z) = \delta/q$ , so the Taylor family (13.25) is *not* strongly summable. On the other hand

$$s\Psi_\varepsilon(w_0) = s \sum_{n \geq 0} \varepsilon^n \Psi_n(w_0) \quad (13.29)$$

is a valid Hahn value, and more generally it is defined at  $z = s(w_0 + \eta)$  for positive  $\eta$  by (3.4). There is no contradiction, and the distinction is exactly where the monodromy lives: in (13.29) each coefficient function  $\Psi_n$  is first an ordinary holomorphic function obtained by ordinary integration and continuation, and summation in the Hahn exponent comes *afterward*; in (13.25) summation in spatial Taylor degree is required to be strongly summable as a family of Hahn numbers, and it fails precisely on this boundary. The boundary value is a *coefficientwise analytic continuation*, not an illegal regrouping of a strong sum; on the inner disk both procedures are admissible and agree by finite-word Taylor identities.

For  $V_q = w(1 + w^q)$  every nonzero zero satisfies  $a^q = -1$  and  $V'_q(a) = -q$ , so all these zeros share the exact exponent

$$\rho_q(\varepsilon) = \frac{\text{Log}(1 + \varepsilon)}{\text{Log}(1 - q\varepsilon)} = -\frac{1}{q} + A_q\varepsilon + B_q\varepsilon^2 + O(\varepsilon^3), \quad (13.30)$$

$$A_q = \frac{q+1}{2q}, \quad B_q = \frac{(q+1)(q-4)}{12q}, \quad C_q = \frac{(q+1)(2q+1)}{12q}, \quad (13.31)$$

the ratio being legitimate because numerator and denominator both start in degree one with the denominator's coefficient  $-q \neq 0$ .

**Theorem 13.24** (All-orders factorization and finite-cover obstruction for the model). *The rescaled coordinate has the all-orders form*

$$\Psi_\varepsilon(w) = w(1 + w^q)^{\rho_q(\varepsilon)} \text{Exp}_H(P_\varepsilon(w)), \quad P_\varepsilon \in \varepsilon^2\mathbb{C}[w][[\varepsilon]], \quad P_\varepsilon(0) = 0. \quad (13.32)$$

At every nonzero fixed point the monodromy multiplier  $M_q = \text{Exp}_H(2\pi i\rho_q)$  satisfies

$$M_q^q = 1 + \pi i(q+1)\varepsilon + O(\varepsilon^2), \quad (13.33)$$

and has infinite order. The leading coordinate becomes single-valued after finite ramification, but no finite branched cover makes  $\Psi_\varepsilon$  single-valued at these points.

*Proof.* Apply corollary 13.9: all nonzero-root exponents coincide and  $\prod_{a^q=-1}(1 - w/a) = 1 + w^q$ , the branches of the logarithms agreeing near 0 where each is normalized to vanish, so their sum is  $\log(1 + w^q)$ ; this gives (13.32) with the same polynomial part. The coefficient at zero in  $q\rho_q$  is  $-1$  and the coefficient at  $\varepsilon$  is  $(q+1)/2$ , so exponentiation gives (13.33); no positive power is 1 by lemma 13.10 since  $A_q \neq 0$ . The leading exponent  $-1/q$  is rational, so a  $q$ -fold local ramification removes its branching, and theorem 13.11 shows that the positive part of the full exponent survives every finite ramification.  $\square$

A circuit repeated  $q$  times thus returns the leading branch to itself and still changes the full answer by a nonzero infinitesimal amount: the monodromy group records information lost under passage to the standard part. This is the phenomenon whose linear-ODE ancestor is already recorded by the companion report *Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers* (docs/surcomplex/differential-equations/); see section 22.1 for the exact inheritance.

**Proposition 13.25** (Explicit second-order expansion). *With  $L(w) = \log(1 + w^q)$  normalized by  $L(0) = 0$ , on any admissible common domain*

$$\Psi_\varepsilon(w) = w(1 + w^q)^{-1/q} \left[ 1 + A_q\varepsilon L(w) + \varepsilon^2 \left( B_q L(w) + \frac{A_q^2}{2} L(w)^2 - C_q w^q \right) + O(\varepsilon^3) \right], \quad (13.34)$$

equivalently  $P_\varepsilon(w) = -C_q\varepsilon^2 w^q + O(\varepsilon^3)$ .

*Proof.* With  $V = V_q$  and  $\mu = 1$ , expansion using (13.10) gives  $[\varepsilon^2]\Omega_\varepsilon = (1/3 - V'/4 - (V')^2/12 - VV''/12)/V$ ; substituting  $V = w(1 + w^q)$  and collecting powers of  $w^q$  yields

$$[\varepsilon^2]\Omega_\varepsilon = -\frac{(q+1)(2q+1)}{12}w^{q-1} + \frac{(q+1)(q-4)}{12}\frac{w^{q-1}}{1+w^q},$$

whose first summand is the polynomial part with normalized primitive  $-C_q w^q$  and whose second has normalized primitive  $B_q L(w)$ ; the coefficient at  $\varepsilon$  integrates to  $A_q L(w)$  and the coefficient at zero gives  $\Psi_0$ .  $\square$

For  $q = 1$  this reads

$$\Psi_\varepsilon(w) = \frac{w}{1+w} \left[ 1 + \varepsilon \log(1+w) + \frac{\varepsilon^2}{2} (\log^2(1+w) - \log(1+w) - w) + O(\varepsilon^3) \right]. \quad (13.35)$$

For  $q = 4$  the coefficient  $B_q$  happens to vanish, but the first-order logarithm and hence the infinite-order monodromy do not: an accidental cancellation at one higher order cannot repair the obstruction.

*Remark 13.26* (Three representations of one inner-disk coordinate). The formal Taylor series (13.25) supplies exact valuations and the sharp disk; the common-domain coefficient series  $\Psi_\varepsilon = \sum \varepsilon^n \Psi_n$  supplies continuation at the rescaled boundary; and the factorization (13.32) isolates monodromy in a single Hahn constant with a single-valued polynomial-coefficient factor left over. They answer different questions: to evaluate very close to 0 the recurrence (13.24) is efficient and directly summable; to determine branching near a boundary fixed point the exact ratio of logarithms avoids computing many individual Taylor coefficients; and to see why algebraicity fails even when the leading function is algebraic, the first correction or the monodromy orbit suffices.

*Warning 13.27* (Transcendental, but differentially algebraic). Transcendence in theorem 13.15 is relative to the specific base  $\mathcal{B}$ . Source 04 volunteers the counterweight:  $H_\varepsilon$  is *differentially* algebraic in a very explicit way, satisfying the first-order linear equation  $H'_\varepsilon = \Omega_\varepsilon H_\varepsilon$  with  $\Omega_\varepsilon \in \mathcal{B}$ . So “transcendental” must not be upgraded to “differentially transcendental”. Nor is it a claim that any individual value  $H_\varepsilon(w_0)$  lies outside  $\mathbf{No}[i]$ ; the statement concerns a *function* in an ambient meromorphic Hahn field and is explicitly consistent with  $\mathbf{No}[i]$  being algebraically closed.

Everything above stays true of  $H_\varepsilon$ , and the sentences above are not retracted by what follows. What the companion report *Tail-Spans and Differential Transcendence in Surreal and Surcomplex Hahn Fields* (docs/surreal/tail-spans-and-differential-transcendence/) shows is that the refusal is about  $H_\varepsilon$  and not about the workspace: its Theorem `tail:thm:mixed` exhibits explicit functions  $F_A(z, t) = \sum_{n \in A} t^n \sqrt{1 - z/2^n}$ , with the branch equal to 1 at  $z = 0$ , one for each infinite  $A \subseteq \mathbb{N}_{>0}$ , all of whose mixed jets under  $\partial_z$  and  $t d/dt$  are algebraically independent over  $\mathbb{C}(z)((t^\mathbb{Q}))$  — the  $\Gamma = \mathbb{Q}$  case of a base of the same shape as  $\mathcal{B}$  — so that these functions satisfy no algebraic differential equation of any finite order over that field. Its Theorem `tail:thm:continuumanalytic` does this for continuum many such functions on the single fixed disk  $|z| < 1$ , with no shrinking of the disk, matching the fixed-domain discipline of section 14. Those functions are not solutions of equation (13.18) and nothing about them transfers to  $H_\varepsilon$ ; the pointer records that “transcendental but differentially algebraic” is a property of  $H_\varepsilon$ , not a ceiling on what a base of this shape permits. Its Remark `tail:rem:dynupgrade` states the same relation from the other side.

## 14 Parabolic dynamics II: time forms, periods, and moduli

This section is also under PAR and  $\text{SD}_\emptyset$ , and its maps are near-identity, hence unipotent: there are *no* eigenvalue ratios and *no* Diophantine hypotheses (warning 2.8). The single defining choice is the

coefficient algebra: everything happens in  $\mathcal{A} = \mathcal{O}(D)((t^\Gamma))$  for *one* fixed ordinary connected  $D \subset \mathbb{C}$ , the fixed-polydisk row of [section 3.2](#), and **no shrinking of  $D$  is permitted in any theorem**. The Hahn monomials are constants, not functions of  $z$ ; the only derivation is  $\partial_z$  applied coefficientwise. Write also  $\mathcal{B} = \mathcal{M}(D)((t^\Gamma))$ , a field, used only for division steps;  $\mathcal{A}$  itself is a commutative integral domain and generally not a field, with  $K = K_\Gamma$  embedded as the constants. All integrals below are *ordinary* complex line integrals taken coefficient by coefficient over ordinary curves in  $D$ .

**Warning 14.1** (These periods are not surcomplex contour integrals). A period below is an ordinary coefficientwise contour integral. It is **not** a surcomplex contour integral, **not** a residue at an arbitrary surcomplex point, and **not** a fine-topological notion. Source 05 cites the companion report *Contours and Stokes: Jordan separation, Cauchy theory, and Stokes calculus on the surcomplex plane* ([docs/surcomplex/contours-and-stokes/](#)) as background and deliberately does *not* use surcomplex contours; it carries a remark forbidding exactly that identification. Furthermore, if one allows an unrelated choice of primitive on each monad one has enlarged the function class and can destroy the global obstruction: the theorems below are about a single element of  $\mathcal{O}(D)((t^\Gamma))$ , not a collection of unrelated infinitesimal germs.

### 14.1 Primitives, time forms, and pullback exactness

**Lemma 14.2** (Support-preserving primitives). *For  $g \in \mathcal{A}$ , a primitive  $P \in \mathcal{A}$  with  $P' = g$  exists if and only if  $\oint_\gamma g(z) dz = 0$  (coefficientwise) for every ordinary closed piecewise  $C^1$  curve in  $D$ . If it exists, the primitive with  $P(z_*) = 0$  is unique and introduces no new support exponents.*

*Proof.* Necessity holds coefficientwise by the ordinary fundamental theorem for line integrals. Conversely define each coefficient primitive by path integration from  $z_*$ ; vanishing periods make this path-independent, and on an ordinary disk the result is an ordinary holomorphic primitive, hence holomorphic on all of  $D$ . Use zero for a coefficient that was zero, so the support is a subset of  $\text{supp } g$ . Two primitives differ by an element of  $K$ , removed by the normalization.  $\square$

Assume now that  $W = W_F = X_F z$  has a nowhere-zero leading coefficient. By [lemma 13.2](#) the *time form*

$$\Omega_F = \frac{dz}{W_F}$$

lies in  $\mathcal{A} dz$ : “holomorphic” refers to all its ordinary coefficient functions, and its *valuation may be negative*. Its significance is elementary: in a local primitive coordinate  $P$  with  $P' = 1/W$  the vector field  $W\partial_z$  becomes a unit translation field; the issue is whether such local primitives fit together globally, and whether one can compare the primitives of two maps without leaving the common coefficient domain. For  $H = z + h \in \mathcal{G}_+(D)$  the pullback of a one-form is  $H^*(g dz) = (g \circ H)H' dz$ .

**Lemma 14.3** (Explicit exactness of an infinitesimal pullback). *For  $g \in \mathcal{A}$  and  $h \in \mathcal{A}_{>0}$  the series*

$$Q_h(g) = \sum_{n \geq 1} \frac{h^n}{n!} g^{(n-1)} \tag{14.1}$$

*is strongly summable and satisfies*

$$(z + h)^*(g dz) - g dz = dQ_h(g). \tag{14.2}$$

*Consequently every coefficientwise period is unchanged by a near-identity pullback.*

*Proof.* For  $S = \text{supp } h$ , the support in (14.1) lies in  $\text{supp } g + (S^* \setminus \{0\})$  with finitely many contributions per coefficient, so termwise differentiation is legitimate and gives

$$(Q_h(g))' = \sum_{n \geq 1} \frac{h^n}{n!} g^{(n)} + h' \sum_{n \geq 1} \frac{h^{n-1}}{(n-1)!} g^{(n-1)} = g(z+h) - g(z) + h'g(z+h) = g(z+h)(1+h') - g(z),$$

which is (14.2). Exact coefficient forms have zero ordinary periods by lemma 14.2.  $\square$

Lemma 14.3 is a support-controlled version of the familiar fact that an infinitesimal change of coordinate acts trivially on one-dimensional holomorphic de Rham cohomology. It is included as infrastructure; the homotopy principle itself is *not* a novelty claim, and there is no assertion that  $s \mapsto z + sh$  is a fine-continuous path in the surreal topology — formula (14.1) is the construction and its support certificate is what licenses it.

**Proposition 14.4** (Time forms under conjugacy). *If  $W_{F_1}, W_{F_2}$  are invertible in  $\mathcal{A}$  then for  $H \in \mathcal{G}_+(D)$ ,  $H \circ F_1 = F_2 \circ H$  iff  $H^* \Omega_{F_2} = \Omega_{F_1}$ ; in particular conjugate maps have identical time-form periods on every ordinary closed curve in  $D$ .*

*Proof.* Equality of time forms reads  $H'/W_{F_2}(H) = 1/W_{F_1}$ , equivalent to proposition 4.7; period invariance is lemma 14.3.  $\square$

## 14.2 The discrete equation as a primitive problem

Let  $F \in \mathcal{G}_+(D)$  be nonidentity,  $W = W_F$ ,  $\mathcal{D} = W\partial_z$ , and

$$E(X) = \frac{e^X - 1}{X} = \sum_{n \geq 0} \frac{X^n}{(n+1)!}, \quad B(X) = \frac{X}{e^X - 1} = \sum_{n \geq 0} \frac{B_n}{n!} X^n, \quad (14.3)$$

with  $B_0 = 1$ ,  $B_1 = -1/2$ . Substitution of  $\mathcal{D}$  into either series is strongly summable, and these are inverse automorphisms of the underlying strongly linear space  $\mathcal{A}$ ; *no* multiplicativity of  $B(\mathcal{D})$  or  $E(\mathcal{D})$  is asserted. Write  $\Delta_F \psi = \psi \circ F - \psi$ .

**Theorem 14.5** (Discrete equation as a primitive problem). *For nonidentity  $F \in \mathcal{G}_+(D)$ ,*

$$\ker \Delta_F = K, \quad \Delta_F(\mathcal{A}) = W\partial_z \mathcal{A}. \quad (14.4)$$

*For  $g \in \mathcal{A}$  the equation  $\psi \circ F - \psi = g$  has a solution  $\psi \in \mathcal{A}$  if and only if the meromorphic Hahn quotient  $g/W$  belongs to  $\mathcal{A}$  and*

$$\oint_{\gamma} \frac{g(z)}{W(z)} dz = 0 \quad \text{for every ordinary closed curve } \gamma \subset D. \quad (14.5)$$

*If  $P' = g/W$ , all solutions are  $\psi = B(\mathcal{D})P + C$  with  $C \in K$ . When the leading coefficient of  $W$  is nowhere zero, the membership  $g/W \in \mathcal{A}$  is automatic.*

*Proof.* By theorem 4.3 and the scalar identities in (14.3),

$$\Delta_F = e^{\mathcal{D}} - I = \mathcal{D}E(\mathcal{D}) = E(\mathcal{D})\mathcal{D}, \quad (14.6)$$

and  $E(\mathcal{D})B(\mathcal{D}) = I = B(\mathcal{D})E(\mathcal{D})$  follows coefficientwise, corollary 3.5 permitting every rearrangement; so the image equals  $\mathcal{D}\mathcal{A}$ . The kernel of  $\mathcal{D}$  is  $K$  since  $W \neq 0$ ,  $\mathcal{A}$  is a domain and  $\ker \partial_z = K$ ,

and both  $E(\mathcal{D})$  and its inverse act as the identity on constants, so (14.6) also gives  $\ker \Delta_F = K$ . Since division by  $W$  is available in  $\mathcal{B}$ , membership in  $W\partial_z \mathcal{A}$  is equivalent to  $g/W$  being the derivative of an element of  $\mathcal{A}$ , which is lemma 14.2. Finally  $\Delta_F B(\mathcal{D})P = \mathcal{D}P = WP' = g$ , and the kernel statement gives uniqueness up to  $K$ ; the last assertion is lemma 13.2.  $\square$

*Remark 14.6* (Why the uncorrected forcing supplies the periods). One might instead multiply by  $B(\mathcal{D})$  and integrate  $B(\mathcal{D})g/W$ . This gives the same obstruction: for  $n \geq 1$ ,  $\mathcal{D}^n g/W = (\mathcal{D}^{n-1} g)'$  is exact, so all Bernoulli correction terms have zero periods. The factorization (14.6) makes the uncorrected quotient  $g/W$  and its obstruction especially transparent.

**Corollary 14.7** (Exact global Abel criterion). *Suppose the leading coefficient of  $W_F$  is nowhere zero. Then  $\psi \circ F = \psi + 1$  has a solution in  $\mathcal{A}$  if and only if all the periods of  $\Omega_F$  vanish, in which case any primitive  $P' = 1/W_F$  already satisfies the equation and all solutions differ from  $P$  by a constant in  $K$ .*

*Proof.* Apply theorem 14.5 with  $g = 1$ ; also  $\mathcal{D}P = 1$  and  $\mathcal{D}^2 P = 0$ , so  $e^{\mathcal{D}} P = P + 1$ , while the Bernoulli formula gives  $B(\mathcal{D})P = P - 1/2$ , differing only by a constant.  $\square$

*Warning 14.8* (An Abel function is not a globally injective coordinate). The identity  $P' = 1/W_F$  guarantees a nonvanishing derivative on the halo under the leading-unit hypothesis, but it does **not** make  $P$  globally injective on a multiply covered domain: even in ordinary complex analysis a locally univalent primitive need not be globally univalent. Corollary 14.7 is an Abel-function theorem, not an unconditional global conjugacy with translation on a subset of  $K$ .

**Theorem 14.9** (Complete finite-dimensional discrete obstruction). *Let  $D_r = \mathbb{C} \setminus \{a_1, \dots, a_r\}$  with distinct ordinary points and positively oriented small ordinary circles  $\gamma_j$  about  $a_j$  enclosing no other puncture, and put*

$$(\mathcal{P}_F g)_j = \frac{1}{2\pi i} \oint_{\gamma_j} \frac{g}{W_F} dz. \quad (14.7)$$

*If  $F \in \mathcal{G}_+(D_r)$  has a nowhere-zero leading displacement coefficient, then*

$$0 \longrightarrow K \longrightarrow \mathcal{A} \xrightarrow{\Delta_F} \mathcal{A} \xrightarrow{\mathcal{P}_F} K^r \longrightarrow 0 \quad (14.8)$$

*is an exact sequence of  $K$ -vector spaces, with explicit right inverse*

$$(q_1, \dots, q_r) \longmapsto W_F(z) \sum_{j=1}^r \frac{q_j}{z - a_j}, \quad (14.9)$$

*so that every forcing decomposes as*

$$g = \Delta_F \psi + W_F \sum_{j=1}^r \frac{(\mathcal{P}_F g)_j}{z - a_j}, \quad (14.10)$$

*with  $\psi$  determined up to a constant once the second term is fixed.*

*Proof.* The initial kernel statement is theorem 14.5. Every ordinary closed curve in  $D_r$  has, against a holomorphic one-form, the same periods as the integer combination of small puncture circles given by its winding numbers: remove small disks about the punctures inside a large disk containing the curve and cut the remaining planar region along arcs joining the boundary components, so that the cut region is simply connected and the integrals along the two sides of each cut cancel.

Applying this ordinary argument coefficientwise shows that vanishing of the  $r$  components of (14.7) is equivalent to (14.5), so middle exactness follows from theorem 14.5. The differential  $dz/(z - a_j)$  has integral  $2\pi i$  around  $\gamma_j$  and zero around the other chosen circles, so (14.9) is a right inverse; subtracting it from  $g$  and applying exactness gives (14.10).  $\square$

No identification of the infinite-dimensional  $\mathcal{A}$  with  $K \otimes_{\mathbb{C}} \mathcal{O}(D_r)$  is used: the finite-dimensional quotient is computed directly by coefficientwise periods and explicit representatives.

**Example 14.10** (A local obstruction where the leading coefficient vanishes). The quotient condition in theorem 14.5 must not be dropped outside the nonsingular stratum. On a disk containing 0 take  $W = \varepsilon z^2$  and  $F = z/(1 - \varepsilon z)$  with  $\varepsilon = t^\delta$ ,  $\delta > 0$ ; this  $F$  is the exponential of  $W\partial_z$  in  $\mathcal{G}_+(D)$ . The equation  $\psi \circ F - \psi = 1$  has *no* solution in  $\mathcal{A}$ , because  $1/W = \varepsilon^{-1}z^{-2}$  is not holomorphic at 0 — equivalently  $F$  fixes 0 and evaluating the equation there would give  $0 = 1$ . On a punctured disk, by contrast,  $-1/(\varepsilon z)$  is an Abel function. This separates a local divisibility obstruction from the global period obstruction.

### 14.3 The complete global conjugacy classification

Fix  $\delta > 0$ ,  $\varepsilon = t^\delta$ , and  $a \in \mathcal{O}(D)$  nowhere zero, and define the nonsingular stratum

$$\mathcal{G}_{\delta,a}(D) = \{F \in \mathcal{G}_+(D) : F = z + \varepsilon a + r, \ v(r) > \delta\} = \{F : W_F = \varepsilon(a + u), \ u \in \mathcal{A}_{>0}\}, \quad (14.11)$$

with normalized time form

$$\beta_F = \varepsilon \Omega_F = b_F(z) dz, \quad b_F = a^{-1} + \text{positive Hahn terms}. \quad (14.12)$$

The leading data  $(\delta, a)$  are a conjugacy invariant under  $\mathcal{G}_+(D)$ : the identity  $W_2(H) = H'W_1$  of proposition 4.7 changes neither the leading coefficient nor the valuation, because  $H = z + \text{positive}$  and  $H' = 1 + \text{positive}$ . It is therefore natural to classify one stratum at a time.

**Lemma 14.11** (Implicit displacement lemma). *Let  $b = b_0 + u \in \mathcal{A}_{\geq 0}$  with  $b_0 \in \mathcal{O}(D)$  nowhere zero and  $u \in \mathcal{A}_{>0}$ . For each  $p \in \mathcal{A}_{>0}$  there is a unique  $h \in \mathcal{A}_{>0}$  satisfying*

$$bh + \frac{b'}{2!}h^2 + \frac{b''}{3!}h^3 + \cdots = p. \quad (14.13)$$

*Every coefficient of  $h$  is holomorphic on  $D$ ; if  $S = \text{supp } u \cup \text{supp } p$  then  $\text{supp } h \subseteq S^* \setminus \{0\}$ ; and if  $p(z_*) = 0$  then  $h(z_*) = 0$ .*

*Proof.* If  $S = \emptyset$  take  $h = 0$ . Otherwise let  $M = S^*$ , well ordered by lemma 3.4; set the coefficient of  $h$  at 0 to zero and determine  $h_\gamma$  for  $\gamma \in M \setminus \{0\}$  in increasing order. The coefficient of  $h_\gamma$  on the left of (14.13) is precisely  $b_0$ ; any other linear term comes from  $u_\alpha h_\nu$  with  $\alpha > 0$  and  $\nu = \gamma - \alpha < \gamma$ , and in a term with  $n \geq 2$  factors of  $h$  every factor exponent is positive and strictly smaller than  $\gamma$ . The coefficient is a finite sum (expand the factors into words in  $S$  and apply lemma 3.4), so the recursion has the form  $h_\gamma = b_0^{-1}(p_\gamma - R_\gamma)$  with  $R_\gamma$  a finite expression in earlier coefficients and ordinary derivatives of  $b$ , defining a holomorphic function on  $D$ . For unrestricted uniqueness, the difference of the two left sides for two solutions  $h_1, h_2 \in \mathcal{A}_{>0}$  factors as  $(h_1 - h_2)(b_0 + v)$  with  $v \in \mathcal{A}_{>0}$  — factor each difference of powers, the resulting sum being strongly summable over the finite union of the positive supports of  $u, h_1, h_2$  — and the second factor is a unit by lemma 13.2. At  $z_*$ , evaluation gives the same scalar implicit equation over  $K$  with linear leading coefficient  $b_0(z_*) \neq 0$ , so if  $p(z_*) = 0$  then zero is a solution and the same factorization shows it is the only positive one.  $\square$

*Warning 14.12* (Where the nowhere-zero hypothesis does its work). [Lemma 14.11](#) is exactly the point at which the classification proof stops applying when  $a$  has zeros. Its common-domain conclusion is stronger than solving a separate germ problem at each point: *every stage divides only by the same nowhere-zero ordinary function  $b_0$*  and uses derivatives of functions already holomorphic on  $D$ , so no radius is lost at any Hahn exponent. Source 05 states outright that the zeros case needs local zero divisors, multiplicities and singular normal-form obstructions combined with global periods, and that its classification is not claimed there.

**Theorem 14.13** (Complete period invariant and marked conjugacy). *For  $F_1, F_2 \in \mathcal{G}_{\delta,a}(D)$  with  $a$  nowhere zero, the following are equivalent:*

- (i) *there is  $H \in \mathcal{G}_+(D)$  with  $H \circ F_1 = F_2 \circ H$ ;*
- (ii)  *$\oint_\gamma \Omega_{F_1} = \oint_\gamma \Omega_{F_2}$  for every ordinary closed piecewise  $C^1$  curve  $\gamma \subset D$ ;*
- (iii)  *$\beta_{F_1} - \beta_{F_2} = dp$  for some  $p \in \mathcal{A}_{>0}$ .*

*When they hold, for every specified  $z_* \in D$  there is a unique conjugacy  $H$  as in (i) with  $H(z_*) = z_*$ , obtained by taking the normalized primitive  $p(z_*) = 0$  in (iii) and solving*

$$\sum_{n \geq 1} \frac{b_{F_2}^{(n-1)}}{n!} h^n = p, \quad H = z + h. \quad (14.14)$$

**No shrinking of  $D$  is permitted anywhere in this statement.**

*Proof.* (i) $\Rightarrow$ (ii) is [proposition 14.4](#). Multiplication by the nonzero constant  $\varepsilon$  shows (ii) equivalent to the vanishing of all periods of  $\beta_{F_1} - \beta_{F_2}$ ; both normalized time forms have the same leading coefficient  $a^{-1}$ , so their difference has positive support, and the normalized primitive of [lemma 14.2](#), when it exists, also has positive support — this is the equivalence of (ii) and (iii) including the normalization of  $p$ . Assume (iii): the leading coefficient of  $b_{F_2}$  is  $a^{-1}$ , nowhere zero, so [lemma 14.11](#) gives a unique positive  $h$  in (14.14), and [lemma 14.3](#) gives  $H^* \beta_{F_2} - \beta_{F_2} = dp = \beta_{F_1} - \beta_{F_2}$ , i.e.  $H^* \Omega_{F_2} = \Omega_{F_1}$ , so  $H$  is a conjugacy by [proposition 14.4](#); its inverse lies in  $\mathcal{G}_+(D)$  by [theorem 4.3](#) and  $H(z_*) = z_*$  by [lemma 14.11](#). If  $\tilde{H} = z + \tilde{h}$  is any conjugacy fixing  $z_*$ , the same pullback identity yields  $d(Q_{\tilde{h}}(b_{F_2}) - p) = 0$ , so that expression is a constant in  $K$  with value zero at  $z_*$ , whence  $\tilde{h}$  satisfies the same implicit equation and uniqueness in [lemma 14.11](#) gives  $\tilde{H} = H$ .  $\square$

**Corollary 14.14** (Simply connected domains). *If  $D$  is simply connected, any two maps in  $\mathcal{G}_{\delta,a}(D)$  are conjugate by a unique near-identity change of variable fixing a prescribed ordinary basepoint, and each is conjugate to  $\exp(\varepsilon a \partial_z)z$ .*

*Proof.* Every ordinary holomorphic coefficient one-form on  $D$  has a primitive, so (iii) of [theorem 14.13](#) holds.  $\square$

For  $a = 1$  the reference map is  $z + \varepsilon$ , and this is a genuine global near-identity conjugacy to that translation on  $D^\#$ ; for general  $a$  the reference map is the indicated vector-field exponential and no univalence of a primitive of  $dz/a$  is required.

*Warning 14.15* (Three hypotheses that do actual work).  $a$  must not vanish, because inversion and the implicit displacement recursion divide by  $a$  or  $a^{-1}$  ([warning 14.12](#)). The conjugating map must have identity ordinary part, because otherwise it can act nontrivially on ordinary homology and on the leading form — source 05 records that extending the classification to a larger group would require the ordinary automorphism action on the leading form and its periods, and that equal period vectors in a fixed basis would then no longer be the correct equivalence relation by themselves. And

all coefficient functions must be defined on the same  $D$ : replacing global functions by independent germs removes the meaning of a single period comparison. The theorem is not claimed after any of these hypotheses is erased.

#### 14.4 Explicit moduli

**Theorem 14.16** (An explicit moduli space). *On  $D_r = \mathbb{C} \setminus \{a_1, \dots, a_r\}$ , for the fixed stratum  $(\delta, a)$  define*

$$q_j(F) = \frac{1}{2\pi i} \oint_{\gamma_j} \left( \beta_F - \frac{dz}{a(z)} \right) \in \mathfrak{m}_K. \quad (14.15)$$

*Then  $F \mapsto (q_1(F), \dots, q_r(F))$  induces a bijection*

$$\mathcal{G}_{\delta,a}(D_r) / \text{conjugacy by } \mathcal{G}_+(D_r) \longleftrightarrow \mathfrak{m}_K^r, \quad (14.16)$$

*and for  $\mathbf{q} \in \mathfrak{m}_K^r$  a representative is*

$$b_{\mathbf{q}}(z) = \frac{1}{a(z)} + \sum_{j=1}^r \frac{q_j}{z - a_j}, \quad W_{\mathbf{q}} = \frac{\varepsilon}{b_{\mathbf{q}}}, \quad F_{\mathbf{q}} = \exp(W_{\mathbf{q}} \partial_z) z. \quad (14.17)$$

*No countability, finite generation or rank-one condition is imposed on the supports of the  $q_j$ .*

*Proof.* The finite union of the supports of the  $q_j$  is positive and well ordered, so  $b_{\mathbf{q}} \in \mathcal{A}_{\geq 0}$  with nowhere-zero leading coefficient  $a^{-1}$ , invertible by [lemma 13.2](#); hence  $W_{\mathbf{q}} = \varepsilon(a + \text{positive})$  and  $F_{\mathbf{q}}$  lies in the stated stratum with normalized time form exactly  $b_{\mathbf{q}} dz$ , and the simple-pole representatives give  $q_j(F_{\mathbf{q}}) = q_j$ . Conjugate maps have the same vector by period invariance; conversely equal vectors give equal periods around each puncture, hence, as in [theorem 14.9](#), equality of every ordinary period, and [theorem 14.13](#) supplies a conjugacy.  $\square$

The parametrization is algebraic over the Hahn constants in its displayed one-form representatives. Its basis is the *specified* puncture loops; arbitrary changes of ordinary coordinates are not quotiented out.

#### 14.5 Nonconjugate dynamics beyond every one-scale jet

Fix  $\delta > 0$ ,  $\varepsilon = t^\delta$ , and define the *flat ideal*

$$J_\delta = \{0\} \cup \{c \in K : v(c) > n\delta \text{ for every integer } n \geq 1\}, \quad (14.18)$$

an ideal of the valuation ring  $K_{\geq 0}$ , nonzero *precisely* when some  $\eta \in \Gamma$  exceeds all the  $n\delta$  (the forward implication by taking the valuation of a nonzero member, the reverse from  $t^\eta \in J_\delta$ ). In particular  $J_\delta = 0$  when the multiples of  $\delta$  are cofinal, as in any Archimedean value group. For coefficient functions the corresponding ideal is

$$I_\delta(D) = \bigcap_{n \geq 1} \varepsilon^n \mathcal{A}_{\geq 0} = \{0\} \cup \{f \in \mathcal{A} : v(f) > n\delta \ \forall n \geq 1\}, \quad (14.19)$$

the equivalence of weak and strict inequalities following by replacing  $n$  with  $n + 1$ .

*Warning 14.17* (The ideals must be formed in  $\mathcal{A}_{\geq 0}$ ). It is essential that (14.18), (14.19) are formed in  $K_{\geq 0}$  and  $\mathcal{A}_{\geq 0}$ , **not** in  $K$  or  $\mathcal{A}$ , where  $\varepsilon$  is a *unit* and its powers generate everything. Consequently the map to one-scale jets

$$\mathcal{A}_{\geq 0} \longrightarrow \varprojlim_n \mathcal{A}_{\geq 0} / \varepsilon^n \mathcal{A}_{\geq 0} \quad (14.20)$$

has kernel exactly  $I_\delta(D)$ , and in the noncofinal case it loses nonzero information. The next theorem shows it can lose a complete family of global conjugacy obstructions, not merely some irrelevant constants.

**Theorem 14.18** (Flatly indistinguishable but globally nonconjugate). *Let  $r \geq 1$ ,  $D_r = \mathbb{C} \setminus \{a_1, \dots, a_r\}$ , and suppose  $J_\delta \neq 0$ . For  $\mathbf{c} \in J_\delta^r$  put*

$$R_{\mathbf{c}}(z) = \sum_{j=1}^r \frac{c_j}{z - a_j}, \quad W_{\mathbf{c}} = \frac{\varepsilon}{1 + \varepsilon R_{\mathbf{c}}}, \quad F_{\mathbf{c}} = \exp(W_{\mathbf{c}} \partial_z) z. \quad (14.21)$$

*All these maps lie in  $\mathcal{G}_{\delta,1}(D_r)$  and satisfy:*

(a)  $F_{\mathbf{0}} = z + \varepsilon$  and  $F_{\mathbf{c}} - F_{\mathbf{0}} \in I_\delta(D_r)$  for every  $\mathbf{c}$ ; if  $\mathbf{c} \neq \mathbf{0}$  and  $\eta = \min_{c_j \neq 0} v(c_j)$  then exactly

$$v(F_{\mathbf{c}} - F_{\mathbf{0}}) = 2\delta + \eta; \quad (14.22)$$

(b) *their time forms are exactly*

$$\Omega_{F_{\mathbf{c}}} = \varepsilon^{-1} dz + \sum_{j=1}^r \frac{c_j dz}{z - a_j}; \quad (14.23)$$

(c)  $F_{\mathbf{c}}$  and  $F_{\mathbf{d}}$  are conjugate by an element of  $\mathcal{G}_+(D_r)$  if and only if  $\mathbf{c} = \mathbf{d}$ ;

(d) a global Abel function in  $\mathcal{A}$  exists for  $F_{\mathbf{c}}$  if and only if  $\mathbf{c} = \mathbf{0}$ .

*Proof.* The finite union of the supports of the  $c_j$  is well ordered and positive, so  $R_{\mathbf{c}} \in \mathcal{A}_{>0}$ , and the denominator in (14.21) has leading coefficient 1 and is a unit, so  $W_{\mathbf{c}} = \varepsilon +$  terms of valuation greater than  $\delta$ ; [theorem 4.3](#) places  $F_{\mathbf{c}}$  in the stated stratum, and for  $\mathbf{c} = \mathbf{0}$  the vector field is constant with exponential exactly  $z + \varepsilon$ . If  $\mathbf{c} \neq \mathbf{0}$  then  $v(R_{\mathbf{c}}) = \eta$ , since a nontrivial linear combination of the  $(z - a_j)^{-1}$  cannot vanish identically — ordinary residues at the punctures recover its coefficients. The geometric expansion gives

$$W_{\mathbf{c}} - \varepsilon = -\varepsilon^2 R_{\mathbf{c}} + \varepsilon^3 R_{\mathbf{c}}^2 - \dots, \quad v(W_{\mathbf{c}} - \varepsilon) = 2\delta + \eta. \quad (14.24)$$

Put  $V = W_{\mathbf{c}} - \varepsilon$  and  $\lambda_0 = 2\delta + \eta$ . In the exponential expansion of  $(\varepsilon \partial_z + V \partial_z)^n z$  for  $n \geq 2$ , the word with no factor  $V \partial_z$  vanishes because  $(\varepsilon \partial_z)^2 z = 0$ , and every other word has valuation at least  $\lambda_0 + (n-1)\delta > \lambda_0$  since derivatives do not lower Hahn valuation; all such words are summable by [lemma 3.4](#). So the leading difference comes from  $V$  itself, proving (14.22), and  $\eta > n\delta$  for all  $n$  puts the difference in (14.19). The logarithm of  $F_{\mathbf{c}}$  is  $W_{\mathbf{c}}$  by the inverse identity in [theorem 4.3](#), and taking its reciprocal gives (14.23); hence  $(2\pi i)^{-1} \oint_{\gamma_j} \Omega_{F_{\mathbf{c}}} = c_j$ , and for the normalized form the corresponding value is  $\varepsilon c_j$ . Equality of periods is therefore equivalent to  $\mathbf{c} = \mathbf{d}$ , so [theorem 14.13](#) proves (c) and [corollary 14.7](#) proves (d).  $\square$

*Remark 14.19* (A failure of one-scale information, not of Hahn equality). All the maps of [theorem 14.18](#) have the same image under (14.20). They do *not* have the same full Hahn coefficients: the missing coefficients occur at exponents larger than every  $n\delta$ . The result therefore contradicts neither uniqueness of Hahn expansions nor separation by arbitrary value-group cutoffs. It shows exactly why replacing arbitrary-rank support control by a single “small-parameter completion” can erase a global obstruction. This is the interpretive reading source 05 attaches to its own theorem, and is a framing argument supported by the theorem rather than itself a theorem.

**Example 14.20** (A concrete surcomplex instance). Take  $\Gamma = \mathbb{Q} \oplus \mathbb{Q}$  lexicographically,  $\delta = (0, 1)$ ,  $\eta = (1, 0)$ , so that  $\eta > n\delta$  for every integer  $n \geq 1$  because comparison is decided by the first coordinate; realize this inside the surreals as  $\mathbb{Q}\omega + \mathbb{Q}$  via  $(q, r) \mapsto q\omega + r$ . Under (3.1),  $\varepsilon = \omega^{-1}$  and  $c = t^\eta = \omega^{-\omega}$ . On  $D = \mathbb{C}^*$  the field

$$W_c(z) = \frac{\omega^{-1}}{1 + \omega^{-(\omega+1)}/z} \quad (14.25)$$

defines an explicit surcomplex map  $F_c = \exp(W_c \partial_z)z$  with the same truncation as  $z + \omega^{-1}$  modulo every positive integer power of  $\omega^{-1}$ , yet  $(2\pi i)^{-1} \oint_{|z|=1} \Omega_{F_c} = \omega^{-\omega} \neq 0$ : it is not globally near-identity conjugate to that translation and has no single-valued Hahn-coherent Abel function on the punctured plane. This is a global effect at a strictly smaller *scale*, and is explicitly **not** a claim about ordinary exponentially small asymptotics of complex numbers —  $\omega^{-\omega}$  is an exact Hahn monomial in a rank-two workspace. A nonzero ordinary multiple of it already gives a different member of the family, and the full parameter space  $J_\delta$  is much larger.

## 14.6 Worked parabolic examples and the iterative residue

For a map written in powers of one positive monomial,  $F = z + \varepsilon v_1 + \varepsilon^2 b + \varepsilon^3 c + O(\varepsilon^4)$  with  $v_1, b, c \in \mathcal{O}(D)$ , direct expansion of (4.4) gives

$$W_F = \varepsilon v_1 + \varepsilon^2 \left( b - \frac{1}{2} v_1 v_1' \right) + \varepsilon^3 \left( c - \frac{1}{2} (v_1 b' + b v_1') + \frac{1}{12} v_1^2 v_1'' + \frac{1}{3} v_1 (v_1')^2 \right) + O(\varepsilon^4), \quad (14.26)$$

obtained from  $\Delta z = h$ ,  $\Delta^2 z = h h' + \frac{1}{2} h^2 h'' + \dots$ ,  $\Delta^3 z = h (h')^2 + h^2 h'' + \dots$  with  $h = F - z$  and the logarithm coefficients  $1, -\frac{1}{2}, \frac{1}{3}$ . Here  $O(\varepsilon^4)$  means membership in  $\varepsilon^4 \mathcal{O}(D)[[\varepsilon]]$  and is used only in examples already specified as one-parameter series; it is never a substitute for arbitrary-rank support conditions. If  $v_1$  is nowhere zero, the time form begins

$$\Omega_F = \left[ \frac{1}{\varepsilon v_1} + \frac{v_1'}{2v_1} - \frac{b}{v_1^2} + O(\varepsilon) \right] dz, \quad (14.27)$$

so already at the next order the ordinary logarithmic derivative of the leading vector field can create a nonzero period.

**Proposition 14.21** (Exact residue of the Euler time form). *On  $D = \mathbb{C}^*$  let  $F_p(z) = z + \varepsilon z^{p+1}$ ,  $p \geq 1$ . Then*

$$W_{F_p}(z) = \sum_{n \geq 1} c_{p,n} \varepsilon^n z^{np+1}, \quad c_{p,1} = 1, \quad c_{p,2} = -\frac{p+1}{2}, \quad (14.28)$$

$$\frac{1}{W_{F_p}(z)} = \varepsilon^{-1} z^{-p-1} + \frac{p+1}{2} z^{-1} + \sum_{n \geq 1} d_{p,n} \varepsilon^n z^{np-1}, \quad (14.29)$$

so that

$$\text{Res}_{z=0} \Omega_{F_p} = \frac{p+1}{2} \quad (14.30)$$

exactly, with no omitted positive-valuation correction. Hence there is no global Abel function for  $F_p$  in  $\mathcal{O}(\mathbb{C}^*)((t^\Gamma))$ .

*Proof.* Taylor substitution by  $F_p$  preserves the grading in which  $\varepsilon^n z^{np+1}$  has grade  $n$ , since  $(z + \varepsilon z^{p+1})^{kp+1} = z^{kp+1} (1 + \varepsilon z^p)^{kp+1}$ ; every term of the finite difference and its iterates therefore has the

displayed form, and taking the logarithm preserves it, the first two coefficients following from (14.26). Factor  $W_{F_p} = \varepsilon z^{p+1} C_p(\varepsilon z^p)$  with  $C_p(x) = 1 - (p+1)x/2 + \dots$  and invert  $C_p$  to obtain (14.29); for  $n \geq 1$  the exponent  $np - 1$  is nonnegative, so none of the last terms has a  $z^{-1}$  coefficient, and (14.30) is exact. The period about 0 is nonzero, so [corollary 14.7](#) forbids a global Abel function.  $\square$

The leading field  $\varepsilon z^{p+1}$  itself has zero period on  $\mathbb{C}^*$  and its exponential has Abel function  $-1/(p\varepsilon z^p)$ ; so passing from that exact flow to its Euler map changes the global obstruction even though their first displacements agree. For  $p = 1$ ,

$$W_{F_1} = \varepsilon z^2 - \varepsilon^2 z^3 + \frac{3}{2}\varepsilon^3 z^4 - \frac{8}{3}\varepsilon^4 z^5 + \frac{31}{6}\varepsilon^5 z^6 - \frac{157}{15}\varepsilon^6 z^7 + O(\varepsilon^7). \quad (14.31)$$

**Warning 14.22** (These coefficients are known, and nothing is claimed about their convergence). The coefficients (14.31) agree, after multiplication by the step size, with the standard modified equation in Hairer's worked example [22]; they are **known examples, not new coefficients**, and are not proposed as discoveries. Nothing is asserted about convergence of the ordinary step-size series in a real step size; Hairer's warning that it generally diverges is cited and accepted. The constant (14.30) is the familiar iterative-residue phenomenon, and the assertion here is its exact interpretation in the stated common-domain Hahn category, not the discovery of that classical invariant. The *same* degree-six expansion appears independently in source 03 as the iterative logarithm of  $z + tz^2$  ([section 21.2](#)); it is one calculation and is printed once, in (14.31).

**Corollary 14.23** (A global coherent normal form for the Euler family).  $F_p$  is conjugate on  $(\mathbb{C}^*)^\#$ , by a near-identity Hahn-coherent change of variable, to the time-one map of

$$V_p(z) = \frac{\varepsilon z^{p+1}}{1 + \frac{p+1}{2}\varepsilon z^p}, \quad (14.32)$$

the conjugacy fixing  $z = 1$  being unique. It is **not** a conjugacy of  $F_p$  to the time-one map of  $\varepsilon z^{p+1}$ .

*Proof.* The time form of (14.32) is  $\varepsilon^{-1} z^{-p-1} dz + \frac{p+1}{2} dz/z$ , with the same period as (14.29); their difference consists of nonnegative powers of  $z$  coefficientwise, hence is exact on  $\mathbb{C}^*$ , so [theorem 14.13](#) applies. The last assertion holds because the leading vector-field flow has zero period while  $F_p$  does not.  $\square$

On any simply connected slit domain in  $\mathbb{C}^*$  the normal form has the local Abel function  $-1/(p\varepsilon z^p) + \frac{p+1}{2} \log z$ , whose logarithmic monodromy is precisely the obstruction to a single-valued global Abel function on  $\mathbb{C}^*$ .

**Example 14.24** (A difference equation with an explicit Bernoulli correction). Take  $W = \varepsilon z$ , so  $F = e^\varepsilon z$ , on  $D = \mathbb{C}^*$ . The forcing  $g = z^m$  with nonzero integer  $m$  has zero period after division by  $W$ , and  $\psi(z) = z^m/(e^{m\varepsilon} - 1)$  is a solution; the denominator has valuation  $\delta$  so its reciprocal is a legitimate Laurent–Hahn constant, and expanding gives  $\psi(z) = (\frac{1}{m\varepsilon} - \frac{1}{2} + \frac{m\varepsilon}{12} - \dots)z^m$ , exactly the Bernoulli solver of [theorem 14.5](#) applied to  $P = z^m/(m\varepsilon)$ . For the constant forcing  $g = 1$ , by contrast,  $g dz/W = \varepsilon^{-1} dz/z$  has nonzero period and no global solution exists, displaying the one-dimensional cokernel for the punctured plane.

**Example 14.25** (Local straightening does not erase global periods).  $F(z) = (1+\varepsilon)z$  and  $G(z) = e^\varepsilon z$  have the same leading displacement  $\varepsilon z$  on  $\mathbb{C}^*$ , with logarithmic vector fields  $\log(1+\varepsilon)z$  and  $\varepsilon z$  and periods  $2\pi i/\log(1+\varepsilon)$  and  $2\pi i/\varepsilon$ , which differ — so there is no global near-identity conjugacy there. On a simply connected domain with a branch of  $\log z$  there is one:

$$H(z) = z \exp\left(\left[\frac{\varepsilon}{\log(1+\varepsilon)} - 1\right] \log z\right),$$

the bracketed coefficient having positive valuation, so the exponential is a strongly summable Hahn expansion with coefficients holomorphic on the chosen branch domain, and direct substitution gives  $H \circ F = G \circ H$ . The failure to make that formula single-valued on  $\mathbb{C}^*$  is exactly what the period theorem detects.

## 14.7 Ordinary slow time on one fixed holomorphic flow chart

**Theorem 4.3** concerns an infinitesimal vector field. One may also ask what happens at the larger time scale on which the leading ordinary vector field moves a finite distance; a direct exponential is generally not a strongly summable Hahn expression there. The correct construction solves the leading ordinary flow first and then lifts all positive Hahn coefficients along it. Write

$$W = \varepsilon X, \quad X(z) = a(z) + \sum_{\beta \in S} t^\beta X_\beta(z), \quad (14.33)$$

with  $S \subset \Gamma_{>0}$  well ordered and all coefficient functions holomorphic on  $D$ ; in this subsection *a may* have zeros. Let  $\Sigma \subset \mathbb{C}$  be an ordinary disk centered at 0,  $D_0 \subset D$  an ordinary open domain, and assume a holomorphic ordinary flow chart

$$\phi : \Sigma \times D_0 \rightarrow D, \quad \partial_s \phi(s, z) = a(\phi(s, z)), \quad \phi(0, z) = z \quad (14.34)$$

is *specified*, with  $J(s, z) = \partial_z \phi(s, z)$  nowhere vanishing on that chart.

**Theorem 14.26** (Common-chart Hahn lifting of the slow flow). *There is exactly one series*

$$\Psi(s, z) = \phi(s, z) + \sum_{\gamma \in S^* \setminus \{0\}} t^\gamma \psi_\gamma(s, z) \quad (14.35)$$

*with coefficients holomorphic on the same product domain  $\Sigma \times D_0$  satisfying*

$$\partial_s \Psi = X(\Psi), \quad \Psi(0, z) = z, \quad (14.36)$$

*$X(\Psi)$  meaning Taylor expansion at the ordinary point  $\phi(s, z)$ . Uniqueness holds among all positive-Hahn perturbations of  $\phi$  on this product domain, not only those with the displayed support bound. Each coefficient is obtained by one ordinary integral,*

$$\psi_\gamma(s, z) = J(s, z) \int_0^s J(u, z)^{-1} R_\gamma(u, z) du, \quad (14.37)$$

*with  $R_\gamma$  a finite expression in previously determined coefficients and in ordinary derivatives of the  $X_\beta$  and  $a$  along  $\phi$ . Finally Taylor evaluation at  $s = \varepsilon$  is strongly summable and*

$$\Psi(\varepsilon, z) = \exp(W \partial_z) z \quad (z \in D_0^\#), \quad (14.38)$$

*and more generally, for  $c \in K$  with  $c = 0$  or  $v(c) + \delta > 0$ , evaluation at  $s = c\varepsilon$  is defined and gives  $\Psi(c\varepsilon, z) = \exp(cW \partial_z) z$ .*

*Proof.* Let  $M = S^*$  and seek  $q = \Psi - \phi$  with support in  $M \setminus \{0\}$ . Expanding  $X(\phi + q) = a(\phi) + a'(\phi)q + \sum_{n \geq 2} a^{(n)}(\phi)q^n/n! + \sum_{\beta \in S} t^\beta \sum_{n \geq 0} X_\beta^{(n)}(\phi)q^n/n!$  — all compositions with  $\phi$  being holomorphic on the specified product because its image lies in  $D$  — the coefficient equation at  $\gamma \in M \setminus \{0\}$  is

$$\partial_s \psi_\gamma = a'(\phi) \psi_\gamma + R_\gamma, \quad \psi_\gamma(0, z) = 0, \quad (14.39)$$

every factor in  $R_\gamma$  involving a positive exponent smaller than  $\gamma$  except for specified coefficients of  $X$ , with finitely many contributions by lemma 3.4 exactly as in lemma 14.11. Differentiating (14.34) in  $z$  gives  $\partial_s J = a'(\phi)J$ ,  $J(0, z) = 1$ , so variation of constants solves (14.39) by (14.37); since  $J$  is nowhere zero and  $\Sigma$  is simply connected the integral is independent of its path in  $\Sigma$  and is jointly holomorphic in  $(s, z)$ , locally by integrating the holomorphic power series in  $s$  with coefficients holomorphic in  $z$  and patching the uniquely normalized primitives. There is therefore *no* successive domain shrinkage and *no* requirement of a uniform bound on these coefficients as  $\gamma$  varies. The support lies in  $\mathbf{M}$  and the expansion of  $X(\Psi)$  is strongly summable by lemma 3.4, so the construction solves (14.36). For uniqueness, at the least exponent where two positive-Hahn perturbations of  $\phi$  satisfying that equation differ, their difference solves the homogeneous version of (14.39) with zero initial condition and is zero; this argument does not require either support to lie in  $\mathbf{M}$ . For the last assertion expand every coefficient at  $s = 0$  and substitute  $s = \varepsilon$ : the possible exponents lie in the monoid generated by  $S \cup \{\delta\}$  and a fixed exponent has finitely many word representations, so the evaluation is strongly summable. Repeated differentiation of (14.36) gives  $\partial_s^n \Psi(0, z) = (X\partial_z)^n z$ , whence  $\Psi(\varepsilon, z) = \sum_n \varepsilon^n (X\partial_z)^n z / n! = \exp(W\partial_z)z$ . For  $s = c\varepsilon \in \mathbf{m}_K$  replace  $\{\delta\}$  by the well-ordered positive support of  $c\varepsilon$ .  $\square$

*Warning 14.27* (Exactly what the slow-time theorem licenses). At an ordinary time  $s_0 \in \Sigma$ , evaluating (14.35) produces an element with leading ordinary part  $\phi(s_0, z)$  and positive Hahn corrections, the ordinary coefficient integrals having *already been performed*: that is a separate analytic operation from an unregrouped Hahn exponential. For example, with  $X = z^2$  and  $W = \varepsilon z^2$  the slow flow is the ordinary rational function  $\phi(s, z) = z/(1 - sz)$  on any product chart avoiding  $sz = 1$ , whereas at a nonzero ordinary  $s$  the raw formal terms of  $\exp(sz^2\partial_z)z = \sum_{n \geq 0} s^n z^{n+1}$  all have Hahn exponent 0 with infinitely many nonzero, and are *not* a strongly summable Hahn family. The rational function comes from ordinary complex analysis on its chart, not from declaring that family a Hahn sum. Theorem 14.26 therefore licenses one particular passage to the time scale  $\varepsilon^{-1}$  — solve the leading ordinary motion, then integrate each positive correction along it — and licenses **no** arbitrary exchange of ordinary with Hahn summation, **no** global continuation past an ordinary singularity of the supplied chart, and **no** interpretation of an arbitrary surreal-valued time as an actual continuous trajectory. It is also explicitly *not* analytic convergence of the modified series in a real step size.

## 15 Hamiltonian normal forms, integrability, and certified domains

This section is under  $\neg\text{FR}$  and, according to the case,  $\text{SD}_0$  or  $\text{SD}_{\text{fr}}$ . The frequencies are *ordinary complex constants*, never surreal or Hahn quantities;  $\Gamma$  needs neither divisibility nor algebraic closure; and the two coefficient algebras

$$\mathcal{P}_n = \mathbb{C}[z_1, \dots, z_n], \quad \mathcal{E}_n = \mathcal{O}(\mathbb{C}^n), \quad \mathcal{H}_\Gamma(\mathcal{P}_n), \quad \mathcal{H}_\Gamma(\mathcal{E}_n) \quad (15.1)$$

are the load-bearing distinction: **no** uniform norm bound as  $\gamma$  varies in the entire case, **no** uniform degree bound in the polynomial case, and results about actual finite-halo evaluation are **not** transferred to  $\mathcal{H}_\Gamma(\mathbb{C}\{z\})$ , whose coefficients are unrelated convergent germs. In  $2d$  phase variables  $q_1, \dots, q_d, p_1, \dots, p_d$  use the fixed sign convention

$$\{F, G\} = \sum_{j=1}^d (F_{q_j} G_{p_j} - F_{p_j} G_{q_j}), \quad \text{ad}_\chi F = \{\chi, F\}, \quad (15.2)$$

and write  $I_j = q_j p_j$ ,

$$H_0 = \sum_{j=1}^d \omega_j I_j, \quad \omega \cdot k \neq 0 \quad (0 \neq k \in \mathbb{Z}^d). \quad (15.3)$$

## 15.1 Supported substitutions and Lie maps

**Proposition 15.1** (Positive near-identity maps). *Let  $U \in \mathcal{H}_\Gamma(\mathcal{E}_n)_{>0}^n$ . Then  $\Phi(z) = z + U(z)$  has a unique inverse  $z + V(z)$  with  $V \in \mathcal{H}_\Gamma(\mathcal{E}_n)_{>0}^n$ ; both are bijections of  $\mathcal{O}_\Gamma^n$  preserving standard parts; and if  $U$  has polynomial coefficients, so does  $V$ .*

*Proof.* Let  $S$  be the union of the supports of the components of  $U$  and  $\mathbf{M} = M(S)$ . The inverse equation is  $V(z) = -U(z + V(z))$ ; expanding the right side by Taylor's formula, its term independent of  $V$  is  $-U(z)$  and every other term has at least one positive input from  $U$  and at least one from  $V$ , so at exponent  $\gamma > 0$  all occurrences of coefficients of  $V$  have exponents strictly smaller than  $\gamma$ , and recursion on the well-ordered  $\mathbf{M}$  determines  $V_\gamma$  uniquely — including at limit positions of  $\mathbf{M}$ , where *no topological limit is requested*. At a fixed exponent only finitely many Taylor terms and input words occur by lemma 3.4, their coefficients using only derivatives, products and finite sums of the coefficients of  $U$ , hence entire, or polynomial in the polynomial case. This gives a right inverse; the same construction for a left inverse, with associativity of supported substitution or the least-exponent uniqueness argument, shows the two agree, and proposition 3.11 gives the actual bijections.  $\square$

**Proposition 15.2** (Supported Hamiltonian exponential). *If  $\chi \in \mathcal{H}_\Gamma(\mathcal{E}_{2d})_{>0}$  then for every  $F \in \mathcal{H}_\Gamma(\mathcal{E}_{2d})$  the family  $(\text{ad}_\chi^n F/n!)_{n \geq 0}$  is strongly summable; its sum  $T_\chi F = e^{\text{ad}_\chi} F$  is an algebra and Poisson automorphism with inverse  $T_{-\chi}$ ; and with  $\Phi = (T_\chi q_1, \dots, T_\chi q_d, T_\chi p_1, \dots, T_\chi p_d)$  one has  $T_\chi F = F \circ \Phi$ , the realized  $\Phi$  being a symplectic bijection of the finite halo. Polynomial coefficients are preserved.*

*Proof.* With  $A = \text{supp } F$  and  $S = \text{supp } \chi$ , each term of  $\text{ad}_\chi^n F$  has support in  $A + nS$ , differentiation does not change Hahn weights, and a fixed exponent receives contributions for finitely many words in  $S$  attached to an exponent in  $A$ : lemma 3.4 gives strong summability and every regrouping used below. Leibniz gives  $\text{ad}_\chi^n(FG) = \sum_j \binom{n}{j} (\text{ad}_\chi^j F)(\text{ad}_\chi^{n-j} G)$ , so after division by  $n!$  and supported summation  $T_\chi$  is multiplicative; the Jacobi identity gives the identical binomial rule for the bracket, so  $T_\chi\{F, G\} = \{T_\chi F, T_\chi G\}$ ; and the binomial identity for  $\exp(\mathcal{E})\exp(-\mathcal{E})$  gives the inverse. For an ordinary entire function the formal chain rule for the derivation  $\text{ad}_\chi$  identifies  $T_\chi F$  with Taylor substitution by  $\Phi$  — check this at each power of an auxiliary formal scalar multiplying  $\chi$ , where it is the usual formal flow identity, then evaluate that scalar at 1 using the proved word summability — and applying it coefficientwise gives the assertion for general  $F$ . The coordinate Poisson relations are preserved, proving symplecticity, and  $\Phi = \text{id} + \text{positive terms}$ , so propositions 3.11 and 15.1 give the actual maps.  $\square$

**Definition 15.3** (The finite-expression certificate). Label each input perturbation coefficient occurrence by its positive Hahn exponent. A term in a normalizing coefficient is built by a *finite expression tree* whose leaves are labeled inputs and whose internal operations are Poisson brackets, linear projections, scalar division, or fixed linear solution operators; its weight is the sum of its leaf weights. For a fixed total weight there are finitely many leaf words by lemma 3.4; every genuinely nonlinear internal step has at least two positive inputs, linear operators only decorating those steps or the leaves, so a fixed leaf word admits only finitely many relevant tree shapes. This is a finite

certificate for every coefficient even when the set of smaller weights is infinite, and it is also the reason a later reordering of weights under rescaling is legitimate ([theorem 15.21](#)).

## 15.2 The homological operator and a sharp divisor threshold

Define  $LF = \{F, H_0\}$ , so that

$$L(q^\alpha p^\beta) = \omega \cdot (\alpha - \beta) q^\alpha p^\beta \quad (15.4)$$

and  $\neg\text{FR}$  makes the kernel exactly the monomials with  $\alpha = \beta$ . For a Taylor expansion at the origin set

$$\Pi\left(\sum_{\alpha,\beta} c_{\alpha\beta} q^\alpha p^\beta\right) = \sum_{\alpha} c_{\alpha\alpha} I^\alpha, \quad \Pi + Q_\Pi = 1. \quad (15.5)$$

**Lemma 15.4** (Entire action projection).  *$\Pi$  maps  $\mathcal{E}_{2d}$  to functions  $g(I)$  with  $g \in \mathcal{E}_d$ , uniquely; it preserves polynomial coefficients, does not decrease phase vanishing order, and commutes with  $L$ . On  $Q_\Pi \mathcal{P}_{2d}$  the inverse  $L^{-1}$  is always defined and polynomial, with no arithmetic estimate.*

*Proof.* For entire  $f$ , Cauchy's estimate on a phase polydisk of radius  $R$  gives  $|c_{\alpha\beta}| \leq C_R R^{-|\alpha|-|\beta|}$ , so  $|c_{\alpha\alpha}| \leq C_R (R^2)^{-|\alpha|}$ ; as  $R$  is arbitrary the action series is entire, and distinct action monomials are distinct phase monomials, giving uniqueness. The remaining assertions follow termwise, and for a polynomial the inverse divides finitely many nonresonant coefficients by nonzero complex constants.  $\square$

One may also realize  $\Pi$  by averaging over the ordinary torus action  $q_j \mapsto e^{i\theta_j} q_j$ ,  $p_j \mapsto e^{-i\theta_j} p_j$ ; no surcomplex contour integration is needed for that observation.

**Theorem 15.5** (Sharp homological criterion). *For nonresonant  $\omega$  the following are equivalent: (i)  $\Theta(\omega) < \infty$ ; (ii) the coefficientwise  $L^{-1}$  maps every nonresonant entire function to an entire function; (iii) the coefficientwise  $L^{-1}$  maps every nonresonant convergent germ at the origin to a convergent germ, with a possible loss of radius. If these fail, there is a nonresonant entire  $F$  of phase order at least three for which  $L^{-1}F$  has no positive radius of convergence.*

*Proof.* If  $M_D \leq CA^D$  and the coefficients of  $F$  obey  $|f_{\alpha\beta}| \leq C_R R^{-D}$ , those of  $L^{-1}F$  obey  $|(L^{-1}F)_{\alpha\beta}| \leq CC_R (A/R)^D$ ; the number of monomials of total degree  $D$  in  $2d$  variables is  $\binom{D+2d-1}{2d-1}$ , which grows polynomially, so this bound proves absolute convergence on every polydisk of radius less than  $R/A$  — one admissible  $R$  for a germ, arbitrarily large  $R$  for an entire function. Conversely if  $\Theta = \infty$ , select distinct degrees  $D_j \rightarrow \infty$  with  $D_j \geq \max(j, 3)$  and nonresonant monomials  $m_j = q^{\alpha_j} p^{\beta_j}$  of those degrees with  $\delta_j = |\omega \cdot (\alpha_j - \beta_j)|$  and  $\delta_j^{-1/D_j} \geq j^4$ , and put

$$F = \sum_j \delta_j^{1/2} m_j. \quad (15.6)$$

On a polydisk of radius  $R$  the  $j$ -th term has modulus at most  $(R/j^2)^{D_j}$  and the tail is bounded by a geometric series since  $D_j \geq j$ , so  $F$  is entire; in the putative inverse the absolute coefficient of  $m_j$  is  $\delta_j^{-1/2}$ , whose  $D_j$ -th root is at least  $j^2$ , violating every Cauchy bound on a positive-radius polydisk. Since  $F$  is itself a germ, both (ii) and (iii) fail, and its order and nonresonance were built into the selection.  $\square$

*Warning 15.6* (What  $\Theta$  measures, and what it must not be advertised as). A polynomial lower bound on  $|\omega \cdot k|$  implies  $\Theta = 1$  but is far stronger than necessary here. Finite exponential growth is the exact condition for this *single coefficientwise homological operator* to preserve all entire functions. The nonlinear Hahn theorem below applies that operator repeatedly and requires *no* uniform norm estimate across Hahn layers; this is precisely why the criterion must **not** be advertised as an ordinary complex-analytic convergence criterion of Bruno or KAM type. Sources 06 repeats this disclaimer in three separate places, and the merge preserves it.

### 15.3 Arbitrary-support Hamiltonian normalization

**Theorem 15.7** (Exact positive-Hahn normal form). *Let  $\omega$  satisfy (15.3) and let*

$$H = H_0 + R, \quad R = \sum_{s \in S} t^s R_s(q, p), \quad S \subseteq \Gamma_{>0} \text{ well ordered, } \text{ord}_0 R_s \geq 3. \quad (15.7)$$

*Assume either (a) every  $R_s$  is a polynomial, with no restriction on  $\omega$  beyond nonresonance (tag  $\text{SD}_0$ ), or (b) every  $R_s$  is entire and  $\Theta(\omega) < \infty$  (tag  $\text{SD}_{\text{fr}}$ ). Then there is a unique positive generator  $\chi$  in the corresponding coefficient algebra with*

$$\Pi\chi = 0, \quad \text{ord}_0 \chi_\gamma \geq 3, \quad e^{\text{ad}_\chi} H = H_0 + B(I), \quad (15.8)$$

*where  $B$  has positive Hahn support and coefficients of action vanishing order at least two, and  $\text{supp } \chi, \text{supp } B \subseteq M(S) \setminus \{0\}$ . The coordinate map  $\Phi = e^{\text{ad}_\chi}(q, p)$  and its inverse are symplectic bijections of  $\mathcal{O}_\Gamma^{2d}$ , fixing the origin, tangent to the identity there, preserving standard parts everywhere, and satisfying the actual identity*

$$H^\# \circ \Phi^\# = (H_0 + B(I))^\#. \quad (15.9)$$

*All statements commute with enlargement of the Hahn workspace.*

*Proof.* Expanding the Lie exponential,

$$e^{\text{ad}_\chi}(H_0 + R) = H_0 + R + L\chi + \mathcal{B}(\chi, R), \quad \mathcal{B}(\chi, R) = \sum_{n \geq 1} \frac{\text{ad}_\chi^n R}{n!} + \sum_{n \geq 2} \frac{\text{ad}_\chi^n H_0}{n!}, \quad (15.10)$$

every term of  $\mathcal{B}$  involving at least two positive factors, counting occurrences of  $R$  and  $\chi$ . Since  $\Pi L = 0$  and  $\Pi\chi = 0$ , the nonresonant normal-form equation is exactly

$$\chi = -L^{-1}Q_\Pi(R + \mathcal{B}(\chi, R)). \quad (15.11)$$

Put  $\mathbf{M} = M(S)$ : at weight  $\gamma \in \mathbf{M} \setminus \{0\}$  the coefficient of  $\mathcal{B}$  depends only on  $\chi_\mu$  with  $0 < \mu < \gamma$ , since an occurrence of  $\chi_\gamma$  would be accompanied by another positive factor and so have total weight greater than  $\gamma$ . Well-founded recursion on  $\mathbf{M}$  therefore determines

$$\chi_\gamma = -L^{-1}Q_\Pi(R_\gamma + [t^\gamma]\mathcal{B}(\chi_{<\gamma}, R)), \quad (15.12)$$

a formula that remains meaningful when  $\gamma$  has infinitely many predecessors, because [lemma 3.4](#) allows only finitely many of them to participate in a word of total weight  $\gamma$ . More explicitly, each coefficient is a finite sum of the labeled trees of [definition 15.3](#): their leaves are input coefficients  $R_s$ , their total weight is  $\gamma$ , and their internal nonlinear nodes have at least two positive inputs, so the finitely many words of total weight  $\gamma$  bound the number and choice of leaves and there are finitely

many tree shapes and bracket orderings. Hence every coefficient uses finitely many applications of derivatives, multiplication,  $Q_\Pi$  and  $L^{-1}$ : the claimed support bound holds and no unproved infinite operation occurs within a Hahn coefficient. In case (a) these operations preserve polynomials by lemma 15.4; in case (b) they preserve entire functions by theorem 15.5. Every application of  $L^{-1}Q_\Pi$  gives  $\Pi\chi_\gamma = 0$ , and the estimate  $\text{ord}_0\{F, G\} \geq \text{ord}_0 F + \text{ord}_0 G - 2$  shows inductively that each  $\chi_\gamma$  has order at least three while terms in  $\mathcal{B}$  have order at least four; hence  $B = \Pi(R + \mathcal{B}(\chi, R))$  has phase order at least four, since a resonant monomial has even phase degree and  $R$  has order at least three, and by lemma 15.4 it is an entire or polynomial function of the actions at every Hahn weight, of action order at least two. Equation (15.11) removes the nonresonant part of (15.10), so (15.8) holds as a Hahn identity, and proposition 15.2 defines the exponential. For uniqueness, at the least exponent  $\gamma$  of the difference of two positive gauge-normalized generators, all terms coming from  $\mathcal{B}$  have weight strictly greater than  $\gamma$ , leaving  $L(\chi_\gamma - \tilde{\chi}_\gamma) = 0$  with a nonresonant difference, on which  $L$  is injective by (15.4); this also rules out additional support outside  $M$ . Finally propositions 3.11 and 15.2 give the symplectic maps, their inverse, and (15.9), and no construction used an exponent outside  $M(S)$ .  $\square$

*Remark 15.8* (What has become exact). In a classical analytic problem the generator can be formal in the phase variables with zero radius of convergence. In case (a) each Hahn coefficient of the generator is instead a *finite polynomial*: its coefficients may be enormous ordinary complex numbers, but they are finite constants at their Hahn layer, and an arbitrarily large ordinary number cannot cancel the positive valuation of  $t^\gamma$  (remark 2.3). Evaluation on the finite halo is therefore controlled by supports rather than by a numerical norm of the normalizing coefficients. In case (b) infinitely many phase monomials occur within a single Hahn layer, so their ordinary convergence must be settled *before* Hahn summability can be invoked; the homological threshold does exactly that, and requires no common numerical bound on the resulting entire coefficients because the common domain is already all of  $\mathbb{C}^{2d}$ .

**Corollary 15.9** (Finite surcomplex phase space). *For polynomial layers, and for entire layers satisfying the threshold, theorem 15.7 realizes the normalization at every finite point of  $\mathbf{No}[i]^{2d}$ , and the maps obtained in different set-sized workspaces agree on their common domain.*

*Proof.* Given a phase tuple, enlarge the workspace to contain all of its exponents and those of the input and apply theorem 15.7; its coefficient formulas and the Taylor evaluation are unchanged by the enlargement.  $\square$

*Warning 15.10* (Uniqueness has a gauge). The unique object in theorem 15.7 is the *single exponential* generator with  $\Pi\chi = 0$ , and the normal form it produces in that gauge. Arbitrary symplectic changes of variables normalizing a Hamiltonian need **not** be unique: in particular transformations generated by functions of the actions can preserve a normal form. Source 06 declines to suppress this freedom by an unstated uniqueness claim, and section D distinguishes the single-exponential gauge recursion from the different algorithm that composes successive exponentials without taking a logarithm.

## 15.4 Exact integrability and the whole Poisson centralizer

Write  $T = e^{\text{ad}_\chi}$ ,  $N = TH$ , and  $J_j = T^{-1}I_j = e^{-\text{ad}_\chi}I_j$ .

**Theorem 15.11** (Exact integrability). *The  $J_1, \dots, J_d$  are entire-coefficient Hahn functions — with polynomial coefficients in case (a) — and*

$$\{H, J_j\} = 0, \quad \{J_j, J_k\} = 0, \quad H = N(J_1, \dots, J_d). \quad (15.13)$$

They are functionally independent wherever, in the normalized coordinates, none of the pairs  $(q_j, p_j)$  is  $(0, 0)$ . These hold as realized identities on the finite halo.

*Proof.* The  $I_j$  commute with one another and with every function of the actions, including  $N$ ; applying the Poisson automorphism  $T^{-1}$  gives the first two identities and the algebra substitution  $T^{-1}$  applied to  $N(I)$  the third. The  $d$  differentials  $dI_j = p_j dq_j + q_j dp_j$  have disjoint coordinate pairs and are linearly independent exactly when each is nonzero; pullback by the invertible differential of  $\Phi^{-1}$  preserves that rank, and [proposition 3.11](#) respects all these identities.  $\square$

**Theorem 15.12** (Poisson centralizer). *In  $\mathcal{H}_\Gamma(\mathcal{E}_{2d})$ ,*

$$\text{Cent}(H) := \{F : \{F, H\} = 0\} = \{G(J_1, \dots, J_d) : G \in \mathcal{H}_\Gamma(\mathcal{E}_d)\}, \quad (15.14)$$

*the representing  $G$  being unique. Its Hahn support need **not** be positive. Thus (15.14) describes all first integrals that admit an entire-coefficient Hahn representation on the finite halo.*

*Proof.* It suffices to classify the centralizer of  $N(I) = H_0 + B(I)$ , since  $T$  is a Poisson automorphism. Let  $\tilde{F} = TF$  with  $\{\tilde{F}, N\} = 0$  and apply  $\Pi$  coefficientwise: by [lemma 15.4](#),  $\Pi\tilde{F} = G(I)$  for a unique  $G \in \mathcal{H}_\Gamma(\mathcal{E}_d)$ , and every action function commutes with  $N$ , so  $U := \tilde{F} - \Pi\tilde{F}$  is coefficientwise nonresonant and also commutes with  $N$ . If  $U \neq 0$ , let  $a_0$  be the least exponent of its support: the coefficient at  $t^{a_0}$  in  $\{U, B\}$  is zero, all exponents of  $B$  being positive and all of  $U$  at least  $a_0$ , so the coefficient at  $t^{a_0}$  in  $\{U, N\} = 0$  is  $LU_{a_0} = 0$  with  $U_{a_0}$  nonresonant and nonzero, contradicting (15.4). Hence  $U = 0$  and  $\tilde{F} = G(I)$ , so  $F = G(J)$ . Conversely every  $G(J)$  commutes with  $H$ ; its definition is legitimate even for arbitrary well-ordered support of  $G$  — write  $J = I + \Delta$  with positive  $\Delta$ , use the ordinary Taylor expansion of each entire coefficient of  $G$  at  $I$ , and apply [lemma 3.4](#). Uniqueness follows by applying  $T$  or by the injectivity of  $g(I) \mapsto g$ ; faithfulness of [proposition 3.11](#) means no additional identity appears on passing to finite-halo functions.  $\square$

**Warning 15.13** (The coefficient class is essential). [Theorem 15.12](#) does **not** classify first integrals defined only near a particular point, multivalued functions, or arbitrary functions that happen to be locally constant in a fine topology, and does not say that all first integrals have polynomial coefficients. The entire-coefficient class is large enough to contain arbitrary entire functions of the actions and rigid enough for the leading nonresonant coefficient argument to work; that is exactly the balance the theorem exploits.

**Proposition 15.14** (Exact ordinary-time evolution). *Let  $\Omega_j(I) = \partial N / \partial I_j = \omega_j + \Delta_j(I)$  with  $v(\Delta_j) > 0$  (positive Hahn support). For ordinary complex  $\tau$  put*

$$q_j(\tau) = e^{\tau\omega_j} \text{Exp}_H(\tau\Delta_j(I))q_j(0), \quad p_j(\tau) = e^{-\tau\omega_j} \text{Exp}_H(-\tau\Delta_j(I))p_j(0). \quad (15.15)$$

*These give an exact flow of  $N$  on the finite halo for every  $\tau \in \mathbb{C}$ ; conjugation by  $\Phi$  gives the flow of  $H$ ; the coefficients depend holomorphically on  $\tau$ ; and the flow law holds as an identity of Hahn-coherent functions.*

*Proof.* The actions are constant along (15.15) because the two factors for each  $j$  are inverse, so  $\Delta_j(I)$  is constant along the curve, and coefficientwise differentiation in ordinary time gives  $\dot{q}_j = \Omega_j(I)q_j = \partial N / \partial p_j$  and  $\dot{p}_j = -\Omega_j(I)p_j = -\partial N / \partial q_j$ . The Hahn exponentials are strongly summable by positivity, each coefficient is entire in  $\tau$  and the phase variables, and their constant Hahn terms are nonzero ordinary exponentials, so finite phase points remain finite; the exponential identities, justified coefficientwise by finite Hahn regroupings, give the flow law, and symplectic conjugacy transports the equations to  $H$ .  $\square$

**Warning 15.15** (This is a coefficientwise holomorphic-time identity). [Proposition 15.14](#) is not an assertion about continuous ordinary-time paths in the fine topology of the proper-class field. No infinite surreal time, no global surreal exponential of an arbitrary surcomplex argument, and no intrinsic coefficient-field derivation is used ([warning 3.2](#)). “Analytic” in this section means exactly the entire-coefficient Taylor-compatible realization of [proposition 3.11](#).

## 15.5 Failure for entire layers: an exact converse

**Theorem 15.16** (Sharp universal normalization criterion). *For fixed nonresonant  $\omega$  the following are equivalent: (i)  $\Theta(\omega) < \infty$ ; (ii) every positive Hahn perturbation with entire coefficients of phase order at least three admits the entire-coefficient gauge-normalized generator and finite-halo realization of [theorem 15.7](#), for every set-sized ordered value group; (iii) every one-parameter Hamiltonian  $H_0 + \varepsilon_0 F$  with  $F$  entire of order at least three admits a symplectic formal change  $\Psi = \text{id} + O(\varepsilon_0)$  with convergent-germ coefficients at the origin transforming it into a function of the actions. If (i) fails, failure of (iii) is already forced at the coefficient of  $\varepsilon_0$ , and there is no positive gauge-normalized entire Hahn generator for the same input in any workspace containing its scale.*

*Proof.* (i) $\Rightarrow$ (ii) is [theorem 15.7](#); (ii) $\Rightarrow$ (iii) follows by taking  $\Gamma = \mathbb{Z}$  and  $\varepsilon_0 = t$ . Suppose  $\Theta = \infty$  and take the entire nonresonant  $F$  of [theorem 15.5](#). If a symplectic change as in (iii) exists, write  $\Psi(z) = z + \varepsilon_0 V(z) + O(\varepsilon_0^2)$  with  $V$  a holomorphic vector-field germ; the first-order symplectic condition says  $V$  is a symplectic vector field, so its contraction with the constant symplectic form is a closed holomorphic one-form, and a closed holomorphic one-form germ is exact (a primitive is given on a small star-shaped polydisk by the radial homotopy integral). Hence, with the sign matching ([15.2](#)), there is a holomorphic germ  $\chi_1$  with  $V(G) = \{\chi_1, G\}$  for every holomorphic germ  $G$ . The coefficient of  $\varepsilon_0$  in  $(H_0 + \varepsilon_0 F) \circ \Psi$  is  $F + L\chi_1$ , and if the transformed Hamiltonian is a function of the actions its nonresonant part must vanish:  $L(Q_\Pi \chi_1) = -F$ . The projection  $Q_\Pi$  preserves convergent germs by the same Cauchy estimate as in [lemma 15.4](#), and coefficientwise injectivity of  $L$  on nonresonant series forces  $Q_\Pi \chi_1 = -L^{-1}F$ , which is not a convergent germ. For the final assertion, a putative positive gauge-normalized entire Hahn generator for  $H_0 + t^\eta F$  cannot have a least nonzero coefficient below  $\eta$  (its normal-form equation there would be  $L\chi_\gamma = 0$ , impossible in the gauge), and at weight  $\eta$  the nonlinear terms have larger weight, so again  $\chi_\eta = -L^{-1}F$ , not entire. This argument does not depend on the rank or divisibility of the enlarged value group.  $\square$

**Proposition 15.17** (An explicit super-Liouville obstruction). *Define integers by*

$$a_1 = 1, \quad a_{n+1} = 10^{4a_n}, \quad Q_n = 10^{a_n}, \quad \alpha = \sum_{n \geq 1} 10^{-a_n}, \quad (15.16)$$

*put  $P_n/Q_n = \sum_{j=1}^n 10^{-a_j}$  (so  $P_n \in \mathbb{Z}$ ) and  $D_n = P_n + Q_n < 2Q_n$ . Then*

$$0 < \alpha Q_n - P_n < 2Q_n 10^{-a_{n+1}} = 2Q_n 10^{-Q_n^4} < e^{-D_n^2}, \quad (15.17)$$

*$\alpha$  is irrational, and  $\omega = (1, \alpha)$  is nonresonant. In two degrees of freedom,*

$$F(q_1, q_2, p_1, p_2) = \sum_{n \geq 1} e^{-D_n^2/2} q_1^{P_n} p_2^{Q_n}, \quad H = I_1 + \alpha I_2 + \varepsilon_0 F, \quad (15.18)$$

*where  $F$  is entire of phase order at least three ( $D_1 = 11$ ), the Hamiltonian has no symplectic normalization  $\text{id} + O(\varepsilon_0)$  with convergent-germ coefficients to an action-only normal form, and its first nonresonant homological solution has zero radius of convergence.*

*Proof.* The first bound in (15.17) dominates the very rapidly decreasing tail by twice its first term; for the last,  $Q_n \geq 10$  and  $\log(2Q_n) - Q_n^4 \log 10 < -4Q_n^2 < -D_n^2$ . If  $\alpha = A/B$  were rational then  $\alpha Q_n - P_n$  would be at least  $1/B$  for every  $n$ , contradicting (15.17); hence  $\omega$  is nonresonant. On a phase polydisk of radius  $R$  the  $n$ -th monomial of  $F$  is bounded by  $e^{-D_n^2/2} R^{D_n}$  and the distinct integers  $D_n$  tend to infinity, so these bounds are summable for every finite  $R$  and  $F$  is entire. By (15.4) the homological divisor of the  $n$ -th monomial is  $P_n - \alpha Q_n$ , so the absolute coefficient after division exceeds  $e^{D_n^2/2}$  by (15.17) and its  $D_n$ -th root tends to infinity, excluding every positive analytic radius; the first-order argument of theorem 15.16 then applies.  $\square$

*Remark 15.18* (The obstruction is one homological inverse, not a long iteration). Both  $F$  in (15.18) and its formal homological solution depend only on  $q_1$  and  $p_2$ , and any two functions of those variables have zero Poisson bracket. Therefore in the purely formal phase category  $\chi = -\varepsilon_0 L^{-1} F$  removes the perturbation with *no* higher Lie corrections at all. The obstruction is thus not divergence of a long nonlinear iteration; it is already the failure of a single homological inverse to preserve the specified analytic coefficient class. This observation is specific to source 06.

*Remark 15.19* (Why polynomial layers escape the same example). Assign the same monomials distinct increasing positive Hahn weights, for instance  $\hat{H} = H_0 + \sum_{n \geq 1} t^n e^{-D_n^2/2} q_1^{P_n} p_2^{Q_n}$ : then every Hahn layer is a polynomial and theorem 15.7(a) applies, despite the *same* frequency vector, because dividing an individual coefficient by its tiny ordinary divisor still gives an ordinary complex number rather than a failure of a phase Taylor series within that layer. This is no contradiction:  $\hat{H}$  is a *different* Hamiltonian, with a different distribution of monomials among the infinitesimal scales, and source 06 records explicitly that “which scale carries which monomial” is an input choice with mathematical consequences and not a physical fact. The divergent formal generator of proposition 15.17 also remains evaluable at infinitesimal phase points — its ordinary complex Taylor coefficients are summed against positive phase supports by lemma 3.4 — so the counterexample does not exclude normalization on an infinitesimal monad; its obstruction is to the ordinary analytic coefficient *germs* required by the global finite-halo theorem. This is the Hamiltonian instance of proposition 6.14.

## 15.6 Support-and-degree control beyond the finite halo

Assume polynomial coefficients in (15.7), split each nonzero coefficient into its finitely many monomials, and regard a monomial occurrence as an input *label*  $\ell = (\gamma, \alpha, \beta)$  with coefficient  $c_\ell \neq 0$  and  $d_\ell = |\alpha| + |\beta| \geq 3$ . The label set is a set; infinitely many labels are allowed. For  $\rho \in \Gamma$  put

$$w_\rho(\ell) = \gamma + (d_\ell - 2)\rho, \quad D_\rho = (t^\rho \mathcal{O}_\Gamma)^{2d}. \quad (15.19)$$

**Definition 15.20** (Admissible rescaling). The input monomial family is *admissible at*  $\rho$  if (1)  $w_\rho(\ell) > 0$  for every label; (2) the set  $W_\rho = \{w_\rho(\ell)\}$  is well ordered; and (3) each weight in  $W_\rho$  comes from only finitely many labels. These are exactly strong summability of the individually rescaled monomials into a positive Hahn series with polynomial coefficients.

The point  $\rho = 0$  is always admissible; a *negative*  $\rho$  gives a domain containing phase points of *infinite* magnitude. For infinitely many input monomials, condition (1) alone implies neither (2) nor (3) (examples 15.25 and 15.26). The Hamiltonian rescaling operator is

$$\mathcal{S}_\rho F(z) = t^{-2\rho} F(t^\rho z), \quad (15.20)$$

which is a Poisson *Lie* algebra homomorphism wherever its monomial expansion is summable,

$$\{\mathcal{S}_\rho F, \mathcal{S}_\rho G\} = \mathcal{S}_\rho \{F, G\}, \quad (15.21)$$

but is **not** a homomorphism for ordinary products: the factor  $t^{-2\rho}$  is exactly what compensates the two phase derivatives in a bracket. It fixes  $H_0$ , preserves resonant monomials, and commutes with  $L$  and  $\Pi$ .

**Theorem 15.21** (Compatible support-certified domains). *Suppose the polynomial input of [theorem 15.7](#) is admissible at  $\rho$ . Then its canonical normalizing map and inverse have strongly summable monomial realizations on  $D_\rho$  and are mutually inverse symplectic bijections of that domain, and the Hamiltonian identity and the commuting action identities hold there. If the normalizing construction is performed independently after rescaling at two admissible values, the realized maps agree on the overlap after undoing the scalings: these domains carry one compatible normalization, not a choice of unrelated local conjugacies.*

*Proof.* Admissibility makes  $H_\rho := \mathcal{S}_\rho H = H_0 + \sum_\ell c_\ell t^{w_\rho(\ell)} q^{\alpha_\ell} p^{\beta_\ell}$  a positive Hahn perturbation with polynomial coefficients, so [theorem 15.7\(a\)](#) gives  $\chi_\rho, \Phi_\rho$  on the finite halo, and undoing the phase scaling gives

$$\Psi_\rho(z) = t^\rho \Phi_\rho(t^{-\rho} z) \quad \text{of } D_\rho, \quad (15.22)$$

symplectic in the original bracket because an equal scalar dilation in all phase coordinates is conformally symplectic and the dilation and its inverse cancel. That this is the *same* normalization uses the labeled trees of [definition 15.3](#) with each input coefficient expanded into its monomial labels: every nonzero monomial contribution to a generator or Hamiltonian coefficient with leaves  $\ell_1, \dots, \ell_r$  has phase degree

$$2 + \sum_{j=1}^r (d_{\ell_j} - 2), \quad (15.23)$$

and every contribution to a nonlinear coordinate coefficient has phase degree

$$1 + \sum_{j=1}^r (d_{\ell_j} - 2), \quad (15.24)$$

both inductively, since a Poisson bracket subtracts two from the sum of phase degrees while  $L$ ,  $L^{-1}$  and  $\Pi$  preserve degree, the coordinate construction starting at degree one. Under Hamiltonian scaling the exponent of the tree in (15.23) becomes precisely  $\sum_j \gamma_{\ell_j} + \rho \sum_j (d_{\ell_j} - 2) = \sum_j w_\rho(\ell_j)$ , and for coordinates the same sum is obtained using  $t^{-\rho} \Phi(t^\rho z)$ . Since  $W_\rho$  is positive and well ordered, [lemma 3.4](#) gives finitely many words at a fixed rescaled weight, the finite-fiber hypothesis gives finitely many choices of original labels per word, and there are then finitely many trees: the rescaled generator, Hamiltonian and coordinate expressions are strongly summable *before* any regrouping. Hence (15.21) applies to every finite tree and may then be summed; the rescaled generator is  $\mathcal{S}_\rho \chi$ , satisfying the same nonresonant gauge and normal-form equation as  $\chi_\rho$ , and uniqueness identifies them, so on coordinates  $\Phi_\rho(z) = t^{-\rho} \Phi(t^\rho z)$ . The same argument applies to the inverse exponential and to the actions, and Taylor evaluation at finite rescaled points is [proposition 3.11](#). For two admissible rescalings both maps arise from the same labeled trees, and at a point of the overlap those tree evaluations are strongly summable by each certificate with every regrouping agreeing coefficientwise — which proves compatibility *without* assuming that rescaling preserves the order of the original Hahn exponents; in general it does not.  $\square$

**Warning 15.22** (Scope of the extension). [Theorem 15.21](#) extends the *polynomial*-coefficient normalizer, Hamiltonian and constructed actions only. It does **not** license evaluating an arbitrary entire function at an infinite phase or action argument; a centralizer statement on  $D_\rho$  must instead be made in the corresponding *rescaled* entire-coefficient algebra, and transporting all original entire coefficients to

infinite arguments without a summability certificate would be an unjustified enlargement of the function class. The theorem gives *guaranteed*, not maximal, domains.

**Corollary 15.23** (A degree-dependent radius). *Let  $H = H_0 + t^\eta P_m$  with  $\eta > 0$  and  $P_m$  homogeneous of phase degree  $m \geq 3$ , and write  $\chi = \sum_{n \geq 1} t^{n\eta} \chi_n$ ,  $\Phi = z + \sum_{n \geq 1} t^{n\eta} \Phi_n$ . Every nonzero  $\chi_n$  is homogeneous of degree  $2 + n(m - 2)$  and every nonzero component of  $\Phi_n$  has degree  $1 + n(m - 2)$ , and the normalizer and its inverse are defined on every  $D_\rho$  with*

$$\eta + (m - 2)\rho > 0. \quad (15.25)$$

*Proof.* Every tree at exponent  $n\eta$  has exactly  $n$  leaves, an ordered abelian group being torsion-free and  $\eta$  the only input weight; apply (15.23)–(15.24). The finitely many input monomials all have the same shifted weight  $\eta + (m - 2)\rho$ , whose positivity supplies all three admissibility conditions.  $\square$

## 15.7 A sharp cubic boundary, and two independent summability failures

**Example 15.24** (The cubic boundary, with a coordinate-independent obstruction). In one degree of freedom take

$$H(q, p) = qp + \varepsilon_0 q^2 p, \quad \varepsilon_0 = t^\eta, \quad \eta > 0, \quad (15.26)$$

where there is no arithmetic issue at all because  $\omega_1 = 1$ . With  $\chi = -\varepsilon_0 q^2 p$  one has  $\text{ad}_\chi q = \varepsilon_0 q^2$ ,  $\text{ad}_\chi p = -2\varepsilon_0 qp$ , and the coordinate exponentials give

$$\Phi(q, p) = \left( \frac{q}{1 - \varepsilon_0 q}, p(1 - \varepsilon_0 q)^2 \right), \quad H \circ \Phi = qp, \quad (15.27)$$

verifiable also by direct rational substitution; its Jacobian determinant is one, so it is symplectic, and  $\Pi_\chi = 0$  makes it the canonical normalizer of [theorem 15.7](#). For  $\eta + \rho > 0$  the expansion  $q/(1 - \varepsilon_0 q) = \sum_{n \geq 0} \varepsilon_0^n q^{n+1}$  is strongly summable on  $D_\rho$ ; at the boundary point  $q = \varepsilon_0^{-1}$ ,  $p = 0$ , every term of that family has exponent  $-\eta$  and the rational function has a *pole*. Hence replacing the strict inequality in (15.25) by a non-strict one cannot be valid in general.

There is also a *coordinate-independent* obstruction on the whole boundary polydisk  $D_{-\eta}$ : from  $H_p = q(1 + \varepsilon_0 q)$  and  $H_q = p(1 + 2\varepsilon_0 q)$ , (15.26) has exactly two critical points there,  $(0, 0)$  and  $(-\varepsilon_0^{-1}, 0)$ , whereas  $qp$  has exactly one. An everywhere invertible symplectic change on that domain with  $H \circ \Psi = qp$  would give a bijection between these critical-point sets by the chain rule, which is impossible. The boundary failure is therefore not merely a defect in one geometric-series proof.

**Example 15.25** (Positive shifted weights, but no well-order). Over  $\Gamma = \mathbb{Q}$  consider the valid polynomial-coefficient perturbation  $R(q, p) = \sum_{n \geq 2} t^{n+1/n} q^{n+2}$ , whose original exponents strictly increase. At  $\rho = -1$  the shifted weights are  $1/n$ , all strictly positive, but forming a *descending* sequence; at the rescaled point  $q = 1$  the rescaled perturbation would be  $\sum_{n \geq 2} t^{1/n}$ , which is not a Hahn series. Positivity alone is therefore insufficient even to *define* the rescaled input.

**Example 15.26** (Positive well-ordered weights, but infinite fibers). Take  $R(q, p) = \sum_{n \geq 2} t^{n+1} q^{n+2}$ . At  $\rho = -1$  all shifted weights equal 1: the weight set is positive and well ordered, but infinitely many labels contribute to that one weight. The rescaled expression is  $t \sum_{n \geq 2} q^{n+2}$ , whose evaluation at the ordinary point  $q = 1$  is defined neither by finite coefficient addition nor by ordinary convergence; it cannot supply a polynomial-coefficient Hahn perturbation on the whole rescaled finite halo.

[Examples 15.25](#) and [15.26](#) show why none of the three conditions of [definition 15.20](#) can be dropped from a general theorem of this form. They do **not** assert that finite fibers are necessary for every possible analytic resummation in some larger, differently specified class.

## 15.8 A genuinely multiscale worked example

Take  $\omega = (1, \sqrt{2})$ ,  $\kappa = 2 - \sqrt{2}$ , and  $A = q_1^2 p_2$ ,  $B_h = p_1^2 q_2$ ,  $C = I_1 I_2$ . For arbitrary positive scales  $\eta, \theta \in \Gamma$  put

$$H = I_1 + \sqrt{2}I_2 + u(A + B_h) + vC, \quad u = t^\eta, \quad v = t^\theta, \quad (15.28)$$

nonresonance following from the irrationality of  $\sqrt{2}$ . The two scales need *not* be Archimedean-comparable; the lexicographic example of [warning 3.6](#) allows  $n\eta < \theta$  for every  $n$ . Retaining  $u, v$  as independent *labels*, and writing  $O_{\text{lab}}(3)$  for terms involving at least three input labels — **not** terms above a valuation cutoff — direct homological calculation gives

$$\chi = \frac{u}{\kappa}(B_h - A) + \frac{uv}{\kappa^2}(A - B_h)(2I_2 - I_1) + O_{\text{lab}}(3), \quad (15.29)$$

$$N = I_1 + \sqrt{2}I_2 + vI_1 I_2 + \frac{u^2}{\kappa}(I_1^2 - 4I_1 I_2) + O_{\text{lab}}(3), \quad (15.30)$$

the coefficients of  $v$ ,  $u^2$  and  $v^2$  in  $\chi$  vanishing and those of  $uv$  and  $v^2$  in the displayed normal form vanishing. The key bracket is  $\{A, B_h\} = 4I_1 I_2 - I_1^2$ ; since  $LA = \kappa A$  and  $LB_h = -\kappa B_h$  the first generator is  $u(B_h - A)/\kappa$ , and the second normal term is  $\frac{1}{2}\Pi\{u(B_h - A)/\kappa, u(A + B_h)\} = \frac{u^2}{\kappa}(I_1^2 - 4I_1 I_2)$ . Also  $\{A, C\} = A(2I_2 - I_1)$  and  $\{B_h, C\} = -B_h(2I_2 - I_1)$ , which gives the mixed generator by  $-L^{-1}Q_\Pi$ ; these identities fix both the sign convention and the placement of the inverse transformation.

To first label degree the coordinate map and actions are

$$\begin{aligned} \Phi(q_1) &= q_1 - \frac{2u}{\kappa}p_1 q_2 + O_{\text{lab}}(2), & \Phi(p_1) &= p_1 - \frac{2u}{\kappa}q_1 p_2 + O_{\text{lab}}(2), \\ \Phi(q_2) &= q_2 + \frac{u}{\kappa}q_1^2 + O_{\text{lab}}(2), & \Phi(p_2) &= p_2 + \frac{u}{\kappa}p_1^2 + O_{\text{lab}}(2), \\ J_1 &= I_1 + \frac{2u}{\kappa}(A + B_h) + O_{\text{lab}}(2), & J_2 &= I_2 - \frac{u}{\kappa}(A + B_h) + O_{\text{lab}}(2), \end{aligned}$$

the full supported series satisfying  $H = N(J)$  and commuting exactly, the displayed remainders serving only to show a manageable initial segment. There are two input degrees — the  $u$  terms cubic, the  $v$  term quartic — so every generator or Hamiltonian coefficient with label  $u^a v^b$  is homogeneous of phase degree  $2 + a + 2b$  while the corresponding coordinate coefficient has degree  $1 + a + 2b$ , and a sufficient domain certificate is simply

$$\eta + \rho > 0, \quad \theta + 2\rho > 0. \quad (15.31)$$

The input label set being finite, positivity already supplies well-ordering and finite fibers here. In a higher-rank group, writing these as valuation inequalities rather than numerical radii avoids introducing a nonexistent real-valued absolute value. If two label weights coincide, for instance  $2\eta = \theta$ , their terms are added at that Hahn exponent and the construction remains valid; label-degree truncation and Hahn-weight truncation are still different operations, and the verification code deliberately uses the former.

## 15.9 A discrete one-variable parallel

The same support mechanism is visible in a simpler setting. It is included as a bridge to the established non-Archimedean literature [\[20\]](#), *not* as a claim that polynomial perturbative linearization is new in rank one.

**Proposition 15.27** (Polynomial-layer linearization). *Let  $\lambda \in \mathbb{C}^\times$  not be a root of unity and let  $f(z) = \lambda z + R(z)$  with  $R \in \mathcal{H}_\Gamma(\mathcal{P}_1)_{>0}$  and  $\text{ord}_0 R_s \geq 2$ . There is a unique  $h(z) = z + U(z)$  with positive polynomial Hahn coefficients,  $\text{ord}_0 U_\gamma \geq 2$ , satisfying*

$$f \circ h = h \circ (\lambda \text{ id}), \quad (15.32)$$

*it is a bijection of the finite halo, and (15.32) holds there as an actual identity. No bound on  $|\lambda^n - \lambda|^{-1}$  is required.*

*Proof.* On  $z^n$  with  $n \geq 2$  the operator  $L_\lambda U = U(\lambda z) - \lambda U(z)$  is multiplication by the nonzero constant  $\lambda^n - \lambda$ , hence invertible on polynomials of order at least two, and (15.32) becomes  $U = L_\lambda^{-1} R(z + U)$ . At a fixed Hahn exponent every nonlinear occurrence of  $U$  on the right is accompanied by a positive coefficient of  $R$ , hence uses strictly smaller exponents of  $U$ , so the supported recursion and finite-expression argument of proposition 15.1 apply; each coefficient remains a polynomial, the least-difference argument proves uniqueness, and propositions 3.11 and 15.1 give the actual conjugacy.  $\square$

For  $f(z) = \lambda z + \varepsilon_0 z^2$ , writing  $h(z) = \sum_{n \geq 1} b_n \varepsilon_0^{n-1} z^n$  with  $b_1 = 1$  gives the exact recurrence

$$(\lambda^n - \lambda) b_n = \sum_{j=1}^{n-1} b_j b_{n-j} \quad (n \geq 2), \quad (15.33)$$

with one phase monomial per Hahn layer; at finite  $z$  it is the *support*, not the complex growth of the  $b_n$ , that makes this a valid Hahn identity. This is the same quadratic Schröder recursion that sources 02, 03, 04 and 05 each also compute, in four different normalizations; section 21.1 prints it once with the translation between them.

## 16 Quasi-periodic divisors: a finite flag for a surreal frequency

Everything up to this point has been *germ and fixed-point* dynamics: a map  $F$  with a distinguished fixed point, its multiplier, its return iterates, its periods, or a Hamiltonian at the origin of a finite phase halo. Source 07 is the other classical half of small-divisor theory. Its object is a *vector field on a real torus* whose frequency vector has surreal components, its divisors are the infinitely many scalars  $k \cdot \Upsilon$  indexed by  $k \in \mathbb{Z}^d$ , and its equation is the cohomological equation. None of the six germ sources contains a resonance lattice, a cohomological equation, a Fourier divisor or a conjugacy to a constant vector field; conversely nothing in this part concerns a multiplier, a return iterate or a period. What the two halves share is the machinery of section 3 — the set-sized workspace, strong summability, the positive-support word lemma, halo evaluation — which is therefore not restated here, and the architecture of remark 2.2, which is instantiated a fourth time in section 17.

The headline of this part has no counterpart in the other six. A frequency vector with  $d$  surreal components can carry a transfinite hierarchy of scales, and there are infinitely many Fourier divisors; yet only **at most**  $d$  distinct divisor valuations are arithmetically visible, and a canonical descending flag of integer lattices identifies them. That finite reduction is what turns an apparently transfinite problem into a finite one, and it is the reason this material occupies three sections rather than a remark.

## 16.1 The torus, the strip, and the notation of this part

Before anything else, read [warnings 2.9 to 2.11](#). They record that this report now carries *two* frequency vectors of different kinds, that  $\kappa_\Upsilon$  below is an element of  $\Gamma$  and not a rate, that  $\Upsilon$  is never the exponent  $\Omega$  of [example 3.7](#) nor the time form  $\Omega_F$  of [section 14](#), and that the finitely many visible divisor valuations are not a spectrum in any sense this collection uses.

Fix  $d \geq 1$  and write  $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$ . Fix  $\rho > 0$  and let

$$\mathbb{T}_\rho^d = \{z \in \mathbb{C}^d / (2\pi\mathbb{Z})^d : |\operatorname{Im} z_j| < \rho \text{ for all } j\}, \quad \mathcal{O}(\mathbb{T}_\rho^d) \quad (16.1)$$

be the *open* complex strip of half-width  $\rho$  around the real torus and the algebra of  $2\pi\mathbb{Z}^d$ -periodic functions holomorphic on it. Functions in  $\mathcal{O}(\mathbb{T}_\rho^d)$  need not be bounded at the boundary of the strip. With  $|k| = |k_1| + \cdots + |k_d|$ , write

$$f(\theta) = \sum_{k \in \mathbb{Z}^d} \widehat{f}(k) e^{ik \cdot \theta}, \quad \|f\|_r = \sum_{k \in \mathbb{Z}^d} |\widehat{f}(k)| e^{r|k|} \quad (0 < r < \rho). \quad (16.2)$$

A periodic holomorphic function lies in  $\mathcal{O}(\mathbb{T}_\rho^d)$  exactly when all these seminorms are finite: boundedness on a narrower closed strip and Cauchy's Fourier estimate give exponential decay there, and inserting an intermediate strip gives the weighted  $\ell^1$  bound; conversely those bounds give locally uniform convergence of the Fourier series and its termwise derivatives on every narrower strip. Hence  $\mathcal{O}(\mathbb{T}_\rho^d)$  is closed under finite sums, products and all orders of differentiation, and Fourier projection onto *any* subset of  $\mathbb{Z}^d$  preserves it, since it only deletes terms from  $\|f\|_r$ .

The torus derivatives  $\partial/\partial\theta_j$  act coefficientwise and annihilate  $K_\Gamma$ : they are external derivations, and [warning 3.2](#) applies verbatim — they are not the Berarducci–Mantova derivation, and  $\partial_\theta \sin \theta = \cos \theta$  here says nothing about whether a surreal  $y$  solves  $\partial^2 y + y = 0$  for that internal derivation. Source 07 states this disclaimer independently and in its own words; it is the same disclaimer, printed once.

## 16.2 The fixed-strip coefficient category

**Definition 16.1** (The common-strip Hahn algebra).  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d)) = \mathcal{O}(\mathbb{T}_\rho^d)((t^\Gamma))$  is the Hahn algebra over the coefficient algebra [\(16.1\)](#): elements are  $f = \sum_{\beta \in B} t^\beta f_\beta$  with  $B$  well ordered and every  $f_\beta \in \mathcal{O}(\mathbb{T}_\rho^d)$ . Fourier coefficients and the torus mean are taken coefficientwise,

$$\widehat{f}(k) = \sum_{\beta} t^\beta \widehat{f_\beta}(k) \in K_\Gamma, \quad \operatorname{av}(f) = \widehat{f}(0), \quad \mathbf{Q}_0 f = f - \operatorname{av}(f),$$

and  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))_{>0}$  denotes positive support, zero included. For a vector the support is the union of the component supports and the valuation their minimum.

This is the *fixed-domain* row of [section 3.2](#) in its torus instantiation: one and the same  $\rho$  serves every Hahn exponent  $\beta$ , and separate analytic germs for separate coefficients do not suffice. The distinction is stated once, in [convention 3.3](#), and is not re-erected here; source 07 imports it from the collection's catalogue explicitly, as a hypothesis of each of its main theorems rather than as a convention. As in the entire row of [section 3.2](#), **no** bound on  $\|f_\beta\|_r$  uniform in  $\beta$  is imposed — a deliberate choice, and the reason nothing here specializes  $t$  to a nonzero real number ([section 18.7](#)).

Elements of  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$  are genuine functions on infinitesimal thickenings: for ordinary  $z_0 \in \mathbb{T}_\rho^d$  and  $\xi \in \mathfrak{m}_\Gamma^d$ , [\(3.4\)](#) applies verbatim and [proposition 3.11](#) makes  $f \mapsto f^\#(z_0 + \xi)$  a faithful  $K_\Gamma$ -algebra homomorphism into **SC**. No continuity from an ordinary torus into the fine topology of **No**[ $i$ ] is claimed.

**Lemma 16.2** (Subexponential Fourier multipliers). *Let  $m : \mathbb{Z}^d \rightarrow \mathbb{C}$  satisfy  $|m(k)| \leq C_\varepsilon e^{\varepsilon|k|}$  for every  $\varepsilon > 0$ . Then  $\widehat{Tf}(k) = m(k)\widehat{f}(k)$  maps  $\mathcal{O}(\mathbb{T}_\rho^d)$  into itself, and so does any finite product of such multipliers with polynomial functions of  $k$ .*

*Proof.* For  $r < r' < \rho$  apply the estimate with  $\varepsilon = r' - r$  to get  $\|Tf\|_r \leq C_{r'-r} \|f\|_{r'}$ . A fixed polynomial in  $k$  is bounded by a constant times  $e^{\eta|k|}$  for every  $\eta > 0$ ; split an arbitrary exponent budget among finitely many factors.  $\square$

**Lemma 16.3** (Taylor substitution on the torus, with its Lipschitz estimate). *Let  $h \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d$  have positive valuation. Then*

$$S_h f = f \circ (\text{id} + h) := \sum_{\beta} t^\beta \sum_{\alpha \in \mathbb{N}^d} \frac{\partial^\alpha f_\beta}{\alpha!} h^\alpha \quad (16.3)$$

*is well defined in  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ , with  $\text{supp}(S_h f) \subseteq \text{supp}(f) + C^*$  for  $C \supseteq \text{supp } h$  well ordered and positive. Substitution is a ring homomorphism, obeys the chain rule, and is associative for infinitesimal substitutions; these are the torus instances of [proposition 4.1](#) and are not reproved. In addition, for positive-valuation  $h, g$ ,*

$$v(S_h f) \geq v(f), \quad v(S_h f - S_g f) \geq v(f) + v(h - g). \quad (16.4)$$

*Proof.* Well-definedness, the support bound and the algebraic identities are [lemma 3.4](#), [corollary 3.5](#), and [proposition 4.1](#) applied with  $U = \mathbb{T}_\rho^d$ : at each exponent the word lemma permits only finitely many Taylor monomials and finitely many decompositions involving  $\text{supp } f$ , and every resulting coefficient is a finite sum of products of derivatives of functions in  $\mathcal{O}(\mathbb{T}_\rho^d)$ , hence lies in  $\mathcal{O}(\mathbb{T}_\rho^d)$ . For the second estimate in (16.4) subtract corresponding Taylor monomials and telescope the products: every surviving term carries one factor from  $h - g$  and all its other factors have positive valuation, and the multi-index-zero terms cancel.  $\square$

[Equation \(16.4\)](#) is the one part of the torus substitution calculus that is not already in [proposition 4.1](#); it is what forces uniqueness in [theorem 18.1](#) and controls the remainder in [theorem 18.4](#).

**Warning 16.4** (Two summations, never interchanged). The Fourier series of an ordinary coefficient is an ordinary complex analytic sum; the series in  $t^\Gamma$  is a Hahn sum. Every argument below groups all Fourier modes *inside each analytic coefficient* first. It is nowhere asserted that the unrestricted two-index family of Fourier and Hahn terms is Hahn-summable, and indeed a family with infinitely many nonzero terms at one Hahn exponent violates [definition 3.1](#) however well its complex coefficients converge. This is the torus form of the discipline of [definition 3.1](#).

### 16.3 The flag, and one support for every reciprocal divisor

Let  $\Upsilon \in F_\Gamma^d \setminus \{0\}$  and collect the component Conway/Hahn normal forms into

$$\Upsilon = \sum_{\beta \in S} t^\beta \varpi_\beta, \quad \varpi_\beta \in \mathbb{R}^d \setminus \{0\}, \quad S \text{ well ordered}, \quad (16.5)$$

so the *exact resonance lattice* is

$$\mathbf{N}_\Upsilon = \{k \in \mathbb{Z}^d : k \cdot \Upsilon = 0\} = \bigcap_{\beta \in S} \ker(k \mapsto k \cdot \varpi_\beta). \quad (16.6)$$

This is *exact vanishing of a Hahn scalar*, never smallness; that is the content of the tag QR.

Set  $L_0 = \mathbb{Z}^d$ . While  $L_{j-1} \neq N_\Upsilon$ , let  $\gamma_j$  be the *least* exponent of  $S$  whose coefficient form does not vanish identically on  $L_{j-1}$ , and put

$$\mathfrak{a}_j(k) = k \cdot \varpi_{\gamma_j}, \quad L_j = \{k \in L_{j-1} : \mathfrak{a}_j(k) = 0\}, \quad \Sigma_j = L_{j-1} \setminus L_j. \quad (16.7)$$

The construction stops as soon as the current lattice equals  $N_\Upsilon$ . The  $\Sigma_j$  are the *nonresonant strata*.

**Theorem 16.5** (Finite flag and one common reciprocal support). *The construction (16.7) terminates after*

$$r \leq d \quad (16.8)$$

*steps, and:*

- (i) *each  $L_j$  is a saturated subgroup of  $\mathbb{Z}^d$  whose rank strictly decreases at each step,  $\gamma_1 < \dots < \gamma_r$ ,  $L_r = N_\Upsilon$ , and the strata partition  $\mathbb{Z}^d \setminus N_\Upsilon$ ;*
- (ii) *for  $k \in \Sigma_j$ ,*

$$k \cdot \Upsilon = t^{\gamma_j} \mathfrak{a}_j(k)(1 + \varrho_k), \quad v(\varrho_k) > 0, \quad v(k \cdot \Upsilon) = \gamma_j, \quad (16.9)$$

*with  $\varrho_k = 0$  permitted; in particular the divisor valuation takes at most  $d$  distinct values on  $\mathbb{Z}^d \setminus N_\Upsilon$ ;*

- (iii) *writing*

$$E_j = \{\beta - \gamma_j : \beta \in S, \beta > \gamma_j\}, \quad E = \bigcup_{j=1}^r E_j, \quad T_\Upsilon = \bigcup_{j=1}^r (-\gamma_j + E^*), \quad (16.10)$$

*the set  $E$  is well ordered and positive,  $T_\Upsilon$  is well ordered, and*

$$\text{supp}((k \cdot \Upsilon)^{-1}) \subseteq T_\Upsilon \quad \text{for every } k \notin N_\Upsilon. \quad (16.11)$$

*With  $\kappa_\Upsilon = \gamma_r$  one has  $T_\Upsilon \subseteq [-\kappa_\Upsilon, +\infty)$ .*

*Proof.* Each  $L_j$  is a common kernel of real additive homomorphisms on  $\mathbb{Z}^d$  and is therefore saturated: if a nonzero integer multiple of  $k$  lies in the kernel, so does  $k$ . At a nonterminal step  $\mathfrak{a}_j$  is a nonzero homomorphism on the finitely generated free abelian group  $L_{j-1}$  with torsion-free nonzero image, so  $\text{rank } L_j < \text{rank } L_{j-1}$ ; at most  $d$  such decreases occur, which is (16.8).

By the minimal choice of  $\gamma_j$ , every coefficient form below  $\gamma_j$  vanishes on  $L_{j-1}$ , and the form at  $\gamma_j$  vanishes on  $L_j$ ; hence the next selected exponent is strictly larger. Termination occurs exactly when every coefficient form vanishes on the current lattice, i.e. at  $N_\Upsilon$ . The same observation identifies the leading term for  $k \in \Sigma_j$ : explicitly

$$\varrho_k = \sum_{\substack{\beta \in S \\ \beta > \gamma_j}} \frac{k \cdot \varpi_\beta}{\mathfrak{a}_j(k)} t^{\beta - \gamma_j}, \quad (16.12)$$

whose support lies in  $E_j$  and is positive. This is (i) and (ii).

A tail of a well-ordered set translated by a fixed element is well ordered, and a finite union of well-ordered subsets of a linear order is well ordered; this gives  $E$ . Lemma 3.4 then gives  $E^*$  and  $T_\Upsilon$ . Finally

$$\frac{1}{k \cdot \Upsilon} = \frac{t^{-\gamma_j}}{\mathfrak{a}_j(k)} \sum_{n \geq 0} (-\varrho_k)^n \quad (k \in \Sigma_j) \quad (16.13)$$

is a Hahn identity by lemma 3.4 — not a limit of partial sums (definition 3.1 and warning 3.6) — with the asserted support. Every element of  $E^*$  is nonnegative and  $\gamma_j \leq \kappa_\Upsilon$ , giving the lower bound.  $\square$

*Remark 16.6* ( $r$  counts visible scales, not monomials). The number  $r$  counts *arithmetically visible* scales, not the monomials of  $\Upsilon$ . Exponents of  $S$  whose forms vanish on the current lattice are skipped, possibly across a long transfinite interval. The theorem does **not** say that only  $d$  monomials of  $\Upsilon$  matter to the eventual solution: every later coefficient can still move higher Hahn coefficients of the inverse, through (16.12). What is finite is the list of divisor valuations and the list of real forms on which all small-divisor arithmetic is tested.

**Proposition 16.7** (Coefficient structure of the inverse). *Fix  $\eta \in \Gamma$  and a stratum  $\Sigma_j$ . On that stratum the function  $k \mapsto [t^\eta](k \cdot \Upsilon)^{-1}$  is a finite sum of expressions*

$$c \frac{\prod_{\ell=1}^n (k \cdot \varpi_{\beta_\ell})}{\mathfrak{a}_j(k)^{n+1}}, \quad \sum_{\ell=1}^n (\beta_\ell - \gamma_j) = \eta + \gamma_j, \quad \beta_\ell > \gamma_j, \quad (16.14)$$

with  $c$  real and empty products allowed at  $n = 0$ . The number of terms and the denominator powers depend on  $\eta$  but are finite for each fixed  $\eta$ .

*Proof.* Expand (16.13) and extract the coefficient. All letters lie in  $\mathbf{E}_j$ , and lemma 3.4 permits finitely many words of all lengths at a fixed weight;  $\beta \mapsto \beta - \gamma_j$  is injective, so the coefficient choices are also finitely many.  $\square$

*Remark 16.8* (A common support is not summability over the modes). The family  $((k \cdot \Upsilon)^{-1})_{k \notin \mathbf{N}_\Upsilon}$  is in general **not** a Hahn-summable family: infinitely many leading terms may occupy the exponent  $-\gamma_j$ . Theorem 16.5 supplies a *common support*, not a Hahn sum over  $k$ , exactly as definition 3.1 requires. Proposition 16.7 is what permits ordinary Fourier summation at each Hahn coefficient; that is the additional analytic step, and it is where arithmetic enters.

**Proposition 16.9** (An arithmetic-free valuation bound). *Let  $\mathcal{C}_\Upsilon = \Upsilon_1 \partial_{\theta_1} + \cdots + \Upsilon_d \partial_{\theta_d}$ . If  $h \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$  has no Fourier modes in  $\mathbf{N}_\Upsilon$  and  $\mathcal{C}_\Upsilon h = g$ , then  $v(h) \geq v(g) - \kappa_\Upsilon$ . No small-divisor hypothesis is used.*

*Proof.* For  $h \neq 0$  put  $\beta = v(h)$ . The nonzero  $h_\beta$  has a nonzero Fourier coefficient at some  $k \notin \mathbf{N}_\Upsilon$ , so  $v(\widehat{h}(k)) = \beta$  and  $v(\widehat{g}(k)) = \beta + v(k \cdot \Upsilon) \leq \beta + \kappa_\Upsilon$  by (16.9). Every Fourier coefficient of  $g$  has valuation at least  $v(g)$ ; rearrange. The zero case is immediate.  $\square$

## 17 The fixed-strip solvability threshold, and an arbitrary-order lifting obstruction

### 17.1 The stratified condition, and the category table instantiated a second time

**Definition 17.1** (Stratified subexponential condition  $\text{SD}_{\text{ss}}$ ).  $\Upsilon$  satisfies  $(\text{SD}_{\text{ss}})$  if for each flag stratum  $\Sigma_j$  and every real  $\varepsilon > 0$  there is a real  $C_{j,\varepsilon} > 0$  with

$$\frac{1}{|\mathfrak{a}_j(k)|} \leq C_{j,\varepsilon} e^{\varepsilon|k|} \quad (k \in \Sigma_j), \quad (17.1)$$

equivalently  $\limsup_{|k| \rightarrow \infty, k \in \Sigma_j} \log^+(1/|\mathfrak{a}_j(k)|)/|k| = 0$ .

*Remark 17.2* (The one invariant, a fourth divisor family, and two instantiations of one table). Remark 2.2 defines one quantity — the exponential rate of the reciprocals of a graded family of nonzero ordinary complex divisors — and evaluates it on three families.  $(\text{SD}_{\text{ss}})$  is that same quantity on a *fourth* family, evaluated stratum by stratum:

| Divisor family                                                                                      | Grading              | Rate                                                |
|-----------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------|
| $\lambda^n - \lambda$ (drifted multiplier)                                                          | $n$                  | $\sigma$ ( $\text{SD}_{\text{dr}}$ )                |
| $\lambda^\alpha - \lambda_j$ (exact multiplier)                                                     | $ \alpha $           | $\tau$ ( $\text{SD}_{\text{ex}}$ )                  |
| $\omega \cdot (\alpha - \beta)$ (ordinary frequency)                                                | $ \alpha  +  \beta $ | $\log \Theta$ ( $\text{SD}_{\text{fr}}$ )           |
| $\mathfrak{a}_j(k) = k \cdot \varpi_{\gamma_j}$ (leading real forms of $\Upsilon$ , on $\Sigma_j$ ) | $ k $                | 0 required on every $j$ ( $\text{SD}_{\text{ss}}$ ) |

So ( $\text{SD}_{\text{ss}}$ ) is the vanishing-rate condition, imposed on each of the finitely many strata; it is the exact analogue of  $\tau = 0$  and of  $\sigma = 0$ , and it is *not* the analogue of  $\tau < \infty$ . That is consistent with the category table of [remark 2.2](#), which is stated there once and is instantiated twice in this report:

| Row of <a href="#">remark 2.2</a>                       | Germ instantiation, divisors<br>$\lambda^\alpha - \lambda_j$ or $\lambda^q - 1$                                       | Quasi-periodic instantiation,<br>divisors $k \cdot \Upsilon$                                                                       |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| one fixed unchanged domain,<br>rate 0                   | fixed polydisk $\mathcal{O}(\mathbb{D}_R^d)((t^\Gamma))$ ,<br>$\tau = 0$ (row one of<br><a href="#">theorem 6.1</a> ) | fixed strip $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ , ( $\text{SD}_{\text{ss}}$ )<br>( <a href="#">theorem 17.3</a> ) |
| polynomial / formal<br>coefficients, no arithmetic      | <a href="#">theorem 6.1</a> rows four and<br>five; $\text{SD}_0$                                                      | <b>not established here:</b> source<br>07 proves no polynomial or<br>formal row for the torus, and<br>none is claimed              |
| radius-free germs / entire<br>coefficients, rate finite | <a href="#">theorem 6.1</a> rows two and<br>three; <a href="#">theorem 15.5</a>                                       | <b>not established here;</b> the<br>fixed strip is the only category<br>source 07 classifies                                       |

Only the first row is instantiated on both sides. Reading across the other two rows, in either direction, would assert theorems no source proves.

## 17.2 Universal solvability on a fixed strip

Let  $\Pi_{\mathbf{N}_\Upsilon}$  denote coefficientwise Fourier projection onto  $\mathbf{N}_\Upsilon$ ; it preserves  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$  because arbitrary Fourier projection preserves  $\mathcal{O}(\mathbb{T}_\rho^d)$ .

**Theorem 17.3** (Universal fixed-strip solvability). *For a nonzero  $\Upsilon \in F_\Gamma^d$  the following are equivalent.*

- (i)  $\Upsilon$  satisfies ( $\text{SD}_{\text{ss}}$ ).
- (ii) For every  $\rho > 0$  and every  $f \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$  with  $\Pi_{\mathbf{N}_\Upsilon} f = 0$  there is a unique  $u \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$  with  $\mathcal{C}_\Upsilon u = f$  and  $\Pi_{\mathbf{N}_\Upsilon} u = 0$ .

When these hold, the solution operator is

$$\widehat{\mathcal{G}_\Upsilon f}(k) = \begin{cases} \widehat{f}(k)/(i k \cdot \Upsilon), & k \notin \mathbf{N}_\Upsilon, \\ 0, & k \in \mathbf{N}_\Upsilon, \end{cases} \quad (17.2)$$

and it satisfies

$$\text{supp}(\mathcal{G}_\Upsilon f) \subseteq \text{supp}(f) + \mathbf{T}_\Upsilon, \quad v(\mathcal{G}_\Upsilon f) \geq v(f) - \kappa_\Upsilon. \quad (17.3)$$

Without the normalization the kernel consists exactly of the elements of  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$  with Fourier support inside  $\mathbf{N}_\Upsilon$ . If ( $\text{SD}_{\text{ss}}$ ) fails, there is already an ordinary real analytic mean-compatible forcing, supported at a single Hahn exponent and on one stratum, for which a forced coefficient of any putative solution is not a distribution on the real torus.

*Sufficiency.* For  $\eta \in \mathbf{T}_\Upsilon$  define the scalar multiplier  $m_\eta(k) = [t^\eta](i k \cdot \Upsilon)^{-1}$  for  $k \notin \mathbf{N}_\Upsilon$  and  $m_\eta(k) = 0$  on  $\mathbf{N}_\Upsilon$ . On each stratum [proposition 16.7](#) writes  $m_\eta$  as a finite sum of expressions with polynomial numerator in  $k$  and a finite power of  $\mathbf{a}_j(k)$  in the denominator; [\(SD<sub>ss</sub>\)](#) and [lemma 16.2](#) make each a subexponential multiplier. There are finitely many strata — this is where [\(16.8\)](#) is used — and projection onto a stratum does not affect the estimate, so  $m_\eta$  defines  $G_\eta : \mathcal{O}(\mathbb{T}_\rho^d) \rightarrow \mathcal{O}(\mathbb{T}_\rho^d)$ .

Write  $f = \sum_{\beta \in B} t^\beta f_\beta$  and put

$$u_\gamma = \sum_{\substack{\beta \in B, \eta \in \mathbf{T}_\Upsilon \\ \beta + \eta = \gamma}} G_\eta f_\beta. \quad (17.4)$$

By [lemma 3.4](#) this sum is finite at each  $\gamma$  and  $B + \mathbf{T}_\Upsilon$  is well ordered, so  $u = \sum_\gamma t^\gamma u_\gamma$  lies in  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ : the common domain and the common support are *proved*, not inferred from the modewise formula. Fourier extraction gives [\(17.2\)](#), and multiplying modewise by  $i k \cdot \Upsilon$  gives the equation. Uniqueness and the kernel description follow from the same modewise calculation,  $K_\Gamma$  being a field. The support estimate is the construction and  $\mathbf{T}_\Upsilon \geq -\kappa_\Upsilon$  gives the valuation estimate.  $\square$

*Necessity.* Suppose [\(SD<sub>ss</sub>\)](#) fails on  $\Sigma_j$ . Choose  $\varepsilon > 0$  and modes  $k_n \in \Sigma_j$ , no two equal up to sign, with  $|k_n| \rightarrow \infty$  and  $|\mathbf{a}_j(k_n)|^{-1} > e^{2\varepsilon|k_n|}$ ; this is possible after decreasing  $\varepsilon$  if necessary. The real function

$$f(\theta) = 2 \sum_{n \geq 1} e^{-\varepsilon|k_n|} \cos(k_n \cdot \theta) \quad (17.5)$$

lies in  $\mathcal{O}(\mathbb{T}_\varepsilon^d)$ , has no resonant modes and sits at Hahn exponent zero. Any solution must have  $\hat{u}(k_n) = e^{-\varepsilon|k_n|}/(i k_n \cdot \Upsilon)$ , so by [\(16.9\)](#) the coefficient of  $t^{-\gamma_j}$  has Fourier coefficients of modulus greater than  $e^{\varepsilon|k_n|}$ . Fourier coefficients of a distribution on a compact torus grow at most polynomially — apply its finite-order continuity estimate to the Fourier characters — so that forced coefficient is not a distribution, and in particular not holomorphic on any strip of positive width. Adding resonant modes cannot alter these nonresonant coefficients.  $\square$

**Corollary 17.4** (The divisor loss  $\kappa_\Upsilon$  is attained). *For any  $k \in \Sigma_r$  and any  $s \in \Gamma$  the solution for  $f = t^s e^{ik \cdot \theta}$  has valuation  $s - \kappa_\Upsilon$ . Hence the valuation loss in [\(17.3\)](#) cannot be improved uniformly.*

*Proof.* Its unique nonzero Fourier mode is divided by a scalar of valuation  $\kappa_\Upsilon$ .  $\square$

*Remark 17.5* (A stratified Diophantine sufficient condition). A convenient sufficient hypothesis is a stratified Diophantine bound  $|\mathbf{a}_j(k)| \geq c_j(1 + |k|)^{-\tau_j}$  on  $\Sigma_j$ . A term of length  $n$  in [\(16.14\)](#) then has a multiplier growing at most polynomially with exponent  $n + (n + 1)\tau_j$ , so every individual coefficient operator has explicit finite loss estimates between narrower-strip seminorms. It is **not** asserted that those losses stay bounded as  $\eta$  varies. The fixed-strip theorem is available precisely because the category demands a common open domain and not a single bounded analytic norm for every coefficient — the same mechanism as in [warning 9.3](#).

**Corollary 17.6** (Higher tails do not change the arithmetic test). *If  $v(\tilde{\Upsilon} - \Upsilon) > \kappa_\Upsilon$  then  $\tilde{\Upsilon}$  has the same flag, the same leading real forms, and — whenever  $\mathbf{N}_\Upsilon = \{0\}$  — the same exact resonance lattice; in that nonresonant case  $\tilde{\Upsilon}$  satisfies [\(SD<sub>ss</sub>\)](#) exactly when  $\Upsilon$  does.*

*Proof.* No coefficient at or below  $\kappa_\Upsilon$  changes. Each nonzero integer vector already has a nonvanishing leading divisor coefficient at one of those levels, so no resonance can appear and its leading form is unchanged.  $\square$

The nonresonance qualification is not decoration: when  $\mathbf{N}_\Upsilon \neq \{0\}$  a new higher tail can split some exact resonances and introduce additional flag levels, and the criterion must then be applied to the extended flag.

### 17.3 Arbitrarily long analytic jets with no analytic lift

The next construction is a failure that no valuation-only argument can detect, and it lives entirely at integer exponents: the obstruction is not caused by exotic uncountable supports.

Define integers and a real number by

$$b_1 = 2, \quad b_{j+1} = 10^{2b_j}, \quad \mathfrak{l} = \sum_{j \geq 1} 10^{-b_j}, \quad (17.6)$$

and put  $Q_j = 10^{b_j}$ ,  $P_j = Q_j \sum_{\ell \leq j} 10^{-b_\ell}$ ,  $\mathfrak{d}_j = Q_j \mathfrak{l} - P_j$ ,  $k_j = (-P_j, Q_j)$ ,  $K_j = |k_j|$ . Here  $P_j$  is an integer with  $0 < P_j < Q_j$  and

$$0 < \mathfrak{d}_j \leq 2Q_j 10^{-Q_j^2}, \quad Q_j \leq K_j < 2Q_j, \quad (17.7)$$

by bounding the tail of the rapidly increasing sequence of decimal places; the same estimate proves  $\mathfrak{l}$  irrational, since a fixed denominator  $q$  would force  $|\mathfrak{d}_j| \geq 1/q$ . Consider the infinitesimally detuned frequency and operator

$$\Upsilon = (1, \mathfrak{l} + t), \quad \mathcal{C}_\Upsilon = \partial_x + (\mathfrak{l} + t)\partial_y = L_0 + t\partial_y, \quad L_0 = \partial_x + \mathfrak{l}\partial_y. \quad (17.8)$$

The real leading vector  $(1, \mathfrak{l})$  is already nonresonant, so there is one flag level, at  $\gamma_1 = 0$ , and  $(\mathbf{SD}_{\text{ss}})$  fails very strongly.

**Theorem 17.7** (Arbitrary finite analytic lifting, but no full lift). *For every integer  $N \geq 1$  and every  $\rho > 0$  there is an ordinary real entire periodic  $f_N$  such that*

- (i)  $\mathcal{C}_\Upsilon u = f_N$  has a mean-zero solution modulo  $t^{N+1}$  with coefficients  $u_0, \dots, u_N \in \mathcal{O}(\mathbb{T}_\rho^d)$ ;
- (ii)  $u_0, \dots, u_{N-1}$  are entire periodic functions;
- (iii) the forced coefficient  $u_{N+1}$  is not a distribution on the real torus, so there is no solution in  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_{\rho'}^2))$  for any  $\rho' > 0$ , and none after an order-preserving extension of the value group.

One may take

$$f_N(x, y) = 2 \sum_{j \geq 1} \mathfrak{d}_j^{N+1} e^{-\rho K_j} \cos(k_j \cdot (x, y)). \quad (17.9)$$

*Proof.* By (17.7) the Fourier coefficients in (17.9) decay faster than  $e^{-RK_j}$  for every fixed  $R > 0$ , so  $f_N$  is entire, real and of zero mean. At the mode  $k_j$  the divisor is  $\mathfrak{d}_j + Q_j t$ ; expanding its inverse as a Hahn geometric series (legitimate by lemma 3.4) gives

$$\widehat{u}_n(k_j) = \frac{(-1)^n}{i} Q_j^n \mathfrak{d}_j^{N-n} e^{-\rho K_j} \quad (n \geq 0), \quad (17.10)$$

with complex conjugates at  $-k_j$  and no other nonzero modes; set every mean to zero. For  $n < N$  the remaining positive power of  $\mathfrak{d}_j$  gives superexponential decay, so  $u_n$  is entire. For  $n = N$  the coefficients are the polynomial factor  $Q_j^N$  times  $e^{-\rho K_j}$ , whose weighted Fourier sums are finite on every strip of width  $r < \rho$ , so  $u_N \in \mathcal{O}(\mathbb{T}_\rho^d)$ . The scalar identities give  $L_0 u_0 = f_N$  and  $L_0 u_n = -\partial_y u_{n-1}$  for  $1 \leq n \leq N$ , which is the finite-jet statement in  $\mathcal{O}(\mathbb{T}_\rho^d[[t]]/(t^{N+1}))$ . At the next coefficient

$$|\widehat{u_{N+1}}(k_j)| = Q_j^{N+1} \mathfrak{d}_j^{-1} e^{-\rho K_j} \geq \frac{1}{2} Q_j^N 10^{Q_j^2} e^{-2\rho Q_j},$$

which exceeds every polynomial in  $K_j$  along the sequence and so is not a distributional Fourier coefficient sequence.

For the value-group clause: at each nonzero Fourier mode the scalar equation has the *unique* solution  $\widehat{f_N}(k)/(i k \cdot \Upsilon)$  in the Hahn field, and the inverse of that divisor is the same geometric series in any larger ordered-group Hahn field. Therefore the coefficient of  $t^{N+1}$  remains exactly (17.10), and admitting negative or noninteger Hahn exponents in a proposed solution changes nothing; extra constants affect only the zero Fourier mode.  $\square$

*Warning 17.8* (The robustness under value-group extension is argued, not assumed). This report's standing hazard is the temptation to assert that some norm or estimate is invariant under enlarging  $\Gamma$ . [Theorem 17.7\(iii\)](#) does **not** do that. It cites no invariance of any norm across a value-group extension; it argues from *uniqueness of the modewise Hahn inverse* in the larger field, which is an algebraic statement about a field, together with the fact that the same geometric expansion computes it. Source 07 states the argument in exactly that form, and the corresponding invariance in the positive direction is [proposition 4.14](#), which is likewise proved rather than assumed.

This is a finite-versus-infinite lifting obstruction in a precise analytic ring. It does not say that the scalar algebraic inverses fail to exist — they all exist — but that infinitely many modewise inverses do not assemble into even one analytic coefficient at a specified order. The same operator (17.8) works for every  $N$ ; only the entire forcing changes. In particular **no test consisting of a fixed finite number of analytic coefficient equations can replace the arithmetic hypothesis** in a general lifting theorem. The construction adapts the classical Liouville mechanism [18], which is not claimed as new; what is offered is the arbitrarily prescribed order at which it first bites.

## 18 Torus normal form: conjugacy to a constant vector field

We pass from the scalar equation to vector fields. Multiplying the whole field by a scalar monomial normalizes

$$\gamma_1 = 0, \quad 0 \leq \kappa\Upsilon = \gamma_r, \quad (18.1)$$

which subtracts  $\gamma_1$  from both the frequency and the perturbation valuations, so the eventual condition  $v(\mathcal{V}) > \kappa\Upsilon$  is unchanged when translated back. Throughout this section  $\Gamma$  is nontrivial,  $N_\Upsilon = \{0\}$  (tag  $\neg\text{QR}$ : no exact resonances) and  $(\text{SD}_{\text{ss}})$  holds. The operator  $\mathcal{G}_\Upsilon$  then inverts  $\mathcal{C}_\Upsilon$  on all mean-zero elements and is applied componentwise. Let

$$\mathcal{X}(\theta) = \Upsilon + \mathcal{V}(\theta), \quad \mathcal{V} \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d, \quad s_0 = v(\mathcal{V}) > \kappa\Upsilon. \quad (18.2)$$

**Theorem 18.1** (Hahn-analytic constant normal form). *Under the preceding assumptions there is a unique pair*

$$h \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d, \quad \mathfrak{c} \in K_\Gamma^d, \quad v(h) > 0, \quad \text{av}(h) = 0,$$

such that, with  $\Phi_h = \text{id} + h$ ,

$$D\Phi_h(\theta) (\Upsilon + \mathfrak{c}) = \mathcal{X}(\Phi_h(\theta)). \quad (18.3)$$

Moreover:

- (i)  $v(h) \geq s_0 - \kappa\Upsilon$  and  $v(\mathfrak{c}) \geq s_0$ ;
- (ii) with  $B = \text{supp } \mathcal{V}$  and

$$P_0 = \bigcup_{j=1}^r (B - \gamma_j), \quad P = E \cup P_0, \quad M = P^*, \quad (18.4)$$

- these are positive generators in a well-ordered set, and  $\text{supp } h \subseteq M \setminus \{0\}$ ,  $\text{supp } \mathbf{c} \subseteq B + M$ ;
- (iii)  $\Phi_h$  has an infinitesimal Taylor inverse on  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ , and all nonzero integer divisors of  $\Upsilon + \mathbf{c}$  have the same leading real coefficients and the same valuations as those of  $\Upsilon$ ;
- (iv) if  $\mathcal{V}$  has real values coefficientwise on the real torus, then  $h$  and  $\mathbf{c}$  are real in the same sense, with  $\mathbf{c} \in F_\Gamma^d$ .

The case  $\mathcal{V} = 0$  is included, with  $h = \mathbf{c} = 0$ .

## 18.1 Reduction to a fixed-point equation

Taking the torus mean of (18.3) and using  $\text{av}(Dh) = 0$  gives

$$\mathbf{c} = \mathbf{c}(h) := \text{av}(\mathcal{V} \circ (\text{id} + h)), \quad (18.5)$$

and the remaining equation becomes

$$h = \mathcal{T}(h) := \mathcal{G}_\Gamma\left(Q_0(\mathcal{V} \circ (\text{id} + h)) - Dh \mathbf{c}(h)\right). \quad (18.6)$$

The argument of  $\mathcal{G}_\Gamma$  has zero mean, since  $\mathbf{c}(h)$  is constant and every column of  $Dh$  has zero mean, so the inverse is used only where it is defined.

All terms of  $B - \gamma_j$  are positive because  $s_0 > \kappa_\Gamma \geq \gamma_j$ . The sets in (18.4) are well ordered and lemma 3.4 applies. If  $h$  has positive support in  $M$  then lemma 16.3 gives  $\text{supp}(\mathcal{V} \circ (\text{id} + h)) \subseteq B + M$  and the same for  $\mathbf{c}(h)$ ;  $Dh \mathbf{c}(h)$  adds  $M$  once more; and applying an inverse coefficient at  $-\gamma_j + \mathbf{E}^*$  shifts these supports into  $B - \gamma_j + M \subseteq M$ . Every resulting term contains at least one letter from  $P_0$ , so no constant exponent appears. This proves preservation of a support certificate. Existence still requires proof that infinitely many iteration steps assemble into a Hahn series.

## 18.2 Existence by marked support words

*Existence in theorem 18.1.* Put  $h^{(0)} = 0$  and  $h^{(n+1)} = \mathcal{T}(h^{(n)})$ . Each iterate lies in  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d$ , has zero mean and positive support in  $M$ , its coefficients obtained by finitely many analytic operations at each exponent.

Call a generator in  $P_0$  a *marked* letter and a generator in  $\mathbf{E}$  *unmarked*; an exponent lying in both keeps both labels. There are only finitely many origins for a given letter —  $j$  ranges over finitely many values and the translation determines the original exponent of  $B$  or  $S$  — so the finite-word conclusion of lemma 3.4 is unchanged by the labels.

A term of  $\mathcal{T}(h)$  consists of a coefficient of  $\mathcal{V}$ , an inverse coefficient, and finitely many factors of  $h$  or its derivatives. The combined exponent of the  $\mathcal{V}$  coefficient and the leading shift  $-\gamma_j$  of that inverse supplies one marked letter  $\beta - \gamma_j$ ; the higher inverse terms supply unmarked letters in  $\mathbf{E}$ ; all other letters come from the factors of  $h$ . By induction, the difference

$$h^{(n+1)} - h^{(n)} \quad (18.7)$$

can have a nonzero coefficient only at an exponent representable by a word with at least  $n + 1$  marked letters. For  $n = 0$  there is at least one  $\mathcal{V}$  factor. For the induction step, subtract the two Taylor expressions and telescope the finite products at each coefficient: every difference term contains a factor from  $h^{(n)} - h^{(n-1)}$  or its derivative together with one additional  $\mathcal{V}$  coefficient. The term  $Dh \mathbf{c}(h)$  has the same property, by writing its difference as  $D(h - g)\mathbf{c}(h) + Dg(\mathbf{c}(h) - \mathbf{c}(g))$

and expanding the second difference by the same telescoping. Applying  $\mathcal{G}_\Upsilon$  creates one new marked shift and unmarked inverse-tail letters, exactly as above.

At a fixed exponent  $\gamma$ , [lemma 3.4](#) permits only finitely many labelled word representations, so their numbers of marked letters are bounded; hence (18.7) has zero coefficient at  $\gamma$  for all large  $n$ . The coefficients  $[t^\gamma]h^{(n)}$  therefore stabilize *exactly*, not approximately. The union of the difference supports lies in the well-ordered  $M$  and each exponent occurs in finitely many differences, so the differences form a Hahn-summable family in the sense of [definition 3.1](#); let  $h$  be their sum. Each stabilized coefficient lies in  $\mathcal{O}(\mathbb{T}_\rho^d)$ ; no new analytic limit has been taken. At a fixed exponent  $\mathcal{T}(h)$  uses only finitely many certified word decompositions and finitely many stabilized input coefficients, so passing to the stabilized coefficients gives  $h = \mathcal{T}(h)$ . Define  $\mathfrak{c}$  by (18.5); (18.6) and (18.5) give (18.3).

Since substitution does not lower valuation,  $v(\mathfrak{c}) \geq s_0$ ; the right side inside  $\mathcal{G}_\Upsilon$  has valuation at least  $s_0$  because  $Dh$  has positive valuation; and [proposition 16.9](#) or (17.3) gives  $v(h) \geq s_0 - \kappa_\Upsilon$ . The support statements are the construction.  $\square$

*Warning 18.2* (The word argument is the existence step, and the obvious substitute is invalid). The weaker estimate  $v(h^{(n+1)} - h^{(n)}) \geq (n+1)(s_0 - \kappa_\Upsilon)$  does **not** suffice in a general ordered group: that sequence of lower bounds need not be cofinal. With  $\Gamma = \mathbb{Z} + \mathbb{Z}\Omega$  and  $v(h) = 1$  the bounds never pass  $\Omega$ , so a sequential valuation argument would prove nothing at or beyond that exponent, and [example 18.9](#) exhibits a solution with coefficients exactly there. No Banach fixed-point theorem and no sequential completeness of  $\mathbf{No}[i]$  is used. This is the same refusal as [warning 3.6](#), reached independently by source 07 and implemented by a *labelled* refinement of the word certificate — counting marked letters — which is the one mechanism in this report that [corollary 3.5](#) does not already supply.

### 18.3 Uniqueness and the inverse coordinate map

*Uniqueness in [theorem 18.1](#).* Let  $h, g$  be positive-valuation mean-zero solutions with  $h \neq g$  and put  $\delta = v(h - g) > 0$ ; their drifts are forced by (18.5). By (16.4),  $v(\mathfrak{c}(h) - \mathfrak{c}(g)) \geq s_0 + \delta$ , and subtracting the expressions inside (18.6) gives valuation at least  $s_0 + \delta$ , since  $\mathfrak{c}(h)$  has valuation at least  $s_0$  and differentiation does not lower valuation. Hence  $v(\mathcal{T}(h) - \mathcal{T}(g)) \geq s_0 + \delta - \kappa_\Upsilon > \delta$ , contradicting  $h - g = \mathcal{T}(h) - \mathcal{T}(g)$ . Uniqueness holds among *all* positive-valuation mean-zero solutions, not only those carrying the constructed support certificate.  $\square$

**Lemma 18.3** (Infinitesimal coordinate inverses on the torus). *Every  $\Phi_h = \text{id} + h$  with  $v(h) > 0$  has a unique inverse  $\text{id} + g$  of positive valuation in the Taylor sense, inducing an automorphism of  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ .*

*Proof.* Solve  $g = -h \circ (\text{id} + g)$  by iteration from zero: the same marked-word stabilization, now marking each occurrence of a coefficient of  $h$ , gives coefficient stabilization inside the positive monoid generated by  $\text{supp } h$ . Uniqueness follows from  $v(h \circ (\text{id} + g_1) - h \circ (\text{id} + g_2)) \geq v(h) + v(g_1 - g_2) > v(g_1 - g_2)$ , which is (16.4). For two-sidedness write  $\Psi \circ \Phi_h = \text{id} + q$ ; associativity gives  $\Phi_h \circ (\text{id} + q) = \Phi_h$ , hence  $q + h \circ (\text{id} + q) - h = 0$ , and if  $q \neq 0$  the second difference has valuation strictly greater than  $v(q)$ , which is impossible. This is the torus counterpart of [lemma 4.2](#) and [proposition 4.1](#), proved here by the labelled word certificate rather than by the triangular recursion, because the unknown is a full Hahn-analytic coefficient family rather than a Taylor coefficient list.  $\square$

*Completion of theorem 18.1.* Lemma 18.3 gives the inverse assertion. Since  $v(\mathfrak{c}) \geq s_0 > \kappa_\Upsilon$ , corollary 17.6 applies and no leading coefficient of a nonzero integer divisor changes; for complex data the same leading real coefficients persist although some higher coefficients can be complex. Complex conjugation together with  $k \mapsto -k$  preserves all scalar Fourier inversions for a real frequency, and Taylor substitution, mean and differentiation preserve coefficientwise reality, so real  $\mathcal{V}$  gives real  $h, \mathfrak{c}$ ; alternatively reality follows from uniqueness.  $\square$

## 18.4 Necessity, and the sharp scale boundary

**Theorem 18.4** (Non-linearizable perturbations beyond any prescribed valuation). *Assume  $\gamma_1 = 0$ ,  $N_\Upsilon = \{0\}$  and  $\Gamma$  nontrivial. If  $(\text{SD}_{\text{ss}})$  fails, then for every prescribed  $H \in \Gamma$  there are  $s > H$ , a real  $\rho > 0$  and a real perturbation  $\mathcal{V} \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d$  with  $v(\mathcal{V}) = s$  such that  $\Upsilon + \mathcal{V}$  admits no positive-valuation mean-zero Hahn-analytic conjugacy to a constant vector field, even on a narrower positive-width strip. One may require  $s > 2\kappa_\Upsilon$  and take  $\mathcal{V}$  with a single Hahn exponent.*

*Proof.* Choose  $s > \max\{H, 2\kappa_\Upsilon\}$ , possible in a nontrivial ordered abelian group. Take the real  $f$  of (17.5) on a failing stratum  $\Sigma_j$  and put  $\mathcal{V} = t^s f e_1$ . Suppose  $\Phi_h = \text{id} + h$  were such a conjugacy on some strip; restrict to a common narrower strip if necessary. Taking means still forces  $\mathfrak{c} = \text{av}(\mathcal{V} \circ \Phi_h)$  with  $v(\mathfrak{c}) \geq s$ , and since  $v(h) > 0$  the equation  $\mathcal{C}_\Upsilon h = \mathcal{Q}_0(\mathcal{V} \circ \Phi_h) - D h \mathfrak{c}$  has right side of valuation at least  $s$ , so proposition 16.9 gives  $v(h) \geq s - \kappa_\Upsilon$ . Subtracting the uncomposed forcing, write  $\mathcal{C}_\Upsilon h = t^s f e_1 + R$  with  $R = \mathcal{Q}_0(\mathcal{V} \circ \Phi_h - \mathcal{V}) - D h \mathfrak{c}$ ; Taylor telescoping (16.4) gives  $v(R) \geq s + v(h) \geq 2s - \kappa_\Upsilon$ . At every nonzero Fourier mode the scalar equation therefore gives

$$\hat{h}(k) = \frac{t^s \hat{f}(k) e_1}{i k \cdot \Upsilon} + \frac{\hat{R}(k)}{i k \cdot \Upsilon}, \quad v\left(\frac{\hat{R}(k)}{i k \cdot \Upsilon}\right) \geq 2s - 2\kappa_\Upsilon > s.$$

For  $k = k_n$  in the failing stratum the coefficient of  $t^{s-\gamma_j}$  in the first component of  $h$  is therefore forced to be  $e^{-\varepsilon|k_n|}/(i \mathfrak{a}_j(k_n))$  — here  $s - \gamma_j \leq s$ , so the remainder cannot contribute — and those coefficients grow faster than  $e^{\varepsilon|k_n|}$  and cannot belong to a distribution, exactly as in the necessity proof of theorem 17.3.  $\square$

**Corollary 18.5** (Exact universal infinitesimal normal-form criterion). *For normalized  $\Upsilon$  with  $N_\Upsilon = \{0\}$  over nontrivial  $\Gamma$ ,  $(\text{SD}_{\text{ss}})$  is equivalent to the assertion that for every  $\rho > 0$  every  $\mathcal{V} \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))^d$  of valuation greater than  $\kappa_\Upsilon$  admits the normalized conjugacy of theorem 18.1. When  $(\text{SD}_{\text{ss}})$  fails, restricting perturbations to arbitrarily higher valuations does **not** rescue a universal theorem.*

**Proposition 18.6** (Failure exactly at the threshold, with no zero of the slow speed). *Let  $\kappa_\Upsilon > 0$  lie in  $\Gamma$  and take*

$$\Upsilon = (1, t^{\kappa_\Upsilon}), \quad \mathcal{V}(\theta_1, \theta_2) = (0, \tfrac{1}{2} t^{\kappa_\Upsilon} \cos \theta_2).$$

*Then  $\Upsilon$  satisfies  $(\text{SD}_{\text{ss}})$  and its last flag level is  $\kappa_\Upsilon$ , yet this perturbation of valuation exactly  $\kappa_\Upsilon$  admits no positive-valuation mean-zero conjugacy to a constant vector field.*

*Proof.* The two strata have leading forms  $k_1$  and  $k_2$ , so  $(\text{SD}_{\text{ss}})$  is immediate. Suppose a conjugacy existed. The mean equation forces  $\mathfrak{c}_1 = 0$ ; since substitution by a positive-valuation  $h$  does not change the coefficient at exponent  $\kappa_\Upsilon$  of  $\mathcal{V}$ , that coefficient is  $(0, \frac{1}{2} \cos \theta_2)$ , of mean zero, so  $v(\mathfrak{c}_2) > \kappa_\Upsilon$  (with  $\mathfrak{c}_2 = 0$  allowed). At exponent  $\kappa_\Upsilon$  in the second component of (18.3),  $t^{\kappa_\Upsilon} \partial_2 h_2$  contributes nothing because  $[t^0] h_2 = 0$ , and neither do the drift or  $D h \mathfrak{c}$ . What remains is  $\partial_1([t^{\kappa_\Upsilon}] h_2) = \frac{1}{2} \cos \theta_2$ ; averaging in  $\theta_1$  makes the left side zero and leaves the right side unchanged.  $\square$

The slow component of this field is  $t^{\kappa\Upsilon}(1 + \frac{1}{2}\cos\theta_2)$ , whose leading real factor is everywhere positive: the example does **not** rely on introducing a stationary point. It says that a perturbation of the *same* size as a slow frequency may require a coordinate correction of order one rather than an infinitesimal one. The strict threshold  $v(\mathcal{V}) > \kappa\Upsilon$  is therefore sharp *as a uniform sufficient bound*; particular perturbations at the boundary can of course still be linearizable.

## 18.5 A canonical invariant density

The claim here concerns coefficientwise integration of analytic functions on the torus. It is **not** a countably additive **No**-valued probability measure on arbitrary subsets, and not a real-time ergodic theorem; [warning 14.1](#) and [warning 15.15](#) record the same refusal elsewhere in this report.

**Lemma 18.7** (Formal change of variables). *For  $\Phi_h = \text{id} + h$  with  $v(h) > 0$  and every  $f \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$ ,*

$$\text{av}((f \circ \Phi_h) \det D\Phi_h) = \text{av}(f). \quad (18.8)$$

*Proof.* Introduce an ordinary formal auxiliary variable  $z$  and  $\Phi_z = \text{id} + zh$ . At each Hahn exponent Taylor substitution and the determinant produce a *polynomial* in  $z$ , there being finitely many support words at that exponent ([lemma 3.4](#)). With  $A = D\Phi_z$ , the finite determinant identities and the chain rule give

$$\frac{d}{dz}((f \circ \Phi_z) \det A) = \text{div}_\theta((f \circ \Phi_z) \text{adj}(A)h) : \quad (18.9)$$

use  $A \text{adj}(A) = (\det A)I$  for the derivative of  $f$ , the derivative-of-determinant formula for the  $Dh$  terms, and the elementary Piola identity — the remaining cofactor derivatives cancel in pairs because mixed partials of  $\Phi_z$  commute. Each coefficient on the right of (18.9) is a divergence of a periodic analytic vector field, hence of mean zero, so the mean on the left is independent of  $z$ ; evaluate the coefficient polynomials at  $z = 0$  and  $z = 1$ . There is no infinite integration in  $z$  and no ordinary convergence of a Hahn sum is assumed.  $\square$

**Theorem 18.8** (Invariant density and its uniqueness). *Let  $\mathcal{X}$ ,  $\Phi_h$  and  $\mathfrak{c}$  be as in [theorem 18.1](#) and put  $\Psi = \Phi_h^{-1}$ . Then*

$$\mu_{\mathcal{X}} = \det D\Psi \in 1 + \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))_{>0} \quad (18.10)$$

*is the unique density in  $\mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$  with  $\text{div}(\mu_{\mathcal{X}}\mathcal{X}) = 0$  and  $\text{av}(\mu_{\mathcal{X}}) = 1$ , and the  $K_\Gamma$ -linear functional*

$$I_{\mathcal{X}}(f) = \text{av}(f \circ \Phi_h) = \text{av}(f\mu_{\mathcal{X}}) \quad (18.11)$$

*satisfies  $I_{\mathcal{X}}(1) = 1$  and  $I_{\mathcal{X}}(\mathcal{X} \cdot \nabla f) = 0$ . For real data  $\mu_{\mathcal{X}}$  is real, with positive surcomplex values on the real torus, being one plus an infinitesimal.*

*Proof.* The inverse is infinitesimal, so its Jacobian determinant is one plus an element of positive valuation, and the chain rule gives  $(\mu_{\mathcal{X}} \circ \Phi_h) \det D\Phi_h = 1$ . Applying [lemma 18.7](#) to  $f\mu_{\mathcal{X}}$  gives the equality in (18.11), and  $f = 1$  gives the normalization. Conjugacy and the chain rule yield  $(\mathcal{X} \cdot \nabla f) \circ \Phi_h = (\Upsilon + \mathfrak{c}) \cdot \nabla(f \circ \Phi_h)$ , whose mean is zero; coefficientwise integration by parts against every Fourier character gives  $\text{div}(\mu_{\mathcal{X}}\mathcal{X}) = 0$ . For uniqueness let  $\tilde{\mu}$  be a normalized solution and set  $g = (\tilde{\mu} \circ \Phi_h) \det D\Phi_h$ ; the same Jacobian identities transport its divergence equation to  $(\Upsilon + \mathfrak{c}) \cdot \nabla g = 0$ , and since  $\Upsilon + \mathfrak{c}$  has no nonzero integer resonances ([theorem 18.1\(iii\)](#)), Fourier uniqueness makes  $g$  constant in  $\theta$ . Then (18.8) gives  $\text{av}(g) = \text{av}(\tilde{\mu}) = 1$ , so  $g = 1$  and  $\tilde{\mu} = \mu_{\mathcal{X}}$ . Reality and pointwise positivity follow from [theorem 18.1\(iv\)](#) and the positive-valuation form of (18.10).  $\square$

The identity  $I_{\mathcal{X}}(\mathcal{X} \cdot \nabla f) = 0$  is the precise invariance assertion. No real-time flow taking values continuously in the fine topology of  $\mathbf{No}[i]$  is constructed, and no ergodic theorem for such a flow is claimed.

## 18.6 Worked examples with genuinely different scales

**Example 18.9** (Three visible scales and a reciprocal support of order type  $\omega^2$ ). Take  $\Gamma = \mathbb{Z} + \mathbb{Z}\Omega$  with  $\Omega$  greater than every ordinary positive integer — the same use of the symbol  $\Omega$  as in [examples 3.7](#) and [12.3](#), an element of  $\Gamma$  and not a frequency ([warning 2.9](#)) — and

$$\Upsilon = (1, t, t^\Omega). \quad (18.12)$$

The flag is  $\mathbb{Z}^3 \supset \{k_1 = 0\} \supset \{k_1 = k_2 = 0\} \supset \{0\}$  with levels  $0, 1, \Omega$  and leading forms  $k_1, k_2, k_3$  on their strata. Each nonzero leading form is a nonzero integer, so its reciprocal has modulus at most one and  $(\text{SD}_{\text{ss}})$  holds with no delicate ordinary arithmetic at all. Nevertheless the mode  $k = (1, 1, 1)$  already has

$$\frac{1}{1+t+t^\Omega} = \sum_{n,m \geq 0} (-1)^{n+m} \binom{n+m}{n} t^{n\Omega+m}, \quad (18.13)$$

whose support has order type  $\omega^2$  — the ordinal  $\omega$ , not a frequency — since every exponent of the  $n = 0$  layer precedes every exponent of the  $n = 1$  layer, and so on. Formula [\(18.13\)](#) follows by grouping words according to their numbers of letters 1 and  $\Omega$ , and is confirmed by the finite recurrence  $C_{n,m} + C_{n-1,m} + C_{n,m-1} = \delta_{(n,m),(0,0)}$  for  $C_{n,m} = (-1)^{n+m} \binom{n+m}{n}$  (negative index meaning zero), which the verification program checks.

On the second stratum  $(k_2 t + k_3 t^\Omega)^{-1} = \frac{t^{-1}}{k_2} \sum_{n \geq 0} (-k_3/k_2)^n t^{n(\Omega-1)}$ , and on the last stratum the inverse is simply  $t^{-\Omega}/k_3$ : a large support coexists with a three-level divisor classification. The forcing  $f = t^{\Omega+1}(\sin \theta_1 + \sin \theta_2 + \sin \theta_3)$  has exact mean-zero solution

$$u = -t^{\Omega+1} \cos \theta_1 - t^\Omega \cos \theta_2 - t \cos \theta_3, \quad (18.14)$$

of valuation  $1 = v(f) - \Omega$ , so the last scale really does control the worst inverse loss ([corollary 17.4](#)). A vector perturbation of valuation  $\Omega + 1$  falls under [theorem 18.1](#) with a coordinate correction of valuation at least one. This is also the example that exposes the invalidity of a sequential contraction proof ([warning 18.2](#)): corrections of order  $1, 2, 3, \dots$  never pass  $\Omega$ , while the coefficient-stabilization proof constructs the exact transformation including its coefficients at and beyond that scale.

**Example 18.10** (Real arithmetic on the leading stratum, and an arbitrary tail). Take  $\Upsilon = (1, \sqrt{2}, t)$ . Its first stratum is all  $(p, q, r)$  with  $(p, q) \neq (0, 0)$ , with leading form  $p + \sqrt{2}q$ ; its second is  $(0, 0, r)$  with  $r \neq 0$ , at level one. Since  $|p + \sqrt{2}q|^{-1} = |p - \sqrt{2}q|/|p^2 - 2q^2| \leq |p| + \sqrt{2}|q|$  for  $(p, q) \neq (0, 0)$ , the first stratum is Diophantine and the second has the integer leading form  $r$ ; the universal conjugacy threshold is  $v(\mathcal{V}) > 1$ . One may now append an arbitrary set-supported real vector tail of valuation greater than one: it can have transfinite support and change every higher coefficient of the solution, yet by [corollary 17.6](#) it changes neither the arithmetic test nor the threshold. This separates ordinary approximation properties from surreal support complexity.

**Example 18.11** (An explicit slow-circle normal form). Let  $s > \kappa_\Upsilon > 0$  and consider  $\mathcal{X}(\theta_1, \theta_2) = (1, t^{\kappa_\Upsilon} + t^s \sin \theta_2)$ , and put  $\varrho = t^{s-\kappa_\Upsilon}$ . The mean-zero conjugacy has the form  $\Phi_h = (\theta_1, \theta_2 + h(\theta_2))$  and carries  $\mathcal{X}$  to  $(1, \nu)$ , where

$$(1 + h'(\theta))\nu = t^{\kappa_\Upsilon} + t^s \sin(\theta + h(\theta)), \quad (18.15)$$

and the new frequency is exactly

$$\nu = t^{\kappa\Upsilon} \sqrt{1 - \varrho^2} = t^{\kappa\Upsilon} \left( 1 - \frac{1}{2}\varrho^2 - \frac{1}{8}\varrho^4 - \frac{1}{16}\varrho^6 - \dots \right), \quad (18.16)$$

the square root being the Hahn binomial branch with constant term one. To see this, use the inverse coordinate in the second variable: the inverse equation gives  $\Psi'(\theta) = \nu t^{-\kappa\Upsilon} (1 + \varrho \sin \theta)^{-1}$ , whose mean is one, so  $t^{\kappa\Upsilon} / \nu = \text{av}((1 + \varrho \sin \theta)^{-1}) = \sum_{m \geq 0} \binom{2m}{m} \varrho^{2m} 4^{-m} = (1 - \varrho^2)^{-1/2}$ , using the geometric expansion, the vanishing of odd sine moments and  $\text{av}(\sin^{2m} \theta) = \binom{2m}{m} 4^{-m}$ . All of these are Hahn identities in the positive monomial  $\varrho$ . The first terms of the unique mean-zero  $h$  are

$$h(\theta) = -\varrho \cos \theta - \frac{\varrho^2}{4} \sin 2\theta + \varrho^3 \left( \frac{1}{12} \cos 3\theta - \frac{1}{4} \cos \theta \right) + O_H(\varrho^4), \quad (18.17)$$

where  $O_H(\varrho^4)$  means that the remaining integer powers of  $\varrho$  have exponent at least four and is **not** an assertion about convergence in a real parameter. In particular the lower bound  $v(h) \geq s - \kappa\Upsilon$  of [theorem 18.1\(i\)](#) is attained, and the first frequency drift occurs later, at valuation  $2s - \kappa\Upsilon$ . The invariant density of [theorem 18.8](#) is equally explicit,

$$\mu_{\mathcal{X}}(\theta_2) = \frac{\sqrt{1 - \varrho^2}}{1 + \varrho \sin \theta_2}, \quad (18.18)$$

with mean one, making  $\mu_{\mathcal{X}} \cdot (t^{\kappa\Upsilon} + t^s \sin \theta_2)$  independent of  $\theta_2$ . The verification program reconstructs  $h$  and  $\nu$  independently by exact Fourier arithmetic and checks (18.15) through degree ten in  $\varrho$ .

## 18.7 What this part does not reach

Three delimitations belong here rather than only in [section 25](#), because they bound the reading of every statement above.

First, all of it is a *coefficientwise Hahn* theory. A formal series  $\sum_{n \geq 0} t^n h_n(\theta)$  lies in  $\mathcal{H}_{\Gamma}(\mathcal{O}(\mathbb{T}_{\rho}^d))$  whenever every  $h_n$  is holomorphic on the same strip, *however fast*  $\|h_n\|_r$  grows — faster than every exponential in  $n$  is permitted — so substituting a nonzero real number for  $t$  need not be meaningful. Nothing here improves classical real analytic KAM or Brjuno arithmetic: those theorems must control Fourier decay and the accumulation of losses through infinitely many nonlinear steps simultaneously [17, 19], whereas (SD<sub>ss</sub>) only guarantees that each *individual* coefficient operation stays holomorphic, the Hahn support argument handling how the coefficients are assembled. This is the torus instance of **N55**, **N56** and [warning 25.1](#).

Second, the torus derivatives are external derivations annihilating  $K_{\Gamma}$  ([warning 3.2](#)); nothing above speaks to internal surreal differential equations, and source 07 says so in order to keep itself separate from the companion report on differential equations ([section 22.1](#)).

Third, the nonlinear theorem requires  $\mathbf{N}_{\Upsilon} = \{0\}$ . The linear theorem allows  $\mathbf{N}_{\Upsilon} \neq \{0\}$ , but the resonant nonlinear case is open, and source 07 warns explicitly that replacing the mean projection by  $\Pi_{\mathbf{N}_{\Upsilon}}$  is *not* justified, since the retained normal form is then no longer a constant vector field and creates additional terms in the conjugacy equation ([question 24.9](#)). The flag itself is finite but not automatically computable: detecting exact integer relations among arbitrary named reals may be unavailable, so existence of the flag is separated from decidability of its entries ([question 24.11](#)).

## 19 Finite-order multipliers and resonant splitting

Nonresonance fails for multipliers of finite order. There is nevertheless an exact normal form retaining precisely the resonant dynamics, with *no* small-divisor estimates anywhere (tag SD<sub>0</sub>).

Note that the hypothesis  $\Lambda^q = I$  here is the multiplier *itself* of finite order, which is neither  $\text{RES}_q$  — where  $\lambda$  is explicitly not a root of unity — nor mere MR ([warning 2.1](#)).

**Theorem 19.1** (Same-domain finite-order decomposition). *Let  $\Lambda$  be diagonal unitary with  $\Lambda^q = I$  for an integer  $q \geq 1$ , and let  $F = \Lambda z + f$  with  $f \in \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{D}_R^d))_+^d$  vanishing coefficientwise to order at least two. There exist a normalized positive coordinate change  $H$  and a positive vector field  $Y$  vanishing to order at least two, both on the same  $\mathbb{D}_R^d$ , with*

$$H \circ F \circ H^{-1} = \Lambda \circ \Phi_Y, \quad Y(\Lambda z) = \Lambda Y(z), \quad (19.1)$$

*the rotation and flow in (19.1) commuting. Every monomial  $z^\alpha e_j$  of  $Y$  satisfies  $\lambda^\alpha = \lambda_j$ . All coefficients have finite support-word certificates from  $\text{supp } f$ , and the construction commutes with workspace enlargement. Moreover*

$$F^{\circ q} = \text{id} \iff Y = 0 \iff F \text{ is positively conjugate to } \Lambda. \quad (19.2)$$

*The same statement holds for radius-free germs, entire coefficients, polynomials and formal coefficients.*

*Proof.* The reduction of  $P = F^{\circ q}$  is the identity, it fixes zero and has exact Jacobian  $I$  there, so  $X = \log C_P$  is a positive vector field whose coefficients vanish to order at least two ([theorem 4.3](#)). Since  $F$  commutes with  $P$ , its pullback commutes with  $\log C_P$ , so  $F$  preserves  $X$  and commutes with all its admissible ordinary flow times. Set  $A_f = F \circ \Phi_{-X/q}$ ; the factors commute, so  $A_f^{\circ q} = P \circ \Phi_{-X} = \text{id}$ , and its ordinary reduction is  $\Lambda$ . Define the finite average

$$H(z) = \frac{1}{q} \sum_{j=0}^{q-1} \Lambda^{-j} A_f^{\circ j}(z). \quad (19.3)$$

The leading term of every summand after multiplication by  $\Lambda^{-j}$  is  $z$ , so  $H = \text{id} +$  positive with value zero and derivative  $I$  at zero, with a positive inverse on the same coefficient domain; reindexing the finite sum and using  $A_f^q = \text{id}$ ,  $\Lambda^q = I$  gives  $H \circ A_f = \Lambda H$ . With  $(H_* X)(w) = dH(H^{-1}(w)) a(H^{-1}(w))$  for  $X = a \cdot \partial$ , Taylor substitution shows  $H_* X$  is positive of the same analytic category and that  $H \circ \Phi_X \circ H^{-1} = \Phi_{H_* X}$ , either by the coefficientwise chain rule or by conjugating pullback operators. Put  $Y = \frac{1}{q} H_* X$ : since  $A_f$  preserves  $X$ , the conjugated map  $\Lambda$  preserves  $Y$ , which is exactly  $Y(\Lambda z) = \Lambda Y(z)$  and gives the monomial resonance condition; and  $F = A_f \circ \Phi_{X/q}$  gives (19.1) with its commuting factors. The operations used are finite compositions with the ordinary rotation, positive logarithms and exponentials, substitution inverses and the finite sum (19.3), each preserving the original coefficient domain and the generated support monoid. Finally  $P = \text{id}$  iff  $X = 0$  iff  $Y = 0$ ; if  $Y = 0$  the displayed normal form is a conjugacy to  $\Lambda$ , and conversely such a conjugacy forces  $F^q = \text{id}$ .  $\square$

In one variable with  $\lambda$  of exact order  $q$ , the resonant vector field has the form  $Y(z) = \sum_{n \geq 2, n \equiv 1 \pmod{q}} a_n z^n \partial_z$ , interpreted coefficientwise when the  $a_n$  are themselves Hahn series; the coefficient *family*, rather than an unrestricted rearrangement in  $n$ , remains the defining object. For  $q = 1$  the theorem reduces to [theorem 4.3](#).

**Warning 19.2** (Credit and non-uniqueness). The finite-factor/flow splitting follows the classical semisimple–unipotent pattern [6], and neither the finite averaging idea nor that abstract pattern is claimed new; what is established here is that the entire construction is valid for arbitrary positive Hahn support and retains the same ordinary coefficient domain. Uniqueness of resonant normal forms is **not** claimed: further coordinate changes commuting with  $\Lambda$  can transform  $Y$ , and formula (19.3) specifies the particular normalized coordinate used. For resonant multipliers of *infinite* order no universal analytic Poincaré–Dulac statement is proved; see [question 24.3](#).

## 20 Critical scales: four different things that fail

Four sources each exhibit a scale at which something stops working, and the four failures are *not* the same failure. Collecting them is what prevents a reader from concluding that the report has one scale obstruction with four proofs.

**Proposition 20.1** (Quadratic scaling and the boundary of the halo theorem). *For every non-root-of-unity  $\lambda \in \mathbb{C}^*$  and every  $\eta > 0$ , the map  $F(z) = \lambda z + t^\eta z^2$  has a normalized conjugacy with polynomial Hahn coefficients*

$$H(z) = z + \sum_{n \geq 1} b_n t^{n\eta} z^{n+1}, \quad (20.1)$$

*which evaluates on every finite halo irrespective of the growth of the  $b_n$ . For every nonzero ordinary  $w$ , substituting  $z = t^{-\eta} w$  into the individual terms of (20.1) is not a strongly summable family. If  $\lambda$  is a non-Brjuno irrational rotation, the ordinary series  $w + \sum_{n \geq 1} b_n w^{n+1}$  has zero radius of convergence as well.*

*Proof.* Let  $L_\lambda(w) = w + \sum_{n \geq 1} b_n w^{n+1}$  be the unique formal solution of  $L_\lambda(\lambda w + w^2) = \lambda L_\lambda(w)$ , whose coefficients nonresonance gives recursively; substituting  $w = t^\eta z$  and dividing by  $t^\eta$  gives exactly (20.1) with  $H \circ F = \lambda H$ . Each Hahn coefficient is polynomial, the support is a subset of  $\{n\eta : n \geq 1\}$ , and evaluation at a finite point is justified by lemma 3.4 with no condition on the ordinary magnitudes. Infinitely many  $b_n$  are nonzero: otherwise  $L_\lambda$  would be a nonconstant polynomial and the degrees of  $L_\lambda \circ Q_\lambda$  and  $\lambda L_\lambda$  would be  $2 \deg L_\lambda$  and  $\deg L_\lambda$ . At the critical scale every nonzero summand becomes  $b_n t^{n\eta} (t^{-\eta} w)^{n+1} = t^{-\eta} b_n w^{n+1}$ , so infinitely many members share the Hahn exponent  $-\eta$ , violating definition 3.1: this is a failure of **local finiteness**, not of a real-valued norm estimate. Finally, if  $L_\lambda$  had a positive ordinary radius it would be a local holomorphic coordinate with derivative one, analytically linearizing  $Q_\lambda$ , contradicting Brjuno–Yoccoz for a non-Brjuno rotation.  $\square$

Even for a Brjuno rotation, ordinary convergence of the coefficient series would not by itself make the repeated-weight family a strong Hahn sum: it would be an additional ordinary analytic summation convention. The non-Brjuno example rules out even that local analytic reinterpretation of the formal linearizer. No assertion is made about sectorial or resurgent summation under additional hypotheses.

**Proposition 20.2** (A precise rescaled-germ obstruction). *For the same family,  $H_t(z) = t^{-1} h_\lambda(tz)$  with  $h_\lambda(w) = w + \sum_{n \geq 2} b_n w^n$  the ordinary normalized formal linearizer of  $p_\lambda(w) = \lambda w + w^2$ , the change of variable  $H_t$  and its inverse are well defined by Hahn sums on the full set*

$$\{z \in K_\Gamma : v(tz) > 0\} \quad (20.2)$$

*and give a conjugacy there. Rescaling to the critical scale  $z = t^{-1} w$  gives  $tH_t(t^{-1} w) = h_\lambda(w)$ , and this formal germ is the Taylor germ of a common-domain Hahn-holomorphic section at  $w = 0$  if and only if  $\text{rad}(h_\lambda) > 0$ .*

*Proof.*  $H_t(z) = t^{-1} h_\lambda(tz)$  is coefficientwise algebra, and  $tF_t(z) = p_\lambda(tz)$  then gives  $H_t \circ F_t = \lambda H_t$  by formal substitution. At any argument with  $tz \in \mathfrak{m}_\Gamma$  the ordinary formal series  $h_\lambda(tz)$  is Hahn summable — scalar coefficients of valuation zero, powers of a positive-support argument, and lemma 3.4 — as is its ordinary formal inverse, and both preserve the infinitesimal monad because their linear term is the identity and higher terms have strictly higher valuation; multiplying by

$t^{-1}$  preserves (20.2). For the last assertion, suppose a section  $A(w) = \sum_{\gamma \geq 0} t^\gamma a_\gamma(w)$  with all  $a_\gamma$  holomorphic on one  $\mathbb{D}_r$  has Taylor series  $h_\lambda(w)$  in  $K_\Gamma[[w]]$ : the coefficient at Hahn exponent zero has ordinary Taylor series  $h_\lambda$ , which must therefore converge at least on  $\mathbb{D}_r$ , all other coefficient germs having zero Taylor series. Conversely, if  $h_\lambda$  converges on a disk, the section with only exponent-zero coefficient  $h_\lambda$  is the desired one.  $\square$

For  $t = \omega^{-1}$  the domain (20.2) includes every finite surcomplex number and many infinite ones, and excludes exactly the critical scale where  $tz$  has nonzero ordinary standard part. This is consistent with the familiar non-Archimedean linearization phenomenon over a trivially valued coefficient field, and is not presented as a replacement for the ultrametric results of Lindahl [20]. Even when  $h_\lambda$  converges ordinarily, substituting a nonzero ordinary  $w$  directly into the displayed Hahn family is *not* a Hahn summation: infinitely many nonzero terms would occupy exponent zero. One must first use the separately justified ordinary holomorphic germ and then its canonical Taylor lift, and this is where an illegitimate exchange of ordinary with Hahn summation would occur.

**Warning 20.3** (The four critical-scale phenomena are distinct). (1) [Proposition 20.1](#) (source 03): at the critical scale *local finiteness* fails — infinitely many terms occupy one exponent — and this is what stops the finite-halo theorems from being all-scale theorems.

(2) [Theorem 13.22](#) (source 04): the sharp disk  $v(z) > \delta/q$  has **two** distinct failure mechanisms, local finiteness *at equality* and *well-ordering* strictly below ((13.28)), and the boundary nevertheless carries a legitimate coefficientwise continuation ([section 13.7](#)) — so here the scale is where one summation stops and a *different* legal operation begins.

(3) [Proposition 20.2](#) (source 02): the failure is at the level of *representability*, an exact iff between positive ordinary radius of the rescaled germ and existence of a common-domain section, plus the observation that a coefficientwise-entire section need not be globally analytic on all of  $\mathbf{No}[i]$  under every spatial rescaling.

(4) [Example 15.24](#) (source 06): the boundary polydisk carries *two* independent obstructions — a pole of an explicit rational normalizer, and a coordinate-independent *critical-point count* — so the failure is not a defect of one geometric-series proof, and [examples 15.25](#) and [15.26](#) separate well-ordering from finite fibers as independent requirements.

Sources 02 and 05 each add the further remark that the existence of a Hahn conjugacy does *not* imply ordinary holomorphic dependence on  $t$  near  $t = 0$ : for a divergent  $h_\lambda$  the formal dependence encoded in (20.1) need not converge as an ordinary two-variable series in  $(t, z)$ ; and source 05's [warning 14.27](#) gives the positive counterpart, one licensed passage to the scale  $\varepsilon^{-1}$ .

## 21 Worked examples and exact coefficient calculations

### 21.1 The quadratic Schröder recursion, once, in four normalizations

Four sources compute the coefficients of the normalized formal linearizer of an ordinary quadratic. They are one calculation; the translation between their normalizations is the following.

For the ordinary equation  $L_\lambda(\lambda w + w^2) = \lambda L_\lambda(w)$  with  $L_\lambda(w) = w + \sum_{n \geq 1} b_n w^{n+1}$ , the first coefficients are

$$\begin{aligned} b_1 &= -\frac{1}{\lambda(\lambda - 1)}, & b_2 &= \frac{2}{\lambda(\lambda - 1)^2(\lambda + 1)}, \\ b_3 &= -\frac{5\lambda^2 + 1}{\lambda^2(\lambda - 1)^3(\lambda + 1)(\lambda^2 + \lambda + 1)}, \end{aligned} \tag{21.1}$$

$$b_4 = \frac{2(7\lambda^3 + 2\lambda + 3)}{\lambda(\lambda - 1)^4(\lambda + 1)^2(\lambda^2 + 1)(\lambda^2 + \lambda + 1)},$$

obtained by equating successive powers of  $w$ , the denominators at stage  $n$  arising from  $\lambda^{n+1} - \lambda$ . In the shifted indexing  $h_\lambda(w) = w + \sum_{n \geq 2} \beta_n w^n$ , the same recursion reads

$$\beta_1 = 1, \quad \beta_n = -\frac{1}{\lambda^n - \lambda} \sum_{m=\lceil n/2 \rceil}^{n-1} \beta_m \binom{m}{n-m} \lambda^{2m-n} \quad (n \geq 2), \quad (21.2)$$

the binomial being the coefficient of  $w^n$  in  $(\lambda w + w^2)^m$ , with  $\beta_2 = -1/(\lambda^2 - \lambda)$  and  $\beta_3 = 2\lambda/((\lambda^2 - \lambda)(\lambda^3 - \lambda))$ ; in the normalization  $h(z) = \sum_{n \geq 1} b_n \varepsilon_0^{n-1} z^n$  of [section 15.9](#) it reads [\(15.33\)](#). Multiplying  $b_n z^{n+1}$  by  $t^{n\eta}$  turns these ordinary formal data into the strongly summable surcomplex linearizer of [\(20.1\)](#), *even when the ordinary series has zero radius* — which is the whole point of [proposition 20.1](#). None of these is a convergence assertion: they are coefficient identities.

## 21.2 The iterative logarithm of a quadratic perturbation

Take  $\Gamma = \mathbb{Q}$ , write  $t = t^1$ , and consider  $F(z) = z + tz^2$  with entire polynomial coefficients. Directly from [\(4.4\)](#),

$$X_F = \left( tz^2 - t^2 z^3 + \frac{3}{2} t^3 z^4 - \frac{8}{3} t^4 z^5 + \frac{31}{6} t^5 z^6 - \frac{157}{15} t^6 z^7 + O(t^7) \right) \partial_z, \quad (21.3)$$

where  $O(t^7)$  refers only to this single-parameter formal expansion. Exponentiating half of this vector field produces the unique positive half-iterate of  $F$ , with polynomial coefficients at every Hahn weight, and the ordinary size and growth of the rational coefficients in [\(21.3\)](#) have no bearing on strong summability in  $t$ . For a general map, with a bookkeeping parameter  $\varrho$  and  $F_\varrho = z + \varrho u$ , expanding the logarithm through degree three gives

$$X_{F_\varrho} z = \varrho u - \frac{\varrho^2}{2} (Du)u + \varrho^3 \left( \frac{1}{12} D^2 u(u, u) + \frac{1}{3} (Du)^2 u \right) + O(\varrho^4), \quad (21.4)$$

since  $\Delta^2 z = \varrho^2 (Du)u + \frac{1}{2} \varrho^3 D^2 u(u, u) + O(\varrho^4)$  and  $\Delta^3 z = \varrho^3 (D^2 u(u, u) + (Du)^2 u) + O(\varrho^4)$ ; with  $u = z^2$  this gives the first three terms of [\(21.3\)](#). [Equation \(21.3\)](#) is [\(14.31\)](#) with  $\varepsilon \mapsto t$  and  $p = 1$ ; it is the same calculation, recorded once in each of the two notations its sources use and identified here ([warning 14.22](#)).

## 21.3 A complete centralizer and a persistent double fixed point

For  $X = tz^2 \partial_z$  the flow has the especially simple expression

$$\Phi_c(z) = \frac{z}{1 - ctz} = z + \sum_{n \geq 1} (ct)^n z^{n+1}, \quad c = 0 \text{ or } v(c) > -1, \quad (21.5)$$

the last series being a strongly summable expansion on finite halos. The time-one map here is  $z/(1 - tz)$ , *not* the quadratic map of [section 21.2](#). By [theorem 4.12](#) the full positive centralizer of this time-one map on the entire-coefficient halo consists exactly of [\(21.5\)](#), and for every nonzero admissible time  $\Phi_c(z) - z = ctz^2/(1 - ctz)$ , whose factor  $ct/(1 - ctz)$  is a unit in the Hahn coefficient algebra. Hence the fixed-point ideal is *exactly*  $(z^2)$  for every such time: the point at zero retains its scheme-theoretic double structure, not merely its location — which is what [theorem 4.10](#) buys over a zero-set statement ([remark 4.11](#)).

## 21.4 Two independent perturbation scales, and a resonant splitting

Let  $t$  and  $u$  represent  $t^1$  and  $t^\Omega$  in  $\Gamma = \mathbb{Q} + \mathbb{Q}\Omega$ , and consider the map with multiplier  $-1$ ,  $F(z) = -z + tz^2 + uz^3$ , illustrating [theorem 19.1](#) with  $q = 2$ . The averaging coordinate and the resonant vector field begin

$$\begin{aligned} H(z) &= z - \frac{t}{2}z^2 + \frac{t^2}{2}z^3 - \frac{5tu}{4}z^4 + O((t, u)^3), \\ Y(z) &= \left(-uz^3 - t^2z^3 - \frac{3}{2}u^2z^5 - \frac{25}{4}t^2uz^5 - \frac{7}{2}u^3z^7 + O((t, u)^4)\right)\partial_z, \end{aligned} \quad (21.6)$$

and  $HF = (-\Phi_Y)H$ . Every displayed monomial of  $Y$  has odd coordinate degree, exactly as  $Y(-z) = -Y(z)$  requires.

*Warning 21.1* ( $O((t, u)^N)$  is a label cutoff, not a valuation cutoff). In [\(21.6\)](#),  $O((t, u)^N)$  means discarded monomials of *total parameter degree* at least  $N$  in an auxiliary two-parameter jet algebra. It is **not** a valuation cutoff: for example  $t^{100}$  has much smaller Hahn valuation than  $u$ , although its total parameter degree is larger. The same caution applies to  $O_{\text{lab}}(\cdot)$  in [section 15.8](#) and to *every* verification program in [section 23](#), all of which truncate by parameter or label degree. Sources 03 and 06 each state this independently, and source 06 adds the sharpest form: if  $\theta > n\eta$  for every  $n$ , infinitely many powers of  $u$  lie below the first  $v$  layer, and no finite label truncation pretends to compute that entire valuation initial segment.

The verification program of source 03 also uses the two-dimensional identity-reducing map  $G(x, y) = (x + tx^2 + uy, y + txy + ux)$ , checking at total parameter depth three the logarithm, the inverse, the exponential, the Leibniz identity and the fixed-point factorization [\(4.11\)](#). That example includes infinitesimal *linear* terms, which are permitted in the identity-reducing flow theorem [theorem 4.3](#) even though they are excluded from the exact-multiplier linearization theorem [theorem 6.1](#) — an instance of the tag distinction of [section 2.3](#) showing up in a computation.

## 21.5 An arbitrary-support example beyond a finite grid

Use  $S$  and  $\Gamma$  from [example 3.7](#) and let  $F = \Lambda z + \sum_{\sigma \in S} t^\sigma f_\sigma(z)$  with each  $f_\sigma$  vanishing to order at least two. Under a valid analytic hypothesis from [theorem 6.1](#), the higher-rank coefficients obey

$$\begin{aligned} h_\Omega &= -\mathcal{L}^{-1}f_\Omega, \\ h_{\Omega+1} &= \mathcal{L}^{-1}\left(D(\mathcal{L}^{-1}f_\Omega)(\Lambda z)f_1(z) + D(\mathcal{L}^{-1}f_1)(\Lambda z)f_\Omega(z)\right). \end{aligned} \quad (21.7)$$

Only the word  $(\Omega)$  contributes in the first line, and only  $(\Omega, 1)$  and  $(1, \Omega)$  can contribute in the second. Infinitely many input coefficients at rational weights below  $\Omega$  enter neither formula. This is a direct use of finite word dependence ([corollary 3.5](#)), not a rank-one approximation to the value group.

## 22 Relation to the companion reports in this collection

Each entry names the report by title and repository path. Companion reports are cited this way and never by their internal labels.

## 22.1 Extends: differential equations over the surreals

*Differential Algebra and Differential Equations over the Surreal and Surcomplex Numbers* (docs/surcomplex/differential-equations/) already contains, in its section on monodromy as an ordinary-domain construction, the phenomenon that [section 13](#) builds on. Its example takes the scalar equation  $\partial_z Y = ((\lambda + \varepsilon)/z)Y$ , obtains the monodromy  $M = e^{2\pi i \lambda} \text{Exp}(2\pi i \varepsilon)$ , and concludes that for  $\lambda \in \mathbb{Z}$  the *ordinary* monodromy is 1 while the full monodromy is not, with leading change exactly  $2\pi i \varepsilon$ : reduction to the ordinary coefficient can erase a genuine global obstruction.

**Inheritance.** The two reports stay separate — fundamental matrices of linear coherent ODEs are not conjugacies of iterated maps, and the algebraicity dichotomy [theorem 13.15](#) over  $\mathbb{C}(w)((t^\Gamma))$  has no counterpart there. But that example is *the* antecedent of the monodromy exponent  $\rho_a$  of [\(13.15\)](#), and is cited as such at the point where  $\rho_a$  is introduced. What is new here is not the invisibility phenomenon but the package around it: the torsion lemma [lemma 13.10](#), the finite-cover obstruction [theorem 13.11](#), the classification of algebraic leading coordinates [proposition 13.17](#) with its already-transcendental first correction, and the dichotomy itself. That report’s cross-category warning is also adopted verbatim, once, as [warning 3.2](#), in place of re-deriving the coordinate-versus-surreal-derivation distinction seven times.

**Source 07 separates itself from that report by name.** Its torus derivatives  $\partial/\partial\theta_j$  are external derivations annihilating  $K_\Gamma$ , and it says in as many words that this convention “keeps the present theory separate from the repository’s internal differential-equation report”. The cohomological equation [\(17.2\)](#) is therefore not a surreal differential equation for the Berarducci–Mantova derivation, and no theorem of either report transfers to the other. That is the same boundary [warning 3.2](#) draws for the germ half, stated once and applying to both. Source 07 also records that a targeted repository code search for “cohomological” returned no match, while stating that this is not an exhaustive audit of every manuscript or every synonym.

## 22.2 Adjacent: surcomplex analysis

*Surcomplex analysis: holomorphic functions on  $K = \mathbf{No}[i]$*  (docs/surcomplex/analysis/) is the foundation that fixes the no-topological-convergence discipline and states the Neumann support material. All seven sources restate that discipline and all seven state the support lemma, with sources 02, 03 and 05 each reproducing a minimal-bad-sequence Higman proof.

**Inheritance.** Inherited, not restated. Exactly one proof of the support lemma is kept — [lemma 3.4](#) with [section A](#), the cleanest finite-word form — and the summability discipline is cited to that report in [definition 3.1](#). The other five statements are deleted. This is also what keeps the present report honest about which parts of its machinery are classical.

## 22.3 Adjacent: infinitesimal analytic geometry

*Infinitesimal Analytic Geometry over the Surcomplex Numbers* (docs/surcomplex/analytic-geometry/) owns the four-ring table and its separating witness. Sources 02, 03 and 04 each re-erect that separation from scratch, with source 02’s  $\sum_{m \geq 1} t^m / (1 - mz)$  and source 04’s  $\sum_{n \geq 1} t^{n\delta} / (1 - nw)$  being that report’s witness with  $\eta = 1$  and  $\eta = \delta$ .

**Inheritance.** Neither the ring table nor the witness is reprinted. Every theorem here states its ring in that report’s names ([convention 3.3](#)), and exactly two rows are added — entire coefficients and polynomial coefficients — with the explicit note that they are new *rows*, not new rings ([section 3.2](#)). Several sources already say they are continuing that distinction; this merge makes the continuation

literal.

## 22.4 Adjacent: contours and Stokes

*Contours and Stokes: Jordan separation, Cauchy theory, and Stokes calculus on the surcomplex plane* (`docs/surcomplex/contours-and-stokes/`) is background for [section 14](#) and is deliberately *not* used there: every integral in that section is an ordinary complex line integral taken coefficient by coefficient over ordinary curves in a fixed  $D \subset \mathbb{C}$ . [Warning 14.1](#) states the resulting prohibition, which source 05 carries as a remark of its own.

## 22.5 Keep separate, with a cross-note: rank-one Berkovich geometry

*Rank-One Non-Archimedean Analytic Geometry inside the Surcomplex Numbers* (`docs/surcomplex/rank-one-berkovich/`) is the one report in the collection where convergence is genuine convergence: it works inside a *fixed rank-one*  $\mathbb{C}((t^{\mathbb{R}}))$  with  $|a| = \exp(-v(a))$  and spherical completeness, where the absolute value is a real-valued one and the field is complete and algebraically closed.

**Cross-note, and it is load-bearing.** This report forbids that reading throughout. It works at arbitrary valuation rank, with a deliberately rank-two example ([example 12.3](#)), and cites Lindahl’s rank-one non-Archimedean linearization as an antecedent it does not reprove. A Lindahl-type ultrametric disk radius and source 01’s maximal centered valuation ball are *different categories* ([warning 2.4](#)), and the exact-ball and shell-periodicity conclusions of [theorems 8.5](#) and [8.7](#) must **not** be transported to  $\mathbb{C}_p$ : source 01 says so itself and cites  $p$ -adic quadratic-like examples whose bounding sphere carries no periodic point ([warning 2.7](#)). Without this note a reader will read the two reports’ “disks” as the same object. No result of this report and no result of that one is transferred in either direction.

## 23 Computational record and reproducibility

The seven verification programs are preserved unmodified in `code/` and their recorded outputs in `data/`. They supply *finite exact symbolic* checks. None of them is a proof of any infinite statement in this report, and no Lean or other proof-assistant development corresponds to any theorem here.

### 23.1 What each suite checks, and what it does not

| Source | Checks | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Explicit disclaimer                                                                                                                                                                                                                                                                                                                                |
|--------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01     | 54     | Nine coefficients of the general cubic second iterate against a closed form; the tuned factorization (12.3) and the degree-3/degree-4 cancellation at $e = 0$ ; the three phase regimes with <i>integer</i> exponent representatives $\delta = 8$ , $\sigma_0 = 2, 4, 6$ (radii $3, 2, 2$ ), their thresholds, shell polynomials and squarefreeness; multiplier ratios by polynomial remainder; the three coherent leading coefficients through their <i>logarithmic derivatives</i> (deliberately avoiding a numeric branch choice of fractional powers); Schröder recurrence to degree 9 for $\lambda = 3/2, -4/3, 5/4, 1 + i$ ; the tangent binomial identity (12.1) for $m = 2..7$ ; $\text{Disc}_Y(1 + \kappa Y - 2Y^2) = \kappa^2 + 8$ ; integrality and residues of $h_2..h_9$ for the normalized tuned cubic at $e = t^4$ , $r = 1$ , including $[X^5]H_0 = 1/2$ and $[X^9]H_0 = 5/8$ . All exact/symbolic, no floating point. | Does not establish the arbitrary-rank support lemma, algebraic closure, maximality for every polynomial, the transfinite coherent lifting, or novelty; not an implementation of general Hahn fields; not proof-assistant verification.                                                                                                             |
| 02     | 1,109  | Reciprocal-jet formula (7.4) versus the inverse recurrence (133) and truncated product of a denominator with its reciprocal (133), both for $2 \leq n \leq 20$ , $0 \leq k \leq 6$ ; ordinary quadratic Schröder identity through degree 12 (13); Hahn-scaled quadratic identity (156); nonlinear recursion with multiplier drift and cubic forcing, $F(z) = (\lambda + t)z + t(z^2 + z^3)$ , checked after each parameter step through $t^4$ and spatial degree ten (154); two-variable multiplicative-detuning reciprocal jets with exact unit phases $(3 + 4i)/5$ and $(5 + 12i)/13$ , total degrees 2..9, both components, jet orders 0..4 (520). Exact in $\mathbb{Q}(i)$ via <code>fractions.Fraction</code> ; test multiplier $\lambda = 3/5 + 4i/5$ , not a root of unity because a root of unity in $\mathbb{Q}(i)$ is an algebraic integer.                                                                                  | Tests algebraic identities, signs, normalizations and truncation only. Does not prove Hahn summability, well-founded recursion, domain preservation, the small-divisor limsup, the exact radii, or novelty. The multivariable test verifies nonzero denominators only in its finite range and does not certify nonresonance for all multi-indices. |

| Source | Checks | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Explicit disclaimer                                                                                                                                                                                                                                                                                                                    |
|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03     | 21     | <p><math>\exp(\log F) = F</math> and two-sided inversion; half and third iterates; time addition; Leibniz; identification of the logarithm with its coordinate vector field; the two-variable fixed-point matrix factorization (4.11); normalized nonresonant conjugacy, its inverse and the degree bound (6.13); the finite-order averaging and resonant flow identities. One-parameter flow checks through degree six; two-parameter multivariable, nonresonant and resonant checks through total parameter degree three. Sparse exact jet algebra over <math>\mathbb{Q}(i)</math> with a symbolic <math>\lambda</math> for the four coefficients (21.1).</p>                                                                                                                                                                                                                                  | <p>Finite polynomial jet identities truncated by <i>total parameter degree</i>, explicitly not a Hahn valuation cutoff in a higher-rank group (warning 21.1). Does not test arbitrary well-ordered supports, analytic radius claims, continued-fraction asymptotics, the universal quantifiers, or priority. No Lean verification.</p> |
| 04     | 2,764  | <p>Koenigs recurrence (13.24) and the leading Pochhammer coefficients for <math>q = 1.8</math>, <math>m \leq 24</math>, through <math>\varepsilon^8</math>; the <i>directly substituted</i> Schröder residual through <math>\varepsilon^9</math> (an independent test, not merely agreement with a displayed formula); the first and second coefficient functions through <math>w^{24q+1}</math>; the logarithm ratio <math>\rho_q</math> of (13.30) at orders 0, 1, 2 for <math>q = 1.8</math>; the derivation identity <math>\log C_g(p) = (\log C_g(w))p'</math> for <math>q = 1.6</math>, <math>p = w^n</math>, <math>0 \leq n \leq 6</math>, through <math>\varepsilon^6</math>; generator coefficients and fixed-point derivatives, reduced using <math>w^q = -1</math> to avoid approximate complex roots. Standard library only, <code>fractions.Fraction</code>, no floating point.</p> | <p>No finite test proves Neumann summability, existence on an arbitrary ordinary domain, compatibility for arbitrary value groups, or the algebraicity dichotomy. Not a symbolic surcomplex CAS and not a numerical evaluator at arbitrary surreal inputs.</p>                                                                         |

| Source | Checks | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Explicit disclaimer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05     | 245    | Euler families $p = 1, 2, 3$ : exp of log, the Julia identity <a href="#">corollary 4.6</a> in the form $W_F(F) = F'W_F$ , inverse via negative time, the homogeneous grading coefficientwise, residues equal $(p + 1)/2$ at degree one and zero elsewhere; a mixed-coefficient map (round trip, log-equals-vector-derivation, Leibniz, Bernoulli difference solver); the flow group law (times 2 and 3 compose to 5); the unique-root formula; an explicit conjugacy with pushforward of the vector field, pullback of the normalized time form and the basepoint marking $h_k(1) = 0$ ; untruncated symbolic identities for the flat family ( $1/W_c = 1/\varepsilon + c/z$ , residue $c$ , leading correction $-1/z$ ); and 50 lexicographic order comparisons. Exact in $\mathbb{Q}(z)[\varepsilon]/(\varepsilon^8)$ . | Transcription and algebra checking only. Proves nothing about arbitrary-rank support finiteness, global primitive existence, the classification, or the slow-time theorem. <b>Fifty of the 245 assertions are finite lexicographic illustrations for <math>n = 1, \dots, 50</math>; the all-<math>n</math> inequality is proved in the text, not inferred from them.</b> The verifier also loses one top coefficient on division by $\varepsilon$ , so reciprocal and pullback comparisons are checked only through degree $N - 1$ . |
| 06     | 119    | Exact arithmetic in $\mathbb{Q}(\sqrt{2})$ , with a hand-rolled bivariate label series ring truncated at total label degree four: the single-exponential gauge recursion for <a href="#">(15.28)</a> ; resonance of every normal coefficient and the nonresonant gauge of every generator coefficient; the four displayed coefficients; all 16 symplectic coordinate brackets; four inverse Lie maps; direct substitution $H(\Phi) = N$ and $J(\Phi) = I$ ; the two first integrals, their involution, and one nonlinear function of the actions being an integral; 51 degree-defect identities; the cubic boundary map <a href="#">(15.27)</a> ; the support counterexample's finite-sample order pattern; and <a href="#">(15.33)</a> for $\lambda = 2$ through $n = 10$ .                                               | Truncates by <i>total label degree</i> in two independent labels, not by Hahn valuation, and only to degree four. Explicitly not a proof of arbitrary-support summability, nor of <a href="#">theorems 15.5 and 15.16</a> , nor of <a href="#">theorem 15.12</a> , nor of the Liouville construction <a href="#">proposition 15.17</a> .                                                                                                                                                                                             |

| Source | Checks | Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Explicit disclaimer                                                                                                                                                                                                                                                                                                            |
|--------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 07     | —      | <p>No printed total; two named checks, every assertion exact. The slow-circle conjugacy <a href="#">example 18.11</a> is reconstructed recursively from <math>(1 + h')\nu = 1 + \varrho \sin(\theta + h)</math> with <math>\nu(0) = 1</math>: at each order the mean is determined first, giving the new coefficient of <math>\nu</math>, then the remaining zero-mean Fourier polynomial is integrated. It asserts at every order that the mean is real and that <math>h</math> is Hermitian, that the full conjugacy residual vanishes through degree ten, that <math>\nu^2 = 1 - \varrho^2</math> coefficientwise through the same degree, and that the three displayed terms of <a href="#">(18.17)</a> are correct; the recorded frequency coefficients are <math>1, 0, -\frac{1}{2}, 0, -\frac{1}{8}, 0, -\frac{1}{16}, 0, -\frac{5}{128}, 0, -\frac{7}{256}</math>. It then checks 117 instances (<math>n \in 0..8</math>, <math>m \in 0..12</math>) of the coefficient recurrence for <a href="#">(18.13)</a>, and classifies all 1,330 nonzero integer modes in <math>[-5, 5]^3</math> into the three strata of <a href="#">example 18.9</a>, obtaining 1,210/110/10. Exact <math>\mathbb{Q}(i)</math> arithmetic by pairs of <code>fractions.Fraction</code>; every check is an <code>assert</code>; no floating point and no tolerance anywhere.</p> | <p>Finite algebraic checks only. They verify neither the irrationality of <a href="#">(17.6)</a>, nor any infinite-mode subexponential estimate, nor the word lemma, nor the infinite-support conjugacy theorem; those rest on the proofs in the text. No Lean or Wolfram verification of any infinite theorem is claimed.</p> |

## 23.2 Independently verified facts about the suites

The following were checked directly against the delivered files and by running the code, and override any contrary statement in a source README.

- (1) **All seven suites pass.** Run on Python 3.14.4 with SymPy 1.14.0 where SymPy is used, every one exits 0. Reported totals where a script prints one: 54 (01), 1,109 (02), 21 (03), 2,764 (04), 245 (05), 119 (06); source 07 prints no total. Source 04's count reproduced the recorded 2,764 exactly; source 05 reproduced its recorded transcript line for line; source 03 reproduced its four recorded example expansions byte for byte; and the regenerated JSON of sources 06 and 07 matched the shipped ones up to line endings only.
- (2) **Four of the seven scripts rewrite their own evidence.** Source 03's `verify.py` writes `data/verification.json` by default; source 04's writes `data/verification.json` and `data/coefficients.csv`; source 06's writes `data/verification.json`, and does so only after all checks pass; source 07's takes `--output` with `default verification.json` resolved against the current working directory and writes it with an unconditional `write_text` — no existence check, no backup, no dry run — and the invocation documented in its own README is exactly the one that overwrites the delivered record. Source 01's script prints to stdout, but its `Makefile`'s `verify` target redirects that stdout into `verification.txt` in the same directory,

overwriting the delivered record. **Every suite must therefore be run on a copy, never in the delivered tree:** a byte-identity check performed after an in-place run compares two equally modified copies and passes while proving nothing.

- (3) **Every suite prints its own scope disclaimer.** That is a real and unusual discipline in this material, and those disclaimers are exactly the limitations carried in [section 25](#).
- (4) Source 04's README lists `SHA256SUMS.txt` among its package contents, and no such file exists in the delivered archive. This is a packaging gap, recorded rather than repaired.
- (5) Source 03's recorded console transcript names its report path as `/mnt/data/surcomplex_dynamics/data/verification.json`, i.e. it was recorded in a different container from the delivered one. The check contents match the article's prose regardless.
- (6) Source 07's README likewise lists `SHA256SUMS.txt` among its package contents, and no such file exists in the delivered archive. Like item (4) this is recorded rather than repaired.
- (7) **The finiteness bound of [theorem 16.5](#) was checked independently of source 07.** Modelling  $\Upsilon_i = \sum_j c_{ij} t^{\gamma_j}$  with  $\gamma_1 < \gamma_2 < \dots$  gives  $k \cdot \Upsilon = \sum_j L_j(k) t^{\gamma_j}$  with  $L_j(k) = \sum_i k_i c_{ij}$ , so  $v(k \cdot \Upsilon)$  is the least  $\gamma_j$  with  $L_j(k) \neq 0$  and  $\{k : \text{level} \geq \gamma_{m+1}\} = \ker L_1 \cap \dots \cap \ker L_m$ , whose saturation has rank  $d - \text{rank}(L_1, \dots, L_m)$ . A new level appears exactly when that rank increases, so the kernel rank strictly drops, and it can drop at most  $d$  times: that is both the bound and the flag. Brute-force testing over rational coefficient matrices with deliberate resonances forced in 40 per cent of the cases gave **zero violations of the bound in 400 random trials**, and a worked instance at  $d = 3$  had kernel ranks  $2 \rightarrow 1 \rightarrow 0$  with achieved levels  $\{0, 1, 3\}$  — three levels, attaining the bound, with two indices skipped exactly where the rank did not move ([remark 16.6](#)). The theorem and its stated mechanism both hold. This is an independent check of a finite mechanism, not a proof of any infinite statement.

### 23.3 Building this report

The report is standalone LaTeX with an internal bibliography and no external `.bib` file, graphics, shell escape or network access:

```
latexmk -pdf -interaction=nonstopmode article.tex
```

Reproducing the seven suites, each *on a copy*, requires Python 3.10 or later and, for four of them, SymPy; the pinned versions recorded by the sources are `sympy==1.14.0`.

## 24 Open questions

The questions below are exactly those the sources leave open, with the categories in which they are open stated explicitly.

**Question 24.1** (Common-domain germs at a finite positive rate, for a fixed multiplier). This is [question 7.10](#), and [warning 7.11](#) is its status: settled negatively in the *drifted* category  $\text{SD}_{\text{dr}}$  by [theorems 7.2](#) and [7.5](#), and **open** in the exact-multiplier category  $\text{SD}_{\text{ex}}$ . An exact characterization would need to control interactions across unbounded word depths rather than a single homological inverse.

**Question 24.2** (The fixed cyclic value group). The negative direction of [theorem 7.2](#) is established over  $\Gamma_* = \mathbb{Q} + \sqrt{2}\mathbb{Q}$ , hence for the universal class of workspaces, and it embeds in any surcomplex workspace containing that ordered subgroup. It is *not* asserted for a fixed cyclic value group such

as  $\mathbb{Z}$ , where different parameter blocks can collide at one Hahn exponent. Resolving the effect of those collisions is a separate problem, not a consequence of the two-independent-exponent proof, and it concerns the existence of *some* smaller common neighborhood for every input, not the weaker statement about retaining the original disk.

**Question 24.3** (Resonant multipliers of infinite order). For a diagonal unitary multiplier with resonances but *without* finite order, find a support-controlled analytic normal form with precise necessary and sufficient coefficient-category hypotheses. Nonresonant monomials can be treated by the homological inverse, but resonant terms create a coupled normal-form problem and remove the global inverse used in (6.5), and a finite cyclic averaging operator is no longer available. The Hamiltonian analogue is open too: for resonant  $\omega$  the first reduction retains a nontrivial resonant Hamiltonian algebra, the gauge equation changes because  $L^{-1}$  has a larger kernel, and whether the dynamics of that retained algebra can itself be integrated under effective support conditions is not answered by the nonresonant centralizer proof.

**Question 24.4** (Higher-dimensional centralizers). In several variables, classification of the positive centralizer reduces exactly to classification of the commuting positive vector fields, and the one-variable meromorphic-ratio argument of [theorem 4.12](#) cannot classify those higher-dimensional Lie centralizers; additional first integrals or commuting geometric structures can occur. Independently, source 05 records that in several variables there is no single reciprocal time one-form encoding a vector field and commuting fields need not be constant multiples, so the period normal form does not transport either.

**Question 24.5** (Moving fixed points, and several variables, for the parabolic theory). Characterize the monodromy of  $w + \varepsilon V(w) + \varepsilon^2 R(w)$  when its fixed points move. An analogue of the local derivative identity [lemma 4.5](#) at moving formal fixed points is a natural intermediate target; translating it into a common-domain continuation theorem would also require controlling those moving singularities. The Euler family *avoids* that difficulty rather than concealing it. In several variables the logarithm-of-substitution construction still suggests a vector field, but the scalar quotient  $\alpha_c/W$  has *no* direct replacement, and resonant homological equations with noncommuting monodromy would replace the one-dimensional integration argument.

**Question 24.6** (The period classification where the leading coefficient vanishes). When  $a$  has zeros, [theorem 14.5](#) still controls the discrete equation, but the conjugacy classification needs more: local zero divisors, multiplicities and singular normal-form obstructions must be combined with global periods. [Warning 14.12](#) identifies the precise point where the proof stops. Separately, extending [theorem 14.13](#) to conjugators with nonidentity ordinary part requires the ordinary automorphism action on the leading form and on its periods, after which equal period vectors in a fixed basis would no longer be the right equivalence relation by themselves.

**Question 24.7** (Maximal domains at infinite scale). [Theorem 15.21](#) gives *guaranteed*, not maximal, domains. [Example 15.24](#) detects a sharp boundary through both a pole and an extra critical point; a general procedure combining support certificates with algebraic or analytic obstructions could yield substantially better maximal-domain information for finitely presented perturbations. Relatedly, for entire layers failing the universal threshold, particular inputs may still normalize because their coefficients vanish at the dangerous monomials; an input-specific necessary and sufficient condition would have to remain stable under the nonlinear brackets created by normalization, and a first-order divisibility test alone need not be enough.

**Question 24.8** (Ordinary analytic realization). Which of the Hahn normal forms constructed here admit compatible *ordinary* analytic realizations? Ordinary analytic specialization of a Hahn or

formal parameter requires growth estimates or an additional summation theory, and this report supplies neither. Sectorial analytic moduli of ordinary parabolic germs are not removed, and the analytic embedding obstruction for ordinary parabolic germs is not resolved. Likewise, adding weighted coefficient norms, Gevrey bounds in the parameter, or ordinary convergence in  $(u, v)$  produces strictly stronger problems whose arithmetic thresholds could differ from the thresholds established here; (5.7) is offered only as a starting point.

**Question 24.9** (The resonant quasi-periodic normal form). [Theorem 17.3](#) allows QR, but the constant normal form [theorem 18.1](#) requires  $N_\Upsilon = \{0\}$ . In the resonant case a natural target is a normal form retaining the Fourier modes in  $N_\Upsilon$ . Source 07 warns explicitly that simply replacing the mean projection  $Q_0$  by  $\Pi_{N_\Upsilon}$  in the proof is **not** justified: the retained normal form is then no longer a constant vector field and creates additional terms in the conjugacy equation. Products and substitutions interact with those modes, and higher perturbations may split the remaining resonances, so the extended flag of [corollary 17.6](#) would have to be carried through the nonlinear argument. This is the quasi-periodic counterpart of [question 24.3](#) and is open for the same structural reason — a larger kernel for the inverse operator — but it is a different problem on a different divisor family.

**Question 24.10** (Quantitative coefficient-growth classes on the torus). One may add bounds linking  $\|f_\beta\|_r$  to a weight on the support. The argument of [section 18](#) does *not* prove preservation of such a weighted class, because inverse denominator powers and derivative losses grow along support words ([remark 17.5](#)). Identifying hypotheses that preserve these weights, and that would permit a meaningful specialization of selected monomials to small real parameters, is open; that is exactly where genuine non-linear convergence arithmetic would have to re-enter, and it is the torus form of [question 24.8](#).

**Question 24.11** (Effective arithmetic certificates for the flag). [Theorem 16.5](#) makes the flag finite but not automatically computable from arbitrary real coefficients: detecting an exact integer relation among arbitrary named real numbers may itself be unavailable. For rational and suitable algebraic coefficient data, explicit lattice and approximation certificates can make the construction effective. The theorems as stated separate *existence* of a finite flag from *decidability* of its entries and its arithmetic bounds, and no complexity claim is made. This is the same distinction as **N64** and [section B](#) draw for the finite-return certificate.

## 25 Limitations and non-claims

This section is the consolidated record required by [convention 1.1\(iv\)](#). It carries **seventy-eight** distinct limitations stated by the seven sources — eleven from source 01, nine from source 02, ten from source 03, fourteen from source 04, twelve from source 05, twelve from source 06 and ten from source 07 — with the originating source named in each case. None has been merged away, softened, or absorbed into a stronger statement. Where two sources state overlapping limitations, both wordings are preserved because their scopes differ.

### 25.1 Status, refereeing, and formalization

- N1.** (01) An AI-assisted, unrefereed research draft; not refereed and not formally verified. Its PDF metadata records the author as “Research manuscript prepared with ChatGPT”. Priority is explicitly not certified and no named published conjecture is claimed solved.

- N2.** (02) An AI-assisted, unrefereed research draft; not refereed, not proof-assistant verified. **No Lean theorem corresponding to the new analytic results was compiled** as part of the package.
- N3.** (03) Self-described AI-assisted, unrefereed, not formally verified. Its literature search is described as targeted and auditable but *not* a priority determination; no named published conjecture is claimed solved; independent expert review of both the proofs and the literature comparison is called appropriate.
- N4.** (04) An unrefereed, AI-generated research draft whose proofs are “offered for scrutiny”, not proofs checked by a proof assistant. No Lean, Wolfram or other proof-assistant formalization is included or claimed.
- N5.** (05) Not independently refereed and not formally verified; none of the central claims is an independently checked Lean theorem.
- N6.** (06) Not peer-reviewed and not proof-assistant verified; no Lean or other formalization is supplied or claimed.

## 25.2 Priority and novelty

- N7.** (01) The repository and literature comparisons are both explicitly *targeted*: three catalogue README files plus a search for “linearization”, not a line-by-line audit of every research package, and the novelty assessment is a “proposed-originality assessment, not an exhaustive priority claim”.
- N8.** (02) Priority is not independently verified. Only three theorems — here [theorems 7.2, 7.5 and 7.6](#) — are named as candidate-new, and the source explicitly concedes that “a specialist could identify an existing formal-family theorem that subsumes part of the present formulation; that possibility should be checked before any publication claim”. The support lemma, the homological inverse, formal nonresonant conjugacy, the quadratic scaling trick and the centralizer argument have classical antecedents and are not offered as independent discoveries.
- N9.** (02) The repository inspection is explicitly *not* a line-by-line audit of every manuscript or every Lean file; only the catalogue and the opening foundation definitions were read.
- N10.** (03) Classical credit is stated plainly and is not claimed as new: formal logarithms, flow embedding and semisimple–unipotent splitting [6]; the small-divisor problem and formal-versus-analytic linearization [11, 13, 14]; and the support lemmas [2, 3, 4].
- N11.** (04) Priority is *not* claimed: the literature search is described as targeted and bounded and “did not certify priority”; no named published open problem is claimed solved; novelty is “to the author’s present knowledge”.
- N12.** (04) Acknowledged precedents, not contributions: Neumann’s support lemma, formal embedding and formal logarithms, and rank-one ultrametric radius theory with the quadratic disk scale [20]. In that source’s own words, “we do not present a disk estimate alone as a new theorem of classical dynamics”.
- N13.** (04) The repository inspection was “a targeted inspection, not an exhaustive line-by-line audit” at one frozen revision.
- N14.** (05) Novelty is assessed by a *targeted*, not exhaustive, literature search; the statements are called candidate new results; an unlocated result may exist in the literature on formal flows, deformation of one-forms, or generalized series; and no named published conjecture is claimed settled.
- N15.** (06) Priority is not established: the contributions are labelled “proposed contributions with proofs”, not certified bibliographic firsts, on the basis of a non-exhaustive literature search; no

named published open conjecture is claimed solved. Classical formal Birkhoff normalization, non-Archimedean linearization and general formal-sum Lie exponentiation are credited rather than claimed [23, 24, 20, 8].

- N16. (04) For the Neumann 1949 reference, the publisher’s direct PDF endpoint was not accessible in the session; the metadata and the standard lemma were cross-checked instead.
- N17. (04) Packaging gap: the source README lists `SHA256SUMS.txt` among the archive contents, and no such file exists in the delivered archive (see [section 23.2\(4\)](#)).

## 25.3 Scope of the hypotheses

- N18. (01) [Theorem 8.5](#) and its package cover nonlinear polynomials in *one* variable, with unit multiplier not a root of unity, over complex Hahn fields of arbitrary ordered-group rank, in equal characteristic zero. Explicitly *not* covered: general infinite Hahn-analytic power series — where a common support certificate, a finite or attained dominant face, and analytic composition would each need separate proof; actual root-of-unity multipliers; attracting or repelling multipliers; and higher-dimensional simultaneous resonances.
- N19. (01) Equal characteristic zero is a *genuine* hypothesis, not cosmetic: the divisor cancellation (8.3) leaves a factor  $n - 1$  in the denominator, a valuation unit here but not in mixed characteristic. The formula and the shell-periodicity conclusion must **not** be transported to  $\mathbb{C}_p$ ; [21, Cor. C] gives  $p$ -adic quadratic-like examples whose linearization bounding sphere contains no periodic point ([warning 2.7](#)).
- N20. (02) Resonant multipliers are excluded *entirely* from [section 7](#); resonant normal forms need a different target equation, not the linear normal form used there.
- N21. (02) The several-variable results are stated only for diagonal unitary  $L$  and *equal-radius* polydisks; no optimal general Reinhardt-domain claim is made or needed.
- N22. (03) The finite-order theorem [theorem 19.1](#) requires  $\Lambda^q = I$ , not merely the presence of resonances, and resonant normal forms are *not* claimed unique: further  $\Lambda$ -commuting changes can transform  $Y$ .
- N23. (04) The most consequential hypotheses of [section 13](#) are not cosmetic: the simple zero at the normalization point permits the regular quotient (13.5); a common ordinary domain permits coefficientwise primitives without an uncontrolled sequence of shrinking radii; positive support makes the operator logarithm strongly meaningful; the polynomial Euler form fixes all zeros and gives the exact factor  $W_\varepsilon = V B_\varepsilon$ , whereas a general perturbation can move zeros or create different coefficient pole patterns; and divisibility of  $\Gamma$  is used to keep the resonant rescaling inside the chosen workspace.
- N24. (05) [Theorems 14.13](#) and [14.16](#) require the leading coefficient  $a$  to be *nowhere zero* on  $D$ . The unit hypothesis in [lemma 14.11](#) is exactly where the proof stops applying, and the zeros case needs local zero divisors, multiplicities and singular normal-form obstructions combined with global periods.
- N25. (05) Conjugators are restricted to identity ordinary part. A conjugacy with nonidentity ordinary part can change the leading vector field and the homology basis, so equal period vectors in a fixed basis would no longer be the right equivalence relation; that extension is not done.
- N26. (05) No converse characterization of the units of  $\mathcal{A}$  is claimed, only the sufficient criterion [lemma 13.2](#).
- N27. (05) The  $\mathcal{A}$ -membership condition  $g/W \in \mathcal{A}$  in [theorem 14.5](#) is indispensable outside the nonsingular stratum; period vanishing alone is insufficient there ([example 14.10](#)).
- N28. (06) Nonresonant frequencies only. The gauge equation changes for resonant  $\omega$ , because the

homological inverse then has a larger kernel; the resonant case is listed as open ([question 24.3](#)).

- N29. (06) No result is claimed for Hahn series with unrelated convergent-germ coefficients, nor for infinite-dimensional Hamiltonian systems.
- N30. (06) Uniqueness is *gauge-relative*: only the single exponential generator with  $\Pi\chi = 0$  and the normal form it produces are unique; general normalizing symplectic changes are not ([warning 15.10](#)).

## 25.4 Exactly what “exact” and “analytic” mean

- N31. (01) “Exact” in [theorem 8.5](#) means the maximal *centered valuation ball* only — not the full pointwise summability locus of the series (exceptional evaluations on the bounding shell are allowed), not a maximal non-ball invariant domain, and not every analytic continuation of the conjugacy.
- N32. (01) The bounding shell  $S_r$  is an algebraic valuation-shell expression: *not* a topological boundary (valuation balls are clopen) and *not* an ordered-modulus circle. The common-domain disk  $D$  is *not* identified with  $B_r$ : all points of the normalized monad have ordinary residue zero while  $D$  contains many nonzero ordinary points ([warning 2.4](#)).
- N33. (01) [Theorem 9.1](#) gives *no* uniform ordinary norm bound on the coefficient family  $(H_\gamma)$ ; these may be arbitrarily large with arbitrarily large derivatives. Only the domain of holomorphy and the support certificate are uniform ([warning 9.3](#)).
- N34. (01) The coherent conclusion is *resonant-only*, and [proposition 10.1](#) shows failure of a *universal* common-domain statement in the nonresonant case — explicitly **not** an if-and-only-if classification, and not every nonresonant case is noncoherent.
- N35. (01) The stability certificate [theorem 8.10](#) is sufficient, *not* necessary: a perturbation may change the first return face and still preserve its maximal slope.
- N36. (02) The positive theorem gives *no* uniform bound on  $\|h_\gamma\|_r$  as  $\gamma$  varies, no norm, no Gevrey or weighted coefficient estimates, and no ordinary holomorphic dependence on  $t$  near  $t = 0$ ; those are strictly stronger problems whose arithmetic thresholds could differ, and ([5.7](#)) is offered only as a starting point.
- N37. (03) “Finite dependence” bounds the number of input *coefficient functions* and analytic operators at a specified weight. It is *not* an unconditional finite-time algorithm: one application of  $\mathcal{L}^{-1}$  to an arbitrary analytic function can involve infinitely many of its ordinary Taylor coefficients, and only polynomial input makes each requested coefficient a finite exact algebraic calculation ([warning 6.12](#)).
- N38. (04) Transcendence is relative to the specific base  $\mathbb{C}(w)((t^\Gamma))$ . The source volunteers the counterweight that  $H_\varepsilon$  is *differentially algebraic* in a very explicit way, satisfying  $H'_\varepsilon = \Omega_\varepsilon H_\varepsilon$  with  $\Omega_\varepsilon \in \mathcal{B}$ , so “transcendental” must not be upgraded to “differentially transcendental” ([warning 13.27](#)).
- N39. (04) “Transcendental” concerns a *function* in an ambient meromorphic Hahn field. It is not a claim that any individual value  $H_\varepsilon(w_0)$  lies outside  $\mathbf{No}[i]$ , and is explicitly consistent with  $\mathbf{No}[i]$  being algebraically closed.
- N40. (04) [Theorem 13.3](#) does *not* assert that the global evaluated map  $H^\#$  is injective on all of  $U^\#_\Gamma$ ; injectivity is proved only on the infinitesimal disk for the resonant model ([warning 13.4](#)).
- N41. (04) The sharp-domain maximality of [theorem 13.22](#) is about *centered valutive balls* under either boundary convention. It does not exclude continuation to a non-ball domain avoiding the obstructing fixed points, and does not assert that every coefficientwise continuation is injective ([warning 13.23](#)).
- N42. (05) An Abel function is not automatically a globally injective coordinate: a locally univalent

- primitive need not be globally univalent ([warning 14.8](#)).
- N43. (06) “Analytic” is a *defined* term meaning the entire-coefficient Taylor-compatible realization of [proposition 3.11](#). The source does not assert fine-topological continuity of nonconstant ordinary-time paths in  $\mathbf{No}[i]$ , and does not identify phase differentiation with an intrinsic derivation of the surreal coefficient field ([warnings 3.2](#) and [15.15](#)).
  - N44. (06) [Theorem 15.21](#) extends the polynomial-coefficient normalizer, Hamiltonian and constructed actions *only*. It does **not** license evaluating an arbitrary entire function at an infinite phase or action argument; a centralizer statement on  $D_\rho$  must be made in the rescaled entire-coefficient algebra ([warning 15.22](#)).
  - N45. (06) Rescaling gives *guaranteed*, not maximal, domains. The cubic example shows a sharp boundary, but the general maximal-domain problem is left open ([question 24.7](#)).
  - N46. (06) [Examples 15.25](#) and [15.26](#) show that none of the three admissibility conditions can be dropped from a theorem of this form. They do *not* assert that finite fibers are necessary for every resummation in some larger, differently specified class.

## 25.5 No global, all-scale, or topological claims

- N47. (02) No global conjugacy on all of  $\mathbf{No}[i]$ , no surreal-time flow, no class-sized analytic construction, and no coherence between arbitrary infinite-scale charts is claimed; [proposition 20.2](#) supplies a concrete critical-scale test that any such extension must pass.
- N48. (03) No all-scale theorem. The common-domain results concern finite halos and the germ/formal results the infinitesimal monad; [proposition 20.1](#) explicitly blocks automatic extension to infinite surcomplex arguments. No global analytic atlas on  $\mathbf{No}[i]$ , no ordinary contour theory on its fine topology, and no fine-topologically continuous complex-time flow is constructed.
- N49. (03) There is no assertion that every set-theoretic or algebraic conjugacy of evaluated functions must arise from a coefficient family; failure in one specified analytic category prohibits no set-theoretic conjugacy, and the formal-coefficient conjugacy always exists under nonresonance and can be evaluated on the monad ([proposition 6.14](#)). No interchange of an infinite ordinary coefficient sum with a non-locally-finite Hahn family is permitted.
- N50. (03) No claim of compatibility with the Berarducci–Mantova surreal derivation or with any global composition operator on  $\mathbf{No}$ ; no claim that these coefficient categories fall outside every framework of Cluckers–Lipshitz analytic structures [\[26\]](#); and no sectorial or resurgent summation is asserted.
- N51. (04) No global Julia or Fatou theory on the proper class  $\mathbf{No}[i]$ , no classification of arbitrary surcomplex dynamical systems, and no ordinary topological convergence in the full surcomplex plane is used anywhere.
- N52. (04) The formal logarithm is *not* identified with a convergent autonomous flow at a numerical parameter; [\[7\]](#) is cited precisely to keep formal embedding and analytic embedding apart ([warning 13.4](#)).
- N53. (05) The slow-time theorem gives nothing past the ordinary singularities of the supplied flow chart, licenses no general exchange of ordinary with Hahn summation, and gives no interpretation of an arbitrary surreal time as an actual continuous trajectory ([warning 14.27](#)).
- N54. (05) No ordinary analytic specialization: growth estimates and a summation theory are neither supplied nor claimed; sectorial analytic moduli of ordinary parabolic germs are not removed; the analytic embedding obstruction for ordinary parabolic germs is not resolved; and which Hahn normal forms admit compatible ordinary analytic realizations is a distinct problem ([question 24.8](#)).

- N55. (06) No ordinary analytic convergence in a complex perturbation parameter is inferred: substituting  $t^n = \varepsilon_0 \in \mathbb{C}^\times$  need not exist on the coefficient algebra and can turn a valid Hahn series into a divergent ordinary series.
- N56. (06) No fixed real-valued norm and no sequential completeness of  $K_\Gamma$  is assumed or available; all control is by supports, and the proper-class field  $\mathbf{No}[i]$  is reached only through compatible set-sized workspaces, never as a single object.

## 25.6 Unresolved problems recorded as such

- N57. (02) The negative common-domain result *requires* multiplier drift  $F'(0) = \lambda + u$  and two rationally independent positive exponents. The source does **not** settle the fixed-multiplier problem when  $0 < \sigma < \infty$  — “eliminating drift removes the direct reciprocal-jet mechanism” — and does not claim the necessity direction for a fixed cyclic value group ([warning 2.6](#) and [question 24.2](#)).
- N58. (03) The intermediate regime  $0 < \tau < \infty$  for *common-domain* germs is explicitly unresolved ([question 7.10](#)), and the radius bound (6.12) is a lower bound only, stated *not* to prove that radius collapse occurs ([warning 6.10](#)).
- N59. (03) No universal analytic Poincaré–Dulac statement for resonant multipliers of infinite order is proved ([question 24.3](#)).
- N60. (03) Higher-dimensional positive centralizers are reduced to commuting positive vector fields but *not* classified; the one-variable meromorphic-ratio argument does not extend ([question 24.4](#)).
- N61. (04) The moving-fixed-point case  $w + \varepsilon V + \varepsilon^2 R$  is open and named as future work; the Euler family “avoids that additional difficulty rather than conceals it”. Several variables is open, the scalar quotient  $\alpha_c/W$  having no direct replacement ([question 24.5](#)).
- N62. (05) Several variables: no single reciprocal time one-form encodes a vector field and commuting fields need not be constant multiples, so [theorem 4.12](#) and the finite-period normal form are explicitly *not* to be transported ([question 24.4](#)).

## 25.7 What the computations do and do not establish

- N63. (01) The 54 finite checks do not establish the arbitrary-rank support lemma, algebraic closure, maximality for every polynomial, the transfinite coherent lifting, or novelty; they are not an implementation of general Hahn fields and not proof-assistant verification.
- N64. (01) The finite procedure of [section B](#) is relative to oracles: equality and valuation of arbitrary Hahn expressions need not be effective, root-of-unity recognition for an arbitrary complex coefficient needs a presentation allowing that decision,  $d^q$  may be very large, and treating a truncated zero as an exact zero would invalidate the maximum. It is an exact characterization and a certificate, not a polynomial-time complexity theorem.
- N65. (02) The finite checks test algebraic identities, signs, normalizations and truncation only. They do *not* prove Hahn summability, well-founded recursion, ordinary analytic domain preservation, an infinite small-divisor limsup, the exact convergence radii, or novelty; and the multivariable test verifies nonzero denominators only in its finite range, not nonresonance for all multi-indices of its sample tuple.
- N66. (03) The 21 checks are finite polynomial jet identities truncated by *total parameter degree*, explicitly not a Hahn valuation cutoff in a higher-rank group. They do not test arbitrary well-ordered supports, analytic radius claims, continued-fraction asymptotics, the universal quantifiers, or priority, and no proof-assistant verification is claimed ([warning 21.1](#)).

- N67. (04) The 2,764 exact assertions are finite consistency checks only. No finite test proves Neumann summability, existence on an arbitrary ordinary domain, compatibility for arbitrary value groups, or the algebraicity dichotomy; the program is neither a complete symbolic surcomplex CAS nor a numerical evaluator at arbitrary surreal inputs.
- N68. (05) `verify.py` is transcription and algebra checking only: it proves nothing about arbitrary-rank support finiteness, global primitive existence, the classification, or the slow-time theorem. **Fifty of its 245 assertions are finite lexicographic illustrations for  $n = 1, \dots, 50$ ; the all- $n$  inequality is proved in the text, not inferred from them.**
- N69. (05) The verifier loses one top coefficient on division by  $\varepsilon$  (its division routine sets the new top coefficient to zero), so reciprocal and pullback comparisons are checked only through degree  $N - 1$ .
- N70. (06) The 119 checks truncate by *total label degree* in the two independent labels, not by Hahn valuation, and only to degree four. They are explicitly not a proof of arbitrary-support summability, nor of [theorems 15.5](#) and [15.16](#), nor of [theorem 15.12](#), nor of the Liouville construction [proposition 15.17](#).

## 25.8 The quasi-periodic source

- N71. (07) Self-declared as proposed original results with detailed proofs, unrefereed, and not formalized in any proof assistant. Its README and title page both state that the exact theorem package was not located in a *focused* literature review and that this does not establish priority; independent mathematical review is called for before correctness or originality is treated as settled. Its classical inputs — ordinary Fourier division with its subexponential criterion, the Liouville construction method, positive-support Hahn inversion, and the finite Jacobian identities — are named as classical and are not claimed [[17](#), [18](#), [2](#), [4](#)].
- N72. (07) A *coefficientwise Hahn* theorem only. It explicitly does **not** assert convergence after substituting a nonzero real parameter for a Hahn monomial, and does not improve classical real analytic KAM or Brjuno arithmetic [[17](#), [19](#)]. No bound uniform in the Hahn exponent on the analytic seminorms  $\|f_\beta\|_r$  is supplied, and growth faster than every exponential in the exponent index is permitted ([section 18.7](#)).
- N73. (07) The torus derivatives are external derivations annihilating the Hahn scalar field, deliberately **not** the Berarducci–Mantova derivation on **No** [[25](#)], so nothing in [sections 16](#) to [18](#) speaks to internal surreal differential equations ([warning 3.2](#)).
- N74. (07) **A support bound is not summability.** The family of reciprocal divisors indexed by the Fourier modes is stated to be in general *not* Hahn-summable; only a common support is proved ([theorem 16.5](#)(iii), [remark 16.8](#)), with ordinary Fourier summation performed inside each Hahn coefficient and never interchanged with it ([warning 16.4](#)).
- N75. (07) No class-sized objects. Everything is a set-sized workspace, with the full surreal field entering only through the embedding ([3.1](#)); no continuity into the fine topology of **No**[ $i$ ] and no class-indexed analytic limit is claimed. The invariant density [theorem 18.8](#) is coefficientwise integration of analytic functions — **not** a countably additive **No**-valued measure on arbitrary subsets and **not** a real-time ergodic theorem.
- N76. (07) The nonlinear constant normal form requires a trivial exact resonance lattice. The resonant case is left open, and the source warns that replacing the mean projection by the resonant projection is not justified, because the retained normal form is then no longer a constant vector field ([question 24.9](#)).
- N77. (07) The flag is finite but *not* automatically computable: detecting exact integer relations among arbitrary named reals may be unavailable, so existence of the flag is explicitly

separated from decidability of its entries ([question 24.11](#)).

- N78.** (07) Three extensions are listed as needing further work — quantitative coefficient-growth classes preserved under the construction, the resonant normal form, and effective arithmetic certificates — and the source states that these arise from the article and are *not* claimed to be catalogued open problems ([questions 24.9](#) to [24.11](#)).
- N79.** (07) The finite checks verify neither the irrationality of the Liouville constant ([17.6](#)), nor any infinite-mode subexponential estimate, nor the word lemma, nor the infinite-support conjugacy theorem.
- N80.** (07) An instance of this report’s standing hazard, but handled: the source cites *no* invariance of any norm across a value-group extension. The robustness of [theorem 17.7](#) under enlarging  $\Gamma$  is argued from uniqueness of the modewise Hahn inverse rather than assumed ([warning 17.8](#)).

The list above contains 80 numbered items for the 78 distinct source-level limitations. The two extra entries arise because source 03’s limitation on resonant normal forms is recorded in two places — **N22**, the finite-order hypothesis  $\Lambda^q = I$ , and **N59**, the openness of the infinite-order case — and because source 04’s hypothesis audit is given its own entry (**N23**) in addition to its fourteen listed non-claims. The distribution of *distinct* source-level limitations is 11/9/10/14/12/12/10 across sources 01 through 07; the corresponding numbered-item counts are 11/9/11/15/12/12/10.

*Warning 25.1* (The one sentence this report refuses). No statement of this report says that Hahn or surcomplex coefficients *abolish* the classical small-divisor problem. What is true is stated in [remarks 2.2](#) and [2.3](#): passing to positive Hahn support *separates* two questions that ordinary analytic convergence forces one to solve at once — algebraic dependence across perturbation orders, and ordinary analyticity within each coefficient — and the sharp thresholds describe exactly that separation. Support alone suffices for polynomial and formal coefficients at arbitrary rank ( $\text{SD}_0$ ); arithmetic enters exactly when an ordinary domain or entirety is demanded ( $\text{SD}_{\text{dr}}$ ,  $\text{SD}_{\text{ex}}$ ,  $\text{SD}_{\text{fr}}$ ,  $\text{SD}_{\text{ss}}$ ); and under PAR there is no divisor family at all ( $\text{SD}_{\emptyset}$ ). The quasi-periodic part makes the same point in its sharpest form: [proposition 16.9](#) is a genuine theorem with *no* arithmetic hypothesis whatever, and [theorem 17.3](#) shows that the moment an analytic strip is demanded the arithmetic becomes not merely sufficient but necessary. The two positions are complementary, not opposed, and stating them without their coefficient categories is what would make them look contradictory.

## A A short proof of the word-order input

We record the special case of Higman’s lemma used in [lemma 3.4](#), so that the support proof can be audited without a separate combinatorial argument hidden in the notation. It is classical [\[3\]](#). Sources 02 and 05 each give the same argument; this is the one copy.

Let  $S$  be well ordered. A word  $(a_1, \dots, a_m)$  *embeds* in a word  $(b_1, \dots, b_n)$  if there are indices  $i_1 < \dots < i_m$  with  $a_j \leq b_{i_j}$  for every  $j$ . A sequence of words is *bad* if no earlier word embeds in a later one. Suppose an infinite bad sequence exists. Choose a bad sequence  $w_0, w_1, \dots$  minimally by word length at each stage: after choosing a prefix, choose a shortest next word among those admitting a bad infinite extension. The choices take place in a set, so ordinary choice suffices, and no  $w_n$  is empty.

Write  $w_n = v_n a_n$ , separating the last letter. Any infinite sequence in a well-ordered set has an infinite nondecreasing subsequence — select a position attaining the minimum of the current infinite tail, then repeat on the tail strictly after that position; the successive minima cannot decrease. So choose  $n_0 < n_1 < \dots$  with  $a_{n_0} \leq a_{n_1} \leq \dots$  and consider

$$w_0, \dots, w_{n_0-1}, v_{n_0}, v_{n_1}, v_{n_2}, \dots$$

This is bad: an earlier  $w_i$  embedding in a later  $v_{n_j}$  would also embed in  $w_{n_j}$ , contradicting the original badness; if  $v_{n_i}$  embeds in  $v_{n_j}$  for  $i < j$ , appending the dominated last letters embeds  $w_{n_i}$  in  $w_{n_j}$ , again a contradiction; and comparisons within the unchanged prefix were already forbidden. The new bad sequence agrees with the chosen one until position  $n_0$  and has the strictly shorter word  $v_{n_0}$  there, contradicting the minimal choice. Hence no bad sequence exists, which is the required word well-quasi-order.

When  $S \subset \Gamma_{>0}$ , a proper word embedding *strictly* raises the sum: either an unused positive letter is present in the larger word, or a matched letter is strictly larger. That observation converts the word theorem into both the well-ordering of  $S^*$  and the finiteness of the words of one weight, as claimed in [lemma 3.4](#).

## B The finite certificate for the resonant threshold

For polynomial inputs with effective coefficient arithmetic, the geometric certificate of [corollary 8.9](#) can be computed without constructing  $H$  at all.

1. Determine  $\alpha = \bar{\lambda}$ . If it is not a root of unity, compute the finite maximum  $\eta(f)$  and return  $r = \eta(f)$ .
2. If its order is  $q$ , compute the finite composition  $g = f^{\circ q}$ ,  $c = \lambda^q - 1$  and  $\delta = v(c)$ .
3. Compute every nonzero nonlinear return coefficient  $b_n$  *including actual cancellations*, and take the maximum [\(8.6\)](#).
4. Scale by  $\rho = t^r$  and reduce [\(8.9\)](#) to obtain  $P$ . Its degree gives the shell multiplicity; factoring  $P$  determines the residue directions; and  $XP'(X)$  gives the leading multiplier ratios at simple roots.

This is a finite algebraic procedure *relative to the required oracles*, and not a claim of universal computability for arbitrary surcomplex inputs: see **N64** of [section 25](#).

## C A coefficientwise recipe for a certified linearizer coefficient

Suppose one wishes to compute  $h_\gamma$  for polynomial input in [theorem 6.1](#), with an effective presentation of the finite word set  $\mathcal{W}_S(\gamma)$ . The following specifies what must be certified.

Collect the finite letter set  $D_S(\gamma)$  and the finite set of partial sums of all words of weight  $\gamma$ : only input coefficient polynomials at these letters can matter. The maximum word length gives the bound  $L = \ell_S(\gamma)$ . In [\(6.5\)](#) retain only  $k = 0, \dots, L - 1$ , and in each Taylor expansion of  $\mathcal{N}_f$  retain only products whose support words can complete an allowed word of total weight  $\gamma$ ; polynomial degrees can be bounded in advance by [\(6.13\)](#). Apply  $\mathcal{L}^{-1}$  to each retained monomial by the exact scalar division [\(5.2\)](#). Nonresonance of the finitely many divisors actually used must be established, or supplied as an exact symbolic condition. Sum the resulting finite expressions and check Schröder's equation at the requested weight; the word certificate proves that discarded expressions cannot affect it. This is a finite exact algorithm *relative to the support certificate and the polynomial coefficient data*.

For analytic input the same support certificate still applies, but  $\mathcal{L}^{-1}$  is an operator on an entire coefficient function, not merely a finite list of Taylor coefficients; a radius certificate from [theorem 6.9](#) is then part of the analytic output. A finite jet calculation alone must *not* be described as computing the entire analytic coefficient, nor as proving its convergence ([warning 6.12](#)).

The parabolic counterpart, for [theorem 14.13](#), is explicit in the same sense. Given  $F_1, F_2$  in one

nonsingular stratum, form their logarithmic vector fields by (4.4), compute the normalized reciprocal forms  $\beta_i = b_i dz$  by the geometric unit expansion of lemma 13.2, and test the periods of  $(b_1 - b_2) dz$ ; on a finitely punctured plane that is a finite list of Hahn periods, though computing an arbitrary coefficient integral is not assumed decidable. When the periods vanish, integrate coefficientwise from  $z_*$  to get  $p$  with  $p' = b_1 - b_2$ ,  $p(z_*) = 0$ ; with  $b_2 = b_0 + u$  and  $S$  the positive union of the supports of  $p$  and  $u$ , the desired  $h = H - z$  has support in  $S^* \setminus \{0\}$  and, for each  $\gamma$  in that well-ordered set,

$$h_\gamma = b_0^{-1} \left( p_\gamma - [t^\gamma] \left[ u h + \sum_{n \geq 2} \frac{b_2^{(n-1)}}{n!} h^n \right] \right), \quad (\text{C.1})$$

the bracket using only earlier coefficients of  $h$ . The number of operations at each exponent is finite, but the total support may be transfinite and need not have a finite presentation. For an ordinary single-parameter expansion  $b_2 = b_0 + \varepsilon b_1 + \varepsilon^2 b_2^{[2]} + \dots$  and  $p = \varepsilon p_1 + \varepsilon^2 p_2 + \dots$  the first coefficients are

$$h_1 = \frac{p_1}{b_0}, \quad h_2 = \frac{1}{b_0} \left( p_2 - b_1 h_1 - \frac{1}{2} b_0' h_1^2 \right),$$

displaying both sources of nonlinear correction: the tail of the target time form, and the change of its ordinary leading coefficient along the displacement. The support certificate is part of the result — it is what makes each coefficient defined even when the multiples of the smallest input exponent are not cofinal — and it is *not* a promise that arbitrary Hahn-series equality, primitive existence or period evaluation can be decided by a finite program.

## D The single-exponential gauge recursion as an algorithm

For a finite collection of input labels  $x_1, \dots, x_k$ , an implementation of theorem 15.7 can work over the ordinary formal label ring through a chosen total label degree  $N$ :

1. set  $\chi = 0$  and retain  $H = H_0 + R$  with independent labels attached to its input layers;
2. for  $n = 1, \dots, N$  compute  $e^{\text{ad}_\chi} H$  through total label degree  $n$ , using the current  $\chi$  of label degree less than  $n$ ;
3. for each coefficient  $E_\nu$  with  $|\nu| = n$ , append  $-L^{-1} Q_\Pi E_\nu$  to the coefficient of  $x^\nu$  in  $\chi$ ;
4. recompute  $e^{\text{ad}_\chi} H$  through the requested order, and compute coordinates by the same exponential and actions by the inverse exponential.

Adding a degree- $n$  generator changes the degree- $n$  transformed Hamiltonian only by  $L\chi_n$ , so step 3 is the *single-exponential gauge* recursion — **not** the different algorithm that composes a list of successive exponential maps without taking their logarithm (warning 15.10). For an infinite input label set this is a conceptual construction rather than a finite executable loop; homogeneous leaf degree still organizes the universal expression trees of definition 15.3, and at a specified Hahn weight lemma 3.4 permits only finitely many leaf words, hence finitely many homogeneous leaf degrees and finite expressions. That is the exact connection between the finite-label algorithm and the arbitrary-support proof. Total label degree is not a valuation cutoff (warning 21.1).

## E Antecedent ledger

For audit, the external inputs and their exact roles. None is claimed as a contribution of this report.

| Input                                                                                                                                                                        | Role here                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Neumann's positive-support lemma [2], via Higman's word well-quasi-order [3], with Poonen's account of Hahn–Mal'cev–Neumann units [4]                                        | <a href="#">Lemma 3.4</a> , <a href="#">corollary 3.5</a> , and <a href="#">section A</a> : the certificate behind <i>every</i> infinite operator expression in this report. Classical; used as a black box for Higman's lemma beyond the reproved special case. |
| Algebraic closure of a maximally complete Hahn field [4, Cor. 4]                                                                                                             | <a href="#">Lemma 8.4</a> and <a href="#">theorem 11.1</a> only. Not needed for <a href="#">section 15</a> .                                                                                                                                                     |
| Conway normal forms and surreal normal-form arithmetic [5]                                                                                                                   | (3.1) and the workspace discipline. No set of all surreals is formed.                                                                                                                                                                                            |
| Formal logarithms, flow embedding, Jordan–Chevalley / semisimple–unipotent splitting [6]                                                                                     | The classical pattern behind <a href="#">theorems 4.3</a> and <a href="#">19.1</a> . What is established here is the arbitrary-positive-Hahn-support, same-coefficient-domain implementation, not the pattern.                                                   |
| Formal interpolation of near-identity maps versus analytic embedding at a numerical parameter [7]                                                                            | Cited exactly to mark the line that <a href="#">warning 13.4</a> draws.                                                                                                                                                                                          |
| Exponentiating strongly summable derivations of generalized-series algebras [8]; automorphisms of the surreals [9]                                                           | The general algebraic mechanism behind <a href="#">theorem 4.3</a> and <a href="#">proposition 15.2</a> . The existence of a formal logarithm is credited, not claimed.                                                                                          |
| Julia's equation and differential transcendence of iterative logarithms [10]                                                                                                 | Background for <a href="#">corollary 4.6</a> and <a href="#">section 14</a> . The parameter-Hahn construction here is a different category; the analytic embedding obstructions for ordinary parabolic germs are not removed.                                    |
| Ordinary homological equation, small divisors, formal-versus-analytic linearization [11, 12]; the Brjuno–Yoccoz quadratic theorem [13, 14]                                   | <a href="#">Section 5</a> ; and Brjuno–Yoccoz used as an <i>external</i> theorem, not reproved, in <a href="#">propositions 10.1</a> and <a href="#">20.1</a> and <a href="#">remark 7.9</a> .                                                                   |
| Regular dependence of ordinary linearization on the multiplier [15]; explicit tree expansions [16]; linear versus nonlinear arithmetic conditions in regularity classes [17] | Natural parameter-regularity context for the reciprocal jets (7.4). No Whitney estimates, no joint analytic parameter convergence, and no novelty claim for tree-organized formal coefficients.                                                                  |

| Input                                                                                                                                                                                                           | Role here                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Equal-characteristic-zero ultrametric linearization, sharp disk radii, and $p$ -adic bounding spheres without periodic points [20, 21]                                                                          | The rank-one antecedent of <a href="#">section 8</a> and the source of the cross-note <a href="#">section 22.5</a> and the characteristic warning <a href="#">warning 2.7</a> .                                                                                                                                                                                             |
| Modified differential equations in backward error analysis [22]                                                                                                                                                 | The origin of the coefficients (14.31), which are known examples ( <a href="#">warning 14.22</a> ); its warning that the ordinary step-size series generally diverges is cited and accepted.                                                                                                                                                                                |
| Convergence and generic divergence of Birkhoff normal forms [23, 24]                                                                                                                                            | Classical context for <a href="#">section 15</a> ; ordinary analytic convergence is never inferred from a Hahn identity.                                                                                                                                                                                                                                                    |
| Transseries as germs of surreal functions [25]; fields with analytic structure [26]; Hahn-meromorphic function theory [27]                                                                                      | Adjacent frameworks, kept distinct: see <a href="#">warning 3.2</a> and <b>N50</b> of <a href="#">section 25</a> . No theorem is transferred between categories without a separate proof.                                                                                                                                                                                   |
| Linear versus nonlinear arithmetic conditions for torus vector fields [17]; the one-frequency cohomological equation and its Liouville obstructions [18]; the Brjuno condition through best approximations [19] | The classical small-divisor background of <a href="#">sections 17</a> and <a href="#">18</a> : ordinary Fourier division and its subexponential criterion, and the Liouville mechanism adapted in <a href="#">theorem 17.7</a> . Neither the ingredient nor any strengthening of ordinary analytic KAM arithmetic is claimed ( <b>N72</b> , <a href="#">section 18.7</a> ). |
| The motivating repository at its pinned snapshot [1]                                                                                                                                                            | Sources 01–06 cite it at commit 39f2be6667ade51bca2b45daa47e289d69c09764 and source 07 at aa846271b4dcae2c055b216126a87210292ec19b, for the coefficient-category distinction only; none uses an unrefereed repository theorem as a premise. The inspections were targeted, not exhaustive ( <b>N9</b> , <b>N13</b> , <b>N71</b> ).                                          |

## F Notation at a glance

| Symbol or condition                          | Meaning and role                                                                                                                                                                                                  |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Gamma, K_\Gamma, t^\gamma$                 | Nonzero set-sized ordered abelian group; $K_\Gamma = \mathbb{C}((t^\Gamma))$ ; $t^\gamma = \omega^{-\gamma}$ for the surcomplex reading (3.1). Divisibility and algebraic closure are assumed only where flagged. |
| $v, \mathcal{O}_\Gamma, \mathfrak{m}_\Gamma$ | $v = \min \text{supp}$ with <i>increasing</i> supports ( <a href="#">warning 2.5</a> ); $v \geq 0$ and $v > 0$ . Trivial on nonzero complex constants ( <a href="#">remark 2.3</a> ).                             |
| $+, > 0$ subscript                           | All nonzero support exponents strictly positive. Not positivity of ordinary complex coefficients.                                                                                                                 |
| $B_r, S_r, D_\rho$                           | Valuation ball, shell and polydisk; larger threshold means <i>smaller</i> domain ( <a href="#">section 2.4</a> ). $S_r$ is not a topological boundary.                                                            |

| Symbol or condition                                                                                                            | Meaning and role                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathbb{D}_R, \mathbb{D}_R^d, D, U$                                                                                           | Ordinary disk, polydisk or open set; larger $R$ means <i>larger</i> domain. A domain for coefficient functions only ( <a href="#">warning 2.4</a> ).                                                                                                                                                                                      |
| $U_\Gamma^\#, D^\#, \mathfrak{m}_\Gamma^d$                                                                                     | Finite halo and infinitesimal monad; evaluation (3.4) is a faithful algebra homomorphism ( <a href="#">proposition 3.11</a> ).                                                                                                                                                                                                            |
| $\text{rad}, \text{rad}_{\text{iso}}$                                                                                          | Ordinary Taylor radius and isotropic Taylor radius of one coefficient function; larger is better.                                                                                                                                                                                                                                         |
| $S^* = M(S), \mathcal{W}_S(\gamma), \ell_S(\gamma), D_S(\gamma)$                                                               | Additive monoid, ordered words of weight $\gamma$ , maximum word length, occurring letters ( <a href="#">lemma 3.4</a> ). $\ell_S(\gamma)$ is finite even when infinitely many support elements precede $\gamma$ .                                                                                                                        |
| $\mathcal{B}_M$                                                                                                                | Uniform-monoid support algebra ( <a href="#">definition 3.8</a> ); the uniformity across Taylor degrees is load-bearing.                                                                                                                                                                                                                  |
| $\mathcal{L}, \mathcal{L}_L, L$                                                                                                | Homological operators (5.1), (15.4); $\mathcal{L}^{-1}$ preserves Hahn exponents and total degree.                                                                                                                                                                                                                                        |
| $\sigma(\lambda), \tau(\Lambda), \Theta(\omega)$                                                                               | The three divisor rates: drifted, exact-multiplier, frequency ( <a href="#">section 2.3</a> ). $\sigma, \tau$ are log-rates in $[0, \infty]$ ; $\Theta \in [1, \infty]$ is the exponential of a rate.                                                                                                                                     |
| $\text{RES}_q, \text{MR/NR}, \text{FR}, \text{PAR}, \text{QR}$                                                                 | The five resonance notions ( <a href="#">section 2.2</a> ). A threshold proved under one is never quoted under another.                                                                                                                                                                                                                   |
| $\text{SD}_0, \text{SD}_{\text{dr}}, \text{SD}_{\text{ex}}, \text{SD}_{\text{fr}}, \text{SD}_\emptyset, \text{SD}_{\text{ss}}$ | The six small-divisor notions ( <a href="#">section 2.3</a> ).                                                                                                                                                                                                                                                                            |
| $\Upsilon, \varpi_\beta, \mathbf{N}_\Upsilon$                                                                                  | The <i>Hahn</i> frequency vector in $F_\Gamma^d$ , its real coefficient vectors, and its exact resonance lattice $\{k : k \cdot \Upsilon = 0\}$ . Never $\omega \in \mathbb{C}^d$ of <a href="#">section 15</a> , and never the exponent $\Omega$ of <a href="#">example 3.7</a> ( <a href="#">warning 2.9</a> ).                         |
| $\mathbf{L}_j, \Sigma_j, \gamma_j, \mathbf{a}_j, r$                                                                            | The resonance lattices of the flag, its strata, its visible levels, its leading <i>real</i> forms and its length $r \leq d$ ( <a href="#">theorem 16.5</a> ). $\mathbf{L}_j$ is not the multiplier $\Lambda$ and not the homological $\mathcal{L}$ ; $\Sigma_j$ is not the ordinary time disk $\Sigma$ of <a href="#">theorem 14.26</a> . |
| $\mathbf{E}, \mathbf{T}_\Upsilon, \kappa_\Upsilon$                                                                             | The shifted tail generators, the common reciprocal support $\bigcup_j (-\gamma_j + \mathbf{E}^*)$ , and the last visible level $\gamma_r$ : an <i>element of</i> $\Gamma$ , never a rate ( <a href="#">warning 2.9</a> ).                                                                                                                 |
| $\mathbb{T}_\rho^d, \mathcal{O}(\mathbb{T}_\rho^d), \mathcal{H}_\Gamma(\mathcal{O}(\mathbb{T}_\rho^d))$                        | The fixed open complex strip, its periodic holomorphic functions, and the fixed-strip Hahn algebra: the torus instance of the fixed-domain row of <a href="#">section 3.2</a> , with one $\rho$ for every Hahn exponent and no uniform seminorm bound.                                                                                    |
| $\mathcal{C}_\Upsilon, \mathcal{G}_\Upsilon, \Pi_{\mathbf{N}_\Upsilon}, \text{av}, \mathbf{Q}_0$                               | The cohomological operator $\sum_j \Upsilon_j \partial_{\theta_j}$ , its normalized inverse (17.2), resonant Fourier projection, coefficientwise torus mean, and mean-zero projection. $\mathbf{Q}_0$ is not the Hamiltonian $Q_\Pi$ .                                                                                                    |
| $\mathcal{X}, \mathcal{V}, \Phi_h, \mathbf{c}, \mu_\mathcal{X}$                                                                | The perturbed torus field $\Upsilon + \mathcal{V}$ , its perturbation, the torus coordinate change $\text{id} + h$ ( <i>not</i> the flow $\Phi_X$ or $\Phi_c$ ), the constant frequency correction, and the invariant density (18.10).                                                                                                    |
| $r, q, c, \delta, P, M$                                                                                                        | Finite-return threshold (8.6), residue order, $\lambda^q - 1$ , $v(c)$ , shell polynomial (8.10), its degree.                                                                                                                                                                                                                             |
| $\varepsilon = t^\delta, s = t^{\delta/q}$                                                                                     | A single Hahn monomial of positive valuation; <i>not</i> a small complex number. All $O(\varepsilon^k)$ mean divisibility in a formal series ring.                                                                                                                                                                                        |
| $X_F, W_F, \Phi_X, \Phi_c$                                                                                                     | Logarithmic vector field, its one-variable coefficient, time-one and time- $c$ maps ( <a href="#">theorem 4.3</a> and <a href="#">corollary 4.8</a> ).                                                                                                                                                                                    |

| Symbol or condition                                                                | Meaning and role                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Omega_F = dz/W_F,$<br>$\beta_F = \varepsilon\Omega_F$<br>$J_\delta, I_\delta(D)$ | Time form and its normalization; “holomorphic” refers to the coefficient functions, and the valuation may be negative.<br>Flat ideal and one-scale-jet kernel, formed in $K_{\geq 0}$ and $\mathcal{A}_{\geq 0}$ , <i>never</i> in $K$ or $\mathcal{A}$ ( <a href="#">warning 14.17</a> ). |
| $\mathcal{B} = \mathbb{C}(w)((t^\Gamma)), \mathcal{L}$                             | The algebraicity base with arbitrary rational-function coefficients ( <i>not</i> $K_\Gamma(w)$ ; <a href="#">warning 13.1</a> ) and the meromorphic ambient field ( <a href="#">13.2</a> ).                                                                                                |
| $\rho_a, M_a, \rho_q$                                                              | Exact monodromy exponent and multiplier ( <a href="#">13.15</a> ), and the model’s common exponent ( <a href="#">13.30</a> ).                                                                                                                                                              |
| $\Pi, Q_\Pi, \chi, J_j, \Theta(\omega)$                                            | Action projection and its complement, the gauge-normalized generator, the commuting actions, and the frequency divisor invariant.                                                                                                                                                          |
| $w_\rho(\ell), \mathcal{S}_\rho$                                                   | Shifted monomial weight ( <a href="#">15.19</a> ) and the Hamiltonian rescaling operator ( <a href="#">15.20</a> ), a Poisson Lie but <i>not</i> a ring homomorphism.                                                                                                                      |
| $I_{\text{Fix}(F)}$                                                                | Ideal generated by $F_j - z_j$ , compared by the explicit invertible matrix ( <a href="#">4.10</a> ), not by its zero set.                                                                                                                                                                 |

## G Per-theorem hypothesis audit

| Statement                        | Indispensable hypotheses, and the boundary of the conclusion                                                                                                                                                                                                                     |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">lemma 3.4</a>        | Well-ordered <i>positive</i> support. No Archimedean hypothesis, no rank-one, no countability, no finitely generated grid. Gives finite words, not a valuation-contraction estimate ( <a href="#">warning 3.6</a> ).                                                             |
| <a href="#">proposition 3.11</a> | One common ordinary domain for all coefficients; positive support of the infinitesimal increment. Faithful, but polynomial/entire coefficients do not automatically evaluate at infinite surcomplex points.                                                                      |
| <a href="#">theorem 4.3</a>      | Ordinary reduction the identity; positive perturbative support; arbitrary open $U$ . No ordinary parameter convergence, no uniform coefficient bound.                                                                                                                            |
| <a href="#">theorem 4.10</a>     | Positive vector field; nonzero admissible time. Ideals compared by ( <a href="#">4.11</a> ). Says nothing about maps whose reduction is not the identity.                                                                                                                        |
| <a href="#">theorem 4.12</a>     | $U \subset \mathbb{C}$ <i>connected</i> , one variable, $F \neq \text{id}$ ; admissible times $v(c) + \alpha > 0$ . Not a classification in $d > 1$ ( <a href="#">remark 4.13</a> ); the centralizer can <i>grow</i> under workspace extension ( <a href="#">warning 4.15</a> ). |
| <a href="#">theorem 6.1</a>      | NR with the multiplier <i>exactly</i> $\Lambda$ ; positive support; no degree-0 or degree-1 coordinate terms at any exponent; a fixed polydisk of finite positive radius in row one. Rows are not interchangeable; the sixth category is <a href="#">question 7.10</a> .         |
| <a href="#">theorem 6.9</a>      | $\tau < \infty$ . <b>Lower bound only</b> ( <a href="#">warning 6.10</a> ).                                                                                                                                                                                                      |
| <a href="#">theorem 6.11</a>     | Polynomial input. Holds for any invertible diagonal nonresonant $\Lambda$ , without $ \lambda_j  = 1$ .                                                                                                                                                                          |
| <a href="#">theorem 6.3</a>      | NR; drift permitted; well-ordered positive $A$ ; $f_a(0) = 0$ . No uniform bound, no Gevrey estimate, no dependence on $t$ at $t = 0$ .                                                                                                                                          |

| Statement                  | Indispensable hypotheses, and the boundary of the conclusion                                                                                                                                                                                                                                             |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| theorems 7.2, 7.5 and 7.6  | <b>Multiplier drift required</b> , plus two rationally independent positive exponents; several-variable version needs multiplicative detuning and equal-radius polydisks. Not asserted for a fixed cyclic value group ( <a href="#">warning 2.6</a> and <a href="#">question 24.2</a> ).                 |
| theorems 8.5, 8.7 and 12.1 | $\text{RES}_q$ with $\lambda$ <i>not</i> a root of unity; one variable; polynomial $f$ ; <i>equal characteristic zero</i> ( <a href="#">warning 2.7</a> ); divisible $\Gamma$ for <a href="#">lemma 8.4</a> . “Exact” means the maximal centered valuation ball only.                                    |
| theorem 9.1                | $\text{RES}_q$ . Gives one fixed ordinary disk on which $P$ has no zero; no uniform norm bound on $(H_\gamma)$ ; resonant-only, and not an iff ( <a href="#">warning 10.2</a> ).                                                                                                                         |
| theorem 8.10               | Same multiplier $\lambda$ ; inequalities on the <i>return</i> coefficients; squarefree $P$ for the matching. Sufficient, not necessary ( <a href="#">warning 8.11</a> ).                                                                                                                                 |
| theorems 13.3 and 13.15    | PAR; simply connected $U$ containing 0; $V$ nonvanishing on $U \setminus \{0\}$ with a <i>simple</i> zero at 0; polynomial Euler form for the dichotomy; divisible $\Gamma$ . Transcendence over $\mathbb{C}(w)((t^\Gamma))$ only, and not differential transcendence ( <a href="#">warning 13.27</a> ). |
| theorem 13.11              | A <i>simple</i> zero $a$ ; ordinary loops on the ordinary universal cover. Finite order exactly when $V'(a) = \mu$ ; the obstruction is visible already at order $\varepsilon^1$ .                                                                                                                       |
| theorem 13.22              | PAR; the model $f = (1 + \varepsilon)z + z^{q+1}$ ; divisible $\Gamma$ . Maximality for <i>centered</i> valuative balls; two distinct failure mechanisms at and below the threshold ( <a href="#">warning 13.23</a> ).                                                                                   |
| theorem 14.5               | $F \neq \text{id}$ . Outside the nonsingular stratum the quotient $g/W_F$ must still lie in $\mathcal{A}$ ; period vanishing alone is insufficient ( <a href="#">example 14.10</a> ).                                                                                                                    |
| theorems 14.13 and 14.16   | Same leading data $(\delta, a)$ with $a$ <i>nowhere zero</i> ; near-identity conjugator on the <i>original</i> common domain; ordinary coefficientwise periods; the specified puncture-loop basis ( <a href="#">warnings 14.12</a> and <a href="#">14.15</a> ).                                          |
| theorem 14.18              | $\delta$ <i>noncofinal</i> , i.e. $J_\delta \neq 0$ ; jets formed in $\mathcal{A}_{\geq 0}$ ( <a href="#">warning 14.17</a> ). Not a statement about ordinary exponentially small asymptotics.                                                                                                           |
| theorem 14.26              | A <i>specified</i> ordinary holomorphic flow chart with image in $D$ and nowhere-vanishing variational factor $J$ . Nothing past the chart’s ordinary singularities ( <a href="#">warning 14.27</a> ).                                                                                                   |
| theorems 15.5 and 15.16    | Nonresonant <i>ordinary complex</i> $\omega$ . Measures one coefficientwise operator; not a Bruno- or KAM-type convergence criterion ( <a href="#">warning 15.6</a> ).                                                                                                                                   |
| theorem 15.7               | $\neg \text{FR}$ ; positive well-ordered $S$ ; $\text{ord}_0 R_s \geq 3$ ; case (a) polynomial layers with no arithmetic condition, case (b) entire layers with $\Theta < \infty$ . Uniqueness is gauge-relative ( <a href="#">warning 15.10</a> ).                                                      |
| theorem 15.12              | The <i>entire-coefficient</i> Hahn algebra. Does not classify local, multivalued or fine-topologically locally constant integrals ( <a href="#">warning 15.13</a> ).                                                                                                                                     |
| theorem 15.21              | Polynomial input admissible at $\rho$ : positivity, well-ordering <i>and</i> finite fibers, all three independent ( <a href="#">examples 15.25</a> and <a href="#">15.26</a> ). Guaranteed, not maximal, domains ( <a href="#">warning 15.22</a> ).                                                      |

| Statement                                                          | Indispensable hypotheses, and the boundary of the conclusion                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">theorem 19.1</a>                                       | $\Lambda^q = I$ — the multiplier itself of finite order, which is neither $\text{RES}_q$ nor mere MR. Resonant normal forms not unique ( <a href="#">warning 19.2</a> ).                                                                                                                                                                                                                                                                                                                                   |
| <a href="#">theorem 16.5</a>                                       | Finite dimension $d$ ; <i>one</i> well-ordered support for the whole frequency vector. Without finite $d$ the kernel rank need not reach its minimum in finitely many steps; independent unbounded scalar choices do not supply a shared support. Gives a <i>common support</i> , not summability of the reciprocal family ( <a href="#">remark 16.8</a> ); and $r$ counts visible scales, not monomials of $\Upsilon$ ( <a href="#">remark 16.6</a> ).                                                    |
| <a href="#">proposition 16.9</a>                                   | No Fourier modes in $\mathbf{N}_\Upsilon$ . <b>No divisor estimate of any kind</b> ; this is the one quantitative statement of the quasi-periodic part that is arithmetic-free.                                                                                                                                                                                                                                                                                                                            |
| <a href="#">theorem 17.3</a>                                       | ( $\text{SD}_{\text{ss}}$ ) on <i>every</i> nonresonant stratum; <i>all</i> input coefficients holomorphic on <i>one</i> open strip. Without the first, a forced coefficient can fail to be a distribution; without the second, independent germs need not give a common analytic domain ( <a href="#">convention 3.3</a> ). The loss $\kappa_\Upsilon$ is attained ( <a href="#">corollary 17.4</a> ); no loss bound uniform in the Hahn exponent is asserted ( <a href="#">remark 17.5</a> ).            |
| <a href="#">theorem 17.7</a>                                       | Integer exponents only; the explicit Liouville constant ( <a href="#">17.6</a> ). The obstruction is to assembling infinitely many modewise inverses into <i>one</i> analytic coefficient — the scalar inverses all exist. Robustness under enlarging $\Gamma$ is argued from uniqueness of the modewise Hahn inverse, not from any norm invariance ( <a href="#">warning 17.8</a> ).                                                                                                                      |
| <a href="#">theorem 18.1</a>                                       | $\Gamma$ nontrivial; $\neg\text{QR}$ ; ( $\text{SD}_{\text{ss}}$ ); normalization $\gamma_1 = 0$ ; $v(\mathcal{V}) > \kappa_\Upsilon$ <i>strictly</i> — <a href="#">proposition 18.6</a> fails at equality. Existence needs finitely many labelled support words per exponent; increasing valuations alone do <b>not</b> produce a Hahn sum ( <a href="#">warning 18.2</a> ). Uniqueness needs the mean-zero normalization and positive valuation of $h$ ; constant translations remove the normalization. |
| <a href="#">theorem 18.4</a><br>and <a href="#">corollary 18.5</a> | ( $\text{SD}_{\text{ss}}$ ) <i>fails</i> ; $\neg\text{QR}$ ; $\Gamma$ nontrivial. Refutes a <i>universal</i> theorem at arbitrarily high valuation; says nothing about any particular perturbation.                                                                                                                                                                                                                                                                                                        |
| <a href="#">theorem 18.8</a>                                       | Coefficientwise torus mean and finite Jacobian calculus, on the conjugacy of <a href="#">theorem 18.1</a> . No measure on arbitrary subsets and no real-time ergodic theorem follows ( <b>N75</b> ).                                                                                                                                                                                                                                                                                                       |

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